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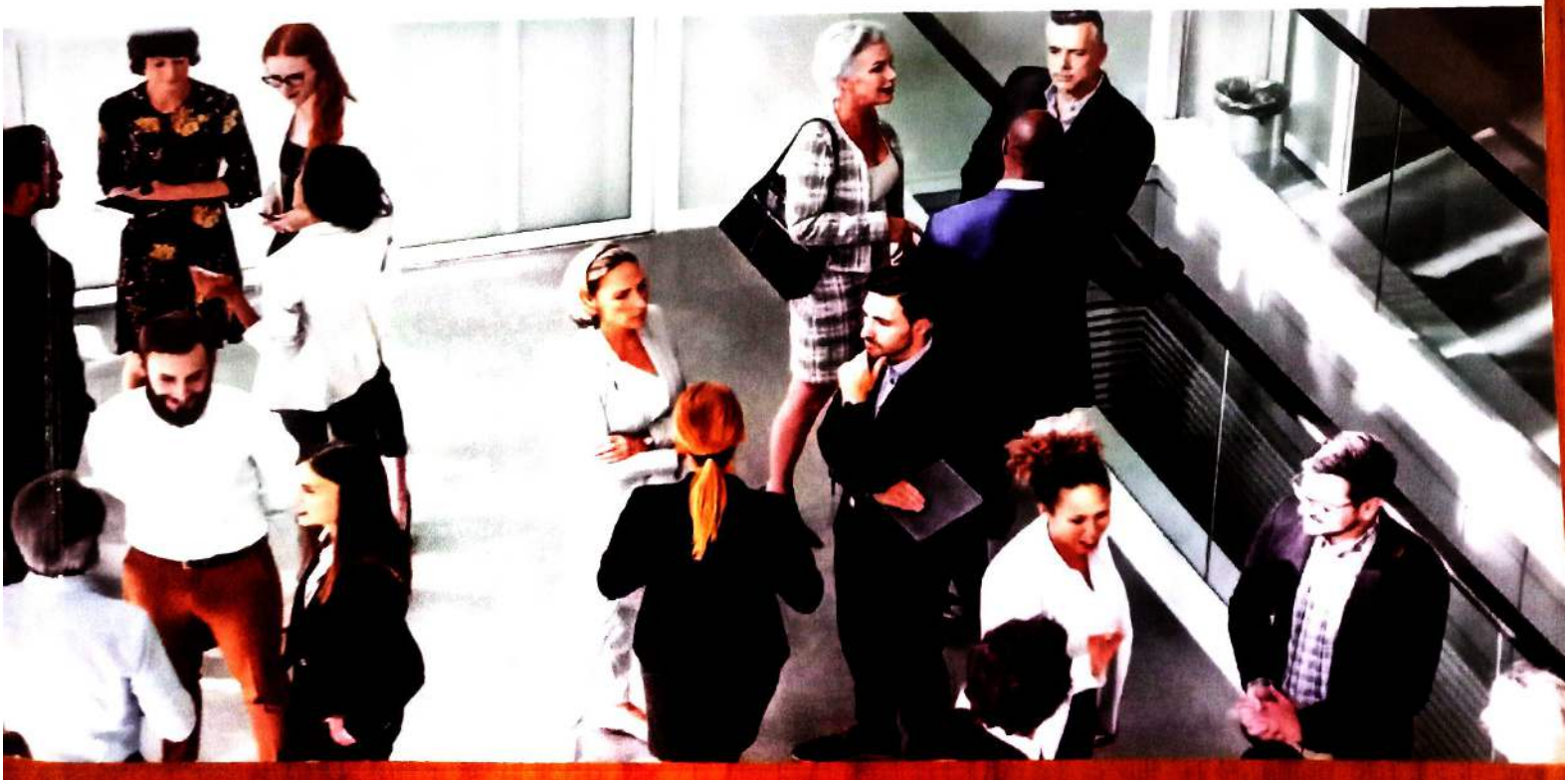

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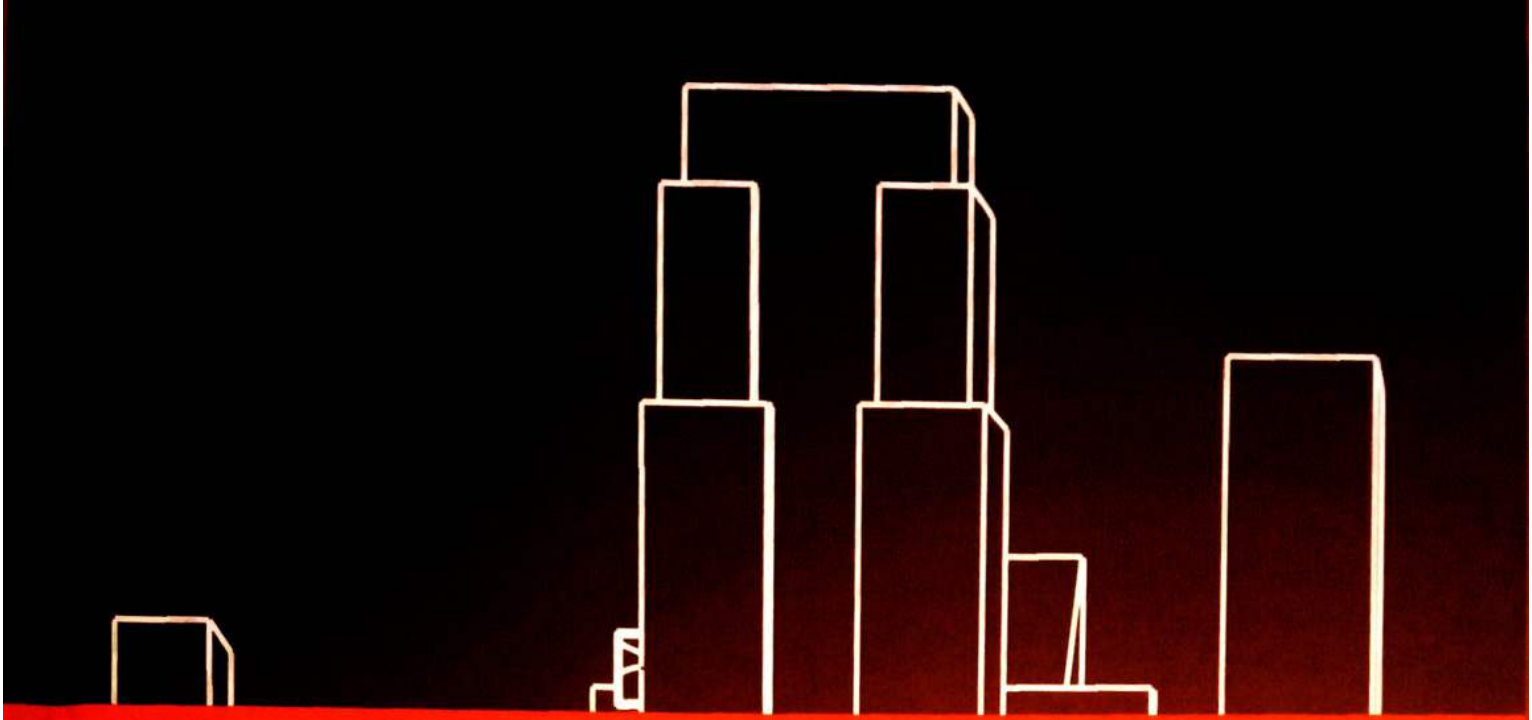

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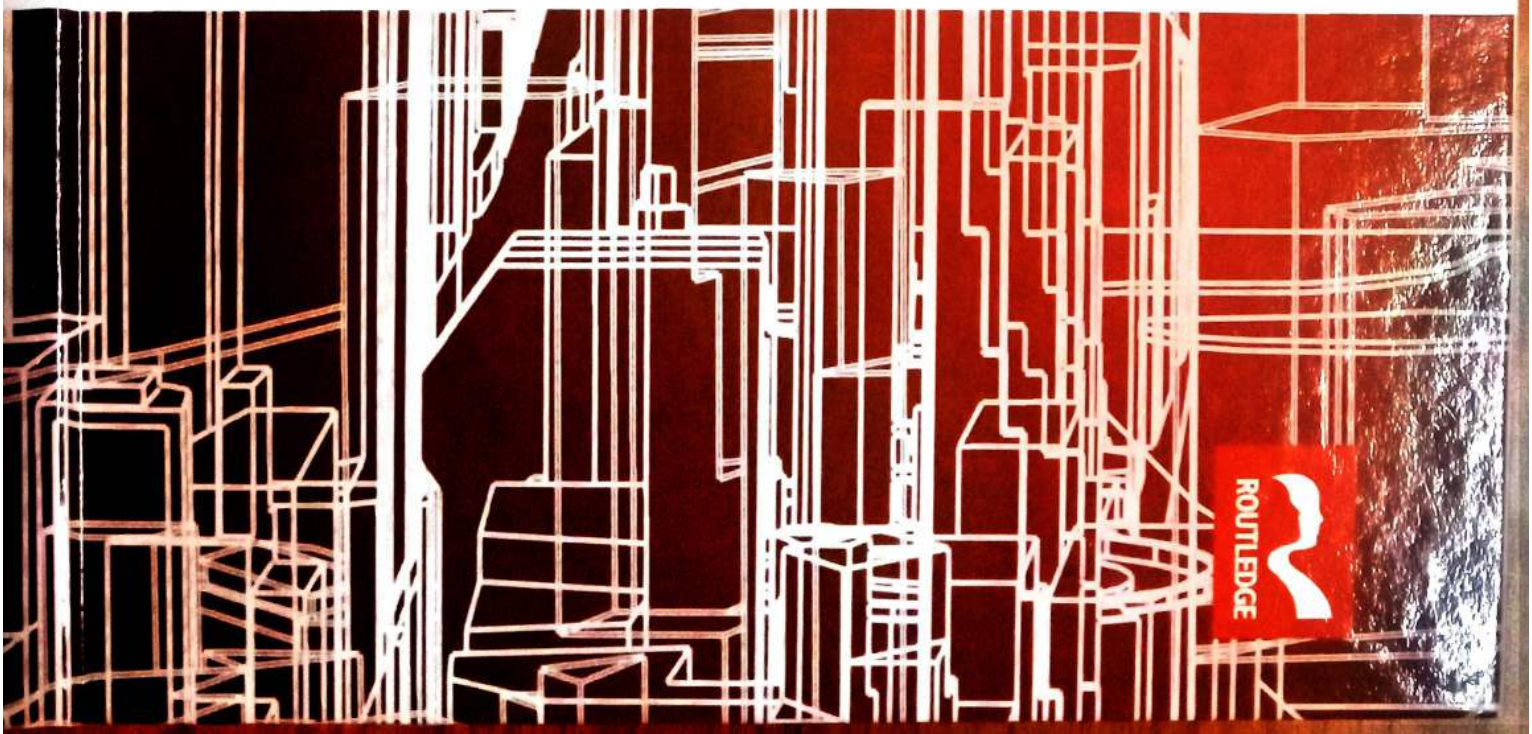

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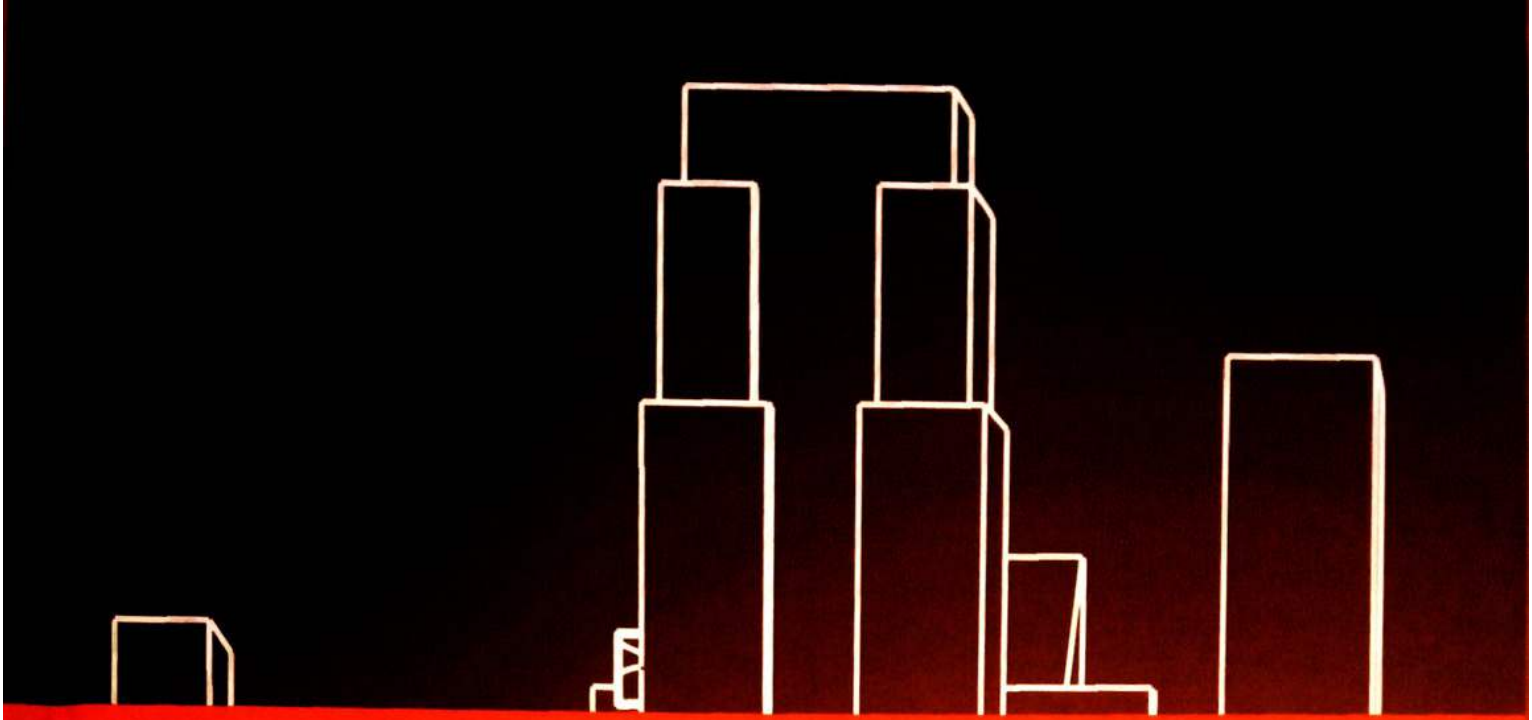

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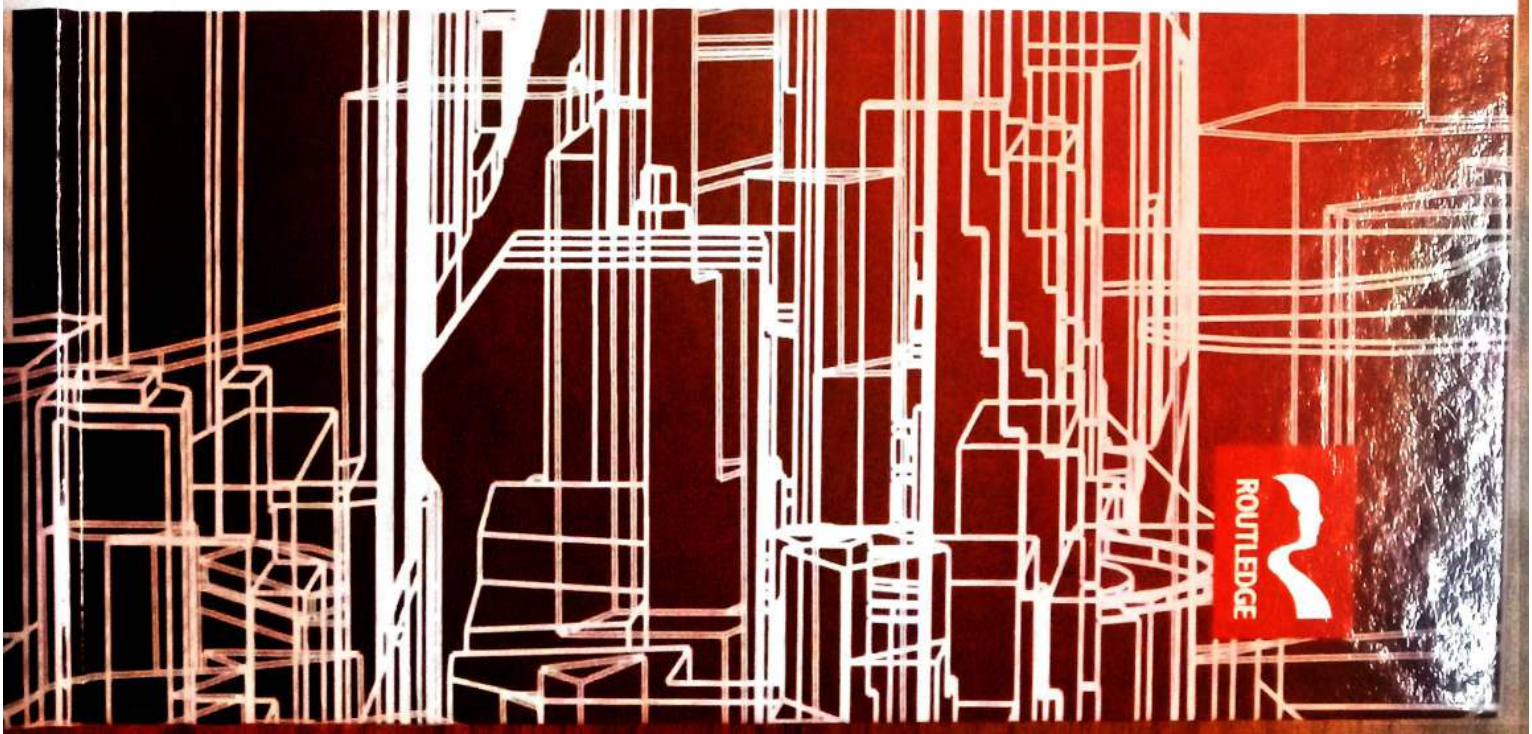

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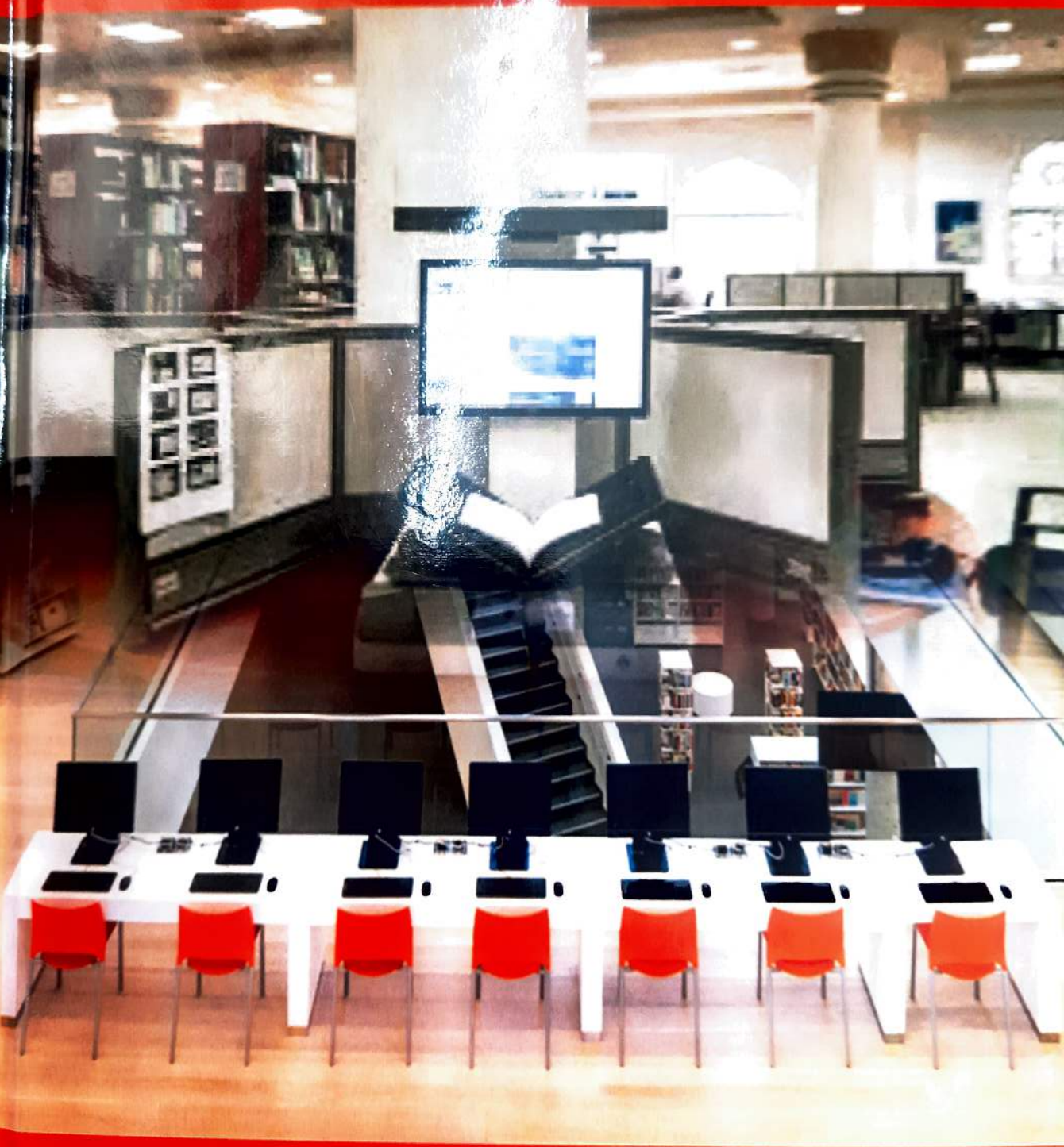

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Exploring Causes and Consequences of Plagiarism in Academic Writing

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Abstract

Plagiarism is a serious form of intellectual misconduct and a violation of academic integrity which calls into question the validity of the work that has been produced.Plagiarism is the practice of claiming the expressions, creations, or ideas of someone else as one's own. without due citation or credit.In India government has already taken different steps to curb plagiarism.The Indian Copyright Act protects original pieces of art, music, theatre, and literature and gives writers the only right to reuse, execute, translate, and modify their creations.The UGC initiated action to educate learners, scholars, and teaching staff about academic integrity and plagiarism. Shodhshuddi, Shodhganga, many plagiarism detection software, and different workshops and training programs for awareness about plagiarism help to curb plagiarism. Inadequate knowledge of citation style, academic writing, lack of research skills are creating knowledge gap between academic integrity. Due to the expectation of keeping a high-grade point average, students may think that they must use tricks to succeed academic. So academic pressure, career pressure, and peer pressure motivates people to plagiarism. Most of the times, because of poor time management, lack of motivation, and difficulties in starting to write any intellectual thing students, scholars, faculties tend to plagiarism. Cultural perspective could lead to confusion or misinterpretation about the importance of originality, which might lead to plagiarism.The demands of academic writing in English may be difficult to learners from non-English speaking

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Development of Mobile Accessibility Guidelines for Disable populace in disruption world

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Abstract

Mobile users with disabilities are having frustrating experience for using touchscreen mobile devices due to poor development of mobile applications user interfaces. Application developers are often hindered for the adaptation of any specific accessibility standards as the available documents of accessibility checklists of specific technologies and mobile web best practices found to be insufficient and difficult for implementation in application development and testing process as it does not provide information of specific compliance.Need is being realized to us to have Comprehensive and widely accepted mobile accessibility standards and guidelines to which application developers will comply their applications.

Keywords – Mobile accessibilities, guidelines,checkpoints, developers, tester etc.

Introduction

It has been found that certain documents of checklists and best practices such as list of accessibility testing checklist and application developers' checklist, Mobile

available which can be referred by the mobile application developing their applications.[1] But mobile application developers challenges to follow any specific guidelines as none of the seems to be complete. Therefore, they have to consider several guidelines as a reference for testing their application accessible and fully with disabilities.[2]Also, it becomes difficult to say that their is compliant to any specific standards or guideline.

Hence, need is being realized to us that there should be a complete and widely guidelines or standards for mobile application to which the application can make their applications compliant.[3]

Problem Statement

To develop a comprehensive general Mobile accessibility guidelines that can be used for conducting access audit of Any mobile applications to make it accessible and usable for persons with disabilities.

Objectives

1. To study various accessibility standards and guidelines to understand their merits and demerits.
2. To understand challenges of mobile accessibility testers, mobile apps developers and accessibility subject matter experts

Need of study

Unavailability of comprehensive mobile accessibility guidelines that can be applicable for all types of mobile applications in all the environment.The available guidelines and standards are technology specific and environment specific such as Android accessibility guidelines, IOS accessibility guidelines.The available guidelines do not have clear checkpoint for auditing the application.

Mobile application developers have to face difficulties in choosing and adaptation of accessibility compliance for their application as none of the compliance documents is complete and widely accepted.Mobile apps can be quickly made accessible up to certain extent for accessibility services like screen readers and magnifiers users if these points will be considered at the time of mobile application development or throughout the development life cycle.

Methodology

Paper is based on secondary data e-sources, refining the newly developed Mobile accessibility guidelines by adapting suggestions from accessibility testers and apps developers. Publishing the newly developed refined Mobile accessibility guidelines from different websites.

Deliverables

General Comprehensive mobile accessibility guidelines document that can be easily testable and applicable for all types of mobile application, Generation of complete and comparative information about accessibility standards and guidelines

General Challenges faced by the mobile application developers:

- Unable to choose specific compliance for making applications accessible.
- Difficulties in addressing multiple accessibility checklists and best practices such as android accessibility developers checklist, Android accessibility testing checklist and MWBP for achieving accessibility.
- Difficulties in understanding and preparation of expertise in multiple accessibility checklists and best practices.
- Unavailability of comprehensive and widely accepted mobile accessibility design guidelines or standards that can be adapted for compliance.
- Unavailability of good accessibility evaluation tools that can ensure high level of accessibility of user interfaces of applications.
- No assurance of complete accessibility of the applications despite of addressing multiple checklists and best practices.
- Difficulty in convincing customers about accessibility of their application as there is no proper base or reference to which customer will compare accessibility of the developed application.
- Difficulty to all the stakeholders to exactly understand what is accessible and what is not accessible.

Solutions:

- A Complete general accessibility design guideline should be made available to the application developers and testers.
- Specific technologies like Android, IOS and Windows should provide standards for their mobile applications which can be referred by application developers as well as customers.

General Accessibility issues and solutions:

We found out some of the important accessibility issues and possible solutions that can be used by the application developers for quickly testing their application and retrofitting accessibility through our experiences. [4]We referred several list of accessibility checklists and best practices such as Android accessibility developers checklist, Android accessibility testing checklist, Mobile Web Best Practises and WCAG 2.0.[5]

List of common Accessibility checkpoints and solutions:

lack of standard UIs: Many languages such as jQuery, PHP etc. are use to make different types of mobile application such as native, hybrid and web application etc. so it is difficult to focus on application accessibility.Labelling UI element: Labelling helps accessibility services to give description of UI control which helps the user to read information in form of images, button and dynamic stuff etc. It helps for blind and cover functional disability, situation disability. This can be achieved by defining ContentDescription attribute.Contrast colour: While making application different colour representation must be avoided. Instead using WCAG AA4.5:1(minimum or 3:1 for large-scale text) and WCAG AAA 7:1(maximum or 4.5:1 for large-scale text) could be best combination.

If colours are used to convey meaning user may perceive information incorrectly which helps partially or colour-blind people. This can be achieved by using block of colour rather than vague outline and shape. Information conveyed with colour must be also available without colour.Minimum text input in the interface: It helps people with motor disability and other situational disability for entering text. This can be achieved by providing alternative means to enter text and avoid free text entry where possible.

Use semantic mark-up: Assistive technology/browser cannot handle invalid mark-up. Hence semantic mark-up helps for assistive technology to navigate from one UI control to another easily. It allows assistive technologies to present more usable interaction model for visually impaired users.Concise page content and page size: It helps blind people, temporary disabled people.

This can be achieved by Using clear and concise languages as mobile users look for quick access to information. Avoiding images larger than screen size. Dividing page into usable but limited size portion. Creating large clickable area. Positioning important thing higher up and less important thing lower down in scrolling view.Keyboard control for touchscreen devices: Focus navigation allows users with disabilities to access UI element over GUI using a directional controller such as a trackball, directional pad (D-pad) or arrow keys, or virtual, such as the Eyes-Free Keyboard etc. by setting focusable and focus order properties to each UI element.Touch target size and spacing: ensuring that touch targets close to the minimum size are surrounded by a small amount of inactive space.

Touchscreen Gestures: Avoiding insert functions and only managing by gesture into mobile application. Always add button /link because most gestures are not intuitive and not recognized by many users.Magnification/Zoom: Zoom features help temporary, partially blind people to enlarge mobile UI and look at content properly on screen and understand.

Use ExploreByTouchHandler: ExploreByTouchHandler is used for supporting virtual view appearing in application, managing accessibility focus and providing event information to accessibility services.

Conclusion

All Data can be collected from various documents, Suggestions can be invited from application developers and accessibility experts and comprehensive widely accepted "Mobile application design guidelines" can be developed for application developers similar to web accessibility guidelines with various levels of compliance.

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Enhancing the Skill Sets and Fulfilling the Industry Requirements

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Abstract

This research study discusses industry-academia collaborations in India with a focus on National Education Policy (NPE) 2020, recent trends, and important challenges related to these collaborations. The economy is supported by two important pillars: academia and industry. The industry will be able to adapt and accommodate to changing market requirements very quickly and cost-effectively thanks to academia's ability to generate and nurture vast numbers of need-based quality manpower with solutions to every challenge. In order to adequately prepare students for the workforce in today's educational system, industry participation in curriculum design and delivery at institutions and universities is crucial. This will close the gap between businesses and educational institutions and prepare students for the workforce. In addition, less time, effort, and money will be expended on the students in the workplace before they begin working on actual projects. The NEP will give the government's Skill India, Start-up India, and Digital India missions even more momentum. The curricula must be compatible with the demands of the industry, thanks in part to regulating agencies like AICTE. Isn't getting a degree necessary only to work in the field? 'Collaboration between business and teachers fosters technological advancement and innovation. trade collaborations are essential for developing analysis and professional personnel. within the current data economy, a fortunate link between business and academe is crucial. The interaction between business and academe helps to fulfil each their necessities and expectations. extramural interactions. instructional establishments should improve their analysis capacities, recognise these business opportunities, and make a subsidiary setting for cooperation. this is able to considerably contribute to India turning into associate degree industrial hub for each tangible merchandise and services.

Keywords - Industry-academia collaborations in India, Changing market requirements, Industry Demands, Skill Development program

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Roadmap to Industry Academia Collaboration

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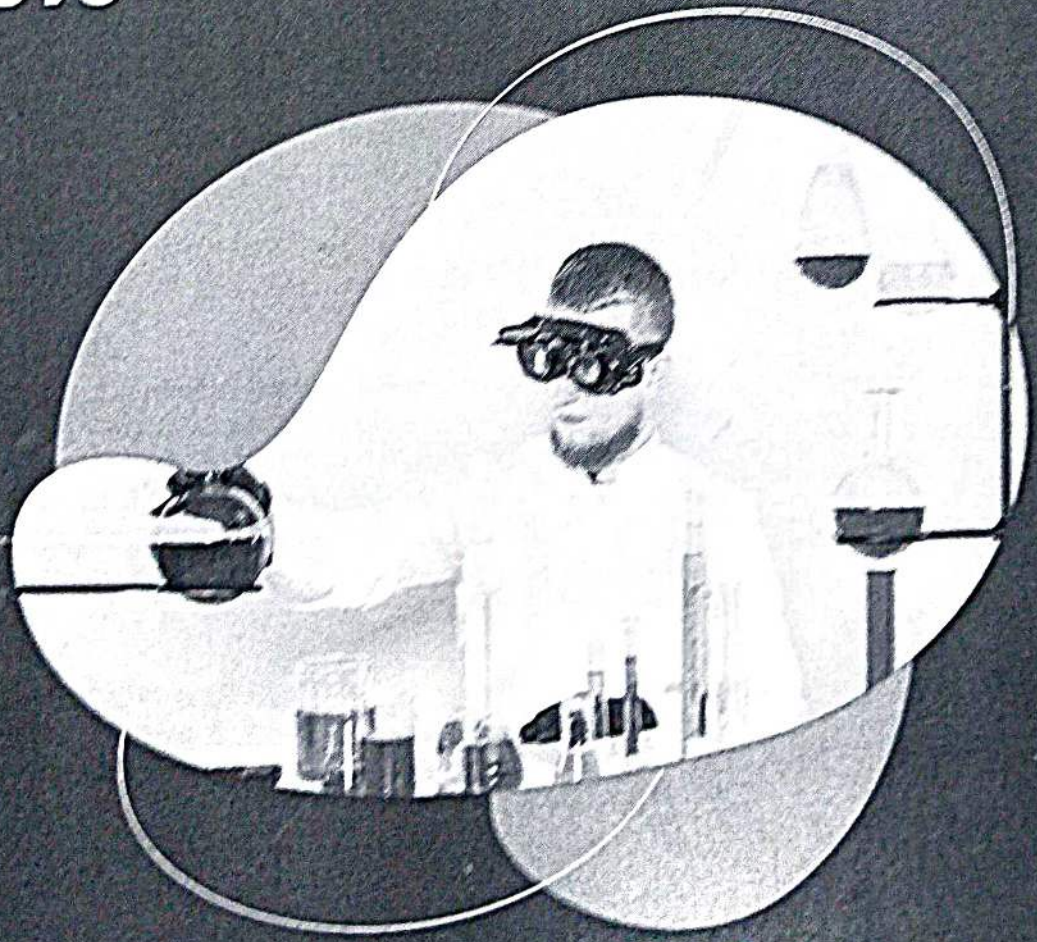
Abstract

The major goal of the study to understand the relationship between entrepreneur development through institute industry interaction. Education and Demonstration of the subject will provide different perspectives for the problem. To respond to the question "Are entrepreneurs born or made?" is a paradoxical, where majority of the people—roughly 50% of them—see themselves as potential business owners and feel that entrepreneurs are created, not born (Andel, 2014). With so many opportunities coming up as the technology is advancing and keeping sustainability in mind, entrepreneurship education if guided from an initial stage can create huge employments and economic group. The education industry mainly lacks the procedure and the guidelines the start-ups should undergo to get the business approved. Having a knowledge of process and documentations through a proper channel will save a huge amount of time and money. Numerous institute industry programmes have produced fruitful research outcomes, however making India the 2nd most entrepreneurial country in the world after China according to global entrepreneur monitor 2018. This is the fruit of the government, industry and academics to take united and decisive actions.

India has a youthful, educated workforce and a thriving economy, and its administration is committed to fostering an environment that will promote economic growth and development. Over the past few years, the "Make in India" campaign has developed a compelling narrative with a larger emphasis on research and innovation. The government supports start-ups with novel concepts, and companies are vying for additional patents in order to gain a competitive edge. The education sector, where institutions put a priority on commercially viable research and patents, is one of the important hubs for fostering the innovative attitude. Young adults at colleges and business schools are thinking large because they have the entire world at their fingertips.

Keywords - Entrepreneurship-Academia Collaboration, youth, Economy, Make in India.

RESEARCH METHODOLOGY IN MULTIDISCIPLINARY SUBJECTS



Dr. Vishnu Wahatule

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E-Learning: Concept, Nature, Advantages, Limitations

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Introduction:

Concept:

A learning system based on formalized teaching but working with electronic resources is known as e-learning.E-learning is a sort of digital learning that takes place online using electronic media.

Most electronic devices, such as a computer, laptop, tablet, or smartphone, can access it, making it a convenient and adaptable way for kids to learn wherever they are. E-learning materials are available in a range of formats, including apps, interactive online platforms, and digital courses.

By using information and communication technology (ICT), or "e-learning," people can learn at any time and from any location-learning is nothing more than the use of technology to link distant professors and students. Multimedia is used in e-learning to improve learning. Training, the provision of knowledge at the right moment, and professional advice can all be provided through e-learning.

E-learning is primarily based on using computers and the Internet, while teaching can also take place in or outside of formal classroom settings.

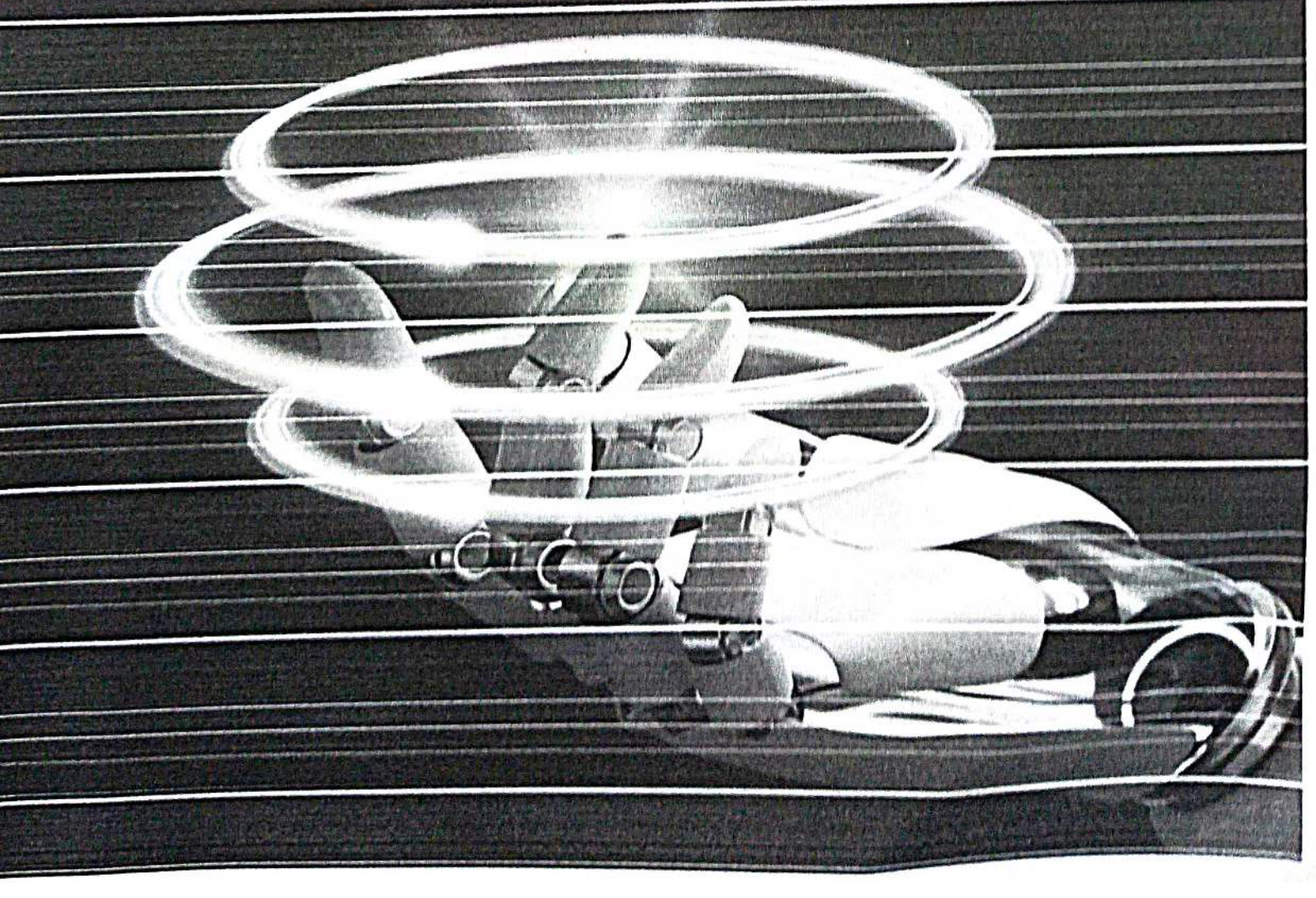
Another name for e-learning is the network-enabled transmission of skills and knowledge that allows for the simultaneous or asynchronous delivery of education to many receivers. Knowledge is available all around us through the e-learning techniques.With the aid of some electronics, such as a computer or mobile computing device like a laptop, tablet, or smart phone, some multimedia support equipment in the form of a microphone, speakers, video camera, LCD projectors, CDROMs/DVDs, touch screens, light pens, smart boards, video conferencing, etc., some communication and telecommunication support equipment's used in addition to Local Area or Wide Area Networks, In addition to these electronic tools, e-learning also requires a strong foundation from a few high-qualitycorrect software.

Depending upon the purpose for which e-learning is used, different names are given such as: Learning Support System (LSS), Learning Management System (LMS), Managed Learning Environment (MLE), Virtual Learning Environment (VLE), Learning Platform (LP) and Course Management System (CMS)Virtual learning environments

Application Of Machine Learning And Internet Of Things

Dr. Imran Baig Mirza

Professor Dr. Shaikh Aftab Anwar



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AN OVERVIEW OF MACHINE LEARNING CONCEPTS AND ITS APPLICATIONS

S. Petricia Leema Roseline
Assistant Professor, Department of Management, Christ College- Pune

ABSTRACT

Machine Learning is a field of artificial intelligence that allows systems to learn and upgrade from experience without being explicitly programmed. It has come an increasingly popular content in recent times due to the numerous practical operations it has in a variety of diligence. ML is one of the most instigative technologies that one would have ever come through. As it's apparent from the name, it gives the computer that makes it more analogous to humans: The capability to learn. Machine learning is laboriously being used moment, maybe in numerous further places than one would anticipate. The function of a machine learning system can be descriptive, meaning that the system uses the data to explain what happed; predictive, meaning the system uses the data to prognosticate what will be; or conventional, meaning the system will use the data to make suggestions about what action to take the experimenters wrote.

Keywords: Diligence, Descriptive, Predictive, Machine Learning, Conventional.

UNDERSTANDING THE CONCEPT OF MACHINE LEARNING

In Computer science field Machine learning is a small branch of Artificial Intelligence (AI). The area focuses on the use of algorithms and data to imitate like humans learn and it gradually increasing its accuracy in learning.

The following figure shows that Deep learning, Neural Networks and Machine learning are all sub-fields of Artificial Intelligence.

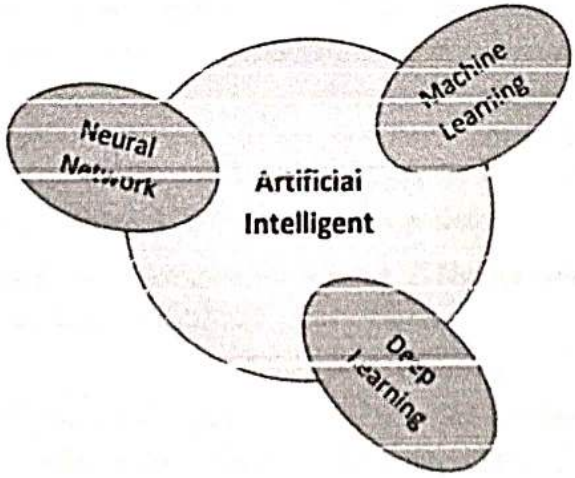

Fig 1: AI Sub-fields

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A Study on Fast Fashion Sustainability: A Qualitative Inquiry

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ABSTRACT

Fashion brands have evolved into digital influencers as a result of their usage of social media platforms to connect with customers and expand their market presence. Instagram, among the multiple social media platforms, has grown in popularity in recent years as a result of its high degree of customer involvement. This research will aid in understanding fashion brands' sustainability consciousness, and the knowledge gained will aid in reducing the environmental impact of fast fashion. The primary research aims to explore fashion brands' awareness of sustainable fashion and to learn how they are adopting and fostering sustainable fashion. The secondary goal is to research fashion brands' business capabilities as they are influenced by fast fashion, as well as how they approach the problem of fast fashion. The exploratory nature of the study necessitates the use of a qualitative technique. A total of fifteen fashion labels were contacted for an interview. The respondents were chosen by a convenient sampling method. To further understand fashion businesses' viewpoints on sustainable and quick fashion, a telephonic interview was preferred. The study's findings imply that consumers are aware of sustainable fashion and are willing to pay a greater price than usual. The study also discovered that brands are growing more environmentally conscious and incorporating sustainable practices into their daily operations.

Index terms: **Fashion, Fashion Brands, Business Capabilities, Sustainable fashion, Fast fashion.**

INTRODUCTION

Fashion has a massive influence on both people and the environment, and fast fashion accounts for a considerable and growing portion of the problem. Adding to the fast-fashion brands' contributions to the environmental problem,(Barnes & Lea-Greenwood, 2006) articulates that fast-fashion chains hurt the environment as most of the products produced under the fast fashion model are made using plastic fibers. Fast fashion businesses have built a name for themselves by letting consumers acquire low-cost, designer-looking clothing for next to nothing. On the other hand, their sales practices significantly impact consumer behavior worldwide. It tries to persuade us that repeating outfits is a bad idea while we know it is a must for sustainability (Darmo, 2020).Fast fashion is a type of retail that regularly releases new products throughout the year and is significantly less expensive than other fashion industry sectors. According to (Shimamura & De Fátima Sanches, 2012),"the phenomenon of fast-fashion, even though it is not a novelty in the sector, draws attention to the high profitability achieved and the continuous expansion" since when the model was adopted by major brands in the years 1990. Fast fashion is currently experiencing unsustainable growth, upsetting the entire economy and causing a significant shift in consumer behavior. The authors (Michaela & Orna, 2015) acknowledge that fast fashion brands use social media to influence the behavior of consumers and greatly influence their purchasing decisions. This has sparked a question about the fashion industry's sustainability. It will be compelling to notice how sustainability can give the fashion industry a fresh perspective. A new perspective is not only by incorporating the most cutting-edge strategies and technologies but also the most legitimate ideas to meet the needs of fashion-conscious customers who are not just keen to buy the changing fad (Binet et al., 2019). The research (Hur & Cassidy, 2019) says that a higher level of awareness among consumers about sustainability would result in more inquiries about eco-friendly products. The lower level of inquiries shows that consumers are not aware of sustainable fashion. Similarly,(Henninger et al., 2016)states an increased interest in sustainable fashion, but awareness about sustainable fashion is still low. Consequently, this paper strives to study the cognition of sustainability in the minds of fashion brands. If fashion brands are aware and keen to spread sustainability to the consumers, they are likely to adopt sustainable fashion and spread awareness about it among consumers. In recent years, the fashion and garment businesses have seen a significant increase in the number of players and a resultant increase in competition, manufacturing volume, and growing concerns about sustainability and environmental implications. Many fashion supply chain organizations have tried various techniques to become more "*green*"("Punishing or Subsidizing? Regulation Analysis

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Goods & Service Tax: A Revolution in Indirect Tax Regime

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Abstract

A single, all-encompassing tax on goods and services consumed in an economy is known as a "goods and services tax." "One Nation, One Tax, and One Market" is the primary goal of the GST. Overall, it is straightforward, effective, and will boost the nation's economic growth by getting rid of a number of state and federal taxes including the VAT, excise tax, and services tax, among others.

With its streamlined tax bands of 0%, 3%, 5%, 12%, 18%, and 28%, the Goods and Service Tax (GST) represents a significant change in the indirect tax system. The goods and services are categorised under a specific subject, and the items are given an HSN number while the services are given a SAC code. (Service Accounting Code, HSN, Harmonized System of Nomenclature)

In this study, we examine the effects of the GST on the economy while concentrating on the issue that business owners are expected to encounter (the fixation of MRP), which is the cornerstone of the Indian economy and accounts for more than 60–70% of GDP. GST was introduced in India, though, in 1999. A committee was formed to develop the GST concept. However, on July 1, 2017, the Indian government reinstituted GST. There was a lot of uproarin support of its execution.

All of the different taxes levied by the federal and state governments were replaced with the GST. The phrase "One Nation, One Tax" refers to the fact that all taxes must be paid in one location nationwide. The abolition of bundled indirect taxes like VAT, CST, Service Tax, CAD, SAD, and Excise is the main benefit of the GST. It tries to eliminate the cascading effects of taxes and promote credit flow.

Key words: GST, GSTN, GST council, tax slab, Indian economy, and Goods and ServicesTax Bill.

Introduction:

GST, or good and service tax, is an acronym. In India, it has taken the place of numerous indirect taxes including excise duty, VAT, service tax, etc. The goods and service tax Act was approved by the parliament on March 29, 2017, and it went into effect on July 1 of that same year. It is assessed on the supply of goods and services that exhibit the traits of a thorough, multi-stage, destination-based tax that is assessed on each value addition.

The following bills became law as a consequence on April 12, 2017:

1. The 2017 Central Goods and Services Tax Act

2. The 2017 Integrated Goods and Services Tax Bill

3. The 2017 Union Territory Goods and Services Tax Act

4. The 2017 Goods and Service Tax (Compensation to States) Bill

It is regarded as one of India's largest taxes reforms. India's economy, which is currently the fifth largest in the world and is expected to grow by 7% by 2023, is one of the fastest- growing in the globe.

Objectives:

• The primary goal of this research paper is to explain the fundamentals of GST and demonstrate how it affects the Indian economy.

• The GST represents a significant economic shift for India. Before the introduction of GST, a number of levies under the indirect tax system had to be paid.

• Double taxation resulted from the VAT and service tax's lack of unification. After the production step, the dealer was not given a CENVAT credit, resulting in double taxation.

• Under the GST framework, the burden of paying taxes and complying with regulations is lessened at every stage of business (in the economy), from the manufacturer to the client.

•Key details on how GST originated and its effects on the Indian economy will be covered in detail in this study paper.

Indian economy:

With a projected growth of 7%, India has a mixed economy (one that is labour intensive). By nominal GDP and PPP, it is the world's fifth-largest economy (purchasing power parity). The three areas of India's economy—primary (agricultural), secondary (industrial), and tertiary— are what will make it the largest expanding economy in the world (service).

The goal of cost reduction and profit growth can be accomplished by a number of techniques, one of which is the allowance of input tax credits at every stage of the supply chain, from manufacturer to retailer. This is what the earlier tax system, which the GST simplifies, lacks.

If a subject matter is eligible to tax, exempt from taxation, subject to a certain tax rate, or was previously subject to another tax system, for example, GST regime categorization is provided full enunciation within the legal framework.

GST tax slab: The rate at which various goods and services are taxed

0%: Milk, eggs, lassi, curd, and unprocessed whole grains unbranded raw honey, unpackaged paneer salt, fresh vegetables,

3%: Articles made of natural or cultured pearls, precious or semi-precious metals, imitation jewellery, and coins make up of the total.

5%: sugar, Coffee, tea, domestic LPG, PDS kerosene, cashew nuts, milk, baby food, fabric, spices, coal, and life-saving medications

12% :Butter, nuts, ghee, fruit juice, and canned coconut water 18% :Soups, pasta, corn flakes, toothpaste, hair oil, and soap

28%: Small autos (+1% or 3% cess), consumer durable like ac units or refrigerator.

Products that are in earlier indirect tax structure:

The federal government has the authority to impose taxes.

1. Crude petroleum
2. High-speed diesel
3. Motor - spirit
4. Natural gas
5. Fuel for aeroplane turbines
6. Tobacco use and related products

State governments have the authority to levy taxes:

1. Crude petroleum
2. High-performance diesel
3. Power-mindedness 4. Natural gas
5. Fuel for aeroplane turbines
6. Drinking alcohol for human consumption

GSTN, or Goods & Service Tax Network, is a section 8 business established under the Companies Act of 2013 and is regarded as the technology support for GST in India. GST requires a robust methodology and IT support because it is a destination-based consumption tax. It is a nonprofit, non-governmental organisation. It offers shared IT services & infrastructure to federal & state governments, together with taxpayers & stakeholders.

The GSTN offers services like registration, return filing, and payment options. In essence, it serves as a liaison between the relevant government and the taxpayers.

GSTN's organisational structure: The GST System Project is a distinctive and difficult IT project. It is distinctive because it aims to establish the first shared IT infrastructure between the Center and States as well as a common taxpayer interface.

The GSTN is 51% owned by business entities, and the remaining 49% is held by the government. 10 crores of rupees ($1.6 million) in authorised capital for the GSTN is held by commercial banks with the remaining 49% held equally by the Central and State governments.

Additionally, a one-time grant of Rs. 315 crores has been approved for the GSTN. Infosys created this extensive and complex technological foundation in September 2015.

Dr. Ajay Bhushan Pandey, an IAS from the 1984 batch who serves as the GSTN chairman, is joined by Shri. Prakash Kumar, the GSTN CEO.

Feature of GSTN:

The GST Network acts as a channel of online communication between the government and company taxpayers and is primarily the front end of the IT ecosystem for taxpayers.

The GSTN processes more than 2 billion invoices in total each month in addition to 65 lakh taxpayer returns and counting. The GST network is accountable for a number of significant key duties, some of which are listed below:

1. Registration

2. Invoice Comparison

3. File Your Return

4. Analysis of taxpayer profiles**website:** https://www.gst.gov.in

GST Council:

The Goods and Services Tax system was implemented after the Constitutional (122nd Amendment) Bill was ratified by both Houses of Parliament in 2016. After it had been ratified by the state assembly of more than 15 Indian states, the President then granted hisassent.

The GST Council, a joint forum of the federal government and the states, was established by the President in accordance with Article 279A (1) of the updated Constitution. Members of the Council include the Union Minister of State for Finance from the Center and the Union Finance Minister (chairperson). Each state has the right to suggest a minister in charge of finances, taxes, or any other minister for membership.

Why was the GST Council founded?

According to Article 279 of the Constitution, the Council is entrusted with "providing recommendations to the Union and the States on significant GST-related matters, such as the commodities and services that may be subject to or excluded from GST, model GST Laws."

Additionally, it establishes the various GST rate slabs.

For instance, a panel of ministers' interim report suggested taxing casinos, online gambling, and horse racing with a 28% GST. A choice will be taken in this regard at the Council meeting on Wednesday.

Making Decisions: The GST Council will provide guidance on the following topics: items and services that may be subject to GST or may be excluded from it; GST will encompass taxes, cesses, and surcharges; the highest threshold amount of turnover for GST purposes, as well as GST laws, levy principles, IGST allocation, and place of supply laws;

Special attention is paid to difficulties in Himachal Pradesh, Jammu and Kashmir, Uttarakhand, the eight northeastern states, and other connected areas.

other difficulties relating to the implementation and management of the GST in India.

Secretariat

The Secretariat oversees the GST Council. The Secretariat is staffed by representatives of the Central and State Governments. In order to establish the Council, the Cabinet and the Council initially met. During the initial meeting for the Council's formation, the cabinet allotted enough money to cover the ongoing and one-time expenses of the Secretariat. The Central Government is in charge of covering the costs associated with maintaining the secretariat.

In a Nutshell: The Effect of GST on the Indian Economy

One could argue that the GST will promote India's economic expansion. The introduction of the GST can help the Indian economy by creating a transparent sales process system.
Additionally, since the GST will do away with the requirement to pay various taxes in several states, conducting business will be made simpler. This enables understanding of the GST's advantageous effects. However, the GST's unfavourable consequences could result in some industries losing money as a result of rising input costs.

With the help of the aforementioned components, it is possible to understand a variety of concepts, including Goods and Services Tax (GST), as well as its effects on the Indian economy, both positive and negative.

A sort of tax that is applicable throughout all of India is known as GST. The Indian economy has been impacted in a number of different ways. First off, it might be argued that GST contributed to the expansion of the country's economy. It helped bring down the cost of goods and services. To understand the impacts of GST on the Indian economy, it can be argued that GST can be useful for enhancing the flow of FDI. GST may eventually help the government generate more money.

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GST may also aid in easing the simplicity of conducting business in India. The GST may also help increase employment opportunities. In spite of all of these problems, the implementation of the GST will increase system transparency to some extent, enabling consumers to comprehend the taxes they are paying and the justifications for them.

GST has the following implications on the National Economy:

• The service sector is expected to be more negatively impacted by the adoption of GST than the commercial sector.

• The GST has increased the cost of e-commerce purchases, online reservations, tobacco, and insurance premiums, while it has cut the cost of essential food staples like edible oil, wheat, rice, lentils, four-wheelers, and electric items.

• Due to the two-tiered one country, one regime, which only applies one tax nationwide, transportation is made easier and investor anxiety is reduced because the Indian market is regulated under one law. • Due to the increased tax collection and less opportunity for bribery due to the large number of people who fell into the tax loop, the GST reduced the incidence of corruption.

•Following the implementation of the GST, the price of the products that the general public purchases will drop, creating demand. Demand increases as output also increases, increasing job opportunities and lowering prices.

The GDP expanded by 7.4% in 2017–2018, which is somewhat higher than the 7.2% growth observed in 2016–2017 but lower than the 7.8% growth seen in 2015–2016, the second year after the GST was implemented.

Methodology: Secondary data have been gathered for the study work. The collection of secondary data and academic journals was sufficient. These were taken and critically examined. The formulation of the study's qualitative and quantitative theory was aided bythis.

Conclusion:

As observed from the aforementioned sections, several topics, such as the Goods and Services Tax (GST), as well as its favourable and unfavourable consequences on the Indian economy, can be understood better. It should be noted that the Indian economy has been significantly impacted by the GST, or goods and services tax. With the help of the aforementioned discussion, it can therefore be said that the idea of the influence of GST onthe Indian economy project can be comprehended.

The secondary data enabled the research study's extensive literature. The first step was to refer to the Institute of Cost Accountants of India website (icmai.in) and extract study materials on the topic of GST. In order to gain a year-by-year summary of the research activities carried out in the area, the dataset was then entered into the programme to build an overlay network visualisation. According to the available space, a detailed discussion of the GST, Goods and Services Tax Bill, Tax Slab, Indian Economy, Cascade Effect of Tax, GSTN, and GST Council is held.

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A DEEP DIVE INTO MACHINE LEARNING

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ABSTRACT
Data mining and machine learning approaches look at data from beginning to end to find hidden patterns in the dataset. A variety of study areas support the establishment of the multidisciplinary discipline of machine learning. The digital era has access to a multitude of data which includes business information, data from social media sites, Internet of Things (IoT) data, cyber security data, cellular data, health data, etc. In deep learning, multiple layers of interconnected neurons are used to process and learn complex patterns in data. These layers allow the network to automatically extract high-level features from raw input data, such as images, speech, or text. Machine learning permits the user to use a computer algorithm for a large amount of data and analyse the same with the help of a computer and make recommendations based on available data and decisions on the base of input data. Machine learning is of four kinds viz. Supervised learning, Unsupervised learning, Semi-supervised learning and Reinforced learning. We have done a deep analysis of techniques of machine learning through this research.
Keywords: Machine Learning, Algorithms, Decision tree, Neural networks, Metadata.

Introduction

Machine learning was created by Arthur Samuel, an American scientist in 1959, when he was working at IBM. According to him, machine learning is a study which gives computers an ability to learn without any systematic programming. Machine learning has a connection with artificial intelligence, which is involved in developing algorithms and statistical models, which enable computers to improve the performance in activities through experience. Machine learning is of four kinds viz. Supervised learning, Unsupervised learning, Semi-supervised learning and Reinforced learning. A recent study showed that machine learning (ML) engineers outpaced all others in salary package, demand as well as growth. In short, ML is a smart career choice for job aspirants. ML is also an important component in areas such as Big Data, Predictive Analytics, Data Mining and Computational Statistics.

The main responsibilities of an ML engineer include the following:

- Study of computer architectures, data structures and algorithms
- Designing machine learning systems
- Creating infrastructure, data and model
- Analyse large, complicated datasets
- Develop algorithms
- Build and maintain machine learning solutions in production etc.

A few examples of machine learning are as under:

- Recognition of speech and image, which help to convert speech into text.
- Google translation
- Prediction
- Extraction
- Extraction

Real-World Examples of Machine Learning (ML) are as under:

- Facial recognition
- Production recommendations
- Financial accuracy social media optimization
- Healthcare advancement
- Predictive analytics

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Review of Literature

Murugan, Sathiyamoorthi (2022) aims to provide an in-depth overview to several commonly applied machine learning techniques and their applications, which are divided into supervised, unsupervised, and reinforcement learning which also serves as a concise manual for prospective researchers of data and machine learning.

Choudhary, Gianey (2017) in their study have provided a general comparison on machine learning algorithms. Unlike static programming methods, which require explicit human instruction, machine learning algorithms are designed to learn from the data and generate predictions.

Dheepak, Vaishali (2021) have provided a brief summary and viewpoint on many machine learning applications by this survey. Duarte, Ståhl (2018) in their study highlights the subdivision of the machine learning field the purpose of which is to make it possible for computers to learn.

Paluszek, Thomas (2016), in the subject of machine learning, past data are utilised to anticipate or respond to future data. It has a close connection to artificial intelligence, computational statistics, and pattern recognition. In fields like facial recognition, spam filtering, and others where it is not practical or even practicable to develop algorithms to execute a task, machine learning is vital.

Rath (2022) in his study has stated that business learning made various techniques. Every financial platform has become safer and more user-friendly for financial transactions as a result of advancements in the internet and digital marketing.

Thakur, Tembhurney & Rane (2021) aims to give readers a mature perspective on the fundamentals of ML, one of the experts' most popular recent study subjects. It outlines the problems and difficulties with machine learning (ML). The selection of features is relevant to feature engineering because feature engineering creates novel features to improve learning objectives while feature selection aims to select the features that are most appropriate.

Objectives

1. To define the study's scope by considering the traits patterns of data, which help in making predictions.
2. To detect and analyse trends which helps to solve problems.
3. To explore the ways in which the solutions based on machine learning might be used in various real-world scenarios.

Secondary Data Analysis

Real-World Data Types

Data is required for the development of machine learning models. There are different types of data, such as structured, unstructured and semi-structured data. The difference is as under:

Structured data: It is organised data, which is easy to search in any database which is rational and mostly categorised as quantitative data, which most researchers and scientists used to work with. It has fixed fields and columns in relational databases and spreadsheets. Examples are names, dates, addresses, credit card numbers and more.

Unstructured data: It has no proper format or pre-defined format and hence it is difficult for collection, formalisation and analysis. It is often qualitative data which cannot be processed and analysed using normal methods and tools. For unstructured data, in order to understand customer buying habits and timings, patterns in purchases and sentiments towards a particular product etc. can be obtained by using data mining techniques.

Semi-structured data: It is a type of structured data which stands at the mid-way between structured and unstructured data. It does not have a rational or tabular data model. Common examples of this are JSON and XML.

Metadata is "data about data," not the usual form of data. Metadata refers to data that describes other data. It provides context and information about the data, such as its format, structure, content, and relationships to other data. In other words, metadata is "data about data."

Metadata: It is used to help organize and manage data, making it easier to search, retrieve, and use. It can be stored in various forms, such as tags, labels, or fields, and can be associated with different types of data, including documents, images, videos, and databases.

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Machine Learning Algorithms

There are different types of machine learning algorithms as under:

(1) **Supervised:** It is a learning function where a pair is taken, and this pair consists of input object and output value. It analyses the training data, which can be utilised for mapping new examples. Both classification and regression problems are supervised learning problems.

(2) **Semi-supervised learning:**, the algorithm is trained to use both small amounts of labelled data and large amounts of unlabelled data. The idea is that the labelled data provides information about the target labels, while the unlabeled data helps the algorithm to learn the underlying structure of the data and generalize better to unseen examples. One common technique for semi-supervised learning is to use the labelled data to train a supervised model, and then use that model to make predictions on the unlabeled data. The predictions can then be used to label some of the unlabeled data, which can be added to the labelled data for further training. This algorithm is mainly used for recognising image and speech as well detection of fraud. Generally it is used where it is very expensive to use labelled data.

(3) **Unsupervised Learning:** It is a type of machine learning algorithm, which is used to draw inferences from sets of data consisting of input data without labelled responses. Unsupervised learning algorithms are not included in observations. Since this is unsupervised, the accuracy cannot be evaluated. This method is used for cluster analysis, which is used for exploratory data analysis.

(4) **Reinforcement input data without labelled responses:** It is the problem of getting an agent to act in the world in order to maximise the rewards.
In machine learning, a learner is not informed as to what action he has to take. He has to discover which technique to use in order to derive the desired results. For example, we can teach a new trick to a dog; and we can use the technique of reward or punishment if it does right or wrong. Same method can be used in the case of computers to do many tasks such as playing chess, job scheduling and controlling robot limbs.

There are several different machine learning techniques, each with its strengths and weaknesses. Here are some comparisons of different techniques:

1. Supervised learning vs. Unsupervised learning:

Supervised learning is a learning function where a pair is taken, and this pair consists of input object and output value. It analyses the training data, which can be utilised for mapping new examples. Both classification and regression problems are supervised learning problems. Whereas Unsupervised Learning is a type of machine learning algorithm, which is used to draw inferences from sets of data consist of input data without labelled responses. Unsupervised learning algorithms is not included in observations. Since this is unsupervised, the accuracy cannot be evaluated. This method is used for cluster analysis, which is used for exploratory data analysis.

2. Decision trees vs. Neural networks:

Decision trees are a simple and interpretable machine learning technique that uses a tree-like structure to make decisions. Neural networks or artificial neural networks are more complex and can model non-linear relationships between variables. Decision trees are useful when the problem is simple and the dataset is small, while neural networks are useful for more complex problems and larger datasets.

3. Logistic regression vs. Random forests:

Logistic model is a statistical model, which takes into consideration the probability of an event taking place by having the log-odds for the event be a linear combination of one or more independent variables. Logistic regression is useful when the problem involves a binary outcome and the relationship between the predictor variables and the outcome is linear, while random forests are useful for more complex problems and datasets with many features.

4. Support vector machines vs. K-nearest neighbours:

Support Vector Machine or SVM is one of the most famous Supervised Learning algorithms, that is used for classification and regression problems. SVMs are useful when the problem involves a large number of features and a clear separation between classes, while KNN is useful when the dataset is small and the relationship between the features and the outcome is non-linear.

5. Clustering vs. Dimensionality reduction:

Clustering is a technique of unsupervised learning that gathers data pointed to on the basis of its similarity. Dimensionality reduction, on the other hand, is a technique that reduces various features in a set of data, when you are retaining so much of the original information as possible. Clustering is useful when exploring the structure of a dataset or when identifying outliers, while dimensionality reduction is useful for reducing noise and simplifying complex datasets.

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Table:1 gives an understanding of the different techniques which is suitable for different scenarios as per the data sets available.

Algorithm	Type	Pros	Cons
Linear Regression	Supervised Regression	Simple to implement and interpret, performs well on linearly separable data.	May underfit or overfit data if not properly tuned.
Logistic Regression	Supervised Classification	Simple to implement and interpret, performs well on linearly separable data.	May underfit or overfit data if not properly tuned.
Decision Trees	Supervised Classification/Regression	Easy to understand and visualize, can handle both categorical and numerical data.	May overfit data if not properly tuned, sensitive to small changes in the data.
Random Forest	Supervised Classification/Regression	Reduces overfitting by combining multiple decision trees, handles both categorical and numerical data.	Can be slow to train on large datasets.
Support Vector Machines (SVMs)	Supervised Classification/Regression	Performs well on high-dimensional datasets, effective in handling non-linearly separable data.	Can be slow to train on large datasets, requires careful selection of kernel function.
Naive Bayes	Supervised Classification	Simple and fast to train and classify, works well with high-dimensional data.	Assumes independence of features, may not work well with highly correlated data.
k-Nearest Neighbors (k-NN)	Supervised Classification/Regression	Simple to implement and interpret, works well with small datasets.	Can be slow to classify new data, sensitive to the choice of k.
Neural Networks	Supervised/Unsupervised Classification/Regression	Can handle complex non-linear relationships, can learn from unstructured data.	Can be difficult to interpret and may require a lot of training data.
Clustering Algorithms (e.g., k-Means, Hierarchical)	Unsupervised	Can identify underlying structure in data, can be used for anomaly detection.	Requires careful selection of the number of clusters, may not work well with high-dimensional data.

Table 1: Pros and Cons of Machine Learning Algorithms subject to the kind of data used.

Machine Learning Algorithms

Classification Analysis

• Binary classification:

• Multiclass classification:

• Multi-label classification:

Regression Analysis

Cluster Analysis

Dimensionality Reduction and Feature Learning

Association Rule Learning

Reinforcement Learning

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In addition to classification analysis, there are other similar algorithms that fall under the broader umbrella of machine learning, including:

1. **Regression Analysis** - This involves predicting a continuous numerical output rather than a categorical one.
2. **Clustering Analysis** - This involves grouping similar data points together based on their features, without predefined labels or categories.
3. **Dimensionality Reduction** - This involves reducing the number of input features while retaining as much information as possible, to make it easier to analyze and visualize the data.
4. **Neural Networks** - These are a class of algorithms that can be used for both classification and regression tasks, and are modelled after the structure of the human brain.

Conclusion

The effectiveness and quality of the dataset of the algorithms of learning are needed for machine learning models in order to be successful. In this study, we have looked closely at machine learning techniques for uses that call for sophisticated data processing. Both supervised and unsupervised machine learning are possible. Unsupervised Learning typically provides superior performance and results for huge data sets, while Supervised Learning is the preferable option if you have a smaller amount of data and clearly labelled data for training. On the other hand, use deep learning approaches when you are in possession of a sizable data collection that is readily available. All of these algorithms have different merits and demerits and the algorithm merits are in line with the specific available problem as well as on the basis of data to be analysed.

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PROPENSITY MODELLING- AN EFFICIENT PREDICTIVE TOOL USING MACHINE LEARNING

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ABSTRACT
Data gains its real importance when used to predict the future rather than just analyzing the past. Propensity modelling looks at the past behavior and uses that information to make predictions of tomorrow and provides a foresight.
Uncertainty has always been an elemental aspect of any business. In almost most of the businesses success or failure depends on human behavior. Through such behavior we can derive patterns which help us to come to various conclusions. Propensity modelling is a technique used to derive patterns and predict such behavior. Propensity modelling predicts what a customer is likely to do with the help of machine learning algorithms which explore patterns in human behavior.
This paper aims to explore the various areas where propensity modelling has been used. It also aims to suggest ways of evaluating the performance of the propensity model.
Keywords: Propensity Modelling, Machine Learning.

Introduction

Propensity modelling is something that was known since 1983 but it's only in recent times that machine learning has gained its true value.

Propensity modelling predicts what a customer is likely to do with the help of machine learning algorithms which explore patterns in human behavior. These models are used for a wide variety of tasks related to prediction in various areas. After building a model it is very important that we evaluate the performance of the model and understand which model is best suited for your organization or business. Various studies provide solutions for businesses to target the right customer and maximize their profit. But very few have been able to predict customers with accuracy.

Organizations are striving hard to reach out to the right customers at the right time. They are in search of models which measure up to one-size-fits-all and bring pieces together to create an effective model that can accurately give targeted solutions.

A model which can evolve with trends, adapts to a large data set, processes real time data and delivers predictions accurately.

Propensity Modelling is one such solution in an age of hyper personalization.

Objectives

1) To understand the importance of Propensity Modelling in Businesses.
2) To explore the different areas where propensity modelling is used.
3) To explore the different challenges involved in implementing propensity models.

Review of Literature

Clarke (2017) a propensity model, monitoring user actions in order to create content as per the propensity of the user. (Tasdighi, Arabi, Harmel, & Line, 2018) in the paper explores the effect of conditioning previous wave data on predictive and discriminative ability of Response propensity Models.

Campisi, Torrisi, Ignaccolo, Inturri, & Tesoriere (2020) the implementation of car sharing service among university students. A propensity model was applied to check the difference between expected and observed frequencies for several combinations of the analysed attributes.

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Kožić, Mikulić, & Krešić (2016) a micro level empirical analysis of the determinants of propensity to travel. (Li, W., Sun, K., Schaub, F., & Brooks, C. ,2021) have investigated students' propensity to consent to opt out of having their data collected and used for learning analytics. Study also suggests that there are various factors such as demographics, personality etc which motivate students to express their consent (Rentsch, A., Schaffner, B. F., & Gross, J. H. ,2019) in their study reveals that there are many demographic variables that are taken into account to predict the turnouts and overreporting in voters.

Zhang (2021), provides solutions for the company's more precise marketing and the discovery of potential customers. Machine learning techniques like Logistic regression and random forest have been used to solve the problem, and compare the prediction effects of the two models. Few studies (Theofilatos, Yannis, Antoniou, Chaziris, & Sermpis, 2017) have contributed to the existing knowledge by utilizing machine learning techniques for predicting accident risk. Findings suggested that accident risk and type depend more on less on the prevailing traffic conditions (i.e) real time traffic. (Barga, Fontama, & Tok, 2015) focusses on buyer propensity models to business processes. Their study applies the key concepts and explores the capabilities of utilizing machine learning for predictive analytics.

Verenich, Kikas, Dumas, & Melnikov (2015) in their research adopts a combination of propensity and influence to provide a solution to targeting users on social media. The study highlights the problem of selecting users on online social networks.

Characteristics of Propensity Models

In order to prevent huge revenue losses and build more safer and targeted marketing campaigns, propensity models need to possess the following characteristics

1.1 Dynamic: Models should be built in such a way that they should adapt to changing trends and should be able to learn and derive from new data as and when it is available.

1.2 Scalable: Models should be created such that they can be used in scenarios which produce a wider area of predictions rather than a single scenario.

1.3 Adaptive: Models should be able to adapt to changes in terms of data validation and deployment.

Types of Propensity Models

1. Propensity to purchase or convert
2. Predicted customer lifetime value
3. Propensity to churn
4. Propensity to engage

Different types of data sources are taken into consideration

1. Information that tells us "Who" the person is.
2. Sources that give us information about "What" a person has purchased in the past.
3. Information about "Why" a person has purchased things.

Steps to implement a propensity model

- Develop a Strategy
- Collect Relevant Data
- Prepare data for modelling
- Create and Test Model
- Deploy a Model

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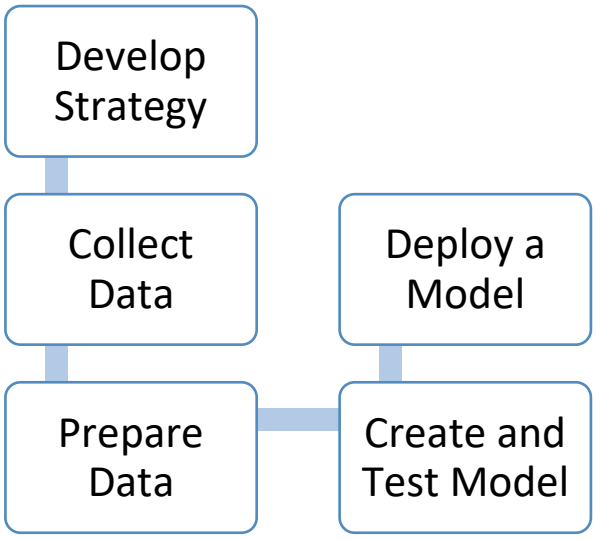

Fig 1: Steps to Implement Propensity Model

Develop a Strategy
It is important to plan out a strategy before you move ahead to implement a model else you would end up investing on resources. The first step is to define the objectives and take a decision as to what conclusions you need to derive out of it.

Collect Relevant Data
Choosing your data depends on the predictions you would make. Data can be derived from first party sources like websites, mobile applications etc or could be from third parties.

Prepare Data
In order to ensure that the data you select is consistent, accurate and complete, the right choice of variables have to be made.

Create and Test Model
After the data is prepared, it has to be trained using different approaches like logistic regression, neural network and other machine learning techniques.

Deploy a Model
The last step is to deploy the model and maintain the model so that results can be improve over time.

Areas where propensity models can be used.
Areas including Insurance, Mortgage, Re-Financing, Credit Cards, Loans, Investments, Satellite TV, Cable, Netflix, Online Banking, Online money management, Voting and Disease are some of the cases where propensity models can be implemented.

Propensity modelling in Travel industry-
Even in the Travel Industry, propensity models can be used to generate meaningful insight from the passenger's behaviour.

A traveller is more likely to go for travel insurance if they are more prone to risk or have purchased insurance in the past.

Likelihood to travel or also termed as propensity to travel depends on the willingness of the person to be a tourist. There are three factors which greatly impact someone's propensity to travel: namely age, education and income (Kožić, Mikulić, & Krešić, 2016).

Propensity models in election campaigns
Propensity models have been used in election campaigns to predict which voters could be encouraged to vote and what campaigning techniques such as flyer, call, door to door campaigning would be best suitable for each voter.

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Through various studies (Rentsch, Schaffner, & Gross, 2019), voter models are built that take into account different variables that help to predict the turnout or overreporting during elections and would make it easy to implement in future elections.

Banking
Key Customers were identified with the help of propensity modelling and thereby helped in designing promotions for the right customers which led to increased productivity.

Banking industry ("How Banks use Product Propensity to Increase Customer Delight and Profitability?," 2018) have benefitted from this modelling with respect to debt collection efforts: propensity to pay. When it comes to collecting debt they do not possess the required expertise which help them to decide which processes should be applied to customers.

In order to reap the maximum benefits of retaining maximum customers at low costs banks use propensity models complemented by customer segmentation and predictive modelling.

Customer Lifetime Value (CLV) is an important dimension that directs banks to generate maximum revenue. More features like customer preferences and usefulness of the product are instilled in order to make sure that the right customers are benefited with the right products.

Healthcare
Propensity in healthcare ("Five best practices in healthcare propensity modeling," 2021) indicates the probability of a patient who is likely to take a particular treatment based on certain characteristics or behaviour.

You are in a better position to manage disease and costs when you are able to identify patients who have a higher likelihood of a particular disease. Preemptive action can be taken to treat a patient and seek better care.

With the help of models which use demographic and other data sources to identify and target those populations which are at higher risks.

Healthcare marketers use a set of features like diagnostics, visit history, socio demographics, socioeconomic, lifestyle for predicting the likelihood of a patient [8].

TV Data company
In order to utilize its maximum potential, marketers should use propensity modelling and use machine learning technologies.

A US based company could manage its spending on advertisements appropriately with the use of AI in propensity modelling and were able to target the right customers. The result was remarkable and the predictions made were accurate to 80% percent.

Airlines
Scandinavian Airlines have made use of machine learning and predictive analytics in propensity modelling. With this they were able to predict the changes of a customer to book a ticket. They were able to target the customers and provide timely offers to frequently bought tickets which led to flights taking off without empty seats.

Mobile Operators
With accurate propensity models' marketers were able to achieve their targets and reduce the customer churn. They could form segments of customers and were able to determine which products were perfectly matched. This resulted in a 30% reduction in customer churn and 2% increase in revenue.

Media and Entertainment Companies:
Advancements and innovations in technology have had a great impact on media and entertainment.
For almost 2 decades, changes in these companies have been tremendous, resulting in higher customer churn and dissatisfied customers.

Propensity modelling ("Can propensity modeling & customer segmentation help media and entertainment companies predict their customer's next move? | Head to Quantzig's article to find out," 2020) has definitely

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proven to tackle the challenges through targeted campaigns which resulted in retaining customers year after year.

Real time cases where Propensity Modelling have been effective

1. In 2012, during Barack Obama's re-election campaign, propensity models were built by a team of data scientists. The use of this analytics helped in winning Obama's re-election.
2. The Bouqs. Co, a US based floral delivery company, invested in developing propensity models to identify most valued customers and provided them with offers.
3. Vodafone, one of the leading mobile operators in Ukraine built a propensity model in order to make better decisions and reduce customer churn which resulted in 30% reduction in customer churn and 2% increase in revenue.

Ways to evaluate a Propensity Model

Propensity models which are effective will provide accurate predictions. In order to evaluate the different types of propensity models we need to divide the data into

- Training Data- which will assist in finding the best model
- Testing Data- which will assist in evaluating the performance of the model which is best suited.

There are three ways for evaluating the quality of the propensity model

Model Training Performance-In this evaluation method, it is assumed that the prediction is most successful when we choose the training data from the same time period as that of the testing data, i.e., training set and testing set are from the same time frame.

Future Model performance- In this method, data upto a certain specific day is considered and then the model is tested using data that appears after the date. Actual activities are tracked in order to see how accurate the predictions are.

Historical Future Model performance- In this evaluation, the model is trained using data up to a specific date and data is tested after that date. Future performance is analyzed and predictions made using past data. In this way, you no longer need to wait for future data.

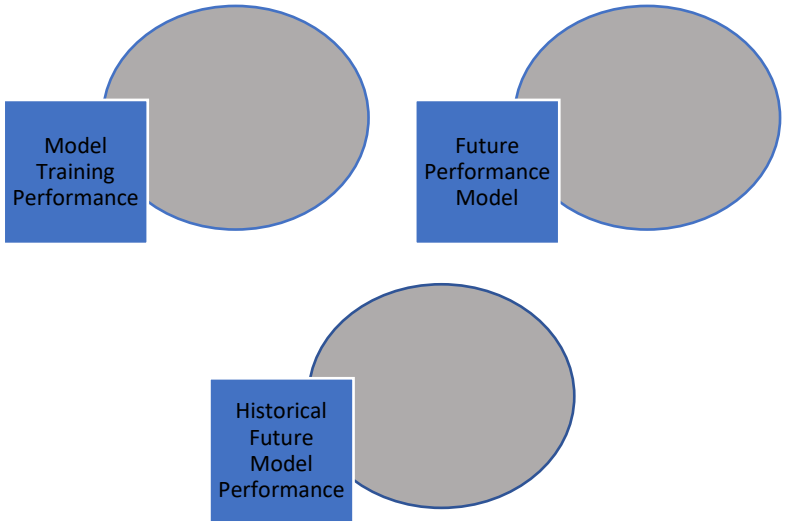

Fig 2: Ways for evaluating the quality of the propensity model

Conclusion

When implementing a certain model for an organization, there are various factors that need to be taken into consideration.

All models will not be actually useful or will not be able to produce the desired benefits.

The intended use of the model needs to be brought into consideration for the expected business impact. In worst scenarios, propensity models developed inappropriately may lead to a negative impact on businesses and marketing tactics. A detailed study of the problems associated or challenges faced have to be undertaken in order to implement the best suited model for an organisation.

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Advertising legislations to ensure compliance of the online advertisement with the associated law

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Abstract

Online advertising is increasing prominence in society at the same time that technology is becoming more and more significant. Online advertising is becoming into a significant economic force in the internet economy as a result of the advertisements' sponsorship of multiple websites and services. False marketing could seriously harm Goodwill due to the market's rivalry and customers' sensitivity.

Investments in advertising have increased significantly, requiring a robust regulatory framework to ensure that all stakeholders are fairly and effectively represented. In India, there is no definite structure or plan in place to regulate advertisements. This is an issue because of the risks associated with advertising. Ads are thus governed by industry-specific legislation, subjecting stakeholders to a range of risks. Although there isn't a single legislation on the subject of advertisements, there are a number of guidelines to make sure that interested parties are informed.

A rapid change in advertising strategies has been brought on by the growth of the Internet and digital media. The rules governing advertising must ensure that these advances are taken into account and that customers are not misled. Government consolidation of several industry-specific rules and regulations into a single, all-encompassing body of law is necessary for the advertising industry.

In this essay, we'll examine the current laws controlling commercials. We'll research the government agencies in charge of enforcing ad laws. We'll also talk about the limitations on online advertising.

Key words: Online media advertising, legal Regulations, ASCI.

WHAT IS MEDIA ADVERTISING ONLINE?

In no Indian legislation is the term "internet advertising" defined. Online advertising is defined by the dictionary as a commercial marketing activity carried out over the Internet to increase consumer awareness of

the benefits, existence, and/or accessibility of a specific service or commodity. The moral and ethical promotion of legal and illegal advertising tactics that aim to ethically represent facts and functions linked to the product or service in question and existing in the jurisdiction is just one of the many standards that regulate online advertising in a number of ways. These laws and standards are put in place to safeguard consumer rights and interests as well as the legal economic rights granted to certain or all business activities and operations.

LEGAL REGULATIONS FOR ONLINE ADVERTISEMENT

There are currently no laws or regulatory bodies in India that deal particularly with online advertising. However, the laws and organisations in charge of overseeing physical advertisements also have the power to control internet advertisements. Let's talk about a few of these laws.

The Consumer Protection Act of 1986 ([1]).

The right to information on quality, potency, quantity, purity, and other attributes is addressed in Section 6 of the Consumer Protection Act of 1986.

To protect trade and unfair practises, Section 2(r) of the Consumer Protection Act lists unfair business practises, such as deceptive advertising and fraud.

Indian Penal Code, 1860 [2].

Terrorism and crime-related advertising are prohibited by the IPC, 1860. Any advertisement that promotes the use of contract assassins, violence, or other unlawful activities is prohibited.

The Cable Regulation Television Act of 1995 (Act No. 3).

According to Section 6 of this Act, an announcement cannot be broadcast on cable or the internet until it complies with the advertisement law's requirements.

Regulations governing advertising are prohibited from endangering people's morality, religious beliefs, or sense of dignity, according to Section 7 of the Cable Television Regulation Act.

the 2003 Cigarette and Tobacco Act ([4]).

This document's Section 5 forbids the promotion of tobacco products and cigarettes of any kind through any media. Even audio advertisements are not permitted. This is forbidden for health reasons, and it also stipulates that a predetermined portion of the container for these substances must bear a warning.

The Food Safety Standard Act of 2006 [number 5]

It states that the use of food that is false, deceptive, or illegal is forbidden and restricts advertising related to grading standards, quantity, quality, or composition.

The Young People (Harmful Publications) Act of 1956 (paragraph six)

This law forbids violent and criminal behaviour toward children under the age of 18. This rule forbids any marketing from mentioning violence towards children or young people in any way.

The essentials of false advertising

In addition to industry-specific regulations, the cost per action expressly restricts the employment of the following marketing strategies to improve brand recognition, product use, or supply:

Untrue claims are made about a good or service's standard, quality, quantity, style, or model.

items that are used, recycled, refurbished, retrofitted, or that are decades old.

It implies that a good or service has support, endorsement, effectiveness, features, utility, or advantages that it does not.

Make erroneous or deceptive claims on the necessity of the use of a good or service.

generically guarantees the performance or effectiveness of a product without doing adequate testing.

makes it possible for adverts to be published.

Advertisement Standards Council of India (ASCI)

For the purpose of policing media content in both printed publications and electronic media, there exist numerous self-regulatory organisations. These groups, which mostly deal with written and visual media, include the Indian Publishing Guild, the Indian Broadcasting Foundation (IBF), and the News Broadcasting Association (NBA). Contrarily, the primary self-regulatory body that manages media advertising is the Advertising Standards Council of India (ASCI).

In responsibility of advertising that increases public trust is the Advertising Standards Council of India, which was founded in 1985. The council promises that the claims and representations made in the advertisement are accurate and sincere. The advertisement must adhere to the standards of public decency and should not insult any particular demographic. An advertising self-regulation code created by ASCI may be applied to problems surrounding the creation of advertisements. Although ASCI does not have a legal status, it has been recognised by numerous Indian laws. The regional and federal laws of India are not replaced or superseded by the ASCI code.

Both advertisements created by Indian businesses and those created outside of India are subject to the Code. The ASCI standards must be followed by all advertisements directed at Indian consumers. It is not needed by law for internet advertising to adhere to the ASCI code. However, the requirements of the course must be met in order to advertise on television. Some restrictions also forbid the marketing of a particular type of content. According to the ASCI code, anyone who believes that the advertisements are deceptive, untrue, offensive, etc. may lodge a complaint with the ad supplier.

The ASCI is a voluntary self-regulatory group that was founded in 1985 to advance ethical advertising and raise consumer trust in advertisements. The goals of the council are:

To assure the veracity and honesty of the claims and representations made in advertisements

To guarantee that advertisements do not violate universally recognised norms of decency

to prevent the indiscriminate use of advertising to promote goods that are thought to be harmful to the general public or to individuals.

To ensure that advertising follow the rules of generally accepted competitive behaviour in business in order to inform consumers about their options in the market.

A Board of Governors and a Consumer Complaints Council make up ASCI. Four representatives from each of the four areas related to the advertising industry make up the Board of Governors:

Advertisers

agencies for advertising

Media (owners of press, television, radio etc) (owners of press, television, radio etc.)

related industries (e.g. outdoor agencies, PR, market researchers, ad producers, business schools)

Self-Regulation of Advertising Using the ASCI Code

The ASCI Code for Self-Regulation in Advertising (also known as the "ASCI Code"), which is applicable to all parties engaged in the commissioning, creation, placement, or publication of advertisements, was adopted by ASCI to govern advertising in India. This ASCI Code is applicable to all advertising that are read, heard, or viewed in India, regardless of where they were originally published or from, as long as they are intended for Indian consumers or are exposed to a sizable number of Indian customers.

The ASCI Code is non-statutory, although it is acknowledged by a number of Indian legislation in addition to being embraced by organisations in the advertising sector. Notably, the ASCI Code states that it is not in conflict with any laws, their regulations, or the means by which they are implemented; as a result, the ASCI Code is intended to control under such laws.

Regulations Concerning the Promotion of Goods and Services

3.1 Alcohol (Beer, Wine, and Spirits)

Any advertisement that promotes the production, sale, or consumption of cigarettes, tobacco products, wine, liquor, or other intoxicants is prohibited by the Cable Television Network Rules, 1994, the Advertising Codes of Doordarshan, the All India Radio, and the Norms for Journalist Conduct published by the Press Council of India. Billboards, signboards, and other forms of outdoor advertising are permitted in some states, but only under tight conditions. Additionally, the ASCI Code forbids the use of children in alcohol product advertisements.

3.2 Professionals such as Chartered Accountants, Company Secretaries & Cost Accountants

These practitioners are barred from soliciting clients or professional work by advertisement. However, they may issue advertisements about their firm or services of their firm, through any mode of transmission, having inter alia details of names of partners, address and website, telephone, mobile, e-mail, fax number of the

member, year of establishment, additional recognised qualifications, languages spoken by the partner(s), honours or awards in the field of teaching, research, authorship etc.

3.3Arms, Ammunition, and Firearms

Such items need a licence from the relevant authorities in order to be sold and purchased. Therefore, under the Arms Act of 1959, advertisements for these products are prohibited in India.

3.4 Food

The Food Safety & Standards Act of 2006 states that no advertisement regarding the standard, quality, quantity, or grade-composition, as well as no representation regarding the necessity of, or utility of, any food, may be made in a way that is misleading or deceiving, or that would be in violation of the provisions of this law or rules and regulations made thereunder.

3.5 Infant Milk Food

3.6 Gaming (gambling, games of chance; differentiate between private-sector and "state" lotteries)3

3.7 Medical Devices

3.8 Medical Services

3.9 Nutritional Supplements: It is regulated under the Food Safety and Standards Act, 2006.

3.10 Occult ("Psychic") Services

3.11 Pharmaceuticals (over-the-counter and prescription medications)

(The above list is not exhaustive)

RESTRICTIONS

In accordance with numerous laws, it is prohibited to promote on the internet for tobacco and its derivatives, alcohol, milk substitutes, narcotics, firearms, lottery, and other products. However, there aren't any particular limitations on online advertising. These restrictions and prohibitions can be considered complete on their own.

LIABILITY

The intermediate act is described in Section 79 of the IT Act. According to this definition, an intermediate is a person who does not actively participate in changing the substance of the advertisement. Since intermediaries merely perform passive labour, they are granted protection from such laws. Only the person who actively promotes the prohibited products or disregards the rules will be held accountable for such actions.

SUGGESTIONS: At the moment, India has no laws governing exclusively online advertising. Online advertising is a practise that is expanding along with technology. Therefore, laws specifically addressing online advertising is required in the modern world.

It is rather troubling that there are so many people present for a single reason of advertisement. These bodies must either be amalgamated or only one dominant body, ASCI, should remain, with the others being

eliminated. Multiple organisations supporting the same cause might be conflict-prone and confusing for producers. Thus, only one body should be present.

CONCLUSION

Numerous legal issues will undoubtedly arise as internet advertising becomes more popular. A consistent principle is useful because it removes the ambiguity brought on by inconsistent laws. Additionally, due to the fact that internet advertising is international, it should not be constrained by national laws in order to protect stakeholders' rights. The regulations created for all jurisdictions would protect the dependability and potential of the market Communications.

The global framework, which may be a company or organisation that focuses on advertising, will make sure that there is supranational authority to carry out decisions. Both consumers and marketers would profit from the increased consistency and legitimacy of internet advertising.

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ARTIFICIAL INTELLIGENCE BASED RISK MANAGEMENT OF CREDIT SALES IN E-COMMERCE

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Abstract

E-commerce, short for electronic commerce, refers to the buying and selling of goods and services over the internet. It encompasses a wide range of commercial activities, including online shopping, electronic payments, digital marketing, and supply chain management. E-commerce has become increasingly popular over the years due to its convenience, accessibility, and the rapid growth of internet and mobile technologies. There are several types of e-commerce models, including business-to-consumer (B2C), business-to-business (B2B), consumer-to-consumer (C2C), and consumer-to-business (C2B). E-commerce platforms can range from small online shops to large multinational corporations and can sell everything from physical products to digital services. E-commerce has revolutionized the way businesses operate, enabling them to reach customers around the world, reduce costs, and streamline operations. However, it also presents new challenges such as security risks, logistical complexity, and the need for innovative marketing strategies. Risk management on credit sales refers to the process of identifying and analyzing potential risks associated with extending credit to customers, and implementing strategies to mitigate those risks. Credit sales are a common practice in many industries, particularly in B2B transactions, where companies may offer credit terms to other businesses to build relationships and facilitate sales. The main risks associated with credit sales include default, late payment, and fraud. To manage these risks, businesses typically perform credit checks and use various risk assessment tools to evaluate the creditworthiness of customers. They may also set credit limits and payment terms, and monitor customer behavior for any signs of potential risk. Other strategies for managing credit sales risks include offering incentives for early payment, using factoring or invoice discounting services, and obtaining credit insurance to protect against bad debt. Effective risk management on credit sales can help businesses avoid financial losses, maintain positive cash flow, and build stronger relationships with customers. This study proposes an artificial intelligence-based risk management system for credit sales in e-commerce. The system utilizes machine learning algorithms to analyze various data sources, including customer transaction history, credit scores, and behavioral patterns, to predict the risk of default or late payment. The proposed system is developed and tested on a large e-commerce dataset, and the results demonstrate its

ability to accurately identify high-risk customers and prevent potential financial losses. The system also provides insights into customer behavior patterns and helps businesses make data-driven decisions for credit management and risk mitigation.

Keyword

Artificial Intelligence, E-Commerce, Credit Management, etc.

1. Introduction

E-commerce, or electronic commerce, refers to the buying and selling of goods or services over the internet. It encompasses a wide range of activities, including online shopping, electronic payments, digital marketing, supply chain management, and more. E-commerce has become increasingly popular over the years due to the convenience it offers, with customers being able to shop from anywhere at any time, and businesses being able to reach customers beyond their physical location. The rise of mobile devices and internet access has made e-commerce more accessible to a wider audience, and it has now become a fundamental part of many industries. There are several different types of e-commerce models, including B2B (business-to-business), B2C (business-to-consumer), C2B (consumer-to-business), and C2C (consumer-to-consumer). E-commerce can range from small online shops to large multinational corporations, and can sell everything from physical products to digital services. E-commerce platforms offer several benefits, such as reduced costs, increased speed and efficiency, and the ability to reach a global audience. However, it also presents challenges such as security risks, logistical complexity, and the need for innovative marketing strategies. Overall, e-commerce is a rapidly evolving field that is transforming the way businesses operate and interact with their customers.

E-commerce based credit risk prediction refers to the use of machine learning and data analysis techniques to predict the likelihood of default or late payment by customers who purchase goods or services on e-commerce platforms. This involves analyzing customer transaction history, credit scores, and other behavioral patterns to identify potential credit risks and make informed decisions about extending credit. The use of data analytics and machine learning in credit risk prediction can improve the accuracy of risk assessments, leading to better credit management and reduced financial losses for e-commerce businesses. By analyzing customer data and identifying patterns, businesses can make data-driven decisions and implement strategies such as setting credit limits, payment terms, or offering incentives for early payment. E-commerce based credit risk prediction can also improve customer relations by providing more tailored credit solutions and reducing the need for manual credit checks, resulting in faster processing times and a better overall customer experience. However, it is important to consider ethical concerns such as data privacy and security when implementing such systems, and ensure that they comply with relevant regulations and guidelines.

Current e-commerce platforms evaluate seller credit by adding up or averaging customer ratings, which can lead to phenomena like "click farming," a form of click fraud where a large group of low-paid workers are hired to click on website elements to boost the status of the website or a product. Additionally, buyer credit ratings do not instantly adjust to new information on the seller's credit standing. We contend that the present approach to assessing credit risk is too simplistic since it fails to take into account important details about the seller's information that may make or break a SME's reputation. Credit risk assessment on existing e-commerce platforms does not take into account these

elements. A study of popular e-commerce sites found that the vast majority of them use the weighted average of buyer reviews to determine a seller's credit risk. These sites, like Tmall, employ a 0–5 rating system. The strategy is simplistic and unchanging, making it harder for shoppers to make informed judgements and enhancing the transaction environment of e-commerce platforms. To effectively and dynamically assess the seller's credit risks on an e-commerce platform, a thorough and responsive credit risk evaluation system is required.

The Internet, big data, and artificial intelligence (AI) have paved the way for the widespread adoption of models for preference prediction, intelligent recommendation, pattern recognition, and other technologies, which have improved the options available to consumers and made it easier for businesses to provide them with accurate, personalised service. Many e-commerce platforms have used big data technology to analyse and predict consumers' shopping behaviours and preferences and offer product recommendations based on the analysis results, thus shifting from traditional transactions to intelligent development and, to some extent, enhancing users' purchasing enthusiasm and customer experience. However, with online shopping, the technology is considerably more user-centric, with intelligent advice based on users' transaction data rather than systematic research of suppliers. Neither the seller's creditworthiness nor the buyer's preferences are being assessed in a mutually beneficial manner through the use of big data or AI technology, and neither party is being matched with the other. However, AI-recommended items have inconsistent quality and are not likely to enhance service quality for customers.

The rapid growth of e-commerce has led to an increase in the number of credit sales transactions conducted online, which presents new challenges for risk management. In this context, artificial intelligence (AI) has emerged as a powerful tool for analyzing customer data and predicting credit risk.The use of AI in risk management for credit sales in e-commerce has the potential to reduce the financial risks associated with extending credit, and improve decision-making processes. This is because AI algorithms can analyze large volumes of customer data and identify patterns that may not be immediately apparent to human analysts, leading to more accurate risk assessments.

The proposed AI-based risk management system in credit sales would utilize machine learning algorithms to analyze various data sources, including customer transaction history, credit scores, and behavioral patterns, to predict the risk of default or late payment. The system would then provide insights into customer behavior patterns and help businesses make data-driven decisions for credit management and risk mitigation. The goal of this study is to evaluate the effectiveness of an AI-based risk management system for credit sales in e-commerce, and to demonstrate how it can be used to reduce financial risk and improve decision-making processes. The results of this study could provide valuable insights for e-commerce businesses looking to adopt new strategies for managing credit risk in an increasingly competitive environment.

2. Related Study

There is evidence to show that customers are wary of doing business with online suppliers due to concerns about their personal information being compromised. Consumers' ability to overcome their fears and doubts about a product or service is largely due to the prevalence of trust in the market. In order for e-commerce to take off, customers need to feel safe disclosing personal information, making purchases, and following the advice of Web vendors. That's why credibility is so important for scientists and clinicians alike. The difficulty in drawing comparisons across studies on e-commerce trust stems from the fact that previous research has employed a

wide variety of definitions of trust. Measures for a cross-disciplinary, multi-dimensional model of e-commerce trust are proposed and validated in this work. The model's four overarching constructs—trust in oneself and others, confidence in institutions, trust in one's own beliefs, and trust in one's own intentions—are broken down into 16 observable, empirically supported sub constructs. Using a fictitious legal advice website, we show that the measures have strong psychometric features. These findings demonstrate the multifaceted nature of trust. Both the internal nomological validity of the proposed links among the trust constructs and the exterior nomological validity of the interactions with three other e-commerce constructs (Web experience, personal innovativeness, and Web site quality) are investigated. Future directions for research and practical ramifications are highlighted.[1]

As Internet connectivity and related technologies have improved rapidly over the past few years, online shopping has rapidly gained popularity. Large-scale sentiment analysis of shopper feedback on e-commerce sites has been shown to effectively boost customer happiness. In light of the sentiment lexicon, the authors of this research suggest a novel model for sentiment analysis called SLCABG. This model combines a convolutional neural network (CNN) with an attention-based bidirectional gated recurrent unit (BIGRU). The SLCABG model is an improvement over preexisting sentiment analysis models of product evaluations because it combines the benefits of a sentiment lexicon with deep learning technology. Combining the best features of a sentiment lexicon with deep learning methods, SLCABG provides optimal results. To begin, we boost the reviews' sentiment characteristics using the sentiment lexicon. The reviews' primary sentiment features and context features are then extracted using the Gated Recurrent Unit (GRU) network and the attention method to calculate weight. Classify the sentiment weight characteristics. This paper crawls and cleans the real-book assessment of dangdang.com, a popular Chinese e-commerce website, to use as a basis for Chinese-language teaching and testing. The data is 100,000 orders of magnitude in size, making it useful for a variety of applications in Chinese sentiment analysis. Evidence from experiments demonstrates the model's potential to significantly enhance the quality of sentiment analysis performed on texts.[2]

This study synthesises existing literature on the topic of AI in e-commerce and suggests avenues along which future research in the field of information systems (IS) may go. Towards this objective, a novel strategy was employed, namely the integration of bibliometric analysis with a comprehensive literature search. A total of 4335 papers were analysed for their bibliometric data, and 229 articles from top IS journals were assessed for their content. The bibliometric study showed that recommender systems are the main topic of study when it comes to artificial intelligence (AI) in online retail. The primary areas of study were determined to be sentiment analysis, trust, personalization, and optimization. This also establishes Chinese academic institutions as the foremost leaders in this field of study. In addition, computer science, AI, business, and management journals have been the primary publishers of research articles on AI's role in e-commerce. The study of prior research sheds light on the most popular areas, methods, and subjects of inquiry among IS academics. Based on these results, recommendations for future study are suggested. For the first time, this paper provides a study that aims to synthesise prior work on the topic of artificial intelligence (AI) in electronic commerce. It provides scholars with new insights on how to further the field. This guide offers professionals a centralised hub of data on how artificial intelligence (AI) might benefit their online business operations.[3]

Many new opportunities for human rights violations have arisen as a result of the widespread shift of retail to the Internet and the replacement of such salespeople with artificial intelligence; the use of AI to process large databases of potential buyers; and the use of chatbots to communicate with consumers. Breaches of the laws governing the collection and processing of personal data, the use of personal information, and interference with privacy fall under the category of violations of citizens' rights and freedoms in the field of artificial intelligence in e-commerce. The purpose of this piece is to explore possible legal mechanisms for regulating such civil relationships. The research allows us to draw the following inferences: To begin, the use of AI is rapidly expanding in the online retail sector. Second, there has been discussion on the topic of developing national and international legal regulation on the use of artificial intelligence in the private sector and at the level of state and supranational entities. Finally, there is no adequate legal framework in Ukraine (to govern these contacts), and local legal theory virtually ignores the subject of legal regulation of relations related to the use of artificial intelligence in e-commerce.[4]

Artificial intelligence, often known as "AI," is a method through which a computer-controlled robot or piece of software may learn to think intelligently, analogous to the way that intelligent humans do. This research examines the effects of AI on online retail. The electronic exchange of goods and services, as well as the related financial and data activities, has led to the development of a number of new technologies. This study's findings and recommendations imply that AI-powered forecasting apps can reliably produce and anticipate E-Commerce trends. This paper describes how AI has changed the face of e-commerce and how it may be used in many sectors of the industry. The study suggests that AI has helped improve the user experience on e-commerce platforms.[5]

3. Methodology

In this research, we use a decision tree to filter potential indexes, and we assess credit using composition algorithms determined by similarities between AI credit models and sellers on e-commerce platforms. Several popular decision tree algorithms include Chi-Square, CART, Quest, and Reduction in Variance. Simplified indices are used for analysis once the most accurate decision tree model is chosen from among numerous candidates. With the help of an AI algorithm, we integrate the data set chosen by the decision tree to create a new model. The decision tree KNN is part of the set of combinatorial algorithms that have been accepted for use in comparing and screening potential models.

3.1. Manifesting the Features of Platform-Based Business Transactions

The chosen credit risk evaluation model must be able to process data at different times, automatically identify the change in seller credit according to the change in data set, and carry out a certain judgement on the seller's credit level in regards to the learning model, all of which are necessary due to the unique characteristics of platform transactions.

3.2. Signs That Can't Be Ignored

The decision tree is used to choose indicators for the e-commerce seller credit risk evaluation system, and these indicators should have some degree of interpretability in order to be consistent with the logic of the e-commerce seller credit risk assessment index system. When assessing a buyer's creditworthiness, the seller places less weight on the removed factors. Seller's sales, review data, and other indications that have a high degree of correlation with seller credit should not be disregarded on principle.

3.3. Objectivity

The model must follow the logic of the credit risk evaluation procedure to choose the indicators that are highly connected with the seller's credit and reject those that have a less significant influence. In addition, there are specific evaluation procedures that produce clear and

understandable data that may be used to effectively assess credit risks. Complying with the logic of the combined algorithms is also required, and the accuracy is enhanced by combining the models, demonstrating that AI algorithms may be enhanced and optimised by using a combination model.

We choose a small number of combination approaches and then use the aforementioned criteria to determine which model provides the highest performance. In the meantime, it may be used to simplify the tangled credit processes of merchants on e-commerce platforms. By using the model in practise, both the e-commerce platform and the buyers may more precisely assess the seller's credit risks, and the resulting indications can be used as a guide for the seller to enhance the quality of their services.

3.4. Dataset and Data Training

In this work, we take advantage of information on Amazon vendors' credit. When it comes to C2C e-commerce in India, With a total of 512 businesses represented, the credit dataset's original data consists of 25 numeric and categorical attributes, such as platform entry indicators (like business licence, regional information, consumer security deposit, and Alipay personal certification), operation level indicators (like number of goods, main business share, buyer credit score, number of evaluators, and number of operating years), and commodity quality indicators (like goods and description compliance). Other numerical qualities, such as buyer credit score and number of evaluators, keep their original data value, while categorical attributes, such as company licence and regional information integrity, are translated into numerical ones. The final criterion distinguishes vendors with good credit from those with poor credit. 275 of the sellers have poor credit, while 541 have excellent credit.

Considering that our goal is to design a model that both increases the accuracy ratio of credit risk predictions and is simple to understand and use, we will construct a hybrid model that combines the decision tree with other AI models. IBM SPSS Modeler models, including support-vector machines (SVM), random forest methods, artificial neural networks, dynamic Bayesian networks, logistic regression, etc., will be chosen for comparison and screening. There will be a three-model cut down based on the accuracy-to-gain ratio and the gain-to-accuracy ratio. Three artificial intelligence models integrating decision trees and chosen models will be developed.

To ensure that the indicators chosen by the decision tree for the initial screening of indices are consistent with the logic of the e-commerce seller credit risk index, they must be interpretable. In theory, the indications having the strongest correlation to the seller's creditworthiness will be chosen, while those with weaker correlations will be discarded. After a first index screening using the decision tree, the reduced characteristics will be chosen to create a new dataset. Our approach partitions the dataset into a training set of 83% and a testing set of 14%. The dataset will be computed using three different hybrid AI models, with five iterations of each model being employed. The model outputs will be contrasted with those obtained from manual index filtering. In addition, we'll check if any model is overfitting by using a cross-validation strategy.

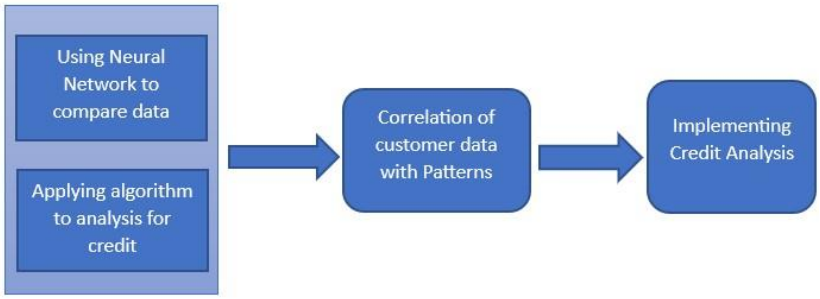

Figure 3.1: Proposed Architecture Model

4. Results and Discussions

KNN (K-Nearest Neighbors) is a supervised machine learning algorithm used for classification and regression tasks. In KNN, the algorithm searches for the K closest data points in the training set to a new input data point, and predicts the output based on the class of the majority of those K neighbors. The KNN algorithm works as follows:Choose a value for K.For each data point in the test set, calculate the distance between that point and all the points in the training set. Select the K data points from the training set that are closest to the test point. Determine the class of the test point by finding the majority class among the K nearest neighbors.

KNN can be used for both classification and regression tasks. In classification, the output variable is categorical and the algorithm predicts the class of the input data point. In regression, the output variable is continuous and the algorithm predicts a value for the input data point.

One of the advantages of KNN is that it is a simple algorithm that can work well on small datasets. However, it can be computationally expensive for large datasets and it requires choosing an appropriate value for K. Additionally, the performance of KNN can be sensitive to the distance metric used to calculate distances between data points.

Table 1: Compression of proposed algorithm with existing:

S.No	Gain	Ratio of Gain	Accuracy
1.	KNN	38	82.54
2.	Decision Tree	33	78.45
3.	SVM	28	72.99
4.	K-means	35	77.23

Here the Table: 1 shows the compression of Ratio gain and the accuracy of the algorithm which is proposed with the existing Artificial Intelligence algorithm. With reference to this KNN has 82.54% of accuracy for calculating and finding the individuals credit score where Decision Tree takes 78.45% of accuracy, SVM returns 72.99% of accuracy and K-Means returns 77.23% of accuracy. Initially our proposed Algorithm gives more accuracy when compared with existing.

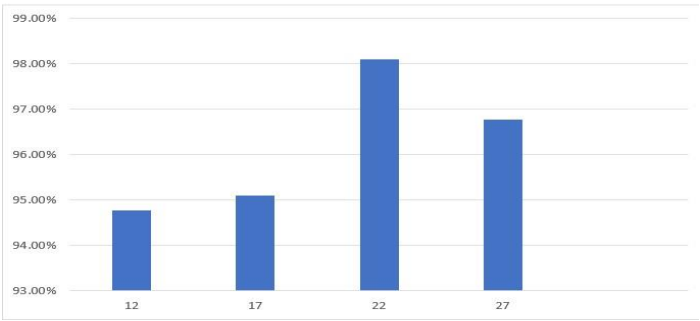

Figure 4.1: Compression of Chi-square parameter taken for decision tree

Figure 4.1 shows the times of Iteration is taken with Chi-square decision tree method with reference to this 12times of iteration gives 94.76% of accuracy and 17times gives 95.10%.Along with that 22 time of boosting iteration gives 98.10% and 27 times gives 96.77%.

6. Conclusion

Artificial intelligence (AI) can play an important role in credit risk management for e-commerce businesses. By analyzing vast amounts of data and identifying patterns and trends, AI can help predict and mitigate credit risk, ultimately leading to better business outcomes. One way that AI can be used for credit risk management is through the use of predictive models. These models can take into account a wide range of factors, such as customer behavior, transaction history, and external economic factors, to predict the likelihood of default. By using machine learning algorithms, these models can continually learn and adapt, improving their accuracy over time.

Another way that AI can be used for credit risk management is through the use of natural language processing (NLP) and sentiment analysis. By analyzing customer feedback and reviews, AI can identify patterns and trends that may indicate a higher risk of default. This can help e-commerce businesses identify potential risks before they become a problem. Overall, the use of AI for credit risk management in e-commerce can provide significant benefits for businesses. By enabling more accurate risk assessments and more effective risk mitigation strategies, businesses can reduce their exposure to credit risk and improve their overall financial performance. However, it is important to note that the use of AI for credit risk management should be accompanied by proper oversight and regulatory compliance to ensure that it is being used ethically and responsibly.

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THE NEW TRENDS AND CHALLENGES TO MANAGEMENT

Abstract

Most assessments from the past few years have indicated that how we work in the future will change. The work will be built on tuning in to the talent, and we will have the ability of a business ready to change, so we must be more adaptable, consider the worth of qualifications seriously, and pay attention to the talent. We are working to fulfil the requirements placed on us in terms of social and environmental obligations. Metrics, which will be crucial to the survival of the organizations, is another important issue that we are now considering. Metrics will become more crucial when organizations shift their focus from survival to proving business value because they will help management stay informed about project performance and how it affects the bottom line and customer service. This chapter examines the impending changes to our working habits and procedures.

Keywords: Talent management, tuning, qualifications, and adaptability.

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might require a whole new generation of managers with alternative conceptual frameworks.

II. CONCLUSIONS

Unless efficacy can be consistently assessed and a discernible increase can be seen, the investment in training and development is of limited value. Its human resources department had to change its job from transactional administration to that of a business partner with an emphasis on value. The goal of communication management is communication. Business is speaking just to itself when its language fails to do that when it protects itself with egotistical but useless babbling. You will have a better chance of assisting others in coming up with practical, effective solutions if you can maintain your composure and professionalism while stating the genuine problems at hand (as opposed to becoming engaged in insulting, inflammatory, or extraneous remarks). Future management trends are those management theories or tactics that need to be actively considered if an organization is to be successfully and profitably managed. The recent attention given to global warming ought to serve as a reminder to all companies to evaluate their hiring procedures. By using audio or video conferencing in place of 50% of their in-person meetings, companies may reduce the amount of time and money spent on meetings while also avoiding needless travel. The managers and their organizations must collaborate to ensure that they are maximizing the value of their education and attaining professional success. Whether or not any of us believe that we were born with the ability to manage, the present business world is going through a such significant change that the only way to remain ahead of the curve is via continual learning. Furthermore, the 100 billion neurons in our brains, which enable us to at least make an effort to adjust to change, should be praised for enabling those of us who were thrown into management by accident.

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ABDC

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ORGANISATION FINANCIAL RISK MANAGEMENT CONTROL SYSTEM USING ARTIFICIAL INTELLIGENCE

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Abstract: Financial risk management aims to detect and manage several risks of any organization. The institutions can match several payment obligations and income streams if the employees can manage the entire system. Artificial intelligence can possess a particular change within its application as financial functions are highly automated. The risks can be easily measured with the help of machine learning and the leaders can solve the problems. Using the .cpp file has helped to mitigate the financial risks in the companies.

Keywords: *Artificial Intelligence, Organization financial risk, Management control, competitive advantage, machine learning*

INTRODUCTION

Financial risk in an organisation is a possible chance of losing money while conducting any business venture. It implies the capability that controls the monetary policy and other debt issues. It can also create a possibility that the cash flow of the company can be insufficient to satisfy its obligations. Risks can be quantified with the profitability of a product and its impact which are represented in numbers [1]. The application of Artificial Intelligence provides the ability to evaluate unstructured information about critical behaviours. It can identify the patterns which are related to previous financial problems. This is how the managers and leaders can transpose it as the risk predators. Financial services are incorporating the methods of AI into the strategies so that the organisation can drive cost efficiencies. The tools of Artificial intelligence are based on the advanced functional and operational advantages and disadvantages. However, organisations should evaluate its use carefully so that they can understand the reward and risks. In this report, various images are going to be provided to explain the situation of the financial management system. The data will be analysed with the help of these graphs and pictures to justify the requirements.

LITERATURE REVIEW

The fin tech companies have started to implement risk management systems with the help of Artificial Intelligence solutions. This helps the employees to facilitate reduce their credit risks and facilitates the decision-making process. The managers need to have enough financial education through which they can manage their risk management practices [2]. This is how the organisation can build a strong position in the industry. The employees should give their attention properly so that they can face their associated challenges. The Technical Committee of the firm can recognise the forces that provide incentives to establish the financial risk control methods. It helps the managers and leaders to restrict activities with the market participants as doubts can arise within the entire system. The organisation should note that the management controls are not a substitute for adequate capital. The financial risks in an organisation need to be accessed carefully as the market participant may have ramifications within the management.

The governance of Artificial Intelligence can help to learn, monitor and govern its adoption. The situation can lead to privacy issues or it can create an unfair outcome [9]. The employees should take proper training so that they can mitigate the risks of AI such as watermarking and differential policy. The application of intelligent algorithms can also be used to mitigate the financial risks in the company. The abilities of AI are embedded in such a process that it can optimise the value chain of the management system. Embedding the methods of Artificial intelligence accurate the implementation of the procedures which can be easily used by the companies. Analysing the C++ code has helped to measure the effectiveness of the systems sources and their quality. Figure 5 states that the frontrunners can create an easy recognition of the AI in the business which can easily motivate the firms [10]. This is why the implementation of these plans helps the company to seek out the cost advantages and entire revenue.

CONCLUSION

The companies are still learning about the technology so that they can improve their business methods. Application of the Artificial intelligence process has helped the employees to save time so that they can execute their work. It has also helped to improve the quality of their services. Using the C++ code has helped to control the financial resources through which have managed to control the system with the machine learning process. The employees need to build their capacity so that they can overcome the risks by using AI. The managers and leaders can only figure out the situations only if they are experienced. Moreover, they should also be responsible while solving the risks.

ACKNOWLEDGEMENT

I would like to thank my teachers who have guided me till the end to complete this research. They have helped me to collect information which is related to this topic. I am also thankful to my friends who have also helped me to execute this project properly. It was impossible to collect the data without their support.

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Personality and Individual differences among Children and Adolescents: Literature in Psychology

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Personality and Individual differences among Children and Adolescents: Literature in Psychology

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***Abstract*- Background: Personality refers to the enduring characteristics and behavior that comprise a person's unique adjustment to life, including major traits, interests, drives, values, self-concept, abilities, and emotional patterns. Literature which intertwines within such fields as history, philosophy, sociology, psychology and so on is a discipline wherein language is used as a medium of expression so as to interpret man, existence and culture, personality and individual differences which have always been studied and discussed by writers, philosophers, artists, psychologists and psychiatrists.**

Objective: The objective of this experimental type research is that how Literature affects the psyche of Children and adolescents leading to the development of personality and individual differences among them

Method: The research had Participants from age group 7 to 18years. The participants were experimented using Observation, Interview and Questionnaire method.

Results: With the above experiment it is clearly seen that children who were shown diverse perspective of Literature and Language turned out to be more responsible, caring, understanding and developed problem solving behavior, whereas those who were not given these privileges were turned out to have just the opposite kind of perspective towards life and people around them.

Discussion: Literature can help children develop positive self-concepts through content and also through activities, which may follow the sharing of a book (Giorgis et al., 2009, p. 179). A child's personality development is part of who they are and who they can become; therefore, selecting the right book for a certain child could help the child deal with a problem or situation they are dealing with at that exact moment.

INTRODUCTION

Personality refers to the enduring characteristics and behavior that comprise a person's unique adjustment to life, including major traits, interests, drives, values, self-concept, abilities, and emotional patterns. The field of personality psychology studies the nature and definition of personality as well as its development, structure and trait constructs, dynamic processes, variations (with emphasis on enduring and stable individual differences), and maladaptive forms.

The word "personality" stems from the Latin word *persona*, which refers to a theatrical mask worn by performers to play roles or disguise their identities. Examples of personality can be found in how we describe other people's traits. For instance, "She is generous, caring, and a bit of a perfectionist," or "They are loyal and protective of their friends."

Types of Personality:

- Type A: Perfectionist, impatient, competitive, work-obsessed, achievement-oriented, aggressive, stressed
- Type B: Low stress, even-tempered, flexible, creative, adaptable to change, patient, tendency to procrastinate
- Type C: Highly conscientious, perfectionist, struggles to reveal emotions (positive and negative)
- Type D: Worrying, sad, irritable, pessimistic, negative self-talk, avoidance of social situations, lack of self-confidence, fear of rejection, appears gloomy, hopeless.

The study of personality and individual differences is an integral field of analysis in psychology. Every single person is unique in the combination of their personality traits and how they behave individually in social situations. Individual differences are related to mental abilities, vocational interests, religious beliefs, political attitudes, and sexuality. These can be predictors of work, education, and broader life outcomes as well as cognitive ability.

Personality psychology is just one aspect of psychological analysis. If we're to consider both nature and nurture, social psychology is the scientific study of how other people's thoughts and behaviors, as well as societal norms, and our environment may also have an effect on our own thoughts and behaviors.

LITRERATURE AND PSYCHOLOGY

Literature which intertwines within such fields as history, philosophy, sociology, psychology and so on is a discipline wherein language is used as a medium of expression so as to interpret man, existence and culture, personality and individual differences which have always been studied and discussed by writers, philosophers, artists, psychologists and psychiatrists. There is a very strong correlation between literature and psychology for the fact that both of them deal with human beings and their reactions, perceptions of the world, miseries, wishes, desires, fears, conflicts and reconciliations; individual and social concerns, by means of varied concepts, methods, and approaches.

An author represents life according to his/her objectives, perceptions, ideologies, and value judgments and opens the doors of the unknown and invisible worlds to readers not only by arousing feelings and emotions but also by helping them to discover the meaning of life and existence

Clearly, literature enables individuals to know and question their identities by raising consciousness and awareness. It is to be noted that man

and existence have always been fundamental elements in most scientific studies, fine arts and literature.

David Lodge in his work titled *Consciousness and the Novel Connected Essay* explains the meaning of literature as follows: "... literature is a record of human consciousness, the richest and most comprehensive we have. Lyric poetry is arguably man's most successful effort to describe quail. The novel is arguably man's most successful effort to describe the experience of individual human beings moving through space and time".

Joseph Conrad in the preface of *The Nigger of the Narcissus*, comments on the significance of the writers and the written texts in human life: "My task which I am trying to achieve is by the power of written word to make you hear, to make you feel- it is before all, to make you see. That and no more, and it is everything" (Preface). Personal history is actually one of the major elements that the reader learns through literature. As for the personalistic knowledge literature provides, it is claimed that "... Even when the ostensible subject of fiction is science itself, it is always a personalistic kind of knowledge that we obtain from it".

The earliest origins of psychology are found in the writings of the ancient Greek philosophers about the nature of life, particularly in the work of Aristotle" who used the term psyche to refer to the essence of life. This term is translated from ancient Greek to mean 'mind,' but it is closely linked in meaning to the word 'breath'" (Lahey). Keeping the ongoing debates in the field of psychology, dating back to Aristotle, in mind, it can be pointed out that "... Modern psychologists study the same actions, thoughts, and feelings that fascinated Aristotle.

The meaning of psychology in literature is explained by Wellek and Warren (1963: 81) as follows: "By 'psychology of literature', we may mean the psychological study of the writer, as type and as individual, or the study of the creative process, or the study of the psychological types and laws present within works of literature, or, finally, the effects of literature upon its readers.

OBJECTIVE

The objective of this experimental type research is that how Literature affects the psyche of Children and Adolescents leading to the development of personality and individual differences among them.

METHOD

- Total Number of participants

50 students participated in this experimental type research.

- Type of Interview

The interview was an Observation and Questionnaire form of interview.

- Population type and survey

The research had Participants from age group 7 to 18years. The participants were experimented using Observation, Interview and Questionnaire method.

The participants were; students from over 3 schools across Pune, India. The experiment took place by dividing the participants into 2 groups, 25 participants in each group. Both the groups were sent in 2 different rooms and for next 3 hours they were being kept engaged.

One group was narrated with value adding story, shown moral videos in mandarin, French, Dutch, Korean and English, and was given a novel-type story book to read.

The second group was sent to the room where they were made to show videos without sound, made them

communicate with their friends in a language the comfortable and do whatever they felt like.

RESULTS

After 3 hours, each of the group were observed by us, the first group's participants had a more graceful and smart walking style, they answered certain questions in a much smarter way than the second group.

This experiment was done on a regular basis for a week and then the Observation was noted in the following week- Total 2 weeks of Observation and question answering

The following are the characteristics shown among the participants of both the group:-

Age group	GROUP 1 (Room A)- With Literature 25 participants	GROUP 2 (Room B)- Without Literature 25 participants
7- 9 years	- Helped friends - Involved in sharing and caring - Better communication	- Didn't involve in helping - Made fun of friends and teachers - Bullied classmates
9-12 years	- Helped their friends with coping with their adolescent changes with body	- Involved in body shamming and - Rude behavior with friends and teachers
12-15 years	- Developed a sense of responsibility for taking up careers and job - Talked in a wise and mature way towards their partners	- Still skeptical about their future - Were not involved in having serious relationships - Had a thinking of mom-dad feeding entire life
15-18 years	- Learnt to control their tantrums - Started being polite with teachers and students - Involved themselves in societal help campaigns like NGOs - Parents thanked for the change seen in their child	- Teenage tantrums and aggressiveness was prevalent and increased - Bullied children and made fun of teachers - Involved in teenage fights and violence.

RESULTS

With the above experiment it is clearly seen that children who were shown diverse perspective of Literature and Language turned out to be more responsible, caring, understanding and developed problem solving behavior, whereas those who were not given these privileges were turned out to have just the opposite kind of perspective towards life and people around them.

Therefore it is clear that how Literature affects the psyche of an individual in order to have a heavy influence upon the Personality and Individual differences among them. So, it's evident that Literature plays a vital role in shaping an individual in a positive way.

DISSCUSSION

Personality development is the development of the organized pattern of behaviors and attitudes that makes a person distinctive (Laberge, 2006). Adults, especially parents and teachers, are principal components of these environments and therefore play a powerful role in helping or hindering children in their personality and cognitive development (Morrison, 2007, p. 99). According to Erikson (2007), children's personalities and social skills grow and develop within the context of society and in response to society's demands, expectations, values, and social institutions, such as families, schools, and other childcare programs (pp. 98-99). Children personality does not occur over night.

Literature can help children develop positive self-concepts through content and also through activities, which may follow the sharing of a book (Giorgis et al., 2009, p. 179). A child's personality development is part of who they are and who they can become; therefore, selecting the right book for a certain child could help the child deal with a problem or situation they are dealing with at that exact moment.

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IMPACT OF GREEN MARKETING ON CONSUMER PURCHASE INTENTION

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Abstract:

The negative impact on the environment is increasing day by day with the rising industrial activities and huge demands across the world. Due to such activities, major problems like global warming, the melting of glaciers, significant decrease in the quality of the air, and abrupt changes in the season are becoming common problems that have taken place that clearly showcase the alarming sign. In order to reduce such diverse impacts on the environment while meeting the demand of the market, the concept of green marketing has evolved. Green marketing, in particular, encompasses a wide variety of actions aimed at reducing the company's environmental impact. Green marketing aims to increase sales of environmentally friendly goods while also inspiring the creation in a more innovative way. It is to be noted that advertising the items is a good way to get the word out there and encourage consumers to make more environmentally friendly purchasing decisions. In this context, this research study is going to showcase the impact of green marketing on customer intention to buy green products and also going to highlight how effective it is to improve the condition of the environment for the betterment of society.

Keywords: *Green marketing, Consumer purchase intention, Environment safety, Consumer satisfaction*

INTRODUCTION

People's awareness of environmental issues has increased during the past decade. The term "green marketing" is used to describe the practice of selling goods or providing services with an emphasis on their positive impact on the natural world. Its origins can be traced back to the latter half of the twentieth century. Increasing numbers of businesses are realizing the potential of green marketing, and more and more shoppers are willing to shell out significant sums to purchase environmentally friendly goods. Until now, little research has been done on how this new market will affect customers and the environment. Green marketing has far-reaching effects on the economy as a whole, leading not only to environmental conservation but also to the development of new markets and the creation of new employment possibilities. Businesses that practice sustainability are likely to attract and retain a large number of eco-conscious clients.

LITERATURE REVIEW

Concept of green marketing

The interest of marketers on ecological marketing and its popularity has grown since 1975 when the American Marketing Association (AMA) introduce an event related to it. Though the term "green marketing" did not appear until 1980 [1]. The American Marketing Association (AMA) defines green marketing as the marketing of products that are presumed to be environmentally safe, and it includes a wide range of activities, including product modification, production process changes, packaging redesigns, advertising strategy shifts, and a greater focus on compliance marketing within industries. Green marketing is defined by the Business Dictionary as "a set of promotional strategies designed to take advantage of consumers' evolving perceptions of a company." More and more, these shifts are being affected by a company's policies and practices that impact environmental quality and reveal the degree to which it cares about its surrounding neighborhood. It can also be seen as the advocacy for items that are less harmful to the environment.

Figure 1: Green marketing and its strategy

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BE ON THE RIGHT PATH: BUILD YOUR E-PORTFOLIO AND BE A PROFICIENT LIBRARIAN

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Abstract:

The librarian is responsible for a diverse variety of ever-changing tasks that are frequently poorly documented. This problem can be resolved for librarians on the tenure-track or who are looking for work by creating an academic portfolio that serves as a record of their work in librarianship, achievements, awards, roles and responsibilities. E-portfolio gives motivation for continuous professional development. E-portfolio is the tool which provides a platform to create self e-identity. This paper outlines how librarians can create a personal professional e-portfolio. Which attributes can be added in the librarian e-portfolio to showcase skills and ability to become proficient librarian.

Keyword: *E-portfolio; Continuous professional development; Librarian.*

Introduction:

We live in an age where we are constantly gaining new knowledge, skills and educational degrees, whether formally or informally. It might be difficult to keep track of all the continuing education opportunities and certificates available. On the other hand, working in a library is a thankless and difficult profession. It can, however, be immensely lucrative. Working in a library is challenging at times, but it can also be rewarding when viewed in a different light. For bridging the gap between achievements and work pressure and prove yourself as a proficient employee we have to take help from technology. There are various resources available to assist librarians in maintaining their abilities, such as an e-portfolio. “The new library will be a place where personnel and facilities are available to support faculty in the creation of digitized teaching materials” (Stoffle & Williams, 1995). An e-portfolio is a digital collection of papers that demonstrate your librarianship skills and expertise. Because it allows you to see yourself from a different perspective, creating an e-portfolio can be advantage to your professional development, career planning, and job hunt. E-portfolio may be used to keep track of your education and training, previous jobs, software skill sets, work experience, résumé, and personal hobbies - basically anything that will help you better understand who you are as a separator and what you bring to the table. Employers, institutes, and different authorities need to know more about candidates' soft skills, such as communication skills, leadership potential, and adaptability under pressure

— characteristics that are difficult to assess or read on paper alone.

What Is E-Portfolio?

A compilation of your work that showcases your talents and abilities is called an e-portfolio. Finding new opportunities depends on it. It's your traditional portfolio in digital form. An e-portfolio is a computer-based version of a paper-based portfolio that includes visuals, audio, video, and interactive components. It gives hiring managers a central location to view all of your work. It can also be a fantastic tool to show potential employers your talents and abilities. "Electronic portfolios are being used by elementary, secondary and college educators to lead their classrooms toward learner-centered rather than teaching-centered learning environments" (Hewett, 2004).

Ravet, S. (2005) mentioned e-portfolio definition , according to the National to the National Learning Infrastructure Initiative (US), an e-portfolio is a "a collection of authentic and diverse evidence, drawn from a larger archive representing what a person or organisation has learned over time on which the person or organisation has learn over time on which the person or organisation has reflected and designed for presentation to one or more audiences for a particular rhetorical purpose". A portfolio is a well-organized collection of materials that illustrates a person's abilities, work, and professional development (Saunders, 2003).

Professional E-Portfolio:

Professional e-portfolios, also known as Career e-portfolios, are representative artifacts that people build to show off their greatest work in order to encourage self-reflection and peer assessment. Through the improvement in professional career, this procedure closes the assessment cycle. This professional e-portfolio is implemented in different fields like medical, arts, teacher education etc. Implementation in the field of library will give a clear picture of library personnel's.

Librarian's E-Portfolio:

The necessity for the e-portfolio determines how the e-portfolio is organised. "Librarians teaching philosophy reflects their beliefs about education and its connection with librarianship" (Lally & Trejo, 2019) The employer's needs should be the main influence on the e-portfolio's content if the librarian is utilising it as a job-hunting tool. However, the portfolio should be organised around the librarian's present, post function and emphasize how effectively the librarian meets his or her job criteria if the librarian is utilising the portfolio to conduct their yearly review evaluation or for their promotion and tenure process. In the field of education mainly teaching, research and service are the main factors in organisation of e-portfolio. But in the library field teaching is not the main job of librarians, so the organisation criteria for librarians, which may include employees' performance, professional service, or research, differ depending on the type of institution and, in some cases, even within the same institution. The librarianship portfolio must therefore detail what each librarian does in their current roles, in a manner similar to that of annual evaluations, and identify the areas of expertise in which each librarian has skills, such as reference, liaison, collection development, instruction, administration/coordination, cataloguing, and acquisitions. While governing all of the librarians' entries, this subdivision also gave them a way to set

themselves apart from not only other academic departments or institutions but also from other librarians working in the same library.

A librarian's time, effort, and contemplation on what they consider to be significant about their accomplishments are all necessary for the building and upkeep of a librarianship portfolio. The librarian must reflect on their effectiveness and how their performance and professional growth develop as a librarian in the library profession.

Functions of Librarian E-Portfolio:

- The portfolio should convey a clear picture of your professional growth as a librarian.
- The portfolio should allow the librarian the chance to explain the part their work had in their professional development, especially if they showed that they advanced in the field of librarianship in a deliberate manner. The librarian should discuss the part their work played in their professional development in their portfolio, especially if they showed.
- The portfolio should serve as a useful tool to demonstrate how you have contributed to the advancement of the library profession as well as your department and college.
- The establishment and upkeep of an academic portfolio can be helpful for librarians as a tool for job searching, as documentation to support performance appraisal or promotion and tenure review, and as a personal reflection tool to stimulate professional development and career progress.

Role of Librarians E-Portfolio:

E-portfolio in continuous professional development:

An excellent tool for continuing professional development is a librarian's e-portfolio. You can keep track of your progress, consider the abilities you have acquired, and record the projects you have worked on. Your e-portfolio can be used to list any training programmes or certificates you have finished. This will enable you to keep tabs on your development and pinpoint your strengths and flaws. It's crucial to monitor your development so you can spot problem areas and implement corrections. An effective tool for showing your abilities and expertise to potential employers is a librarian's e-portfolio. On your e-portfolio, you can list any pertinent projects and educational experiences. The users, colleagues, library professionals, authorities will find your CV to be more fascinating and compelling as a result. Employers appreciate learning that you are self-starting and have kept a record of your efforts. Applying for higher education, projects and scholarships might be facilitated by an electronic portfolio.

E-portfolio as showcase tool:

The Librarian e-portfolio also worked as a showcase tool that helped dispel some falsehoods and challenged the preconceptions of the negative role of the librarian, as a librarian working in a field beset by unjust archaic prejudices. The majority of librarians nowadays are quite tech-savvy and use cutting-edge services to help others. Librarians can plan to use the e-portfolio format as a kind of marketing "technology tool" that would promote not just librarian’s abilities but also some of the technical successes and improvements of the College's library when they chose it as a promotion tool. It exhibited not just technological

abilities but also the library's involvement in a round-the-clock live chat service, the development of online library research courses to help students with their needs, and the use of social media to advertise library services. Librarian's lengthy list of duties as a faculty librarian may be found in a traditional CV or portfolio. These duties were not only highlighted in an e-portfolio, but they were also prominently displayed by incorporating links to video clips, social networking projects, and actual online tutorials.

E-portfolio for self-identity:

Self-e-identity is another crucial application of e-portfolios. It is a tool that enables you to consider your unique identity and qualities. It allows you to keep track of your talents, interests, and ambitions in addition to your passions and interests. Your leadership abilities and skills can be documented in your e-portfolio. It enables you to consider the kind of leader you are and want to be. You can use it to keep track of your extracurricular activities and charity work as well.

Cumulative Role of E-portfolio:

During tenure-track years at the College, e-portfolio functioned as a comprehensive collection of all the requirements librarians have to meet. It also monitors individuals' growth as a librarian and faculty member at the College. This form of electronic portfolio, according to Hewett (2004), is a collection of documentation that demonstrates progress toward meeting predetermined standards. The e-portfolio serves as documentation and a cumulative file to show that the specific competencies required for tenure at the college want to meet. It can detail collection development responsibilities and how, during individuals tenure-track years, librarians worked with teaching staff from various disciplines.

E-portfolio improves your CV:

Your CV can be enhanced by using the e-portfolio. You can provide examples of your work, such as digital samples, pictures, or summaries of initiatives in which you were involved. You can use this to demonstrate your abilities, information, and experience. The content you can put in your CV is frequently rather constrained, whereas the stuff you can add to an e-portfolio is typically not.

E-portfolio for feedback on your work:

You can get feedback on your work using the e-portfolio. You can set up a place where you can get comments on your projects, presentations, and writing. This can be a terrific approach to get input from mentors and peers. Then you can use this criticism to polish your work and advance your abilities.

E-portfolio gives opportunity to learn new abilities and information:

New knowledge and abilities can be discovered via the e-portfolio. The e-portfolio can be used as a repository for samples of your work and pertinent information. When you have more experience and want to attempt something new, you can return to these samples. This can be a fantastic method to pick up new abilities and get ideas for new endeavors.

E-portfolio gives opportunity to connect other professionals:

A person's competence, proficiency, expertise, etc. are reflected in their professional e-portfolio. To solve professional challenges, issues, it can be helpful to seek assistance from other professionals. It offers a forum for talking about and sharing current trends. E-

portfolio demonstrates the library and librarian's areas of expertise.

Outline of librarian e-portfolio:

Personal Information
Educational qualification
Research contribution
Achievements and responsibilities
Information about Library and library services
Personal Interests

Part	It includes
Personal Information	Biographical information with the position of the professional.
Educational qualification	Detailed educational qualification with research related achievements and work experience which describes librarians different roles and responsibilities in different positions and institutions.
Research contribution	Other than just research degree, contribution in research in the form of research projects, guidance in research, writings in research papers, conference papers, even research ID's which indicates research contribution collectively.
Achievements and responsibilities	Different achievements and awards gained during professional tenure and different responsibilities give a platform to librarians to show their abilities for self growth and institutions too.
Information about Library and library services	This area will help to promote library and library services. Which gives an idea about librarian's contribution in library development.
Personal Interests	Other than being a librarian, he or she is a human being with special abilities and skills. Usually which can not be shown in traditional CV, this area provides this platform to showcaseanother aspect of personality.

Conclusion:

For the academic librarian, building a professional portfolio opens up a world of possibilities. It can be utilised in the advancement and tenure processes in addition to job searching. Academic librarians' roles are evolving now more than ever to keep up with new technological developments, educational focuses, and increased standards of quality and competition. In addition, when budgets are slashed and more librarians start to retire, we are now experiencing a personnel and financial crisis. Because of this, it is imperative for us to know exactly what we are capable of in our work and to be able to show proof of it through documentation. The professional portfolio might hold the key to demonstrating who we are and why we matter to people both inside and outside of libraries and how academic librarians resemble other faculty in many respects. E-portfolio will help to give a respected position in the professional community.

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RESEARCH PRODUCTIVITY OF DEPARTMENT OF LIBRARY AND INFORMATION SCIENCE OF SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE (SPPU)

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Abstract:

An institution's academic progress is reflected in its research productivity. The current study attempted to investigate the research contribution of Savitribai Phule Pune University's library and information science department. The Ph.D. thesis records were extracted using the Shodhganga e-theses and dissertation databases. A total of 82 theses were obtained from the SPPU's LIS departments under Shodhganga repository. The study examined all LIS department records available in the Shodhganga repository from 1977 to 2021. MS-Excel was used to extract the data, after which it was analysed using various formulas. The study's major findings revealed that 2008 was the most productive year for the LIS department in terms of research. Dr. Neela J. Deshpande guided 15.85% of Ph.D. theses in the LIS department. Male researchers and guides predominate in the LIS department; 67% of males are research guides and 66% are research scholars. The majority of Ph.D. candidates organised their work into six chapters.

Keywords: *Department of LIS; Ph.D.; Research productivity; Shodhganga; SPPU.*

Introduction:

LIS research in India began in the early 1930s with the initiative of Dr. S.R. Ranganathan, who promoted the library movement (Maity & Hatua, 2015). The direction of LIS research has shifted over time. The evolution of computers and the internet altered the face of libraries and LIS research. "The discipline of Library and Information Science (LIS)has grown more divided into the sub disciplines of Library Science and Information Science" (Lund, 2020). There are several tools available to assess research productivity and trends. An individual, institution, or nation's academic progress is measured by their research productivity (Tomei et al., 2014).The tools used to evaluate the results of research include scientometry, bibliometry, and content analysis. Many Indian universities now conduct high-quality research in the area of library and information science. The evaluation of the institute's research productivity indicates the quantity, quality, growth, and current position of research. It also helps to identify research gaps in each area that need to be addressed in the future.

The research productivity of the institution/university reflects the quality of that institute. There are numerous tools available to assess an institution's

productivity in terms of research. An institution's research output includes research articles, projects, doctoral theses, and dissertations. SPPU is one of Maharashtra's oldest state universities. There are a number of universities in Maharashtra that provide LIS education, but SPPU's department of library and information science is the oldest. In 1958, SPPU launched a diploma programme in LIS, and in 1965, they started a bachelor's programme. The LIS department contributes significantly to research. This department has produced 85 LIS professionals who have earned their Ph.D.s.

Shodhganga:

Shodhganga is a digital repository of Indian Electronic Theses and Dissertations set up by the INFLIBNET Centre. Shodhganga has provided open access to electronic theses of all the universities and centres registered under the UGC. As per the amendment made in the Minimum Standards & Procedure for Award of M.Phil./Ph.D Degree Regulation, 2009 on May 5, 2016, it is mandatory for researchers to submit the electronic copy of theses to the shodhganga through their university. Shodhganga is one of India's largest repositories of electronic theses and dissertations. It also has the ability to capture, index, store, disseminate, and preserve ETDs (Electronic Theses and Dissertations) submitted by the researchers. Shodhganga shows the academic and research progress of each university and department. It also provides a simple and advanced search facility, and researchers can browse the data through universities and departments.

Savitribai Phule Pune University has contributed 11894 theses and dissertations till date. The department of library and information science contributed 85 theses to Shodhganga. The research contribution of the Department of LIS is appreciable as compared to other universities in Maharashtra.

Objectives:

1. To access the most leading research guides in the Department of LIS of SPPU
2. To know the broad area of research of Department of LIS
3. To find out gender wise research contribution in Department of LIS
4. To know the chapter scheme pattern used by researchers in the Department of LIS
5. To know the most productive year in terms of research for the Department of LIS

Methodology:

The current study attempted to analyse a Ph.D. thesis submitted to Shodhganga repository by Savitribai Phule Pune University's department of library and information science. All thesis data was collected from the shodhganga website (https://shodhganga.inflibnet.ac.in/). It shows total 85 records of theses, but there are 3 duplicate entries. So in this study we are considering the department of library and information science submitted 82 theses to the shodhganga electronic thesis and dissertation database. All of the data was extracted into Excel and analyzed using various formulas.

Scope and Limitations:

The current study only included 82 doctoral theses available in Shodhganga e-theses database under the Savitribai Phule Pune University, Pune, department of library and information science. The study

examined all of the theses accessible in Shodhganga under the SPPU's department of library and information science until 2022.

Review of Literature:

(Sheikh & Jan, 2017) explored the "research productivity of LIS schools in Pakistan". According to the study, 87 percent of total Ph.D. thesis were completed in the last five years, indicating that LIS schools' research productivity has increased. (Maity & Hatua, 2015) examined doctoral theses from the databases INDCAT, Vidyanidhi, and INFLIBNET. From 1950 to 2012, a total of 1058 records were extracted. According to the study, research trends in library management have increased since the IT revolution.(Manjunatha,2019) investigated the LIS department's research contribution at Karnataka State Universities. The study concentrated on university-level research outcomes, leading research guides in the LIS community, and the number of theses published by LIS departments each year. The Shodhganga ETD database contained a total of 209 theses, which were analysed and extracted.

(Atapour, Hamdipour, &Shenavar, 2022) assessed the research productivity of the Middle Eastern LIS department. The data was obtained from the Web of Science database, and the analysis was carried out with the help of the journal citation report. According to the study, Ilan University has the highest research productivity.(Satija, 1999) described the beginnings of LIS education and Ph.D. programmes in India, as well as Dr. S.R. Ranganathan's contribution to shaping the LIS profession. The author listed the research contributions of Indian universities in the form of Ph.D. theses. The study also identified the major areas of research in library and information science. The current study examined several research articles. They discovered that while many studies have been conducted on research productivity, the department of LIS at SPPU's remains unstudied. As a result, researchers conducted this study.

Data Analysis:

Table 1: The most leading research guides in the Department of LIS of SPPU

Sr. No.	Research Guide	Num. of Students	Percentage
1	Deshpande, N J	13	15.85
2	Patil, Suresh K	12	14.63
3	Singh Surya Nath	11	13.41
4	Konnur, M B	8	9.75
5	Prasad, A R D	8	9.75
6	Mahajan, S G	6	7.31
7	Ganpule, S R	5	6.09
8	Riswadkar, M R	5	6.09
9	Bansode, Sadanand Y.	3	3.65
10	Panage B M	3	3.65

Table No. 1 reflects the top ten research guides at SPPU's Department of LIS. According to the data retrieved from the Shodhganga theses and dissertation repository, there are a total of 16 research guides. Dr. Neela J. Deshpande secured first place, with 13 (15.85%) students having completed their Ph.D. under her

supervision. While Prof. Dr. S.K. Patil has 12 (14.63%), and Prof. Dr. Singh Surya Nath has 11 (13.41%), research students have received Ph.D. degrees under their supervision.

Table No. 2. Broad area of research in department of LIS of SPPU

Sr. No.	Area of Research	No. of theses
1	Library and Information Services	10
2	ICT	8
3	Knowledge Management System	6
4	Use Study	5
5	User Study	4
6	Information system	3
7	Human resource planning	3
8	Library Networking	3
9	Ontology, Taxonomy	3
10	Public libraries	3

Table No.2 shows the broad area of research which was studied in the LIS department of SPPU. The Highest 10 researchers were carried out their research on library and information services, followed by ICT use in library (8) and Knowledge Management system (6).

Table No.3. Gender wise research contribution in Department of LIS of SPPU

Gender	Research Guides	Researchers	Total
Male	12 (75%)	54 (66%)	66 (67%)
Female	4 (25%)	28 (34%)	32 (33%)
Total	**16**	**82**	**98**

Table No. 3 shows the gender-wise distribution of research contributions at SPPU's Department of LIS. It reveals that male research guides predominate in the Department of LIS (12, 75%). 54 students were guided by male research guides. Only 4 (25%) female research guides guided the 28 students. In terms of research, there are 54 male researchers (66%) and 28 female researchers (34%). There is a 67% male contribution to research and a 33% female contribution.

Table No. 4. Chapter scheme pattern used by researchers in the Department of LIS

Sr. No.	No. of Chapters	No. of Thesis
1	4	1
2	5	11
3	6	38
4	7	25
5	8	2
6	9	1
7	11	2
8	12	1
9	13	1

Table No.3 depicts the chapter scheme pattern used by LIS researchers. There are a minimum of 4 chapters and a maximum of 13 chapters in the thesis. For thesis writing, the majority of researchers (38) used 6-chapter schemes, while 25 used 7-chapter schemes.

Table No.5 The most productive year in terms of research for the Department of LIS of SPPU

Sr. No.	No of Thesis	Ph.D. Completed Year
1	9	2008
2	7	2011
3	6	2014
4	5	2010
5	5	2009
6	5	2005
7	4	2007
8	4	1998
9	3	2017
10	3	2016

Table No. 5 illustrates the Department of LIS's most productive year in terms of research contribution. The Department of LIS completed 82 Ph.D.s over a 45-year period, from 1977 to 2021. In 2008, nine researchers received their Ph.D. 7, 6 and 5 researchers received their Ph.D. degrees in 2011, 2014, and 2010.

Findings:

- It is found that total of 82 research students have completed Ph.D. in Department of LIS of SPPU from the last 45 years.
- It has been noted that the main areas of research in the LIS department at SPPU include ICT and library and information services. 10 Ph.D. candidates conducted research on libraries and information services, and 8 on ICT.
- The analysis shows that Dr. Neela J. Deshpande is the leading research guide in the Department of LIS of SPPU who has guided 13 (15.85%) research students.
- In terms of research contribution, males are dominated in both roles, as research guides and as researchers. In the SPPU department of LIS, the majority of research guides (75%) are male and only 25% are female. On the other hand, 66% of researchers are male, while 34% are female.
- The current study reveals that, more than 46% of researchers have written their Ph.D. theses with six chapters, and 30% have made seven chapters to present their Ph.D. research work. As a result, it has been discovered that a six-chapter scheme is most familiar to the majority of research scholars.
- In terms of Ph.D. research, the 2008 year was the most productive year for the department of LIS at SPPU. Nine research scholars received their Ph.D. degrees in 2008. It is also worth noting that the LIS

department's research contribution has increased since 2005.

Conclusion and Suggestions:

Shodhganga is now a key database for every researcher to browse and explore previous research work related to his or her study. INFLIBNET plays a significant role in improving research quality and services in India. SPPU is one of Maharashtra's oldest universities, and it is well known for providing high-quality higher education and research. According to a study, the Department of LIS's research contribution has been increasing since 2005. The quality of the institution and the department determine an institution's research productivity. Each university in India is responsible for showcasing their research outcomes through Shodhganga.

The UGC has taken the lead and published a draft for all universities and research institutions to submit e-copies of theses to the Shodhganga e-theses and dissertation repository. As a result, each university should update and maintain the data as it is published. The LIS department should increase the intake of research guides and encourage LIS students to pursue their Ph.D.; this will automatically increase the department's research productivity. In addition, the LIS department organises research awareness programmes for students to motivate them to explore new areas of LIS.

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E- Portfolio:A Tool to Showcase Electronic Identity

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Research Scholar,
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Abstract: -

E-portfolio is powerful tool to develop electronic identity. This paperemphasizes on historical development of e-portfolio. It includes evaluation of portfolio from paper portfolio to e- portfolio. There are basic three categories of e-portfolios like Showcase/Professional e-portfolios, Learning e-portfolio, Assessment/ General Education e-portfolio. This defines the content or aim of the e-portfolio. To develop portfolio there are two basic two stages like one is Multimedia development process and second is portfolio development process. Now days use of E- Portfolio is increased in many fields with effective results.

Keyword: -E-portfolio, Showcase/Professional e-portfolios, Development process of e-portfolio, Use of E-portfolio.

Introduction:-

Identity is the deep knowledge of where I come from, where I am going & to whom I belong.

Individual identity is at the heart of who we are. It shapes how we portray ourselves, our expectations for how we interact with others and how they treat us, and our perceptions of our own strengths and potential. Identity is formed after a person completes an educational degree or an intensive training programme, both of which have a significant impact on the creation and shape of a person's professional identity. A person's professional identity is how they see themselves in connection to a profession and their membership in it. Individuals identify themselves in their current or future professional lives by their views and attitudes, values, reasons, and experiences, which form their professional identity.In order to improve our identity, we must offer ourselves in a variety of ways. Like a portfolio format. The most significant aspect of a portfolio is self-reflection. The core aspects of every portfolio are reflections on one's own qualities and competences.

Objective of the Study: -

- To take overview of portfolio and e-portfolio.
- To know the history and development of e-portfolio.
- To identify types of e-portfolio.
- To know e-portfolio development process.
- To Know uses of e-portfolio in higher education.

- **E-portfolio:-**

Origin of word Portfolio:-

According to Wikipedia, Portfolio comes from the Italian words Portare, which means "to carry," and Foglio, which means "leaf or sheet."

Portfolio

A portfolio is a collection of a student's work (such as assignments and tests) that has been accumulated over time and is used to evaluate performance or progress.

Electronic portfolio also called as a Virtual portfolio Web portfolio E-folio, Webfolio.

Definition is e-portfolio

According to Lorenzo, G., Ittelson, J. "An e-portfolio is a digitized collection of artefacts including demonstrations, resources, and accomplishments that represent an individual, group, or institution. This collection can be comprised of text-based, graphic, or multimedia elements archived on a Web site or on other electronic media such as a CD-ROM or DVD."

- **Historical Development of Portfolio to E-portfolio:-**

Originally portfolio evolved from loose paper to electronic format and local network to World Wide Web. Here we divide evolution of portfolio into 5 time periods:-

Origins and early usage: - Portfolios were first used in the 15th century in Italy by artists and architects. portfolio means "Folder or case for carrying loose paper or pictures,". Michellozzo, the architect, submitted a design portfolio for a new hospital to the city in 1440. Leonardo da Vinci's work is another prominent example from European history. His 7000 pages of loose folio containing his ideas, inventions, and studies were compiled after his death. 'Da Vinci Codex Arundel' is the title of this collection.

The 70s and 80s: -Higher education was added to the portfolio in the early 1970s. In 1970, Ford and Larking established portfolio as a method of assessment in the fields of education and English. "The portfolio system requires the disinterested judgement of each student's work, which is collected, like an artist's best representative work, into a "portfolio," they explained. Between 1986 and 1990, the teacher education industry adopted portfolios as a means of demonstrating students' abilities, progress, and accomplishments to audiences**.**

The 1990s: -Portfolio was introduced in numerous sectors, including nursing, medicine, and science, around the beginning of the 1990s. This was the first time an electronic portfolio was created. Helen Barrett introduced e-portfolios in the academic sphere, which produce and store a portfolio utilizing CDs and video compression technologies. During the 1990s, portfolios underwent a transformation into electronic format, moving from a local network to the World Wide Web. "A portfolio is a hypertext," Purves noted in this regard. It is made up of a number of text or artefact areas that the author has developed and structured. In most cases, the author in education is a pupil. Among the artefacts, the learner develops a network."

The 2000s: -Between 2000 and 2010, universities introduced e-learning, social media, virtual learning, Open Education Resources, Videos, blogs, and e-portfolios. In the year 2000, the e-portfolio platform experienced explosive growth. During this time, the use of e-portfolios became more widespread, and many scholars and educators began to apply theoretical studies to higher education.

The 2010s onwards: -In the last decade, there has been a lot of research on the influence of e-portfolios on users. During this time, many modules were produced, and students were assessed using an e-portfolio system. During this time period, an e-portfolio was also employed to collect evidence of facts and maintain paperwork. Academic and co-curricular aspects such as sports, counselling, and various experiences have been included in the e-portfolio, allowing them to broaden their intellectual framework. This decade saw the emergence of revolutionary concepts such as blockchain, MOOCs, and data ethics.

Types of e-Portfolios

According the University of Clemson following are the main three types :-

a) **Showcase/Professional e-portfolios:-** These e-portfolios are mostly used to display (Showcase) the highlights of students' and professionals' academic careers.

b) **Learning e-portfolio :-** These portfolios are often prepared by students as part of a course to demonstrate learning and the process of learning. These portfolios are frequently shared with other students in order to receive comments from their peers. Formative feedback is an important element of the learning process, and learning portfolios reinforce this approach.

c) Assessment/ General Education e-portfolio :- Feedback was given to educational institutes or organisations for formative and summative assessment assessments. On the quality of evidence professionals utilised to exhibit their general education competencies in individual portfolios.

Use of Educational e-portfolio:-

At all levels of education, e-portfolios have proven to be effective. The e-portfolio has evolved into a cutting-edge tool for online learning and professional growth. In academics, an e-portfolio can be used for a variety of purposes, including:

a) Awareness andSetting Objectives: - E-portfolios can be quite useful in raising awareness and comprehension of one's own digital identity development. It gives students the chance to examine their own strong and weak points and reflect on their achievements. It also gives them the ability to develop an e-portfolio as they add new objectives to their collection and revisit previous ones. This record of personal and professional development aids people in analysing their learning over time, increasing knowledge retention, refining short- and long-term goals, and gaining insights they may have missed if their progress had not been kept in such a manageable and visual manner.

b) Academic Guidance and Recommendations: - Students and instructors are the ones who use e-portfolios.

- **Interactivity:** - Ideally at all times, e-portfolios will be available to students and advisors. Therefore, advisors will real time. Students and advisers may also be able to communicate via the chat or comment features of the e-portfolio, depending on the platform.

Performance Tracking: - Sometimes, academic advisory may have only a list of grades to help them review the success of the students in their classes. Through e-portfolios, counsellors can understand more analytically how students are doing on their academic journey.

Planning: - E-portfolio allows for more in-depth feedback from academic advisors to help students achieve their goals. For example, if a student fails, an advisor can review entries in their goals. For example, if a student fails, an advisor can review entries in the e-portfolio to decide which tools would most benefit the student, or if a student performs well, an advisory may be able to point the student to a course that will allow him or her to sharpen those skills.

Portability: - Assuming that students will ideally be able to take their e-portfolios with them from year to year ok even from year to year or even from school to school, academic advisors can use e-portfolios to determine the progress of students even though they have never already communicated with a student.

Institutional Uses: - Use e-portfolios can also benefit University administrators. Schools can use e-portfolios of students to prove that they meet those criteria for accreditation or internal academic standards. Furthermore, department heads can use e-portfolios to assess the weakness and strengths of their students so that they can make curriculum changes if necessary to help them achieve the desired results are better. In this way, e-portfolios provide administrators with

such a strong centralized source to turn to when assessing the performance of students and the effectiveness of their programs.

Career Uses:- For the overall growth and development and for the showcasing your skills e-portfolio plays the vital role in career.

Conclusion: -

E-portfolios are valuable flexible tools with several educational and career benefits. E-portfolio provides opportunity to professionals to improve success, receive feedback and plan their best work to show future employers or clients. Institution can introduce a platform with the right planning and foresight that allows professionals to enhance their professional development. Consistent efforts and self-discipline of professionals, can convert into effective use a portfolio to become advance throughout a career and to make one stand out from the crowd.

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Massive Open Online Course (MOOCs) for Library and Information Science Education

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Abstract:-

In this online learning environment Massive Open Online Courses (MOOCs) is an innovative addition. This innovative and useful landscape growing and spreading fast in different discipline, Library and information science (LIS) is not exception for that. LIS professionals have a better opportunity to learn new skills with the help of MOOC. This paper focused on MOOC concept, Library and information science and MOOC, Benefits of MOOCs, Need of MOOC in LIS and Contribution of SWAYAM in MOOC with reference to Library and information Science.

- **Keywords:-** MOOCs, LIS, SWAYAM

- **Introduction :**

During year 2020, total more than 385.36 lakhs students were enrolled in higher education. To meet the growing demand for higher education the number of distance education programmers is growing up in India. On other hand due to digital movement in India the country has gone through various waves of transformations. The Government initiatives like National Knowledge Commission, Digital India and so on, have encouraged people to move towards digital way of communications, accessing and delivering services on e-platforms. The significant changes in use of the technology in online education has seen emergence of the concept of Massive Open Online Courses (MOOCs). Which the Indian higher education sector has also facing **(Joshi, 2020)**. Term MOOC was coined by Dave Cormier (UniversityofPrinceEdwardIslandIn 2008 University of Manitoba first time delivered Massive Open Online Courses (MOOCs)) **(Yuwan Li and Powell Stephen, 2013)**. The credit course was offered by University for regular student as well as for off-campus students. Since then MOOCs was initiated. MOOC is now favorite trend in the online learning environment.

- **Overview of MOOC :**

MOOC contains the terms like Massive Open Online,and Courses respectively. Platform like Edukart, EdX, Udacity, Coursera are popular now a days. For educating people and to reduce cost of education with consideration of affordability of students, higher educational institutions are implementing MOOCs. MOOCs are online courses designed for large numbers of registrations of participants, the can access their courses from any corner of the world with the help of internet. Most of the courses on MOOCs are used to be delivered free of cost but some course providers charging nominal fees for certificate. Nature of the MOOCs contains audio –video lectures, text documents in the form of notes, discussion forums and time bound assessment of progress in the form of quizzes and descriptive question - answers.

- **List of MOOC Providers (Vijaykumar, 2019)**

S.No	Provider	Type	Headquarters	Founded
1	ALISON	Commercial	Ireland	2007

2	Canvas Network	Commercial	USA	2008
3	Coursera	Commercial	USA	2012
4	Coursmos	Commercial	USA	2014
5	edX	Non-profit	USA	2012
6	Eliademy	Commercial	Finland	2012
7	FutureLearn[1][2]	Commercial	UK	2012
8	iversity[3]	Commercial	EU	2013
9	Kadenze	Commercial	USA	2015
10	Khan Academy	Non-profit	USA	2006
11	Lynda.com	Commercial	USA	1995
12	NPTEL	Non-profit	India	2015
13	OpenClassrooms[4][5]	Commercial	France	2007
14	openHPI[6][7]	N/A	Germany	2012
15	OpenLearning[8]	Commercial	Australia	2012
16	Open2Study[9][10]	Commercial	Australia	2013
17	Peer to Peer University	Non-profit	USA	2009
18	POLHN	Non-profit	Western Pacific Region	2005
19	Shaw Academy	Commercial	Ireland	2013
20	Stanford Online	Non-profit	USA	2006
21	SWAYAM	Non-profit	India	2015
22	Udacity	Commercial	USA	2012
23	Udemy	Commercial	USA	2010
24	WizIQ	Commercial	India/USA	2007

- **Characteristics of MOOCs:-**

1. Openness – Everyone, anytime, form anywhere and n number of people can join.
2. OnlineDelivery of content- with the help of Internet and with the help of audio visual and multimedia.
3. OnlineEvaluation and Assessment
4. Course content videos
5. Evaluation Through Quizzes and Assessment
6. For Doubts clearance availability of discussion forum.

- **Need of MOOCs in Library and Information Science:-**

Library and information science has multidisciplinary approach. Total 105 universities provide bachelor of Library and information science courses, 78 universities provide the Master of library and Information science courses, 45 Universities provide PhD in Library and Information Science (**Joshi, 2019**). 21st century's librarian is updated with ICT Skills, Management skills, social skills etc. We need to go with MOOC for the betterment of academic fraternity and there is need to upgrading the skills. Through the MOOCs platform, LIS professionals and students can play theroles of learner, connector, and collaborator in a self-directed yet social learning experience. MOOC can serve as a transformative environment for professional development **(Bansod, 2019).**

The role of libraries is changing. Traditional libraries were more of a store house of knowledge. But over a period the role has changed from storing to disseminating knowledge and even the means of providing knowledge to peoples. In academic environment the governing bodies like University Grants Commission (UGC) and All India Council of Technical Education (AICTE) have taken wise steps towards making campuses user friendly by implementing policies of use information and communication Technology (ICT) on campuses. The part of it has also promoted E-resources in libraries to make user easy to access knowledge resource.

- **Why LIS should part of MOOCs?-**

a) With simple user registration process user can create user account and can find course according the interest area. Most of the courses are free or available with nominal charge. Users can get valuable course content of renowned institutes at free of cost.

b) Content videos are very pin pointed and shorts strict with the core theme. Learner can pause; rewind and play the video in case of learner miss out any valuable information, which is not possible in regular lectures.

c) Course content, theme, material is combined with user's studies or job because MOOC is very easy as the user can control his pace of learning.

d) MOOCs help to save money and time.

e) MOOC offered courses can be more interesting and interactive because of the multimedia mode. Learner can use any gadgets like computer, tabs, mobile phones etc. so they can assess anytime and anywhere.

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- **Benefits of MOOCS:-**

1) Gaining expert learning for the field.
2) Efficacyofonline learning
3) Downloadable course content/ Material
4) Diversity in teaching- learning.
5) Assessment helps to enhance learning.
6) Helps to increase focus and attention.
7) Peerassistance, out–of–bandlearning
8) EnhancetheeducationandresearchinLIS

- **Areas of Application of MOOCs in LIS Profession**

i. Theory of Library and Information Science
ii. ICT Applications for Libraries
iii. Practices in LIS
iv. Continuing of Education
v. English Communication/General Learning
vi. Students Centric Learning
vii. HelpsinTrainingandMotivation
viii. Best Tool for Board of Studies

- **MOOCs in Indian Scenario:-**

In the beginning IGNOU has developed National Digital Repository i.e. eGyankosh. Ministry of Human Resource Development has developed Sakshat Program for National Mission on education through ICT. For delivery of e-content to faculties of colleges, central and State

Universities Indian Government was developed 'One Stop Education Portal'.Shiksha Program was developed by CBSE for delivering the electronic content to the class XI and XII students. INFLIBNET developed e-PG Pathshala in 2013 for e-learning platform for audio –video content for PG curriculum. Indian Institute of Technology. Bombay developed three courses for EdX platform. IIT Delhi, IIT Kanpur, Birla Institute of Technology and Science (BITS) Pilani, Lady Sri Ram College, Jagga, IISC Banglore, Visvesvaraya Technological University etc is involved in developing MOOCs. Major change happened after MHRD Government of India announced a portal for MOOCs called SWAYAM i.e. "Study Webs of Active-Learning for Young Aspiring Minds".

SWAYAM

For bridging the gap between digital divide SWAYAM was developed to achieve 3 important education policy objectives access to education; equity in education and quality in education. The main aim of the SWAYAM platform is to deliver quality education. Nine national coordinators have been identified who are working and dealing with a particulararea.

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1. AICTE (All India Council for Technical Education) for self-paced courses
2. NPTEL (National Programme on Technology Enhanced Learning) for Engineering
3. UGC (University Grants Commission) for non-technical post-graduation education
4. CEC (Consortium for Educational Communication) for under-graduate education
5. NCERT (National Council of Educational Research and Training) for school education
6. NIOS (National Institute of Open Schooling) for school education
7. IGNOU (Indira Gandhi National Open University) for out-of-school students
8. IIMB (Indian Institute of Management, Bangalore) for management studies
9. NITTTR (National Institute of Technical Teachers Training and Research) for Teacher Training programme. (**SWAYAM Website**)

Currently, the contents of the SWAYAM are made available in two languages English and Hindi, now government is trying to provide courses in regional languages like Marathi, Malayalam etc.

Courses Offered by SWAYAM in 2021-22 in Library and Information Science

Sr No	Course Title	Level	Course Start Date	Duration	Credits	Offered by	Course Type
1	Libraries: An introduction	Diploma	1st Feb 2022	12 Weeks	4	IGNOU	Core
2	Library Information and Society	Postgraduate	1st Feb 2022	12 Weeks	4	IGNOU	Core
3	Communication Skills (Library and Information Science)	Postgraduate	1st Feb 2022	12 Weeks	4	IGNOU	Core
4	Management of Library and Information Center	Postgraduate	1st Feb 2022	12 Weeks	4	IGNOU	Core

5	ICT in Libraries	Postgraduate	1st Feb 2022	12 Weeks	4	IGNOU	Core
6	ICT in Libraries	Diploma	1st Feb 2022	8 Weeks	4	IGNOU	Core
7	Information Sources and Library Services	Diploma	1st Feb 2022	12 Weeks	4	IGNOU	Core
8	Digital Library	Postgraduate	3rd Jan 2022	15 Weeks	5	CEC	Elective
9	KOHA Library Management System	Undergraduate/ Postgraduate	27 April 2020	12 Weeks	2	AICTE	Elective
10	Library Automation and Digitization	Diploma	1st Feb 2022	12 Weeks	4	IGNOU	Core
11	Sr.Secondary: Library &Information Science	School	1st Nov 2021	24 Weeks	10		Core

Conclusion:-

Massive Open Online Course is offering huge opportunities as well as challenges before library and information science professionals. Library professionals must explore these possibilities in most effective way. It is very difficult to provide all aspect and skill of library and information science to students; especially indeveloping countrieslibrary training institutes face certain challenges such as shortage of teachers, funds, skill levels, availability ofresources and infrastructure. In spite of their disruptive nature, it is true that MOOCs will bring in an opportunity forlibrarianship in improving the LIS education and skills of library professionals. So courses like MOOCS will be more effective and cost benefited to all learner to enhance their skills in effective ways.

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EFFECT OF PILATES EXERCISE TRAINING PROGRAM ON FLEXIBILITY VARIABLE OF INTERCOLLEGIATE FOOTBALL PLAYERS

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Abstract:

The purpose of this study was to examine the effect of Pilates exercise training program on flexibility variable of intercollegiate football players. It was an experimental study in which pre-test & post-test non equivalent groups design was used. 30 intercollegiate boys football players were selected as sample by using simple random sampling technique total intercollegiate boys football players (n=30) from Christ college, Pune. They were equally divided into, Experimental group (n=15) & Control group (n=15) boys. Sit & reach test was conducted on both the groups obtained data was analyzed by using Independent sample t-test. Result Pilates exercise training program was useful to improve flexibility variable. This shows the significant effect at 0.05 level thus researcher concludes that there was significant improvement on flexibility performance of experimental group as compared to control group due to the treatment given.

Keyword: Pilates exercise program, Sit and reach test & Intercollegiate Football players.

Introduction:

Physical activity is as old as human life. A primitive man, even watchful of his foes, had to keep himself in a very good physique for his survival. Therefore the primitive man had to be a good footballer, runner, thrower and swimmer for his existence. Movement is the basis for life and growth, which is realized through physical activities, sports and games. It is a natural and an inborn quality of a child to involve himself in physical activities right from the day of his birth. Play is an integral part of our nature. It helps us to develop physically, mentally, and socially.

"Physical fitness can neither be achieved by wishful thinking nor outright purchase."

- Joseph H. Pilates

The football players most of involving themselves in physical exertion every day by sweating hours at ground practice, gyms with conditioning or football matches in clubs or playing some games apart from recreation. In the college life students those participates in the sports in that situation students play double role academic achiever with sports players. In this study researcher selected sample of intercollegiate football player those studied in the Christ college and played intercollegiate matches and regular practice in ground. The footballers need exercise and physical fitness has been extremely emerging as one of the weapon to protect our body from the diseases and to stay healthy and to live a better life. In the recent many new forms of exercise have been discovered for rehabilitation or fitness purpose among them one of the popular exercises is the Pilates exercise. In recent years many athletes have been paying attention to the Pilates exercise for the conditioning purpose and also included in their exercise routine, which helps them to enhance their performance in the competition. Researcher was made Pilates training program for intercollegiate footballer to regularly practice to implement his scheduled. Pilates improves mental and physical well-being, increases flexibility also strengthens muscles through controlled

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THE EFFECT OF SIX WEEKS PRO AGILITY CONDITIONING DRILLS TRAINING PROGRAM ON AGILITY SKILL PERFORMANCE OF COLLEGIATE HOCKEY PLAYERS

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Abstract:
The purpose of this study to determine if six weeks pro agility conditioning drills training program on improve collegiate hockey players agility skill performance. It was an experimental research. In this study true experimental pre-test and post-test equivalent group design was used. The sample of this study was collegiate boys hockey players n=20 among that (Players each 10 experimental and control group) the age group of below 20 years those selected through simple random sampling technique from Arihant college of arts, commerce & science, Pune district. For this study independent variable is pro agility conditioning drills training program and dependent variable is agility skill performance. The selected sample of hockey players both groups experimental and control pre-tested by Illinois agility test was conducted. After pro agility conditioning drills training program implemented for six weeks for experimental group and control group did not part any treatment after accomplished implement training plan post-test conducted on both the groups and data collected. Both groups collected data analyzed by using Independent sample t-test and interpretation drawn. Result shows that from the finding after that post-test significant reduce the time of agility skill performance basis of concluded researcher significant improvement of Illinois agility skill performance of experimental group compared to control group due to the treatment of training program.

Keyword: Pro Agility Conditioning Drill Training Program, Agility Skill & Hockey Players.

Introduction:

Field hockey is a field invasive team game played by both the sexes; it played indoor and outdoor. It is a popular and dynamic team game in which players compete at the same field of action as their opponents; field hockey has a long history but nowadays rules and playing surface are changed. A rule change (means that all matches were played in four 15-minute quarters) and the rules of the game allow unlimited substitutions. That's why modern field hockey demands that each player of the team be able to play in all play positions. To achieve the best possible performance in field hockey we need player readiness in the early age group. Field hockey were requires high specific strength, speed, agility, and flexibility for achieving success at elite level of competitions.

Vital of each Field positions:

Yadav, S. K., Prajapati, S. K., & Mishra, M. K. (2015). Agility of high and low achievers male hockey players of Banaras Hindu University: A comparative. International Journal of Physical Education, Sports and Health 2015; 1 (5): 23-24.(Online).

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Influence of bank-specific and macroeconomic factors in determining nonperforming loans in Africa

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Abstract

This study aims to understand the various bank-specific and macroeconomic factors which might lead to loan default in Africa: Botswana, South Africa, Namibia, Nigeria, and Tanzania. Panel data techniques are used to analyze and quantify the impact of macroeconomic and bank-specific variables on NPLs. The proposed hypothesis has been tested using the correlation coefficient and multiple linear regression. It is suggested that banks consider the macroeconomic conditions, this practice can help the management of the banks to estimate the overall performance of their loans and avoid increasing the risk of default.

Keywords – *Nonperforming loans, loan size, repayment period, lending interest rate, employment rate, GDP growth rate, inflation rate*

1. Introduction

The banking system plays a major role in contributing to the economic growth of a nation. It is known that bank facilitates, and loans are considered to be mutually beneficial, but it is also known that banks bear a major risk of loans becoming non-performing. For developing countries, the role of banks becomes even more important as it would help in the growth and sustainability of the country (Ikram *et al*., 2016). A banks financial stability can be best described through its ability to overcome non-performing loans since it shows that the bank can overcome credit risk, resources allocation efficiency and operational skill. If the payment of the interest due haven't been made in 90 days or more, it is considered to be a non-performing loan. Financial distress refers to a point when a borrower is incapable to meet a payment obligation to lenders and other creditors (Zaki*et al*., 2011). When the economy is showing an upward trend then the asset prices increase, loan recovery rates increase, and overall NPLs decrease (Bardhan and Marjit, 2016). Different banks in the same economy handle their NPLs differently, as it appears that bank-specific factors play a significantly larger role in the creation of non-performing assets (NPAs) over time, as these factors directly influence a bank's health. (Bardhan and Mukherjee, 2016). Loan characteristics like loan decision making process, loan default management, loan recovery processes, risk exposure and most importantly the performance depends on different

banks. All of these elements have varying effects on NPAs. In light of the foregoing, this work has two objectives. Firstly, we will investigate how loan characteristics affect loan default in Africa. To do this we will study the impact that loan amount, loan term and interest rate will have on loan default. Secondly, we will determine the effect of macroeconomic factors on loan default and to do so we will study the impact of employment rate, inflation rate and GDP growth rate on loan default (Climent-Serrano, 2017). To do this we will construct a multiple linear regression model with a sample period from 2011-2020.In this regard, this study attempts to answer the following unanswered questions:

RQ1) Would bank specific factors influence loan default?
RQ2) Would macroeconomic factors determine loan default?

2. Literature Review

2.1NPAs and Inflation | NPLs and GDP growth rate

Bardhan and Mukherjee (2016)examines the duty of the bank-specific variables for describing the complexities of NPAs banks in a panel data approach. The mentioned results have been obtained by managing the factors of macroeconomic like inflation and real GDP. Using Generalized Method of Moments (GMM) approach in dynamic models, it was found that there is considerable time resilience of NPAs in the existing banking system. The conclusion reached signifies those bigger banks have more defaults than smaller banks. Abusharbeh (2020) has examined the impact of the bank-specific variables as well as macroeconomics on the credit quality on Palestine since the last decade. The impact of macroeconomic and financial variables is analyzed and quantified using panel data techniques. Housing prices, bad debt coverage, interest rates, and solvency have been proved to be the variables that have the biggest impact on NPLs, according to the research. This leads to the hypotheses:

H_1 Inflation has a significant influence on Loan default
H_2 GDP growth rate has a significant influence on loan default

2.2 NPLs and Loan Size | NPLs and Employment

Using variables reflecting loan and borrower characteristics, industry and macroeconomic conditions, and many recovery process variables, Khieu et al. (2012) generated a model for bank loan recoveries. Following Acharya et al. (2007), the paper used the ordinary least squares method using Huber White (1980) heteroskedasticity-consistent standard errors, taking into account a linear relationship between recovery rates and explanatory variables. Loan features are more important drivers of recovery rates than borrower characteristics prior to default, according to the findings.. Mota et al. (2018) used a sample of 752 microcredit loans granted in Portugal by the National Association for the Right to Credit, which used individual lending mechanisms and granted loans in collaboration with several credit institutions, to investigate the determinants of microcredit loan repayment. An ordered logistic regression was used in this investigation, with a dependent variable consisting of four discrete meaningful sequentially ordered categories. Loan size, on the other hand, had a statistically minor impact on borrowers' ability to repay their loans. This leads to the following hypotheses:

H_3 Loan size has a significant influence on loan default
H_4 Employment rate has a significant influence on loan default

2.3 NPLs and Lending Interest Rate | NPLs and Repayment period

Olarewaju (2020) sought to investigate the factors firm specific and macroeconomic factors accountable for non-performing loans. From 2010 to 2017, a total of 880 data sets were analysed using the dynamic panel regression analysis on 110 commercial banks from nine countries. Louzis et al. (2010) investigated the factors of nonperforming loans in the Greek banking system, looking at each loan category separately (consumer loans, business loans and mortgages). Ahmed et al. (2021) investigated the macroeconomic and bank-specific determinants of NPLs for commercial banks from 2008-2018. Ikram *et al*.(2016) empirically explored the determinants of NPLs, small and medium enterprises (SMEs) held by the commercial banks. Zhang et al. (2017) did an empirical study to evaluate the factors that influence the likelihood of acquiring a loan in online peer-to-peer lending. Kiros (2020) looked at the elements that influence loan settlement for MSEs sponsored by Somali Microfinance Institutions, taking lender characteristics into account. Kurniawan and Wijaya (2020) investigated whether the loan granted factor influences

lending in Indonesia. According to the findings of binary logistic regression, the loan repayment period, grace period, and loan release timeliness all have a statistically significant impact on loan repayment. This leads to the following hypotheses:

H_5 Lending interest rate has a significant influence on loan default
H_6 Repayment period has a significant influence on loan default

3. Research Methodology

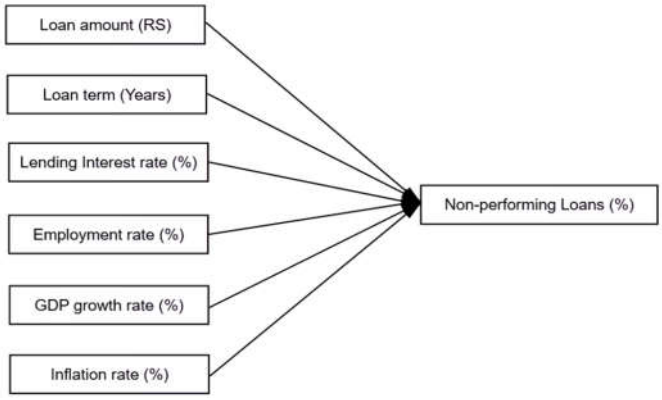
3.1 Data Source

The purpose of this paper is to identify the signification of macroeconomic and loan determinants of NPLs in Africa. In this regard, to enhance the generalizability of the findings, Africa is divided into three clusters, namely: underdeveloped, developing,and developed nations using the Human Development Index (HDI). The result form this paper will help us in understanding the causes of credit risk in Africa (Abusharbeh, 2020). Countries with an HDI score greater than 0.800 are categorised as very high, those with a value between 0.700 and 0.799 are classified as high, those with a value between 0.550 and 0.699 are classified as medium, and those with a value less than 0.550 are defined as low.. In this backdrop, South Africa and Botswana were selected as the developed country, Namibia was selected as developing country, and Nigeria and Tanzania were selected as underdeveloped country. Time series panel data were gathered from the World Bank for a period of 10 years, from 2011 to 2020. The data regarding loan size and repayment period is created hypothetically.

3.3 Model Construction

Most previous studies that addressed the factors causing NPLs have used the multiple linear regression model, considering macroeconomic and bank specific characteristics acknowledging the findings of previous studies, this paper employs a balanced panel data analysis to pool the cross-sectional data over a certain period in order to identify the determinants of credit risk (Abusharbeh, 2020). Further this paper uses correlation and multiple linear regressions with the help of its dependent variable and independent variable.

Figure 1: Conceptual Framework



4. Analysis and Discussion

The statistical test has been done for five different countries that are Botswana, South Africa, Namibia, Nigeria and Tanzania. Table 1 shows the correlation result for 6 countries between the dependent variable (NPLs) and independent variable (GDP growth rate, Inflation rate, Employment rate, Lending Interest rate, Loan size and Repayment period).

4.1 Botswana

In case of Botswana, loan amount has a significant influence on NPLs as the significance value is 0.035 which is less than 0.05. Thus, we can accept H_3. Chaudhary and Ishfaq(2003) argue that the higher loan amounts are, the stricter the terms and conditions attached to loan contracts need to be (e.g., higher interest rates). The p-value is positive thus

suggest a positive relationship between Loan size and NPLs, thus increase in loan size leads to increase in NPLs and vice versa. Loan term has a significant influence on NPLs as the significance value is 0.007 which is less than 0.05. Thus, we can accept H_6. The reason for the same can be if the repayment period is longer the borrower might use the funds for other purposes hoping to repay the loan from later cash flows (Roslan&Mohd, 2009). Thus, it can lead to loan default. Employment rate has a significant influence on NPLs as the significance value is 0.003 which is less than 0.05. Thus, we can accept H_4.

4.2 South Africa & Namibia

Loan size, lending interest rate, and employment rate all have a substantial impact on NPLs in South Africa, with a significance of less than 0.05. Thus, we will accept H_3, H_4, and H_5. Also there exist a positive correlation between loan size, lending interest rate and employment rate and the NPLs. On the other hand, repayment period, GDP growth rate and inflation rate does not have a significant influence on NPLs as the significance is more than 0.05. Thus, we will reject H_1, H_2, and H_6. The situation in Namibia is like that in South Africa, in that loan size, lending interest rate, and employment rate all have a substantial impact on NPLs, with a significance of less than 0.05. As a result, H3, H4, and H5 will be accepted. A positive association exists between loan size, lending interest rate, and employment rate, as well as NPLs. On the other hand repayment period, GDP growth rate and inflation rate does not have a significant influence on NPLs as the significance is more than 0.05. Thus, we will reject H_1, H_2, and H_6. Also there exist a negative correlation between inflation rate and NPLs. Both GDP growth rate and repayment period has a positive correlation with NPLs.

Table 1 – Correlation Coefficient

African Nations		Loan amount (RS)	Loan term (Years)	Lending Interest rate (%)	Employment rate (%)	GDP growth rate (%)	Inflation rate (%)
NPL (%) Botswana	Pearson Correlation	.666*	.789**	-.874**	.829**	-0.221	-.858**
	Sig. (2-tailed)	0.035	0.007	0.001	0.003	0.539	0.002
	N	10	10	10	10	10	10
NPL (%) South Africa	Pearson Correlation	.835**	-0.424	.666*	.789**	0.561	0.174
	Sig. (2-tailed)	0.003	0.222	0.035	0.007	0.092	0.63
	N	10	10	10	10	10	10
NPL (%) Namibia	Pearson Correlation	.835**	0.475	.666*	.789**	0.606	-0.302
	Sig. (2-tailed)	0.003	0.166	0.035	0.007	0.063	0.397
	N	10	10	10	10	10	10
NPL (%) Nigeria	Pearson Correlation	.653*	.740*	0.337	-0.533	-.655*	.843**
	Sig. (2-tailed)	0.04	0.014	0.342	0.113	0.04	0.002
	N	10	10	10	10	10	10
NPL (%) Tanzania	Pearson Correlation	0.434	0.584	.835**	-.632*	-0.159	-.672*
	Sig. (2-tailed)	0.21	0.076	0.003	0.05	0.662	0.033
	N	10	10	10	10	10	10

4.3 Nigeria & Tanzania

In the case of Nigeria loan size, loan term, GDP growth rate and inflation rate has a significant influence on NPLs as the significance is less than 0.05. Thus, we will accept H_1, H_2, H_3, and H_6. Also there exist a positive correlation between loan size, loan term, and inflation rate and NPLs. The pace of GDP growth and the number of nonperforming loans (NPLs) have a negative relationship. However, neither the lending interest rate nor the employment rate have a major impact on NPLs. As a result, we will rule out H4 and H5. There is also a negative relationship between unemployment and NPLs. Furthermore, there is a positive relationship between lending interest rates and nonperforming loans. Because the significance is smaller than 0.05, the lending interest rate, employment rate, and inflation rate all have a substantial impact on NPLs in Tanzania. Thus, we will accept H_1, H_4, and H_5. Also there exist

a positive correlation between lending interest rate and NPLs. Employment rate and Inflation have negative correlation with NPLs. On the other hand, loan size, repayment period, and GDP growth rate do not have significant influence on NPLs. Thus, we will reject H_2, H_3, and H_6. Loan size and repayment period have a positive correlation with NPL, and GDP growth rate has a negative correlation with NPLs. From table 1, we can say that GDP growth rate is the least significant variable that has an influence on NPLs as it has significant influence on NPL only in Nigeria. Out of the five African countries studied, lending interest and employment rate had a substantial impact on NPLs in four of them.

5. Discussion and Conclusion

5.1 Theoretical & Practical Implications

The current study evaluates a variety of categories, including macroeconomic and bank-specific characteristics, to determine the best leading indicators for NPLs in Africa (Botswana, South Africa, Namibia, Nigeria, and Tanzania). The study hypotheses about these types of variables' comparative and predictive performance are established and empirically tested. The study concluded that relying just on macroeconomic variables to forecast the progression of NPLs is ineffective. Therefore, researchers should also consider widening the scope of variables used. Most of the studies done in this area have taken one country but this paper has taken not only 5 countries but also taken high developed, medium developed and low developed countries based on their HDI. This study examined various bank specific and macroeconomic factors that can affect NPLs. The main contribution of this paper is the econometric analysis results, which may help in a better understanding of the variables that influence NPLs. We underline that both bank management at the micro level and indicator management at the macro level are equally crucial for controlling NPLs in banks. Banking regulations play a major effect. It should focus on bank-specific parameters like loan size and lending interest rate.. The accumulation of nonperforming assets (NPAs) might be greatly reduced if bank-specific issues were managed more effectively. The most significant components of credit management have been identified as risk identification and measurement; consequently, more time and effort should be spent assessing how much risk borrowers bring to bank loans. As a result, managers must accurately assess the ability to repay the loan. Furthermore, it is critical to stress the importance of risk management while also attempting to strike a balance between customer acquisition, which comes at the cost of the risk of default and the loss of any interest income the customer could have brought, and the loss of any interest income the customer could have brought.When managing the credit risk of their loans to avoid the possibility of increasing defaults, bank management can refer to the performance of macroeconomic conditions to predict the performance of their bad loans. To maintain stability in banking sector it is important that regulatory authorities establish more effective monitoring process. To detect banks with potential NPLs, regulatory authorities should focus on managerial performance. In order to avert future financial instability, authorities should focus on banks' risk management systems and procedures. In terms of lending policy, banks are advised to apply a strict credit policy to all borrowers. When banks loosen their policies, they lose bargaining power with borrowers. Banks with low capitalization ratios must reassure their managers that they are not taking too many risks.

5.2 Conclusion, Limitations, and future scope

Bad loans are an important indicator of a bank's asset quality, financial health, and thus liquidity and solvency. In Botswana, South Africa, Namibia, Nigeria, and Tanzania, this study looks at some of the important macroeconomic and bank-specific factors affecting non-performing loans. The results of this study indicate that loan amount, repayment period, lending interest rate, employment rate, GDP growth rate, and inflation rate have a significant impact on NPLs. We found out that lending interest rate and employment rate are the most significant factor influencing NPLs in Africa. On the other hand, GDP growth have been the least significant factor influencing NPLs as it was only in the case of Nigeria where GDP growth rate significantly influenced NPLs. The limitations in the paper provide opportunity for further research. The data was collected only for 10 years' time period, as more the time period more reliable the result will be. Loan amount and repayment period data were created hypothetically. The study only examined bank specific and macroeconomic factors influencing NPLs. Future researchers may consider the microeconomic factors in the emerging financial markets. As this paper has studied limited banking indicators as contributing factor of NPLs, future investigators should contemplate more attributes as determinants of the NPLs to perform more in-depth analysis. Moreover, cross-cultural impacts on NPLs can also be examined.

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Impact of Social Media Marketing on the Buying Behavior

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Abstract

The influence of social-media marketing aspects such as entertainment, electronic word-of-mouth (EWOM), brand awareness, and brand image on purchase intention is investigated in this research. The study collects data from social-media users in India via an online survey, as shown by their consumption of various items and companies that they follow on social media. We employed Pearson's correlation and regression, as well as buy intent, to analyze social-media marketing features. The findings show that interaction, electronic word-of-mouth, brand awareness, brand image, and entertainment are the primary components directly impacting customer purchase intention, which are subsequently strengthened. This study adds to the body of knowledge in the field of branding by elucidating the function of social-media marketing aspects in the brand-building process. Social media is a marketing tool that is known for its ability to effectively communicate brand-related information as well as its ability to increase customer brand engagement and understanding. However, it needs to be seen how useful these aspects are for these reasons. This study indicates that key social-media marketing factors, such as EWOM, Brand Image, Brand Awareness, and Entertainment, are essential drivers in the brand growth process in India by experimentally evaluating a theoretical model.

Keywords – *Purchase intent, brand knowledge, brand awareness, brand image, social media marketing*

1. Introduction

Consumer–brand engagement (CBE) is a field of marketing research that generates a lot of buzzes. According to reports, over 80 per cent of marketers will strive to engage customers by creating a connection with them over the next five years, based on the assumption that engaged customers will visit 40% more per year, resulting in a 20% increase in profitability (Pansari & Kumar, 2017). CBE is a multidimensional construct with cognitive, emotional, and behavioral dimensions. It is defined as "consumers' brand-related cognitive, emotional, and behavioral activity related to the point of focus. The importance of CBE for marketing strategy, which includes the building and strengthening of consumer-brand connections, has sparked research attention. CBE is an influential construct for building customer satisfaction, brand love, and brand loyalty (Islam & Rehman, 2016) as well as improving advertising effectiveness and overall business performance (Hollebeek, 2011). It is recognized as a priority area for marketing research. As a result, marketers want to learn more about how CBE is developed, maintained, and preserved (Calderet, 2016); As a result, there have been various calls to supplement the theoretical framework that underpins CBE with empirical evidence (Islam & Rehman, 2016), particularly in terms of its antecedents and consequences. An empirical study on CBE is needed to improve knowledge of the construct, drivers, and approaches, such as marketing communications for recruiting and maintaining customers through improved consumer-brand relationships. Social media marketing (SMM), which is inextricably linked to CBE, entails creating and implementing marketing strategies that use a social media platform to communicate brand-related information and manage customer relationships (Schultz & Peltier, 2013). Indeed, social media marketing (SMM) has a significant impact on customer impressions of companies (Kellar, 2009), and consumers are increasingly accessing social media channels to get information about products and brands (Mangold & Faulds, 2009). As a result, some people believe that social media marketing (SMM) enhances customers' brand understanding more than traditional marketing communications (Bruhn, 2012). The ongoing difficulties marketers confront in developing and implementing SMM strategies that are both engaging and helpful to consumers justifies more research into SMM content and consumer experiences (Schultz & Peltier, 2013). The present request for an empirical study on the techniques and content that lead to CBE and CBE consequences is an example (Brodie, 2011). Brand knowledge, which is thought to generate economic success by influencing brand loyalty, purchase intention, and competitive advantage, is one consequence that has piqued academic attention (Chiang, 2017). The paper investigates the influence of social-media marketing components such as entertainment, electronic word-of-mouth (EWOM) on consumer–brand engagement and brand knowledge are investigated in this research paper. We will then conduct a thorough analysis of the pertinent literature to identify research gaps. As a result, a theoretical model is developed in which SMM aspects operate as drivers of CBE and brand knowledge. The methods utilized and the findings of the empirical study are then discussed, and the implications for theory and practice, limits, and future research prospects.

2. Literature Review

CBE is a new idea in marketing that is gaining traction in both the practical and academic worlds (France et al., 2016). The term refers to “consumers' distinctive level of cognitive, emotional, and behavioral involvement in brand encounters.” Bowden (2009) found that self-brand connection, buying intent, and other factors influence customer behavior. brand loyalty and intent. CBE's antecedents and consequences, as well as the possibility for CBE, are discussed in the literature. CBE and brand knowledge are influenced by social media brand communication. As an example, According to Schultz & Peltier (2013) social media should be an effective medium for sharing information enhancing consumer-brand interaction, hence adding to CBE (Barger, 2016) suggested following Schultz & Peltier (2013) recommendations, pay close attention to information on social media to encourage customers to participate with postings in the brand's social-media-based community, so bolstering CBE Consumers are more likely to engage with brand posts that feature sensations that are less commercial in character, according to (Swami, 2013). According to (Dessart, 2015), interactive and multimodal brand posts have a good impact on CBE, hence increasing brand popularity. In conclusion, through social media platforms, firms may improve CBE and develop consumer–brand interactions (DeVaries & Carlson, 2014). In the marketing realm, the prevailing perspective of CBE advocates for the interactive consumer experiences and customer–brand interactions which is extremely important. in today's world (Hollebeek, 2011), as well as promising CBE.

H1. Entertaining SMM content is related to consumer–brand engagement.

As social media has grown in popularity, so has the number of customers who assess businesses and goods based on EWOM. This, according to (Krishnamurthy & Kumar, 2018), has a favorable impact on customers' evaluations of goods and services, motivating them to put in more cognitive effort while reading EWOM. Given the

perceived trustworthiness of social-media platforms (Islam & Rehman, 2016), the creation and sharing of EWOM among consumers fosters a sense of closeness and emotional relationship between brands and consumers (Brodie, 2011) resulting in positive feelings (DeVaries & Carlson, 2014). As a result, EWOM's availability on social media platforms and utilization aid in the creation of pleasant brand experiences and favorable brand perceptions, emotions for the brand, thus strengthening the consumer–brand relationship.

H2. EWOM is related to consumer–brand engagement.

Brand awareness is a crucial component of brand knowledge (Keller, 2013), referring to the capacity of potential consumers to recognize/recall a brand in their thoughts, hence assisting in the acquisition of new customers connecting brands with items. Assumed to be a prerequisite for brands to exist be taken into account by customers during the decision-making process (Langaroo, 2018), brand awareness is the result of customers' recurrent and memorable interactions with a brand familiarity with the brand This exposure is brought about by the brand aspects such as the brand name, tagline, and packaging (Keller, 2013), as well as advertising in print a variety of formats (Datta, 2017), and interactive experience the CBE procedure (Brodie, 2011).

H3. Consumer–brand engagement is related to brand awareness.

Due to its effect on customers' brand choice (Cobb-Walgren, 1995), brand image is an essential aspect in the brand-building process. This positively correlates with the brand's capacity to charge a premium price (Persson, 2010). As a result, brand image can influence future earnings (Chiang, 2017), as well as brand loyalty. Consumers usually view brands with a strong and favorable brand image as a major choice in their decision-making process, according to (Keller, 2013). This motivates marketers to improve CBE via a variety of channels in order to generate powerful and favorable brand experiences (DeVaries & Carlson, 2014). Brand image refers to the set of connotations associated with a brand in customers' minds, representing how the brand is regarded in their minds and how it varies from rivals' brands (Webster & Keller, 2004). The descriptive aspects and features that characterize a brand, reflecting what people believe about it, are known as brand attributes. Consumers' perceived value (e.g., utilitarian, experiential, and symbolic) associated with brand qualities is referred to as brand benefits (Keller, 2013). Consumers' evaluations and judgements of a brand's features and advantages, as well as the composition of all relevant brand aspects and events in their memory, are referred to as brand attitude.

H4. Consumer–brand engagement is related to brand image

3. Research Methodology

A web-designed questionnaire will be designed for this research purpose by using Google Forms which can be based on a web link, which can be open in all internets operating systems such as internet explores, Google chrome, Firefox, and Mac OS, etc. Primary data were collected for address both dependent and independent variables by based an on-line survey utilizing a structured questionnaire in a web page format circulated through all leading social media platforms like Facebook, Twitter, Instagram, WhatsApp, and LinkedIn. “Moreover, referral mode was also used to increase the reach of the circulation. The research type is deductive, and variables are analyzed using quantitative tools and techniques. "Primary data will be collected through structured questionnaires with closed statements measured with Likert’s scale" (where 1 = strongly disagree and 5=strongly agree), yes/no/may be questions and some direct questions. The research focuses on complex purchases that require extended problem solving, where social media is more likely to be utilized. Respondents were then asked whether or not they had used social media in their decision-making. The questions measured the same concepts in the different contexts and differed only very little in their wording." Buyer decision feature has no dimension of objective and thus its operation is problematic. "The approach to measuring decision quality can be objective or subjective (Aksoy and Cooil, 2006)." Subjective methods are assessments of the choice-maker, taking into account things that is utmost significant for the respondent in regards to the decision. The questionnaire was designed as such so that it can measure the influence of Social Media Marketing on customer behavior on the basis of different parameter like age, income, preferences etc.

4. Analysis and Discussion

The Cronbach’s alpha for the defined variables is mentioned in the above table and the value of Cronbach’s value for each variable is greater than 0.7. The findings showed a constant degree of reliability (>.70) ranging from 0.78 to 0.88. Thus, the variables are reliable. The KMO test is used to see how strong the partial correlation between the variables

is. KMO values near 1.0 are considered optimal, whereas values below 0.5 are considered unacceptable. Most experts now consider that a KMO of at least 0.80 is sufficient for factor analysis to begin. We had a KMO value of 0.962 based on our results. This shows that the variables' degrees of information overlap significantly and the presence of a strong partial correlation. As a result, factor analysis is a viable option. Pearson's correlation coefficient is used to find the correlation level between the variables, and the correlation coefficient's value is shown in above table. We can see that there is a strong positive correlation between all the variables. Electronic word of mouth has the highest and the strongest correlation with the purchase intention while brand image also has a very strong correlation with the purchase intension. Purchase intention is the dependent variable. The hypothesis is tested using linear regression analysis. The R square value in table 1 was .419, and the p-value in table 2 was less than the significant level, i.e., 05 (significant), indicating that the hypothesis (H1) should be accepted. The R square is nearer to 0.5 it means the model is effective, and we can determine the relationship between the dependent and independent variables. As a result, regression analysis shows that Entertainment is related to purchase intention.

5. Discussion and Conclusion

Theoretical implications

SMM features have a significant impact in increasing CBE and brand knowledge, according to the findings of this study. Interaction, EWOM, and trendiness are significant factors in building CBE, which boosts brand awareness and image, according to this study, which largely confirms previous results. As a result, interaction, EWOM, and trendiness should be viewed as ways to increase consumers' cognitive processing, affection, and activation towards the focal brand, as well as being effective in influencing consumers' brand perceptions (Dessart, 2015). The influence of entertainment on CBE was non-significant, contrary to other research' findings (e.g., Barger, 2016; Carlson & DeVaries, 2014), implying that entertaining material on social media is not a brand-building component for a high-involvement product like cell phones. The data also show that CBE has a significant and beneficial influence on brand awareness and image, demonstrating the value of CBE in brand knowledge development (Keller, 2013). Furthermore, the data show that interaction, EWOM, and trendiness have indirect effects on brand awareness and image. This validates the usage of interactive social media content, EWOM, and trendiness postings in the brand-building process (Chiang, 2017). Finally, the non-significant association between personalization and CBE might be explained by the amount of engagement. The average degree of smartphone participation among respondents was 4.97 out of 7, indicating a strong level of involvement. When customers are highly active, they enjoy having more access to information about items and brands, such as extensive product comparisons, and they spend more time creating a brand expectation or notion (Chen & Tsai, 2008). According to (Krishnamurthy & Kumar, 2018) highly involved customers choose generic and complete information over personalised information. Because a smartphone is a high-involvement, standardised product, highly engaged customers want to read detailed information and specifications in their decision-making processes rather than expecting personalised information. As a result, customised SMM material may not have a direct impact.

Managerial Implications

Interaction, EWOM, and trendiness are all important SMM factors for creating CBE and brand knowledge for high-involvement items like cell phones, according to the research. As a result, marketers should gain by combining current and contemporary interactive material as well as supporting positive EWOM. This is predicted to improve consumers' brand understanding by stimulating cognitive processing, affection, and activation. As a result, marketers should let (and, if necessary, assist) customers in finding what they are looking for, such as EWOM facts on specific items (Krishnamurthy & Kumar, 2018), therefore encouraging consumers to distribute EWOM willingly. From a management standpoint, this study encourages marketers to explore using social media to improve CBE and brand awareness. SMM is a crucial tool for helping establish CBE and brand knowledge rather than merely another advertising route to reach customers. Marketers should also aim to influence consumers' emotional responses to brands by effectively managing their social-media activities, possibly by rewarding consumers who actively share their EWOM on various social-media platforms with information comprehensiveness, value-added information, and information understand ability. As a result, marketers should think about utilising a number of social media sites that allow customers to share their experiences and referrals with others. This method has been successful in boosting CBE and purchasing intention in Asian environments.

Theoretical contributions

This paper presents two key contributions in order to solve the mentioned theoretical flaws. First, the research establishes a theoretical framework for examining the significance of SMM is an important part of the brand-building process since it confirms that engagement, EWOM, and word-of-mouth are all important factors. CBE and brand knowledge are both driven by trendiness. Second, the research establishes a distinction between the relative relevance of individual SMM parts in the construction of CBE, so leading to by determining the most essential individual brand-building aspects in SMM literature construct. SMM strategies are thought to play a significant role in forming strong and favourable brand knowledge in consumers' minds, and a growing number of empirical studies have looked into the theoretical and practical implications of using SMM to build brands. However, the impact of various parts of SMM on CBE and brand knowledge is unknown, necessitating both theoretical and empirical research.

Conclusion

Social media nowadays is a powerful tool for marketers and companies to market their products. Social Media is also explored as a tool for customer engagement and feedback. Social networking sites act as forums for the introduction of new brands, items, and the exchange of information. Today's social networking sites act as incubators for innovative company concepts. Brand Image plays a vital role in word of mouth and because of people spending maximum of their time on their mobile phone or laptops it has become necessary for the brands to be active on social media to create brand awareness. Because of changes in the functioning of social media platforms, consumers' engagements with social networking sites have altered. Entertaining content nowadays catches the eye of the viewers and helps in creating brand awareness and also making the consumers buy the products. When purchasing online, customers have a wide selection of possibilities. Companies need to post engaging content and need to get feedback and surveys, so they come to know what people want. In order to maximize the impact of their efforts, firms should incorporate their social media management methods with the customer behavior aspects highlighted in this study. At last, I can conclude that social media is a powerful tool nowadays for not only marketing, but also for generating revenue.

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GOVERNMENT EXPENDITURE ON HEALTH INFRASTRUCTURE SERVICES

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Abstract *- When social welfare became a priority, government spending gained increasing importance. Government expenditure occupies a central place in the development programs of socialist countries. Government corporations are spending huge amounts in various sectors through government activities. This has been a priority in India for nearly four decades since independence. The state governments have also continued to increase their expenditure for various reasons. In the last few years and especially when the program of economic reforms was being implemented, there was a universal discussion about government expenditure again. Can the government take a firmer stance on its spending, particularly in privatization and globalization programs? The answer is mostly yes. Education, health, housing, basic services, and many other things are still justifiable to continue to spend heavily and thus public spending is justified everywhere even in today's economic structure.*

Keywords- Government Expenditure, Mathematics of Health Expenditure, Social Welfare

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Introduction-

Expenditure incurred by Central and State Governments as per budget is Government Expenditure. The government has to incur huge expenses continuously to run the administration. The government has to carry out the projects of maintaining law and order, keeping the judicial system in good condition, implementing various development programs, eradicating poverty, and increasing employment. The government performs these tasks to eliminate economic and social inequality and to protect against foreign invasions. There are varying opinions on public expenditure. In eighteenth-nineteenth century England and Western Europe, the government generally had a policy of non-intervention in the overall economy. At that time, the government strongly believed in the principle that everyone knows how to protect their interests, and thus the interests of the society as a whole will be achieved through individual interests. But this picture changed in the twentieth century. Government expenditure has come to be considered indispensable for fulfilling the responsibilities of economic development and social justice.

Research Hub International Journal

Objectives of the Research Paper –

The researcher has set the following objectives for the research paper.

1. To study the theoretical approach of public expenditure.

2. To study public expenditure and budgetary expenditure on health infrastructure.

Research Methodology –

The type of research presented is descriptive.

Data Collection –

Secondary tools are used while collecting data for the research paper. The published information is used. In this, central and state government budgets, reference books, internet have been reviewed.

Government Expenditure Theoretical Background –

Rising government spending isn't just a coincidence or an economic crisis. The philosophical and theoretical meeting has been presented by many scholars before. Adolf Wagner (1835 – 1917), a German economist, was at the forefront. He showed a direct and clear correlation between growth in the economy and growth in government activities. The growth in government transactions is faster in both of them, he argued. It implies that the relative participation of the public sector in total economic transactions will be increasing. He also explained these previous reasons. E.g. – The major reason behind an increase in public expenditure is expenditure on law and order and war. Expenditure on these matters expands due to persistent war-like situations, persistent instability, and crime in the country. New modern and expensive technology has to be used for this work. Another factor behind the expansion of spending is the growing social responsibilities of the government. Behind this are the arrangements for health, education, and literacy, welfare schemes for the poor and disabled, and projects for the welfare of senior citizens. Child welfare, women's welfare, or other matters of public welfare fall into the third category.

Wiseman and Peacock Approaches -

Both collected general statistical information on public expenditure in England for the period 1890 to 1955. According to his study, the increase in public expenditure in the country is true, but it is not straight and continuous, it is in a shocking manner and at a lower rate. As such, public expenditure is fitted within the limits of the country's gross national income; But if the pressure of public needs increases and the total revenue of the government becomes insufficient, the government has to meet the situation by increasing its expenditure in various ways.

If necessary, the expenditure has to be met by increasing the income through tax or other means. Whichever theory of public expenditure is considered, one thing is common and that is that goods or services are not provided through market mechanisms in modern societies; Their arrangement has to be done by the government deliberately by other measures. For this

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reason, the government inevitably incurs increasing expenditure on health care, education, pollution control, research, and development. In this type of service demand increases.

For example, as income levels rise during development, the demand for education services increases. In the advanced stages of development, research and development, professional education, and expansion of higher education institutions are taking place. They need a lot of money. Hospitals, universal health services, pensions for senior citizens, provision of social services, and urbanization all have to be provided for.

Mathematics of Health Expenditure -

While the government's share of total healthcare expenditure in the country is on the rise, it is encouraging, but as it is still very small compared to the Gross Domestic Product (GDP), there is a lot of room for improvement. The figures released by the National Health Accounts Office for the financial year 2017-18 illustrate the math of health expenditure. As these figures are from three years ago and the increase in public health expenditure in the last two years due to Corona does not make the NHA report out of date. Rather it shows the comparative status of health expenditure. In the financial year 2013-14, government expenditure accounted for 28.6 percent of the total expenditure on the health sector; It went up to 40.8 percent in 2017-18, an increase of over 12 percent in four years. 59.2 percent of the expenditure on health is still borne by the private sector. Only if it is reduced, the burden of health expenditure on the citizens can be reduced and the next step towards a welfare state will be taken. The proportion of individual expenditure on health by citizens has fallen below 50 percent in 2017-18. It was 48.8 percent that year and 64.2 percent in 2013-14. Although this decline is significant, it is higher than the global average (18 percent). Only if the government's share of health expenditure increases will the proportion of individual expenditure decrease. In 2017-18, the government spent 5.1 percent of its total expenditure on health. In 2013-14, it was only 3.7 percent. The need for government spending on both health and education to be significant relative to overall GDP is consistently expressed, but it is seen from the report of 'NHA' that there is a minimal increase in government expenditure. In 2013-14, the government spent only 1.15 percent of the GDP on the health sector. In 2017-18, the ratio improved marginally to 1.3 percent. There is a demand to spend at least three percent of GDP on health; The National Health Policy in 2017 also set a target of 2.5 percent, which is much higher than the same year. Everyone is feeling the huge cost of neglecting the public health sector during the Corona period. At a time when health costs are rising day by day and health insurance is far from common, an increase in the government's share of expenditure will only be healthy for social security.

The latest report of the recently published National Health Account Estimates has revealed that the state government is inexcusably neglecting an important sector like public health facilities. On the one hand, Maharashtra, which is considered to be economically advanced, shows enough concern for public health.

The latest report of National Health Account Estimates has revealed that only 0.7 percent of the total budget is spent on the health of the people of the state. There is a lack of government health facilities in many places in urban as well as rural areas of the state and the

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picture is that citizens have to rely on private doctors and hospitals. The report says that the poor people of the state are suffering the most financial impact.

According to the latest report of National Health Account Estimates, the state government has spent only 0.7 percent of the total budget of the state to provide health facilities to the citizens during the year 2018-19. It means that the state government has spent only 1470 rupees per person on the health facility of the common citizen. In the year 2014-15, this percentage was as low as 0.5 percent. Statistics show that citizens spent a total of Rs 66,703 crore on health facilities in Maharashtra during the year 2018-19. Of this, only Rs 17,934 crore (26.9 percent) was spent by the government. The remaining expenditure of Rs 48,769 crore was borne by the citizens of the state from their own pockets. In 2018-19, the Maharashtra government has spent as little as 0.7 percent of the total budget on health care, but states like Himachal Pradesh have spent 1.7 percent of the total budget on health facilities for the citizens of the state.

Inadequate number of Government Health Centers -

A team from the Asian Development Bank recently surveyed the public health facilities available in Maharashtra. According to this survey, it has been seen that there is a dearth of government health facilities in many places in urban as well as rural areas of the state. There is a total requirement of 2,299 Primary Health Centers in the state and currently, 1,828 Primary Health Centers are functioning. A total of 471 Primary Health Centers are still facing a shortage. There is a total requirement of 14,122 sub-health centers in the state. Out of them, 10,668 sub-centers are functioning and 3,444 sub-centers are lacking. While there is a need for 574 community health centers in the state, community health centers are functioning in 364 places and there is a shortage of 210 centers. A total of 1165 primary health centers are needed in urban areas. But here too only 538 primary health centers are functioning and there is a shortfall of 627 centers.

Conclusion – Government hospitals are seen to be under pressure in many places. Due to this lack of government health facilities, common citizens have no choice but to go to private doctors or hospitals. Currently, available health facilities are lacking specialist doctors, surgeons, or health workers universally.

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EFFECT OF PLYOMETRIC EXERCISE TRAINING PROGRAM ON SELECTED SKILLS OF INTERCOLLEGIATE BADMINTON PLAYERS OF PUNE CITY

Mr. Sagar Dhikale
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Abstract:
The study was an experimental research after going through review of literature discussing with expert and based on observation researcher constructed the six weeks plyometric program on skills of intercollegiate badminton players. The researchers take permission to the principal of Crist college Pune city. Also notified them about the need and importance of the study and took approval of 60 minutes plyometric program in a day for six weeks and also the time table and schedule of the plyometric program was planned in such a manner that the college hours were not be disturbed. In the same way the researcher assembled all subjects and was given to them instructions about the importance, description of the experiment and explanation of Badminton smash test and plyometric program after that the whole population (N=100) of intercollegiate badminton players were tested and selected a total number of 24 players in the age of 18-24 years the help of simple random sampling technique. The selected subjects were divided into two group's i.e. experimental group and control group and six weeks plyometric program which was given six days in week was manipulated only on experimental group but not control group. After plyometric program both the groups i.e. experimental group and control group, were pre & post tested for data collection. After procedure the data was analyzed by independent t-test and results were drawn.

Keyword: Plyometric exercise program, Badminton Smash test & Badminton players.

Introduction:

Badminton is a popular sport in the world. They appeal to different age groups and skill of all men and women to play inside or recreation and competition. The shuttle does not bounce and air hose, work fast game requiring quick reflexes and fitness must be played to some degree. Recreationally and mental benefits of playing social badminton can participate in and implement lean. Played by experts actually shot during a rally on the smash mails per hour to 200 miles per hours on the bullet drop can be very different, it is considered the world's fastest sport. In a recent United States Open men's actually a net rate of 89 shots in the second half as the doubles final match! However, in both singles and doubles play in our lives for physical activity to meet individual needs and abilities can handle. Badminton with its increasing popularity, in addition to the physical conditioning of the demand for highly skilled players has become extremely competitive. Badminton has many attractions. It is simply that there may be a few games are really easy to play. People new to the game but enjoying the long rallies and can be viewed using a

variety of strokes. The exercise of the shuttle to and from hitting the floor to create the picking, but it's also a challenge appears.

The Skills of Badminton:

The practical meaning of the skill, technique, tactics and strategies is important for the badminton player from the performance point view. Skill is a sequential movement, which is the productivity in ratio of all motor abilities. Further, skill, may be understood as an art of coordinated use of hands or any other part or parts of the body to perform a specific job, in playing badminton every stroke/shot is a continuous movement, the perfection depends on the fitness of various motor components. The grace and beauty depends how one has practiced the particular skill.

Required Basic Skills in the Game of Badminton:

- Racket Grip
- Service
- Forehand Strokes
- Lob
- Toss
- Clear
- Drop
- Smash
- Round the head stroke
- Backhand Strokes

The Skill of Smash:

In the Badminton sports smash skill is known as king of the entire stroke. A great stroke, fascinating alike in its sheer power and as a means of uncorking bottle up inhibitions and ill temper. The exciting rally- ending, power stroke of a game generally it is used to hit down any lifted reply falling short of the doubles long service line. Vary strong players will smash from the base-line but weaker ones will find that their stroke lacks sting because of the swift deceleration of a shuttle and they may be caught out of position by a quick, well placed return. It is played like the clear, with the basic throwing action, but the point of impact is 18 inches in front of the head; a powerful wrist action brings the racket head over the shuttle and a strong downward follow through maintains that trajectory and power.

Plyometric Training for Badminton players:

Plyometrics refers to a type of intense training that may be understood by an athlete who wants to improve speed power. This type of training also refers to very fast, explosive excision (normally performed with body weight) to improve power output and neutral activation of the muscles (the ability for a muscle to contract quickly). Basically plyometrics relies on an element of physiology called the stretch-shorten cycle. This stretch-shorten cycle means that the muscle is rapidly stretched and then contracted which increases the force applied on the muscle. Receptors within the muscle called muscle spindles react to this sudden stretching by sending a signal to the brain saying this stretching is potentially dangerous that the brain contracts the muscle to stop the

stretching. This serves to protect the stretch and prevent against any possible injury. The pre-stretch of the sac must be used by 0.2 of a second or otherwise same type of bounce or any possible advantage gained by sac is lost. So recapping, the sac is initiated by the muscle spindles which defects a stretch and responds by causing the muscle to contract.

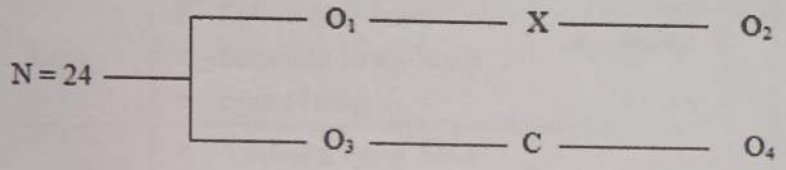
Material and Method:

Method of the study

The present study was an experimental research which was conducted with a purpose to examine effect of plyometric exercise training on selected skills of intercollegiate badminton players of pune city.

Research Design

True experimental design was used for this study to check the hypothesis. i.e, Subjects was divided into two groups; one was experimental group and second was control group. This research was based on pre-test and post-test equivalent group design. Before the plyometric program researcher has taken pre-test of both the groups. After that the researcher gave treatment of six week plyometric program only to experimental group. After the implementation training program researcher has taken post-test of both groups i.e. experimental and control groups.



Method of Sampling

Population:

The population of this study was all the boys' intercollegiate Badminton players between the age group of 18-24 years from Pune city. (N=100)

Sample:

Present research whole populations of boys' intercollegiate Badminton players were tested and a total number of 24 were selected from Crist college Pune city. So in order to drawn a sample from the population a simple random sampling technique was used of this study.

Variables of the Study

Variables are the condition all characterized that are experimental manipulates, controls and observes.

Independent variables:

The Independent variables are the conditions or characteristics that the experimenter manipulates or controls in attempt to ascertain their relationship to observed phenomena.

In this research the Independent variables was plyometric program.

Dependent Variables:

Dependent variables are the conditions or characteristics that appear, disappear, or change as the experimenter, removes, or changes independent variables.

In these research dependent variables were Badminton Smash.

Tools for Data Collection

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HIGHER EDUCATION IN INDIA

PROSPECTS – CHALLENGES – CONCERNS

EDITORS
SONY J. CHUNDATTU, CMI
THOMAS KARIMUNDACKAL, SJ

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Higher Education in India
Prospects – Challenges – Concerns

Editors
Dr. Sony J. Chundattu, CMI
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3

Management Education in India Opportunities and Challenges

Maninder Kaur

Introduction

Business has become a key driver of success of a country's economy. Management education plays a significant role in present dynamic business era. Increasing competition in business world, led by trends of globalization and rapid technological changes, has enhanced the relevance of business studies manifold. It is imperative to update management skills by business executives due to highly vulnerable business external environment. Management education draws knowledge from different disciplines like Commerce, Economics, Statistics, Operation research, Industrial engineering, Accounting, Psychology, Law etc. that helps students to think differently and add value to their existing qualification. A flexible business education and training system is needed to develop operational, managerial, behavioural, inter-personal, decision making and leadership skills. Management education equips business executives with requisite competitiveness and capabilities in business fields such as marketing, finance, human resources, production, accounting etc. enabling them to accept ever rising challenges with confidence at the corporate world. Furthermore, development of innovation and creativity in business processes, technology, products and services enables organizations and nations to become more competitive and demands for a management team capable of prioritizing investments, allocating resources including human capital to achieve strategic milestones. Effective business practices and processes exercised by efficient

5

English Learning and Teaching in India Prospects and Challenges

Priya Wahab

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Introduction

We are living in the age of science and technology which is changing the world at a terrific speed. There are reasons to believe that as a major language of communication, English has played a significant role in proving its importance in the scientific and technological progress made in various corners of the world. English is ruling the world in the present era being spoken by more than half of the world population. Though people in different parts of the world are free to use different languages of their convenience for communication and business in their domain, yet it is a common belief now that the English language shapes the way people wish to see the world. English has now emerged as the global language, the connecting language of the people across the world. With its 'coronation' as the '*lingua franca*' of the modern era, it is the only language favoured by various sectors for communication. English has established its dominance all over the world not out of any imposition but because of the realization that it has certain advantages.

Pooja Singal writes, "Language is not just a medium of communication, it is the medium that helps us to construct our everyday world. There is a close connection between Language and cognitive development" (Singal, 323). English shares the ability to express the multiplicity of ideas and complexity of thoughts that demand expression in our modern civilization.

6

Paperless Education
A March towards E-Learning

Blesson Varghese James

Introduction

In the traditional classroom students used to sit through long teacher presentations. Today the world is changing and both students and educators are benefiting from technological advancements. Educational institutions are progressively implementing paperless teaching solutions to involve a generation of students who are familiar with devices such as computers, smart phones and tablets, as well as to make the classroom more inclusive and interactive.

Paperless education alludes to the full grasp of technological innovations that enables a classroom environment that is no longer dependent on pen and paper to pass on information from instructors to students. New innovation is continually advancing into the classrooms trying to improve the learning experience and numerous colleges and schools are utilizing personal computers and cell phones in the classroom to upgrade the academic performance of the students. Universities, colleges and schools are these days using various methodologies when setting up the paperless classroom.

Going paperless in the academic front has received a great deal of attraction specially during these uncertain times where being socially away is safer. COVID-19, the global pandemic has wreaked havoc on every aspect of the economy. Although it has been marginally easier for MNCs and experts to accept work from home as the new normal and continue business as usual,

10

Higher Education System in India
Need for Developing Global Competence and Research Culture

Puja Jindal

Introduction

"The ability to read, write, and analyse; the confidence to stand up and demand justice and equality; the qualifications and connections to get your foot in the door and take your seat at the table - all of that starts with education," says Michelle Obama. The twenty-first century has brought with it profound challenges to the nature, values, and control of higher education sector. Societal expectations and public resources for higher education are undergoing fundamental shifts. Changes both within and outside the academy are altering its character – its students, faculty, governance, curriculum, functions, and very place in society (cf. Zusman, [n.d.]).

Sustainable growth and development of any country is possible only through quality higher education. It is higher education which helps in character building of a nation by broadening the minds of people and enhancing their rational, logical and analytical thinking. There is no denying that in the coming times, world leaders will be only those countries who have been able to develop quality higher education system. It is higher education that leads to pronounced growth in all aspects such as social, aesthetic, cultural, moral, intellectual, economical and many more.

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Higher Education A Challenge towards Prospective Development of Indian Women

Deepa Sujith

Introduction

It has been universally proven that socio economic success of a nation depends on the success of its educational system. It has the distinguishing characteristic of transforming societies and equipping them to remain successful in a global economy. Indian education system has advanced tremendously in the past seven decades and has grown up to creating one of the largest education systems, being the third largest in the world. India as a developing nation is showing significant progress in the field of education. The fundamental right to education lays down compulsory and free education to all children, and it has brought about a revolution in the education system of the country with a stunning enrolment in primary education. Later with increased number of higher education institutions (HEIs) we could see a drastic increase in the number of students enrolling for higher education too. Despite growing investment in education and increased enrolments, the country still faces lots of challenges in its surge towards becoming a knowledge driven economy. Being one of the rapid growing developing nation, and as a country with huge human resource potential we need to see a massive population who can drive our economy forward. Still we see a large proportion of our resources, especially women are underutilized. Though in recent times we see a ray of

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Role of Ethics and Morality in Higher Education

Sony Chundattu, CMI

Introduction

Education is not simply imparting knowledge in a particular faculty or subject or making one fit for securing jobs or fair well in exams, but at the same time is also a training in logical thinking which helps the coming generations adjust to the ever changing environment. It also means opening the doors of the mind, cleansing the soul and realization of the self. Education should aim at making human life better not only through economic upliftment of individual but also through social, moral and spiritual strengthening. This will not only improve human life but also realize the "higher truth" i.e. "*Tamaso Ma Jyotirgamaya*" from darkness to light. Thus education is not only a way of earning but it also helps to develop human personality with skills, values, morals and enhancement of different attributes of man. So education is a vital means for the potentialities of a human being to emerge in a positive direction so that a man can live in society with full of dignity.

Nowadays, we hear a lot from different news channels regarding the challenges faced by the higher education sector of India. The experts present before us a vivid picture of the various challenges in the field of higher education and point out the reasons for each of them. Demand supply gap, lack in quality of specialized human resources, faculty shortage, shortcomings in the field of research and development etc. are cited as the main challenges of Indian higher education. Apart from all these factors, ethical deterioration

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A Study on the Changing Role of Internet Of Things (IoT) in Businesses and the Various IoT Data Sources Utilised to bring Value Addition in Telecom Industry

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A Study ontheChanging RoleofInternet Of Things (IoT)inBusinesses andtheVarious IoT Data Sources Utilisedtobring ValueAdditioninTelecom Industry

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Abstract

The emergence of IOT has proved to be beneficial to our lives keeping us connected to people and the things around us. The use of such technologies has also changed the way the businesses operate. With the onset of IoT in the telecom industry there seems to a drastic transformation in the way telecom service providers facilitate communication between people and devices. They also implement different models to utilize IOT effectively which in turn adds value to the organization or business. This paper aims to explore the impact of IoT in businesses while focusing more on how IOT data sources can be utilized in the telecom industry. This paper also identifies the different models that can be adopted to utilize IoT more effectively along with the difficulties involved.

Keywords: Internet of Things, IoT, Business Models, Telecom Industry

1. Introduction

IoT has made a huge impact not only on the business operations but also on the methods used to collect and exchange information. IoT devices help organisations to make more sound decisions by recording and transferring data to monitor critical processes, providing better insights and strengthening efficiency. With the help of IoT solutions and by applying big data analysis to the massive volumes of data acquired by telecom companies, valuable insights about the client's behavior patterns can be analyzed, which can be further used to benefit customer engagement. Business leaders have come up with better services for their customers in order to enhance their business processes.

Objectives:

1) To understand the role of IoT in business processes.
2) To explore the various areas where IoT can be utilized in the telecom industry
3) To understand the kinds of IoT data sources that can be utilized.
4) To identify the challenges involved while adopting the models to utilize IoT effectively.
5) To understand the various business models that can be used to implement IoT

2. Review Of Important And Relevant Literature On The Study

a) ***Chan & Hubert*** [1], conducted a study that emphasisedon those organisations which have more than one partners or solution bases in order to substantiate his proposed business framework. Different case studies were adopted to examine the fundamental elements of business model and carry out a qualitative analysis.

b) **Tanim*& M M Zaman*** [2]**,** had conducted a research on *"Internet of Things (IoT) driven new Business Models and the role of IoT in the current businesses in the EU".*

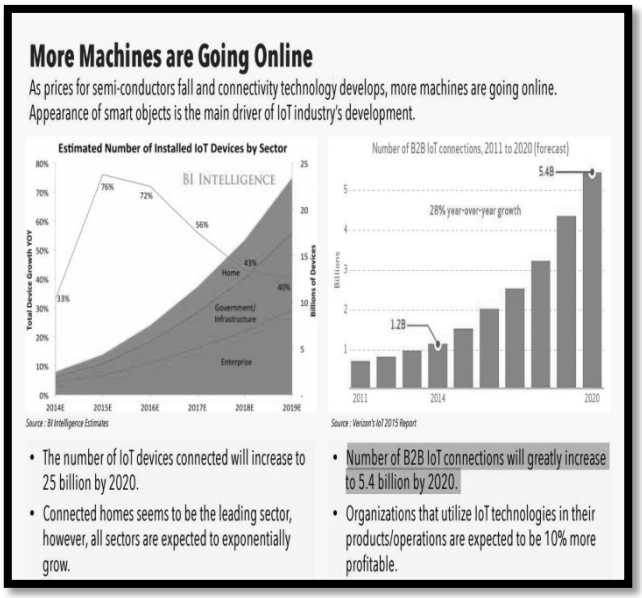
a. The research work focused on finding out new business models from IoT infrastructure and ecosystem and the role of IoT in the existing business

in the EU. This work also emphasized on how the people or customer will get benefitted from IoT.

c) **Manal et al.** [3], this paper highlighted the most important topics and concerns related to Telecom IoT provision which proved to be beneficial for both researchers and operators. The paper also discussed the success components of Telecom IoT along with the elemental strategies that Telecom operators can implement to offer IoT.

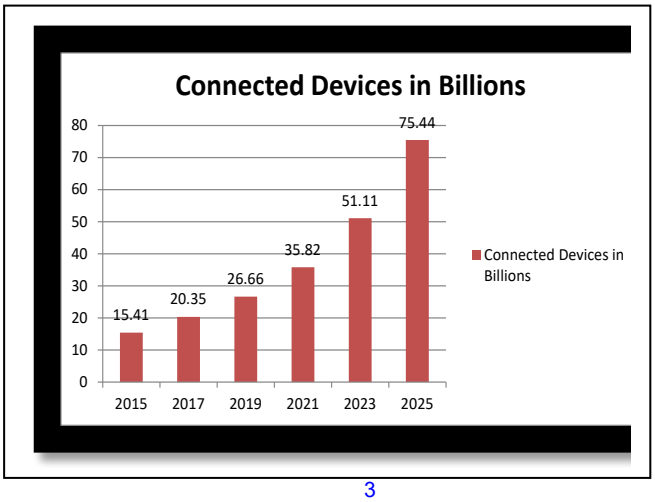
3. Role Of IoT In Business Processes

It a predicted that the IoT market in the country will contribute to USD 15 billion by 2020, and constitute nearly 5% of the total global market. These figures in turn anticipated the increase in IoT devices in India from the year 2016 - 2020 from 200 million units to 2.7 billion units respectively [3].



Source: BI Intelligence Estime, Verizon's IoT 2015 Report

These figures shown above are as per forecast that was done in the year 2015. But the current figures have crossed the predictions done years back.



Source: A Reliable Communication Framework and Its Use in Internet of Things (IoT) - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Internet-of-Things-IoT-connected-devices-from-2015-to-2025-in-billions_fig1_325645304 [accessed 20 Oct, 2020]

Study reveals that by 2025, the total number of devices connected to Internet of Things (IoT) is expected to reach 75.44 billion .The IoT along with the existing prevalent Internet Technology, is the next significant move in transforming the world into a place which would seem more connected.

Implementation of IoT has altered the manner in which data is exchanged and collected to handle business processes. The business realm is revolutionising in the following ways:

Enhanced customer experience and better business insights: If a business understands how its customers use its products they can better fulfil their needs and focus on strengthening the customer experience [4]

Inventory: IoT software and devices have helped in tracking and managing inventory by providing more regulated options which can be operated automatically [5].

Data Sharing: The manner in which data is collected, exchanged and interpreted is completely changing with the inception of IoT in businesses. An IoT device record and track the patterns of consumer interaction with such devices and uses this data for improvement and growth thereby providing a better customer experience.
This data is further used to study demands of consumers, buyer patterns, and improvement areas to come up with innovative methods and ideas for marketing and advertising.
[5].

Traffic monitoring: The IoT applications can track the real-time traffic of customers and requirements and customize the overall experience by improving the business strategies. This data can then be utilized to review the customer behaviour, interests and buying patterns which ultimately will enable the organization to generate better loyalty to customers. [6].

Better Productivity and Efficiency: The productivity of any business can be increased remarkably with enhanced information about the consumers and the market. Immediate impact can be made on the productivity of the businesses when IoT devices are connected to each other and controlled in order to ameliorate efficiency. The augmentation in productivity and efficiency will strengthen the profits to a greater extend.

Remote Work: With the use of IoT technology employees can associate and work remotely. Studies have revealed that employees connecting remotely are content and more productive, which in turn has considerably improved the way businesses operate [5].

Skilled Workers: Businesses have felt the need to recruit workers who are more skilled and can manage IoT technology more effectively and adequately. Negative impact on the productivity of the business will be made when you hire an employee who will not be able to make optimum use of IoT technology and in turn will result into wastage of time and money. If businesses start recruiting such skilled individuals, more individuals will have an inclination toward latest technology, which will also reinforce their value for future endeavours [5].

4. Areas Where Iot Data Can Be Used In Telecom Industry

Business analytics, equipment monitoring, building predictive models, machine to machine communication and monitoring safety at remote sites can be achieved with the implementation of IoT in telecom industries.

i. IoT services integrated into telecom facilitated in offering connections with applications that provided better user interface.

ii. Information related to products and services provided to consumers can be tracked more easily with the help of IoT in telecom industries. Network issues can be detected based on the information which in turn would help in providing better services.

iii. With the help of pre-integrated IoT sensors into products, data collected after the deployment can be used for performance evaluation by the telecom companies.

iv. Telecom operators can monitor the performance level of devices through IoT sensors installed at various workplaces like- workstations, warehouses, factories, and network towers.

v. Predictive analytical models can be built with the help of large volumes of data which can help in deriving desirable outputs.

vi. The telecom can accomplish better machine to machine conversations by permitting IoT enabled block chain systems in communication, through a lower power wide area network.

vii. Improved location-based services can be provided byCellular network-connected devices through proximity sensors in the devices connected to mobile networks.

viii. Autonomous cars can communicate with other autonomous cars in its range through IoT enabled sensors and use of 5G in the mobiles (LTE-V). Telecom and smartphone companies are investigating the potentiality of the LTE infrastructure by creating LTE-V. LTE-V can be used in autonomous cars for machine-to-machine collaboration. [7].

ix. Greater **physical security** can be provided by IoT enabled cameras and beacons in the worksite.

5. How Data Sources Can Be Utilized:

5.1 IoT data source can be utilised to analyse customer's actions and predict their pattern of behaviour in a periodic manner. This can be done by exploring data or by designing a model, analysing data from the past, identifying new data sources, correlating that data with the past, applying appropriate algorithms that will help capture appropriate data and also in forecasting. With the help of new IoT data sources, it will be easier for telecom companies to predict meticulously, improve customer satisfaction and ensure that they come up with more customised campaigns.

5.2 Distributed messaging system, is a form of IoT data source that permits Telecoms to transfer data between processing centres, towers, and subscribers. It is very important to process the data in the proximity of the source so that huge amount of available data can be easily transferred to various centres. This will help predict heavy traffic intervals and avoid congestion.

5.3 Data sources generated from IoT also help in prediction of churn by detecting the customers that have greater inclination to deviate.

5.4 IoT data source facilitate in providing services related to security. They also help in taking immediate actions by collecting and employing summarised data from firewalls, network devices and various end points.

In The Telecom industry has made investment in IoT in the following areas:

1. Planning, monitoring, and maintenance for Telecom
2. IoT infrastructure and services for Telecom customers
3. IoT solutions development

6. Challenges Identified In Adopting IoT

There are various challenges involved while adapting to the implementation of IoT. These challenges are related to performance, identification, authentication, reliability, security, privacy, scalability, precision, interoperability, compatibility, mobility and investment. In order to adopt the new business models, there is a need to resolve the IoT challenges.

A few challenges identified are:

6.1 Compatibility and Interoperability of Different IoT systems [8]**:** With several vendors and service providers, it is very complex to sustain the issues related to interoperability of IoT systems. A lot of issues related to compatibility of systems may also be encountered.

6.2 Identification and Authentication of Technologies: It is indeed a tedious task tobring along a huge number of connected devices under a single platform. It needs proper formalization and system architecture that will make it easy to identify those devices with proper authentication.

6.3 Integration of IoT Products with IoT Platforms: Noteworthy application of IoT will be made possible only if enterprises integrate various IoT connected products

with suitable IoT platforms. As the IoT ecosystem continues to flourish; companies are rediscovering their distinctiveness and revolutionizing their complete process.

7. Business Model

A Business model focuses on delivering value and generating revenue [9]. An organization cannot be successful unless there is a distinct and systematic plan to generate profit and revenue. Traditional business models are not suitable to organizations due to the nature of the IoT structure which requires firms to associate with their competitors. Moreover, due to the agile nature of the market requirements it is essential for companies to quickly accommodate to market challenges in order to be successful. Innovations in business models are becoming "new pathways to competitive supremacy" [10]. IoT opened pathways for new opportunities in businesses which helped companies develop advanced business models and services and explore new revenue streams.
Alexander Osterwalder defines a business model as:
"A model which describes the philosophy or principle of how a business creates, delivers, and captures value."

The different business models which can be utilised to create value addition to the business or organisation:

7.1 Revenue sharing:

This model of revenue sharing is adopted by businesses from automotive to telecom, to maximize revenues from the Internet of Things (IoT). Most industries have already well-established this model of Revenue sharing. The telecommunications industry has been using this model for generations, for example, roaming agreements that prevail between multiple carriers.
Multiple forces are behind the increase of revenue sharing in the IoT. Leading among them are the steady growth in cloud-connected services, M2M computing and 5G. At the core of it are the IoT service providers. They have a focussed more on expanding connected services that generate more value to their customers along with generating a larger share of revenues.

In the arena of IoT, everything seem to be connected, including revenues. Delivering IoT-based services involves team effort which includes multiple third parties. Thus, IoT revenues in businesses must be shared among device makers, application developers, data storage providers, analytics experts and other vendors.

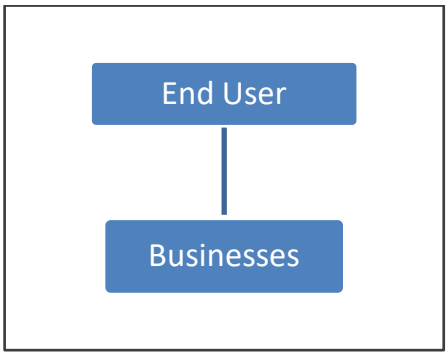


Figure 1: Traditional Business Model

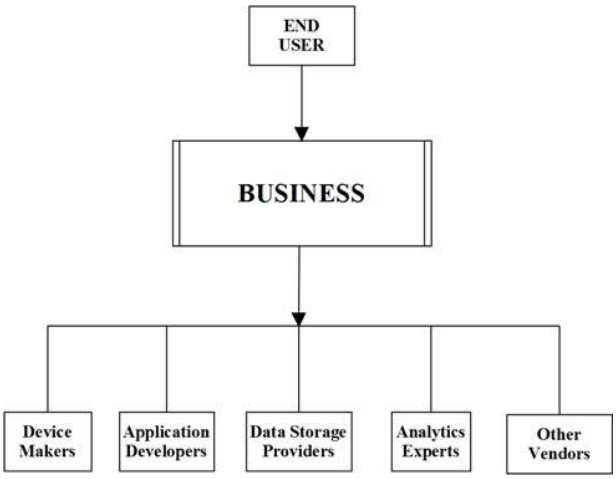


Figure 2: IoT Business Model

7.2 Subscription Model:

According to an article by Regalix. "The elemental transformation in the business model is from product sales to service sales". An IoT subscription model may seem fascinating to most customers.Means of generating revenue is predictable and recurring. Your primary focus can be to explore different creative means to generate recurring revenue [11].

Through this IoT subscription model businesses can generate recurring revenue through connected devices. This model enables the business organisation to build a better relation with the customer by gathering more information about the customer through the devices and providing them features as per their needs.

7.3 Outcome-based Models:

IoT-enabled outcome-based business model alters the way the organization functions. This model focuses more on delivering the result(s) they are seeking for rather than selling a product or service to the client. Through this business model organizations can eliminate the tasks of processing, gathering, analysing and generating the output. They only need to charge for the outcome which they were claiming for [11].

7.4 Asset Sharing Model:

At some point of time in business every organisation might have used a business model that adopts asset sharing. The model will let businesses to fragment their **assets** with different other business units. The idea of asset sharing comes into picture when a business invests on expensive equipment and is not able to use it to the best of its capacity [11]. In this way customers get to use the system at a reduced price.

7.5 IoT-as-a-Service:

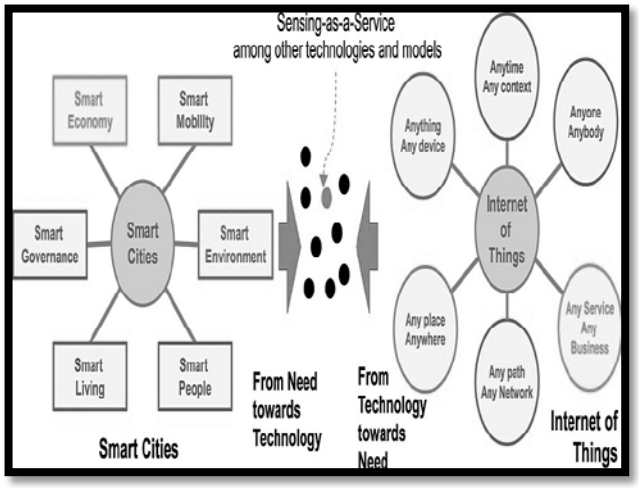
A model which is most widely used in businesses in recent times is the IoT-as-a service. Through this model, a business generates revenue by providing its solution enabled by IoT on hire. This IoT model is not only confined to the software or physical aspect, but it encompasses entities like information and data. The outcome or results generated from the data can also be leased in this model. The **IoT-as-a-Service model** can generate recurring revenue and can also be considered as a component of the subscription model [11].

7.6 Product as a Service:

Business model can also be reformulated by converting a traditional product into a product which is connected `as-a-service. This type of service requires a radical transformation in the way we look at a finished product in a single transaction.
End result is to provide solutions to customer issues and charge them for usage or performance. This paradigm shift will have its own impact on the organization as a whole. Its repercussions will have its effect on processes, people and technology. [12]

7.7 Sensing as a Service:

Sensing as a Service may be evolved as an idea that is governed by the cloud computing concept Every Thing as a Service. Consumers acquire the data which is generated by the IoT devices owned by other units through the middleware .Consumers may be billed on the basis of the quantity of data used by the sensing device [13]. Sensing as a Service (aaS) brings new data monetization possibilities to operators, using existing infrastructure to supply real-time information which is processed and analysed environmental data. Through this model networks are managed through remote monitoring of assets.



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8. Conclusion

In order to adopt a particular model, it is very important to understand how your product will add value to your customer and to your organisation. With the help of IoT enabled business models, a business can maximise its profit and expand its working thereby creating value to the organisation. IoT will lead to adapting new standards and platforms in future. This will pave the way to an overall productivity growth along with an improvement in quality of life. In spite of all the benefits there are still a lot challenges and issues that needs to be considered in order to achieve the full potential of IoT.

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Augmented Analytics and Modern Business Intelligence Adoption to Maximize Business Value

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Abstract: *The business success is critical for any organization irrespective of its size. Understanding business value and advanced technology capabilities in Business Intelligence (BI) solutions for analysis and decision making is the need of every management, researchers and analysts.*

Data is available at every corner, at every platform and it is being produced all the way using digital equipment. These complex data sets for analysis and decision-making can be used on various platforms and are more accessible than ever to find insights and explanations. There is an abundance of datasets and data analysts, but organizations are still unable to leverage its potential to the fullest and this perhaps is the failure of most data analytics initiatives.

This paper covers the need for Augmented Analytics adoption in companies to maximize business value with respect to added advanced analytics capabilities in BI Tools for effective, accurate and timely decision making, and business analysis.

Keywords: *Business Intelligence, Augmented Analytics, Machine Learning, Natural Language Processing, Citizen Data Scientist.*

1. Introduction

This Business Intelligence (BI) is a term that refers to the use of data analysis to design strategies and make decisions so that companies have a competitive advantage. Although it can become a product in the digital age, unless it has the ability to deal with changing business needs and make effective decisions based on advanced analysis techniques. The term BI refers to tools designed together with business applications and strategies to collect, integrates, analyze and interpret business information. Its main purpose is to provide better business decision making. Business analysis is a field that uses statistics and qualitative techniques to obtain knowledge from raw data, while the predictive model is a field that uses data analysis techniques to make effective and evidence-based business decisions [1].

Traditional BI needs to analyze isolated databases to create basic reports. The analysis is performed by specialized data analysts and access to the reports produced is limited to very few. Regular business users do not have access to data. Hence, they cannot make data-wise decisions. The BI tools are made more accessible to business users with the development of self-service technologies. These tools are equipped with intuitive graphical user interfaces and smart dashboards.

With the help of self-service BI tools, business users can more easily obtain the information they need to make better decisions without relying too much on data analysts and IT professionals. Moreover, compared with traditional BI systems, these tools can handle larger numbers, allowing for more in-depth analysis. Most current BI tools and platforms fall into this category.

Self-service BI tools are more insightful and easier to use than traditional BI, but these tools have some limitations Most of the process of data preparation, structuring, and cleaning for accurate analysis is manual. This manual process of turning raw data into useful insights is usually a lengthy process involving several steps.

In addition to being cumbersome, this manual data analysis process in self-service BI is inefficient and prone to human error. Similarly, the quality of insights provided by Self-service BI systems is limited to the types of queries that business users make. If users do not query their BI system in the correct way, they may ignore hidden trends and insights, resulting in failure to resolve missed opportunities or alerts in a timely manner. [2]

Therefore, it is important not only to automate data preparation, but also to automate part of the data analysis and insight discovery process. This can be achieved by using Artificial Intelligence (AI), Machine Learning (ML) and other advanced technologies to power BI.

In the following sections, BI and Advanced data analytics approaches have highlighted along with the Augmented Analytics technology, its benefits, essential elements, business adoption and business requirements to focus the need of it in BI tools to automate data analysis, increase business value and for overall organizational growth.

2. BI and Advanced Data Analytics

The Gartner [3] in their report have estimated that 85 percent and more projects on Big data are failed. Datasets are increasing in size and becoming more complex and dynamic, and traditional BI systems are not able to manage them. The failure of data analysis is very obvious. They either cannot obtain the data, process, prepare the data, or even cannot understand the data.

Traditionally, the processes involved in preparing, browsing, and manipulating data have been mostly manual. With the rapid growth of data volume, the traditional manual data analysis process has become inefficient due to the complexity brought about by the continuous increase of data volume. In addition to manual operations, the data science process requires specialized data scientists, and these scientists are not many. In fact, there are very few data analysts and scientists available in the global market. Even if the organization manages to obtain the right data experts, they do not have or have little business knowledge, so they need comprehensive training in the analysis required by the business.

Modern BI systems with advanced analytics are capable of analyzing large amounts of data. However, the process before analysis, data cleaning, is still very manual and needs to be performed by data scientists. They have to manually develop metadata and ensure

data profiling, quality, modeling and manipulation. Consequently the possibility of human error increases even before data analysis.

With Modern BI and analytics platforms, users can easily find the data they are looking for and visually explore data patterns and relationships. But the limitation is that users may not be able to understand hidden data trends and skews that they do not know, and may affect the business. Users continue to focus on data exploration with their own preconceived notions and past experiences or they will have to manually explore all the potential permutations and combinations. This process is obviously time-consuming and users are likely to miss important insights.

Most Modern BI tools offer smart visualizations and interactive dashboards. However, it requires special skills and understanding of what is really important in the data. The [4] has covered four general data analysis types, including descriptive, diagnostic, predictive and prescriptive analysis that organizations can implement together with BI tools and strategies to make better and more effective decisions. These four types are illustrated in the Figure. 1

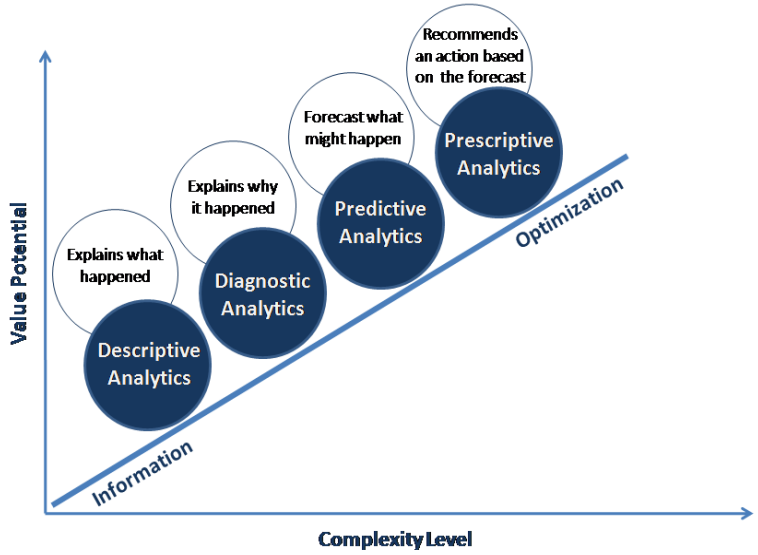

Figure 1. Levels of Data Analytics

Gartner said that Advanced Analytics is a top business priority [5]. In that, Gartner have depicted the four types of Analytics capabilities which are depicted in Fig. 1.

In Descriptive Analytics comprehensive, accurate and past data is processed for business reporting and effective visualization to understand what had happened with the company's business strategies in the past and it explains what was exactly happened with descriptive analytics capability. In case of Diagnostic Analytics, companies can drill down to the root-cause and it can isolates all confounding information to understand what certain things had happened in the past. In order to forecast what will happen in the future, the companies are looking for the Predictive Analytics capabilities to make business strategies that remain fairly consistent over time. In Predictive Analytics, decisions are automated using algorithms and technology. While Prescriptive Analytics, recommends action and business strategies based on competitors testing strategies and outcomes. It an Advanced Analytics technique to make specific recommendations based on the forecast. The prescriptive model utilizes an understanding of previous three models

descriptive, diagnostic and predictive for what happened, why it happened, and a variety of what will likely to happen analysis to help determine the best course of action to take in the future.

Business leaders want their company to become a data-driven enterprise and recognize the need to adjust Advanced Analytics features to make it a digital business. However, in most cases, the company is simply collecting more data, big data, rather than converting that into actionable insights. According to Gartner [6], 34% of companies undertake either diagnostic analytics or explore why KPIs are working in a certain way. Although 72% and 3% have partially or minimally completed their respective tasks, even lower percentages can fully perform predictive and prescriptive analysis.

Augmented analysis has the potential to improve the company's analytical methods; it will simplify root cause analysis, predictive analysis and prescriptive analysis, and further reduce the burden on data scientists. Even, existing information analysts will evolve into citizen data scientists [7] and they will be able to spend less time on data preparation and descriptive analysis, and more on advanced analysis aided by smarter software and modern BI Tools.

3. Augmented Analytics: Future of BI

The Gartner [8] defines Augmented Analytics as a method that uses artificial intelligence (AI), machine learning (ML) and natural language generation (NLG) to automate insights, which will mark the next wave of disruption in the data and analytics market. Organizations should plan to use these supporting technologies to adopt augmented analysis to automate data preparation, integration, insight, and interpretation. By automating many aspects of data analysis, ML, NLG and AI along with BI and associated business strategies, organizations will leverage the potential of it, in making effective decision making and for overall organizational growth.

Gartner have also proposed an Augmented Analytics will be the future of big data and analytics [9] which would further deliver in making improved BI strategies and decision making. Augmented Analytics is an innovative approach to think about data and analytics. It includes AI, ML and Natural Language Processing (NLP), into traditional BI. It is an advanced approach in which AI works in the background to learn and improve the results. It also provides faster access to insights derived from various data. This technology ultimately helps to discover hidden patterns, generate insights, and reduce bias. In addition, companies can collect the most appropriate data model for analysis and understanding of BI insights.

Organizations need to deploy Augmented Analytics capabilities into BI, it will not only democratize the use of the data but also make it simple for business users and experts to make effective decisions based on added advanced analytics functionalities without help from data scientists or business professional. It will also help to go beyond predictions of future business events or situations and obtain unbiased prescriptive analysis to understand what might be done next. That is really the future of BI!

With augmented analytics, business users can:

- Provide unbiased data analysis.
- Easily determine the root cause of the problem.
- Identify unknown business growth opportunities.

- Democratize company-wide insights from BI to enhance overall business performance.
- Turn actionable insights into opportunities.
- Develop rapidly to build an agile enterprise.

The global market is evolving at an unprecedented pace. Due to changes in markets, regulations, demands, and emergencies such as the COVID-19 pandemic, it is difficult to predict [10]. In this way, the most successful organizations will be those that will adapt innovatively to this change. Augmented Analytics is important for organizations to identify trends at the onset, and even predict or prescribe them before they can happen.

4. Benefits of Augmented Analytics

There are various benefits of augmented analytics for an enterprise, including:

- IT support systems can easily focus only on business strategies and special projects.
- ML can effectively analyze big data sources, while Augmented analysis helps to conduct in-depth automated data analysis.
- Simplifies overall data analysis and makes it easier to get to deep insights.
- Users and an enterprise become data-driven.

Augmented analytics will make it easy to handle the complexity and big data scale from in and out of the enterprise to collect, prepare, integrate, analyze and interpret and make it effective for making smarter business decisions well in time.

5. Essential Elements of Augmented Analytics

There are elements of an Augmented Analytics tool which will perform augmented data discovery, data science, ML algorithms, and data preparation.

1. NLP: The technical language skills will mainly benefit BI and citizen data scientists. NLP makes it convenient to interact with the system natural language using text and voice commands.
2. NLG: It is another augmented analytics capabilities that empowers BI tools to recite results to better understand the complexities behind the business data.

In order to deal with any deviations, errors in data and improve the effectiveness of BI tools, augmented analytics should be recommended with the following tasks:

  - most appropriate visuals for data
  - how to enrich and contextualize data for improved analysis and interpretations
  - how to cleanse and prepare data for further use

- Insight Generation: BI tools need to provide insights free from bias and suitable for hypotheses testing. Algorithms used in insight generation need to describe the data, identify KPIs, and draw out the variables that influence the performance. It

also needs to perform outlier analysis to avoid unexpected results than the anticipated one.

- Prediction: In order to predict, these tools should be able to use algorithms to train/test predictive models which are based on common business functions or problems like customer churn, customer behavior, attrition, increase or decrease in sales across various business dimensions.

Integrating Augmented Analytics with AI, ML NLG and NLP into BI are key elements to provide better user experience across business functions and business analytics. The core business processes for data integration, insight generation, and correlations analysis will become efficient with this approach.

It is important from the organizations point of view to adapt augmented analytics platform to extract meaning from complex and ever growing data deluge. The technologies involved in it will make easy to share business findings. To automate several key aspects of business data management including increase in the productivity of data experts, data governance and its accuracy, and reduction in overall costs as well as for many other such business factors Augmented Analytics systems are essential.

6. Augmented Analytics Adoption

The Augmented analytics integrates AI, ML and NLP elements into analysis and BI processes to simplify data preparation, discover insights, visualize and share results with business users.

Augmented analytics functionalities will disrupt the analytics market and its adoption will certainly be game changer for business experts particularly for modern BI, advanced analytics and for data science platforms. The key is to automate insights and embed this technology into enterprise applications which will help business transformation. In order to perform analysis, currently, content authors such as data analysts, citizen data scientists, and business experts repeatedly perform data processing, insight generation, and knowledge representation activities to understand hidden insights. The core steps are:

- Preparing the data
- Finding patterns in the data and building models
- Sharing and operationalizing findings from the data

The augmented analytics rapidly perform accurate insights for business users and augments their analysis by using AI-ML-NLP to automate these three analytic processes used in data discovery and visual platforms are illustrated in Figure. 2. Here, business users often iterate between preparing data and finding patterns in data.

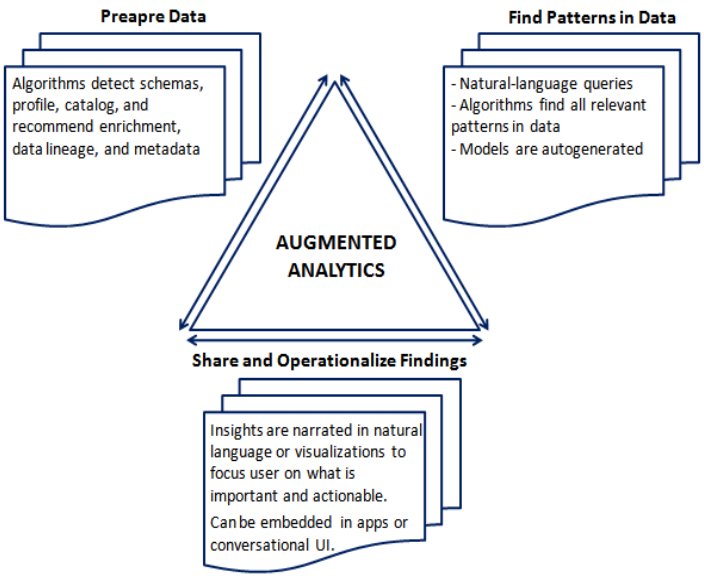

Figure 2. Emerging Augmented Analytics Workflow

6.1. Preparing Data

Data preparation for analysis is the most time-consuming task. Modern BI tools and analytics applications offer basic data preparation functionalities for joining, manipulation and transformation of data.[11] Business users or data scientists have to perform much of the data profiling, maintain quality, its modeling, data manipulation, data enrichment, metadata development and harmonization work. In either way, it creates a bottleneck for users involved in data preparation and also creates risk due to lack of centralization or because of users ability to build analysis content.

One of the components of Augmented analytics is augmented data preparation it performs various tasks to find relationships in data, profiling and recommend the best methods for cleaning, reconciling, enriching, manipulating and modeling data with approaches to capture metadata and lineage for reuse and governance.

6.2. Finding Patterns in Data

Modern BI and analysis tools enable business users to use interactive techniques such as data pivoting, sorting, filtering, linking, grouping, and certain custom calculations to view and explore associations and patterns in data. Compared with traditional BI systems, these methods are very effective in providing insights and can promote business development in a highly competitive market. As data becomes complex, large, and multidimensional, the process becomes time-consuming and laborious to explore hypotheses or to infiltrate and combine to ensure complete and accurate results.

On the other hand, in Augmented Analytics, the algorithms used to detect correlations, clusters, outliers and associations are automated and applied to business data. In this case, only statistically meaningful and relevant results can be better explained to users with the help of advanced visualization functions. This can be done by applying multiple algorithms in parallel to reduce the risk of losing valuable insights in the data.

Most of the Augmented Analytics platforms will make the basic analysis model open for testing and verification by business experts. This is one of the important features to build the confidence and confirm the reliability of automated analytics and insights.

AI, ML, and NLP automation in Augmented analytics are entering the field of data science to simplify functional design, model generation, and prediction. This will make professional data scientists more efficient, and the business models they build are less prone to bias.

In view of the lack of professional data scientists in the market and the growing demand for their skills, a new category of citizen data scientists will be expected to achieve higher productivity and require more analytical work.

6.3. Sharing and Operationalizing Findings From Data

Modern BI and analytical tools have developed intelligent visualization and interactive dashboards to provide collaboration capabilities to assist in presenting patterns and hidden insights.

With addition of ML, NLG, and NLP, augmented analytics can also automate text or voice narration in data to provide actionable insights to business users.

Organizations need to adapt Augmented Analytics capabilities to make better business decisions based on automated analytics using AI, ML, NLP algorithms. The following are some necessary suggestions that organizations or business leaders should follow when implementing these technologies on business data.

- Determine the ML features and algorithms for data preparation and integration.
- Plan for augmented analytics in terms of business outcomes, enhance self-service BI and data scientist's skills.
- Define algorithms to automate pattern detection in data which can simplify insight generation and model building.
- Recognize the expert data scientist to interpret and validate models, insights and business applications, they will still need to perform these tasks.
- In order to improve the skills of citizen data scientists, apply the current deployment model through AI-ML-NLP technology, and implement analytical governance among data engineers, analysts and data scientists.
- Execute augmented analytics initiatives along with existing analytics and decision making processes to understand the similarities and differences.
- Encourage business users to work collaboratively and iteratively with expert data scientists both internal, if available, and external.
- First, lists the business problems that are difficult to solve with traditional BI in terms of time, technology, and data processing complexity, and then apply augmented analytics to evaluate its effectiveness.
- When using Augmented analytics tools in learning and business decision-making, assign responsibilities and roles to business analysts and data scientists, and enable them to interact to better understand the overall business process and augmented analytics capabilities.
- Hire people with data analysis skills in the business area and invest in augmented analytics throughout the organization.

7. Augmented Analytics Risks Factor

The emergence of visual augmented analytics allows a wide range of non-technical users to access easy-to-use, visually interactive, and more advanced tools to prepare and analyze data, thereby democratizing analysis.

Augmented Analytics will eventually become more accessible and will continue to evolve in terms of technologies and features. Business executives should plan to adopt and understand its impact on BI and analysis. They also need to plan to invest in the development of data analysis skills to enable users to gain insights and develop business strategies to assess the impact on traditional or currently supported BI tools and analysis capabilities.

To date, BI tools need to perform a lot of manual data analysis and business user intervention and interpretation. Looking at volume, complexity, and speed at which big data is generating, this approach of data analysis and insight generation is not suitable or scalable in today's business environment.

Augmented analytics is the next big thing in the market and it will disrupt the big data and analytics industry. On the other hand, Analytics leaders need to embrace to establish and maintain a competitive advantage, especially in industries that are undergoing digital transformation. Efforts to adopt Augmented analysis may encounter resistance for several reasons:

- Believing in traditional BI and analytical methods, manual data analysis, and assumptions that it can solve business problems and provide feasible insights.
- Less knowledge of Augmented Analytics tools and their functions in terms of effective implementation.
- Employment is unsafe, Augmented analytics may not require manual intervention due to its automated data analysis function.
- Companies rely on traditional decision-making processes and find it difficult to adapt to strategic changes.
- Business executives think that Augmented Analytics is still some time away. Businesses can still consider the existing BI tools with an advanced analytics approach.

These reasons are challenges to Augmented analytics implementation. This paradigm shift needs to be challenged through the existing processes and strategies of decision-making and business analysis, and needs to prove the similarities and differences between traditional BI and Augmented Analytics. Organizations need to create an environment to support and adapt the Augmented Analytics in an effective way that it is has potential and can create business value. This will further help in identify business issues that cannot be seen using traditional BI, and will also help deliver relevant results within an appropriate time frame. This will lay the foundation for futuristic automated data analysis and decision-making methods in BI.

8. Augmented Analytics Tools

There are many BI systems with overlapping functions in the market, but they are different in terms of providing services to different users. Analytics companies have developed products that can provide advanced analysis and automated data discovery, pattern recognition, smart visualizations and interpretations. For example, Watson Analytics from IBM, Lumira from SAP, Power BI from Microsoft and Visual Analytics from SAS offers advanced analytics capabilities for Augmented Analytics. Some companies have begun to invest in NLG, including Narrative Science, Yseop and Automated Insights. Tableau, Qlik and many other such tools are not behind to invest in Augmented Analytics.

There are major players in the market, such as Datawatch, Paxata, UniFi, Trifacta, Alation, Informatica and Oracle, which provide automated data preparation functions. However, these functions are still in its infancy and adoption is in the early stages. Most of these tools can run advanced analysis algorithms and allow expert data scientists to evaluate the model and provide it for further enhancements, making it a high-level descriptive, predictive and prescriptive model.

A number of AI, ML and data science platforms, Visual Statistics and SAS Visual Data Mining and Machine Learning, RapidMiner and KNIME need little skills to build advanced analytics models. These tools include advanced analytic techniques such as semantic analysis, graph analysis, ML, NLG, NLP, pattern matching, sentiment analysis,

network and cluster analysis, forecasting, smart visualization, advanced statistics, neural networks and other complex features along with the programming languages or packages such as open source R, Python and MATLAB support.

Compared with traditional analysis methods, the Augmented analysis platform is unique because it can automatically execute the analysis process at a faster speed than humans and avoid decision bias [12]. Although Augmented analytics development is in progress and it is still at an early stage compared with disciplines such as BI tools.

Major Software providers offering Augmented analytics services in the market [13] include IBM, SAP, Salesforce, Qlik, Microsoft, Oracle, SAS, TIBCO, dotData, Aible, AnswerRocket, Big Squid, DataRobot, Outlier, Prevedere, Tellius, Evolved Analytics DataStories, H2O.ai, and Sisense.[14]

Augmented analytics is augmented intelligence, it supports AI, ML, NLP, NLG and advanced analytics algorithms including descriptive, diagnostic, predictive and prescriptive. Gartner described it as a technology for humans and machine partnerships [15]. Products such as Siri from Apple and Alexa from Amazon, are great examples of augmented intelligence having AI and NLG technologies embedded in it. [16]

9. Conclusion

The Augmented Analytics provides effortless and automated data collection, preparation, integration, analysis, visualization and interpretation to the business system. The BI vendors have designed tools to adopt Augmented Analytics and Advanced Analytics capabilities to make industry future ready though the development of these tools are in the early stages. The business industry can leverage its full potential unless it is implemented and adopted by moving away with traditional BI and decision making process. The adoption of Augmented analytics will bring business opportunities and consensus among business users, little knowledge and training in the use of these tools can further realized it. In this paper we have documented the need of Augmented Analytics adoption in businesses to maximize business value, it is high time for Data Analytics leaders and Business Industry to shift focus on to this automated tools and techniques for Business Data Analysis. We have also highlighted how Augmented Analytics can be seen as the Future of BI and how organizations can lead in terms of faster, more accurate and automated data analysis processes using AI, ML, NLP, NLG, and using other such important technologies, and tools with Augmented Analytics capabilities. There is a way ahead for full implementation of Augmented Analytics in business industry but the future is not too far to adopt as it has immense value potentials in industry.

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Locomotive Ergonomics: "An Analysis for Effective Stress Management of Indian Railway Loco Pilots"

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ABSTRACT

Work stress is inevitable in human life. People working in any organizational setup experience stress at some point in their work life. Work stress leads to lack of interest in work, creates discontent, increases absenteeism as well as mental health issues, which can be seen as a rising trend these days. Many organizations conduct regular workshops and training sessions for managing work stress. But there are certain factors which are not considered in prime fascia. One such factor is ergonomics. Most of the studies have proven that work stress is caused by the workstation design or workplace environment. An appropriate workstation design supports the need of the employees to work in a conducive work environment and utilize and explore their potential to the fullest. The main discussion of ergonomics workstation design in this paper includes control stand, operating equipment, working chairs, lighting, pollution level, sound proof and air pressure level. This paper presents a study on 42 Loco Pilots of Pune Division Indian Railways, channeling in various directions. Descriptive statistics is used for data analysis. The findings support that the control stand, operating equipment, working chairs, lighting, pollution level, sound proof and air pressure levels are not favorable to loco-pilots. Therefore this paper is focused on the importance of ergonomics with respect to occupational stress and its need for improvement in the cabin interiors of loco pilots. The study is concluded with some effective suggestions after having a discussion with a few respondents.

Keywords: Loco Pilots, Occupational Stress, Stress Management, Ergonomic, Workplace

INTRODUCTION

"**Ergonomics** is also known as human factors. It's the way of disciplined scientific study that helps us to study and understand the human interactions with the products, equipment, environment and systems." (Journal of Ergonomics Research.)

It is an open truth that the ergonomics can make our job easier and keep our work force safer. Safety is an asset which has to be valued more than just a priority. But it's given very little light on. One of the factors that assure safety is Ergonomics. Therefore, it must be incorporated and treated as an asset. Ergonomics began in the United States during World War II, when scientists designed new advanced systems without fully considering the people who would use them. Gradually it became clear that systems and products should be designed to take into account many human and environmental factors if they should be used safely and effectively. This awareness of the physical requirements of the people resulted in the discipline of ergonomics. Safety and ergonomics are the sciences of human factors. Both play a significant role in the success of a design or task. They can work synergistically to enhance each other, improve safety and productivity and also reduce employer's costs.

Both safety risk assessments and ergonomics help to identify hazards in the workplace that are at risk, and determine adequate preventive measures and risk monitoring. Each type of evaluation must be based on a holistic approach considering the total burden on the body while at work.

Loco Pilots means Engine Loco Pilots or any other competent Railway servant for the time being in charge of driving a train under the system of working. This Paper focuses to identify the efficiency of ergonomics of Loco Pilots, Indian Railways.

Loco Pilots are the unsung heroes of the Indian Railways. Their contributions have seldom been appreciated. Working for almost 10 hours a day, Loco Pilots not only have to perform their duties with mere perfection but they also have to ensure the life safety of other people. They cannot afford the slightest errors because the work that they do has always given them stringent punishment for errors, that is life or fatal accidents. The greater injustice is that even though the Loco Pilots may receive salary, allowances, reimbursement, quarters etc. to that of Class 3 employees, the workload as well as stress is preeminent to Class 1 employees. Work pressure is unbearable and cannot be described in words.

The job requires Loco Pilots to be in their full attentive zone, a slight miss of focus or attention could be a serious mistake. Even small mistakes like missing a signal or crossing the speed limit of the track could be attributed as a reason for an accident. In addition, their working conditions with extreme noise pollution and irregular working hours affects their sleeping patterns as well. Most of the Loco Pilots are facing ear related ailments such as hearing loss, vertigo in the later stages of their careers due to their long working hours in the noisy environment. The odd timings and erratic working hours greatly affect their regular activities, which leads to disturbances in their social life/interactions; a reason for rising mental related issues in their profession. Unlike other professions they don't even have a loo or toilet in the loco (cabin) and they are not supposed to take toilet breaks in between their work schedule. It is shocking to note that some of them use tracks to relieve themselves. The small loco cabin with just two people is another addition to their occupational hazard. Many times, the loco pilots witness accidents where animals or humans get run over by the train, whichcreate a lasting impact on them. It greatly affects their mental well-being.

Lack of facilities such as AC in their cabins adds to the discomfort to work in hot, humid or even cold conditions be it day or night. A Loco Pilot job requires them to pull mostly night shifts and work on weekends giving them very little time to spend with their family members. All this in some way or other adds worries to their already stressful life.

Objectives of the Study

1. To study the ergonomics of loco cabins of Loco Pilots.
2. To study the impact of ergonomics of loco cabins on Loco Pilots in their occupational stress.
3. To analyze the need for improvements in the ergonomics of loco cabins.

Hypothesis

H0: Ergonomics of Loco Cabins are appropriately designed for a conducive working environment for a loco pilot.

H1: Ergonomics of Loco Cabins are not appropriately designed for a conducive working environment for a loco pilot.

REVIEW OF LITERATURE

Kian Sek Tee, Eugene Low, Hashim Saim, Wan Nurshazwani Wan Zakaria, Safinaz Binti Mohd Khairuldin, Hazlita Isa, M. I. Awad and Chin Fhong Soon (2017) "A study on the ergonomic assessment in the workplace", This document aims to review the approaches and instruments used by the previous works of the researchers in the evaluation of ergonomics. Ergonomics has earned attention and taken into consideration the workers of different fields of work recently. It has had a great impact on the comfort of workers who directly affect efficiency of work and productivity. The workers have claimed to suffer pain and injuries in their workplace. Musculoskeletal disorders (MSDS) are the most common problem that workers frequently report. This problem occurs due to the lack of knowledge and alertness of workers to ergonomics in their surroundings. The two main methods that are often used for ergonomic evaluation are the rapid assessment of the upper extremities (rule) and the rapid body assessment (REBA). Popular devices are inertial measurement units (IMU) and Microsoft Kinect.

kayla. M. Fewster, Maureen,F, Riddle, Surabhi Kadam and Jack.P.Callaghan(2019),"The need to accommodate Monitor Height changes between sitting and standing", has emphasized on the necessity of sit-to-stand workstations in the workplace. The study has identified a research gap regarding the height adjustments of the desk from sitting to standing, for which there is not much study conducted so far. To overcome the limitation the study was conducted on 16 participants who stood and sat at a sit-to-stand workstation while adhering to current Canadian Standards Association (CSA) Guidelines for Office Ergonomics.

Dennis R. Jones(2015),"The relationship between working conditions and musculoskeletal / ergonomic disorders in a manufacturing facility – a longitudinal research study", the researcher studied the relationship between the working conditions and musculoskeletal / ergonomic disorders in a manufacturing facility. The researcher claims that the biomechanical and psychosocial aspects of work have a significant influence on the individual worker's health and well-being. The data was collected from a large manufacturing unit. The objective of the research was to identify the stressful working conditions and the means to arrest it. The overall aim of this research was to improve the long-term health and well-being of workers in a manufacturing facility. The methodology applied was the Balance Theory Model of Smith & Carayon-Sainfort. The study identified a research gap that more research studies need to be done in order to provide an answer to the existence of the link between biomechanical factors and psychosocial factors, and musculoskeletal disorders.

R. N. Sen and A. K. Ganguli (1982),"An Ergonomic Analysis Of Railway Locomotive Loco Pilots Functions In India" , had conducted a 3-tiered analysis of electric locomotive Loco Pilots functions in India and a task-operation-subsystem. The researcher concluded that there were serious deficiencies in the design of the current cabins of the locomotives of the Indian railroads. These defects made insecure working conditions, and opened the door to "human

error" accidents. Therefore, the role of man, the activities he has to carry out, and the physical and social environment in which they have to do work, all must be given detailed consideration to optimize the relationship of man-machine-environment. Only in this way can efficiency and alertness be ensured, along with both physiological and psychological well-being. It must be remembered that the correct design of the workplace is an essential part of operational security and reliability.

Kate Dobson (2015),"Human Factors and Ergonomics in transportation control systems",The research paper observes the evolution of Human Factors (HF) and ergonomics in the railroad from Operator's point of view. The practical areas for the application of HF at specific points in rail signaling and control systems are briefly described. HF's considerations in advanced train control systems and movement towards automation are discussed, as well as the impact of new technologies in the context of the operation itself. It has been claimed that there is a greater dependence on the operator to be vigilant and react efficiently when the intervention is required in the automation, both within the control room and withinthe environments of the control cabin. This article illustrates some of the human performance concerns for the new transport control systems that are faced today and analyzes how this area of cognitive care, human error and workload is difficult to evaluate and predict.

From the above review of literature, the identified gap is the study related to stress management of Loco Pilots. A very few studies are conducted on this topic. For a long time not much research has been done on this area. Since we are rapidly witnessing revolution in climate, culture and society, it is the utmost need to conduct more such researches to serve the betterment of railways in the fast-progressing society. This paper fulfills one of the purposes to reduce the widening research gap in this area and also provide a scope for future research studies.

RESEARCH METHODOLOGY

In order to fulfill the objectives, the study has adopted the survey method for soliciting the relevant data. The Loco Pilots connected with the train movement who formed the sample representing the universe for the study, were approached personally by the researcher at their workplaces and residences.

The collected data was codified and transferred to summary sheets for tabulation. The codified data was converted into percentages and based on the frequency and percentage the analyzed findings were presented in the form of tables and graphs. Hypothesis is tested and the conclusions were drawn accordingly.

Sample DESIGN
The universe for this study comprises the following personnel.

All Loco Pilots and assistant Loco Pilots connected with the train movements.
The Loco Pilots are groups that are distinct and therefore each group can be considered as a stratum. Stratified random sampling method is adopted for drawing 30% of sample from the total population of **Loco Pilots of Pune division**.
Sample is chosen as follows.

Sr.No	Category	Population	Sample	Percentage
1	Mail LP	39	16	41%
2	Passenger LP	8	3	37.5%
3	Goods LP	36	6	16.67%
4	Asst LP	76	17	22.36%

The total size of the sample is the sum of the samples drawn from the four categories i.e., 16+3+6+17 = 42. Therefore 42 is the sample size for the Loco Pilots.

FINDINGS

The study has the following hypothesis set to be tested.

H0: Ergonomics of Loco Cabins are appropriately designed for the conducive working environment for a loco pilot

H1: Ergonomics of Loco Cabins are not appropriately designed for the conducive working environment for a loco pilot

The study has considered 3 aspects of Ergonomics – Control Stand, Driving Seat and working space. For each of these aspects, the researcher had prepared statements to assess the status. The respondents were given three options for each of the statements – Always, Sometimes and Never. The statements were made in such a way that; if the respondent perceives ergonomics as conducive, he/she will respond as "Always". The tool was administered on 42 loco pilots.
Table 1 below gives the number of responses to a particular category for each of the statements under the aspect "Control Stand"

Table 1a: Number of Responses to Each of the following statements under Control Stand

	Always	**Sometimes**	**Never**	**Total**
Uniform design of control stands	3	6	33	42
Easily accessible	6	30	6	42
Are Gauges and Displays readable during night time on run	8	29	5	42
Is Front visibility clear in all types of Locos	3	37	2	42
Are lookoutglasses clear	5	36	1	42
Are wipers working properly during the raining season	2	39	1	42
Is the head light focus sufficient during Night Time	3	37	2	42
Is speedometer easy to read on run	5	36	1	42
Is digital speedometer more convenient to you	38	4	0	42
Is the location of speedometer convenient to Loco pilots	5	34	3	42
Is the speedometer parallel to the Loco pilot's eye view	1	24	17	42

The above table clearly shows that the number of respondents saying "Always'' is very low, not even 25% of respondents showed content in the statements regarding the control stand. The only aspect of the digital speedometer showed satisfaction; which could be overall interpreted as the loco-pilots are not happy with respect to the control stand aspect.

The assessment was done in one more way. There are 11 statements under "Control stand". Total of "Always" response was calculated. Following table depicts the number of "Always" responses out of 11.

Table 1b: Count of "Always" response out of 11 statements

	Frequency	Percent
0	3	7.1
1	25	59.5
2	4	9.5
3	4	9.5
4	3	7.1
5	1	2.4
6	1	2.4
8	1	2.4
Total	42	100.0

The table 1b shows that, out of 11 statements; maximum of 25 respondents have said "Always" for only one statement.

This again could be interpreted as the control stand aspect is not favorable to loco-pilots.

Table 2 below gives the number of responses to a particular category for each of the statements under the aspect "Driving Seat"

Table 2a: Number of Responses to Each of the following statements under Driving Seat

	Always	Sometimes	Never
Does adequate technical maintenance take place	6	26	10
Height Adjustment	6	28	8
Forward and Backward movement	5	30	7
Side Adjustment(Right to left)	5	30	7
Back Rest	10	25	7
Hand Rest	5	30	7
Proper cushion seat	9	28	5
Foot Rest	6	17	19

The above table clearly shows that the number of respondents saying "Always" is very low, not even 25% for all statements; which could be interpreted as the loco-pilots are not happy with respect to the driving seat aspect. The assessment was done in one more way. There are 8 statements under "Driving Seat". Total of "Always" response was calculated. Following table depicts the number of "Always" responses out of 8.

Table 2b: Count of "Always" response out of the 8 statements

	Frequency	Percent
0	28	66.7
1	4	9.5
2	8	19.0
7	1	2.4
8	1	2.4
Total	42	100.0

The table 2b shows that, out of 8 statements; maximum of 28 respondents have said "Always" for 0 statements; i.e., for none of the statements respondents have said "Always". **This supports the finding that driving seat's aspect is not favorable to loco-pilots.**

Table 3 below gives the number of responses to a particular category for each of the statements under the aspect "Working Space"

Table 3a: Number of Responses to each of the following statements under Working Space

	Always	Sometimes	Never
Enough working space	2	34	6
Effective heating and cooling systems	2	27	13
Are the drivers trained in the use of cabin components	18	23	1
Is Driver cab pollution free	1	3	38
Are drivers'cabin sound proof	0	0	42
Are the horns located far from the driver's cab	0	35	7

The above table clearly shows that the number of respondents saying "Always" is very low, not even 25% for all statements except 8.22; which could be interpreted as the loco-pilots are not happy with respect to the working space aspect.

The assessment was done in one more way. There are 6 statements under "Working Space". Total of "Always" response was calculated. Following table depicts the number of "Always" responses out of 6.

Table 3b: Count of "Always" response out of 8 statements

	Frequency	Percent
0	23	54.8
1	16	38.1
2	2	4.8
3	1	2.4
Total	42	100.0

The table 3b shows that, out of 6 statements; maximum of 23 respondents have said "Always" for 0 statements; i.e., for none of the statements, respondents have said "Always" and 16 respondents have said "Always" for only 1 statement.

This supports the finding that working space aspect is not favorable to loco-pilots.

SUGGESTIONS

The study would like to suggest certain points for the improvement of the conditions of locomotives. This may help in reducing the stress of loco pilots due to poor ergonomics.

Table No: 1a (Control Stand)
The loco pilots are working with various types of diesel and electric locos on a day-to-day basis. The cabs and operating handles are differently designed on each loco which may create stress and confusion among them. Therefore, introducing a uniform cab is very essential. It's been a long-standing demand from the loco pilots.

The regular maintenance and upgrading of the following such as head light focus, proper working of wipers and sanders, visibility of important cab gauges, visibility of speedometer or digitizing it is recommended.

During the summer season, the temperature of the loco cabin is normally higher than the atmospheric temperature for about 5 -7 degrees and vice versa. In the winter season the cab is not air proof to work efficiently. As Loco Pilots efficiency is bound to suffer if the temperature of the cab is either too low or too high, it is essential to have air-conditioning in the cab.

It has been noticed that the Joy Stick type throttle handle provided may be prone to accidental movement while operating, E.g., Jumping to dynamic brake while easing the throttle and vice versa and over jumping the notches. Operating switches and controls shall be clearly marked, so that they are easily identifiable by the Loco Pilot. Critical controls should be positioned or designed (e.g. covered) to prevent accidental operation. Operating switches and controls shall be positioned in such a way as to enable easy navigation of the cab controls.

Table No: 2a (Driver'sseat)
The loco pilots' working chairs should be upgraded with modern features such as cushion seat, height adjustment, side adjustment, forward and backward movement, back rest, hand rest, foot rest etc. It will provide comfort for working long hours on duty. Compressed air suspension seats will be preferred to absorb the vibrations so that back pain is minimized.

Table No: 3a (Working Space)
Prolonged Exposure to loud noise more than permissible may lead to stress and deafness. Various studies proved that noise can cause hearing impairment, hypertension, ischemic heart disease, annoyance, mental stress, sleep disturbance as well as changes in the immune system. Therefore, the loco cab should be sound proof and both the side horns can be shifted far end off the cab. Noise levels in the Loco Pilot's compartments must be kept as low as possible by limiting noise at its source through appropriate measures (acoustic insulation, sound absorption) using the 'State of the art' technology available at the time of installation.

Doors of the Loco Cabin should be wide in such a manner that accommodates different body sizes of Loco Pilots. It has been noticed that internal and external doors of the Loco Cabin are open frequently and distract the attention of the Loco Pilot due to improper latches.

Loco Cab and Running plate floor surfaces should be slip resistant.
It has been noticed that no Emergency Exit doors are provided in all types of Locos. It will play a vital role in case of accidents when the Loco is capsized and normal doors are unable to open.

General Suggestions:
The interview and discussion with the Loco Pilots helped to identify their problems and the requirements that need to be taken care of.

Driving crews must be protected against sudden and major air-pressure fluctuations that might occur - especially at high speeds - when passing other trains and/or travelling through tunnels (both of these may occur at the same time). This system should also be capable of being adjusted manually by the Loco Pilot.

Loco should design and construct cabs which reduce exposure to Electromagnetic Fields to the minimum that can be achieved. The driving cab must be regularly tested to ensure that the exposure to Electromagnetic Fields (EMF) is kept to a minimum. In the electrified section Locos are running constantly with a prolonged time under the 25 KV watt.
There is no proper adjusting windscreen provided in look out glasses to avoid sun glazes.

All indicator lights must be designed and positioned so that they can be read in all variations of natural or artificial lighting especially during periods of bright sunlight. Indicators should be positioned on the desk in such a way that windscreen glare is mitigated during periods of darkness. Indicator lighting behind the Loco Pilot should be kept to a minimum and shrouded to prevent windscreen glare during periods of darkness. When additional lamps are provided (e.g. a lamp for the ALP) these must not dazzle the Loco Pilot. Desk and instrument lights should be provided with a rotary control dimmer so that the lighting levels within the cab can be kept to a minimum to aid vision during times of reduced visibility and darkness.
Toilet facilities must be provided on Locos since Loco Pilots are working long hours on duty, sometimes without stopping for a longer period. It will help to minimize the physical and mental stress of Loco Pilots by holding back the natural calls.

CONCLUSION

The construction and design of the Locomotive is one of the factors highly influencing the performance of the train Loco Pilots in security related matters. The cabin is the daily workplace of the Loco Pilots; the facilities offered are of great importance to them both physically and mentally. Control stand, Driver seat, working space are the three variables taken for the present study. The study found that Loco Pilots face a common problem of lack of uniformity in the control stand, which is similar to a dashboard for a four-wheeler of different Locos. This leads to confusion among them which can cause stress leading to some grave mistakes.

Another major problem is noisy cab since the Engine sound and Horn can cause hearing ailments among the loco pilots. Most Loco Pilots also suggested more comfortable driving seats. Since the working hours exceeds more than 8 hours, it's a necessity to have comfortable seats, or else which could lead to other health problems such as backache etc. Considering all these major issues, the working space should be acoustic and of high standard.

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Sony Chundattu, CMI

Christ College Pune Research Series-1

Responsibility and Accountability to Our Common Home

Caring, Nurturing and Sustaining the Environment

Editors
Thomas Karimundackal, SJ
Sony Chundattu, CMI

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Christ College Pune Research Series-2

ECOLOGICAL CRISES AND RESPONSES

SCIENTIFIC AND SPIRITUAL PERSPECTIVES ON BREATH OF EARTH AND DEPTH OF LIFE

Sony Chundattu, CMI

"'Care for our common home' demands an ecological conversion, a moral rootedness and spiritual depth, which in turn, would lead to religious openness, economic restructuring and political commitment."

Dr. Sony Chundattu, through this book, helps us to rethink about our existence in this universe not as a problem but as a provider of solution and hope. … This book is indeed documentation and rational analysis of the present crisis and how to move forward rationally, ethically and spiritually to exist, co-exist and pro-exist. I congratulate Dr. Sony Chundattu for coming up at this critical juncture with the precious book **"Breath of Earth, Depth of Life"** which will definitely enable us to breathe smoothly and encounter our life in this universe as a custodian rather than an annihilator!

Prof. Dr. Mathew Chandrankunnel, CMI
Director, Ecumenical Christian Centre
Whitefield, Bangalore

Dr. Sony J. Chundattu, CMI is the Principal of Christ College, Pune, and author of the book Breaths - An Elixir for Liver. He is conferred with Life Long Award from IBRF for his outstanding contribution in the field of Academic Administration. He is also the recipient of Golden Award of Human Excellence for Global Service from United Nations University for Global Peace and World Peace Excellence Award 2018 from World Academy Foundation, Kolkata.

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EMPOWERED WOMEN ENNOBLED HUMANS

ENABLING DIGNITY AND EQUALITY

Editors
Thomas Karimundackal, SJ
Sony Chundattu, CMI

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4

Women Empowerment and Empowerment of Society

Sony Chundattu, CMI

Abstract

Women empowerment calls for improving the social, economic, political and legal strength of the women to ensure equal right to women and to make them confident enough to claim their rights. It is to make them self-reliant, independent, and generate confidence to face any difficult situation and to enable them to participate in various socio-political development endeavors. It also means we are engaging them positively and promoting their relevance in the society. Thus, this article deals with the importance of women empowerment, advantages of women empowerment, the situation of women empowerment in India, and various measures to be taken for women empowerment.

Keywords

Women empowerment, self-reliant, independent, self-esteem, employment, equality, gender difference

Introduction

"Countries and nations which do not respect women have never become great nor will ever be in future," says Swami Vivekananda. Women empowerment calls for improving the social, economic, political and legal strength of the women to ensure equal right to women and to make them confident enough to claim their rights. Empowerment of women would mean encouraging women to be self-reliant, independent, have positive self-esteem, generate confidence

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Climate Change
An Existential Threat to Humanity

Sony Chundattu, CMI

Abstract

Climate change is a great threat to humanity, and it may lead to one of the potential global catastrophes. We are aware of how humanity has undermined the natural resources upon which its very survival is based. We are witnessing an accelerating loss of clean air, drinkable water, CO_2 absorbing forests, pollinating insects, protein-rich fish and storm-blocking mangroves etc. and these pose threats to humanity. It is impossible to know clearly what will happen in the future if deforestation and greenhouse effects continue to rise in a massive scale as we have it now. Ours may be a more technologically advanced society, but this gives little ground to believe that we are immune to the threats posed by the climate change. One of the reasons for the refusal to accept the acute reality of climate change is what we call "motivated interference," which occurs when we hold a specific bias to ignore the evidence. One approach that encourages us to take action about climate change is to emphasize simple and ordinary things that each individual can do in his/her everyday life. Thus, this paper shows how climate change poses a real existential threat to humanity.

Keywords

Climate change, existential threat, greenhouse effect, deforestation, global warming, collapse of civilization etc.

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2

The Status of Women in Indian Society Today

Priti S. Khandve

Abstract

The following article is an honest attempt to bridge the gap between the status of women in ancient India and todays India. It contains description of how women were treated in ancient era through various factors. Women had great respect in ancient times. After certain period of time the status of women in the society was degraded. So many people, "I would say angels for women in trouble," fought for returning their respect by all ways and for all possible issues. All these issues are categorised and discussed here along with fighters and their contribution to bring women out of those problems. These efforts are resulted into bringing new and various laws in the society. But as we see today, modern society knows very well to do desired things and playing with the laws. So in conclusion, the status where women stand in Indian society today is questionable.

Keywords

Women empowerment, women with full freedom, women with partial freedom, women with no Freedom, Vedic period, epic period, caste system, politics.

Introduction

Two epics in Hindu Mythology "Ramayana" and "Mahabharata" carry a lot of weight and reverence in our society. Currently, I find that a lot needs to be revisited and understood regarding the status of women in India today. I do believe the interpretation of these two epics needs to be studied in depth

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3

Empowerment of Women The Code for Success

Nilima Shingate

Abstract

This paper is an attempt to analyse the success code or the key factors based on women empowerment. As well as it explains how to achieve success through various ways to empower women like save the girl, education, health, ensure women safety, gender inequality, give her freedom, build self-esteem and confidence, boost decision- making power. Today the empowerment of women has become one of the most important concerns of 21st century. But practically women empowerment is still an illusion of reality. Women empowerment is the vital instrument to expand women's ability to have resources and to make strategic life choices. Empowerment of women is essentially the process of upliftment of economic, social and political status of women, the traditionally underprivileged ones, in the society. This study focuses on number on factors which will make women powerful and independent, and it concludes by an observation that empowerment of women is very much essential to achieve sustainable development in the society.

Keywords

Women empowerment, education, health, gender equality, freedom, independence.

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6

Women Empowerment and Political Participation

Puja Jindal

Abstract

Much has been talked about the empowerment of women in recent years. There have been numerous discussions, surveys, reforms, declarations, amendments at all levels in order to ensure that women get equal rights and equal opportunities to be a part of every arena, be it social, economic, civic or political. Changes have definitely been seen. This article specifically focuses on the role of women as decision makers and policy changers. Women can be very much instrumental in bringing about socio-economic change if they get to walk the corridors of power. In India, after the 73rd Amendment Act, the participation of women at local level political institutions has increased greatly. And in spite of problems being encountered, they are making an impact at grass roots levels, leading to boosted confidence, improved status of women and changed perception about their capabilities as change makers. There still is a long road ahead, but steps are being taken, and that is what makes all the difference.

Keywords

Women empowerment, grassroots level institutions, political participation, sustainable development, elected women representatives

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9

Empowering Women through Self Help Groups

Swati Jadhav

Abstract

The word empowerment means giving power and power means having the capacity and the means to direct one's life towards desired social, political and economic goals or status. Empowerment provides a greater access to knowledge and resources, more autonomy in decision making, greater ability to plan lives, more control over the circumstances which influence lives, and freedom from customs, beliefs and practices. Empowerment of women is not just a goal in itself, but key to all global development goals. Women's empowerment is a much debated topic, which has many aspects to it. It is not only multidimensional in approach, but also helps one understand that, when women are empowered, they gain self-confidence and actualize their own power and capabilities. Self Help Groups (SHGs) have also brought women together from all spheres of life, helped them speak for themselves and fight for their rights wherever needed.

Keywords

Self Help Groups (SHGs), women's participation, women empowerment, self-empowerment.

Introduction

SHGs are small voluntary groups that are formed by people related by an affinity for a specific purpose who provide support for each other. They are

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12

Bridging the Gap
Women, Communication and Technology

Leena Sanu

Abstract

The status of women has been varying ever since the early Vedic period. The emergence of Technology has brought radical changes in the lives of women. Use of Internet has also played an important role in empowering women of today. Measures adopted by various organizations have gone way ahead in bridging the gender gap. Endless opportunities and possibilities for information exchange have been feasible through the various avenues set up for women empowerment. Speeding up the process of change will only be achievable through government policies and coordinated actions within organizations. Communication and Technology have paved the way in narrowing the gap and tackling the gender inequality at different levels. This study aims to explore the different measures and opportunities that will eradicate all barriers to women's full participation in the digital world and thereby empower women and the nation.

Keywords

Women, communication, technology, Internet, empowerment.

Introduction

"The fight is not for woman's status but for human worth"[1] India is striving hard in terms of providing equal opportunities to women which accounts to

1 Dr. Bir Pal Singh, Status of woman in ancient, medieval and modern period (National Law Institu

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16

Empowerment of Women Not to Acquire but to Realize

Sushil Laxman Sawant

Abstract

This article is an attempt to show that women are physically, mentally and emotionally stronger than men using various indicators based on the data from secondary sources. Empowerment of women must leads to control over resources and ideologies. It particularly includes greater self-confidence. It must teach an inner transformation of one's consciousness that enables one to overcome external barriers. However women of India enjoy somewhat lower status than that of men. Gender gap exists regarding access to education and employment. The freedom of movement and power of household decision may vary considerably with their age, education and employment status. Empowering women is to make them realize their strength and change their attitude towards life to achieve the goal.

Keywords

Physical superiority, emotional superiority, competent leadership, life expectancy, COVID 19, brain, communicator, passion

Introduction

While delivering a lecture in my class, I recited a saying, Practice makes MAN perfect. Then I asked, what about WOMEN? Most of the students said that this is true for both. But I said mischievously, No. This is not true for women

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4

Environmental Pollution and Its Solutions

Shagufta Khan

Abstract

Environmental pollution remains a major source of health risk throughout the world, though risks are generally higher in developing countries, where poverty, lack of investment in modern technology and weak environmental legislation combine to cause high pollution levels. Associations between environmental pollution and health outcome are, however, complex and often poorly characterized. Levels of exposure, for example, are often uncertain or unknown as a result of the lack of detailed monitoring and inevitable variations within any population group. Exposures may occur via a range of pathways and exposure processes. Individual pollutants may be implicated in a wide range of health effects, whereas few diseases are directly attributable to single pollutants. Long latency times, the effects of cumulative exposures, and multiple exposures to different pollutants which might act synergistically all create difficulties in unravelling associations between environmental pollution and health. Nevertheless, in recent years, several attempts have been made to assess the global burden of disease as a result of environmental pollution, either in terms of mortality or disability-adjusted life years. About 8–9% of the total disease burden may be attributed to pollution, but considerably more in developing countries. Unsafe water, poor sanitation and poor hygiene are seen to be the major sources of exposure, along with indoor air pollution.

Keywords

Contamination, ecosystem, fossils, disaster, combustion, vulnerable, management etc.

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5

Loss of Biodiversity

Aware Balasaheb Parasram

Abstract

Loss of biodiversity has negative impacts to the functioning of ecosystem. Major factors affecting to the loss of biodiversity are natural calamities and man-made calamities. Ecosystem has capacity to regain loss due to natural calamities. But losses due to man-made calamities are to such large extent that cannot be regained and reproduced. In August 2019 India was affected by flood. Maharashtra and Kerala was mostly affected States due to heavy rain, in spit of destroying human lives it also destroyed trees and animals. Australia bushfire in 2019 had destroyed billions of animals, trees. Overexploiting resources like overfishing results to destroying these resources permanently. Deforestation has negative impacts on ecosystem like reduction in rain, climate change. Pollution also destroy whole ecosystem, for example acidification of water destroys whole underwater ecosystem. Technology has negative impact like emission of radiations from cell phone towers affect to bird population. Governmental and non-governmental organisations should take steps toward conservation of biodiversity for healthy ecosystem.

Keywords

Biodiversity, loss of biodiversity, natural calamities, man-made calamities, ecosystem, conservation, etc.

Introduction

Biodiversity is the variety of species of plants and animals in given specific area. The reduction in the variety or richness of species is known as biodiversity

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6

Alien Species Boon or Curse

Aparna Lal

Abstract

This article deals with alien species, the process of their occurrence, characteristics to understand their existence, effects and use in ecosystem, their economical and biological benefits. Invasive alien species are a huge threat to any ecosystem, and often they can vanish an entire ecosystem. In order to avoid this different measures have been carried out by different organisation. Proper management and control is needed to avoid alien species to be transformed as invasive alien species. *Eichhornia crassipes* is a free floating plant found growing in almost all the aquatic environment of Asia Pacific region. It is edible and widely used as a cleaning agent and source of fibre in textile industries. However it also kills the aquatic life by restricting flow of oxygen into water. Tilapia is globally accepted food source, different food products are made from it, however, it is proven to be one of the hazardous invasive alien species. Eucalyptus, one of the vital medicinal plant has proven to be dangerous for forest due to its flammable character. Present article is helpful in studying the various aspects of alien and invasive alien species.

Keywords

Alien species, invasive alien species, foreign plants in India, Tilapia, Eichhornia etc.

Introduction

Humans like to transport numerous belongings since historic period either for their personal or social gain. They have been transporting animals and plants

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9

Impact of Climate Change and Farmers' Suicides in India – A Case Study

Yogesh Patil

Abstract

Agriculture is the primary source of livelihood for about 70% of India's population. Farmers are the backbone of Indian agriculture. Even though, farmer suicide account for 11.2% of all suicides in India. There are various reasons for farmer suicide but most of them are associated with climate change such as monsoon failure, changing weather patterns, changing precipitation patterns and changing temperature. Indian Agriculture is mainly based on monsoon rainfall. 56% of the country depends on snow-fed rivers for its water and in such situation even marginal change can have catastrophic effects on agricultural production. Economic loss due to the natural calamities like flood, Drought is mainly responsible for farmer's suicides. Productive agriculture is mostly depends on favorable climatic conditions. Because of changing climatic conditions it is very difficult for farmers to take quantitative and qualitative production of crops. To cope with all those situations we need to strengthen the farmers with different protective policies which will help farmers to survive and grow in changing climatic conditions.

Keywords

Climate, farmer suicide, flood, drought, policies etc.

Introduction

Climate change is a scenario in global or regional climate patterns, in particular a change apparent from the mid to late 20th century onwards and attributed

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10

Reduce, Reuse and Recycle

Tresa Antony

Abstract

In the quest of development, man has exploited nature wantonly. Today the earth is facing a lot of environmental issues, one among them being waste disposal. Consumerism has led to a lot of waste generation, more than that which can be decomposed or recycled. In this article the types of waste, their nature and severe effects of waste on the environment have been discussed. It mentions the current scenario of waste disposal in India and their drawbacks. In order to tackle this issue it is necessary to bring about a change in lifestyle so that the amount of waste generated is reduced. It also suggests to adopt the principle of "3R", i.e Reduce, Reuse & Recycle for our daily lives. If we as individuals and as an institute practice this, it would go a long way in solving the problem of waste disposal.

Keywords

Reduce, reuse, recycle, types of waste, waste management,

Introduction

Human beings control our planet today. This is quite different from what it was centuries ago. Man had a cordial relationship with the environment but in the course of time, in order to achieve his selfish goal in the name of development, he started exploiting the nature. The environment had to pay the price for man's progress in the form of deterioration due to overpopulation, landfills, deforestation and consumerism. The Earth is currently facing a lot of

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11

Advantages of Organic Farming

Suvarna Parande (Nade)

Abstract

Organic farming aims for human welfare without harming the environment and follows the principles of health, ecology, fairness and care for all including soil. The modern concept of organic farming combines the tradition, innovation and science. The scientific literature shows that organic farming practices build soil quality, maintain water quality, support biodiversity, and have potential to mitigate global climate change while supporting an economic bright spot. Organic farming is comprised of an integrated suite of practices that provide these benefits in addition to producing food, fibre and feed. The role of organic agriculture, whether in farming, processing, distribution or consumption is to sustain and enhance the health of ecosystem and organisms of soil to human beings. In particular organic farming is intended to produce high quality, nutritious food that contributes to maintenance of human health. Implementation and the marketing of organic food products as well as the increasing demand of the organic products in the domestic as well as international market, there is ample scope for organic food industries to expand and generate revenue for strengthening the economy. This article mainly analyses and evaluates the advantages and benefits of organic farming in Health, Ecological, Economic, National prosperity, environmental protection in human life

Keywords

Organic farming, organic food industry, ecological, human health, environment etc.

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12

Environment and Wireless Communication

David Thomas Vidhate

Abstract

This article is concerned is primarily concerned with environment and wireless communication. Communication through air is the oldest form of communication. All living beings of this planet use some form of frequency level of the atmosphere to communicate with their own species. Today, due to advancement in the network and wireless network technology almost all our houses are connected to the wireless technology through servers and wireless sensor networks. Wireless communication is among technology's biggest contributions to mankind. Wireless communication involves the transmission of information over a distance without help of wires, cables or any other forms of electrical conductors. The transmitted distance can be anywhere between a few meters (for example, a television's remote control) and thousands of kilometers (for example, radio communication). Some of the devices used for wireless communication are cordless telephones, mobiles, GPS units, wireless computer parts, and satellite television. In short, this article reflects upon the use of wireless communication and wireless technology that we have been using for more than five decades.

Keywords

Environment, wireless sensor, wireless communication, smart city, layers of atmosphere etc.

Introduction

The telecommunications sector has grown from the telegraph, where communication took days, to modern mobile technology, where large amounts

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14

Renewable Resources and Eco-Friendly Vehicles

Nisha V. More

Abstract

Since the human civilization has come into existence, energy plays a vital role in the growth and development of the mankind, with advancement in technology the requirement and exploitation of energy has also increased. However as the human population has increased more pressure is exerted on existing energy resources i.e. fossil fuel, and as we are facing the threat of the resources being exhausted which is even threatening human life with the global warming resulting to climate change at an alarming rate. Researchers are continuously engaged in finding out alternative sources of energy, which are named as renewable resources that can be used for cooking, lighting our homes as well as running the vehicles on the roads. Such vehicles are called as eco-friendly vehicles or non-polluting vehicles. This article focuses on various renewable resources and shows, focusing mainly on Indian scenario, how it is beneficial for generating alternative sources of energy for the e-vehicle. Such 'e-vehicle' will be highly benefitting for the future mankind to reduce the greenhouse effect and there by less use of fossil fuel

Keywords

Renewable resources, environment, solar, wind, hydropower, geothermal, biomass-vehicle, vehicle to grid etc.

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18

Smart Cities for a Sustainable Environment

Petricia Leema Roseline

Abstract

Cities are increasingly pressured to change to maintain competitiveness and improve the quality of life of its citizens. Smart city bridges the gap between the two challenges: the growth of urban infrastructure and the sustainable use of resources. Smart cities bring together infrastructure and technology to improve the quality of life of citizens and enhance their interactions with the urban environment.

Keywords

Infrastructure, urban, challenges, Smart City, technology.

Introduction

Smart cities aim to use technology solution to address issues such as air and water pollution, sanitation, and public safety. The initiatives often focus on areas such as energy and water conservation, with cities incorporating smart street lighting and automated water meters. Smart City means different things to different people. The conceptualization of Smart City, therefore, varies from city to city and country to country, depending on the level of development, willingness to change and reform, resources and aspirations of the city residents.

1. What is Smart City?

The Smart Cities Mission of the Government is a bold, new initiative. It means to set examples that can be replicated both within and outside the Smart City,

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Women Empowerment and Microfinance

Jiji Johnson

Abstract

Microfinance is a category of financial services targeting individuals and small businesses that lack access to conventional banking and related services. In this paper we see how microfinance helps the ones who have no or very little access to such financial services, especially women. We see the history of microfinance, empowerment provided to women, how women contribute, the evidence of the empowerment and the challenges and limitations. After taking a lot of factors into consideration we find that even though microfinance has a lot of positives and is a start in leading women in the right direction, it is not sufficient to provide women with the financial independence required to live in this day and age.

Keywords

Microfinance, women empowerment, challenges, limitations.

Introduction

According to the microcredit summit campaign report 2007, over 3,300 microfinance institutions reached 133 million clients with a microloan in 2006. 93 million of the clients were among the poorest when they took their first loan. 85 percent of these poorest clients were women.[1] According to

1 International Labour Office Geneva, Small change, Big changes: Women and Microfinance, 2008.Available from http://www.ilo.org/wcmsp5/groups/public/@dgreports/@gender/documents/meetingdocument/ wcms_091581.pdf (accessed on 01- April 2020).

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5

A Sustainable Empowerment Model for the Underprivileged Women in the Society

Geetha Rajan. N

Abstract

There is no tool for development more effective than the empowerment of women[1]. The key for inclusive development is when all sections of society men, women, poor, can lead a better life that can create sustainability. The different facets of life must come together and work in synergy for growth and development in general. Women are an important part of society and they need to be a part in the labor force if we need sustainable and inclusive development. In many countries women are achieving equality in health outcomes and enrollment rates in primary school but the world has not seen the same kind of progress with regard to gender equality in economic opportunities. Women consistently trail men in formal participation of the workforce, access to credit, savings rates, income levels, entrepreneurial skills, as well as inheritance and property rights. Normally women invest in their families and communities significantly more than men. Thus they spread wealth which results in increasing the quality of life for themselves and their families. Therefore Women empowerment should start from the bottom level of the society. The underprivileged women in the society need to be empowered more to have an inclusive development in the country. When women participate in the economy, everyone benefits[2]. It is only then the agenda of the government of ***Sabka Saath Sabka Vikas*** can be a reality.

1 https://twitter.com/kofiannan/status/442418339840012288?lang=en (accessed May 19, 2020).

2 https://quotefancy.com/quote/828642/Hillary-Clinton-When-women-participate-in-the-economy-everyone-benefits (accessed May 19, 2020).

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8

Homepreneurship
A Key to Women Empowerment

Deepa Sujith

Abstract

Over the past few decades there has been a steady progress and transformation in the status and position of women. Women have proved their potential in various areas like politics, space research, aeronautics and even entrepreneurship. The various policies formulated by the Government after independence along with the developmental programs and support mechanisms initiated by the non-governmental organisations have played a major role in encouraging women. In spite of all these efforts a large number of Indian women still remain incompetent and most of them are unaware of these programs and as such expected level of development is not achieved especially in the rural areas and among underprivileged section. Indian women and Indian society even while following constitutional rules and principles remain in the clutches of social prejudices and only a negligible proportion of the total population can be brought under the category of empowered individual. This study therefore aims to understand the concept of Homepreneurship, as an initial step for women to explore the area of business where their hidden entrepreneurial calibre can be realised and utilised to attain economic self-sufficiency along with simultaneous social and moral development while maintaining work life balance. In the process women will develop their confidence and reach a level of self-esteem which is the ultimate aim of empowerment. Eventually, touching the pinnacle of success they will also contribute towards income generation and sustainable development.

Keywords

Homepreneurship, entrepreneurship, empowerment, economic self-sufficiency, self-esteem

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7

Environmental Laws and Water Act of 1974

Rutuja Purohit

Abstract

Safeguarding the environment and maintaining ecological balance is a task not only of the government but of every human being, corporation and association. The Constitution of India under Article 51 A(g) states 'protection of environment' as a fundamental duty of every citizen. The concept of 'environmental protection' is an old wine in new bottle, as its presence is seen in the Indian traditions since time immemorial. To understand the present scenario of environment protection and conservation of natural resources, it is imperative to revamp the past Indian traditions, cultures etc. of protecting the environment. Until 1970's, there were no precise well recognized environmental laws. However, the Stockholm Conference of 1972 paved way to environment awareness. This article will give an insight of legal framework of environmental laws with special reference to Water Act, 1974. Indian judiciary along with legislature plays a pivotal role in building the legal environmental framework. Various series of landmark judgments have flooded the environmental laws, making provisions for constant amendments.

Keywords

Environment protection, ecology, environment awareness, Stockholm declaration, fundamental duty, environmental laws, Water Act, 1974

Introduction

Safeguarding the environment and maintaining ecological balance is a task not only of the government but of mankind. It is a social obligation. Environmental protection and improvement were explicitly incorporated into the Constitution

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13

The Indian Railways An Emerging Green Leader

Blesson Varghese James

Abstract

Indian Railways (IR) being the largest national transporter, faces tough competition from other modes of transport considered time saving and effective. Nevertheless when it comes to environmental impact specifically with regard to air pollution numerous studies have shown Indian Railways' consistent efforts towards reducing carbon footprint. Taking the case of the Indian Railways this paper attempts to explore and review various initiatives and possibilities of emerging as a green leader. Finally citing the IEA recommendations the paper argues for intensive investment in Indian Railways to ensure a greener and cleaner future.

Keywords

Indian Railways, modes of transport, green leader, electrification, solar energy, green building etc.

Introduction

The wheel was invented way back in 4500 B.C. One of the greatest inventions in the history of mankind, the invention of wheel, became fundamental for transportation, agriculture and economy. Wheel was used first as a potter's wheel by the Mesopotamians and later for transportation which was a nightmare back then. As years passed by, the world witnessed evolution of the wheel in various forms, from potter's wheel to wheel barrow, from carts and wagons to motor cars and trains. Transportation has an impact on the factors

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16

Sustainable Development
A Viable Solution for Environmental Degradation

Maninder Kaur

Abstract

Environmental degradation is drastically impacting natural resources, quality of air, water and soil, plants, animals, micro-organisms, wildlife, biodiversity and ecosystem. Sometimes, destruction is so huge that it cannot be reversed. High level of CO_2 emission, climate change, global warming, ozone layer depletion, acid rain, loss of biodiversity, water degradation, and natural disasters can be listed under dangerous impacts of environmental degradation. In today's world, economic performance gets the highest priority pushing back the concern about environmental sustainability. Global environmental issues are now in limelight bothering the governments and city planners worldwide. Governments, NGOs, corporate houses, local organizations and citizens should come forward and work in harmony to discuss and solve environmental issues and improve its quality and ultimately the quality of human life. This paper focuses on the causes and adverse impacts of environmental degradation. Sustainable development and its three dimensions: Economic sustainability, social sustainability and environmental sustainability has been explained. Implementation of sustainable development has been acknowledged as a strategy to address the emerging issues of increasing environmental degradation. Furthermore, it throws light on a successful sustainable development model implemented in Magarpatta city.

Keywords

Environmental degradation, sustainable development, climate change, global warming, eco-friendly practices, Magarpatta city, etc.

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13

Women Entrepreneurship and Women Empowerment

Ruby Phumale

Abstract

The attitude to create something new, through an activity which creates value in the socio-economic system can be defined as entrepreneurship. Entrepreneurship is an attitude which emerges from one's state of mind. It is the outcome of one's surroundings and experiences. Entrepreneurship is the positive step one takes in view of their career and life. Women in India occupy and play an important role in entrepreneurship. Even though this did not hold true historically in India, the status of women in the society has seen a great deal of evolution with changes likewise in the environment of the country.

Keywords

Woman entrepreneurs, Industry, economy, self-reliance, democratic values, entrepreneurial mindset.

Introduction

A woman entrepreneur is one who desires economic independence, has an eye for creativity and innovation, utmost confidence in herself and the ability to generate employment opportunities for several people. Women Entrepreneurs may be defined as the women or a group of women who initiate, organize and operate a business enterprise. The Best Project No – 2 (2004) published by the European Commission defines a female entrepreneur as a woman who

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2

Global Environmentalism - Issues and Challenges

Assunta Mendonca-Sandhir

Abstract

Since time immemorial, development through globalization and environmental degradation are intertwined in a very subtle manner. They continue to be debated and discussed. Against the backdrop of globalization, this article delves into an overview of global environmentalism and how it is perceived from various schools of thought, from a pre-globalization world to a post-globalization world. It also tries to explore the role of global environmental politics in ensuring global environmental security with the collaborative initiatives of various international organizations, role of nation states as actors, policy makers, scientists, researchers and many more, who have tried to ensure conservation of the planet. It discusses the specific issues and challenges that are faced at the global level and tries to provide a holistic picture of these problems. The destiny of the global environment remains debatable, and calls for concrete action, compliance and agreement for the benefit and coexistence of all.

Keywords

Environmentalism, globalization, green grabbing, climate change, environmental refugees, biodiversity, sustainable development, circular economy, etc.

Introduction

Global environmentalism has become the most debated and important issue today in world politics as environmental problems transcend national boundaries

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15

Women's Crawl towards Empowerment Voicing through their Writings

Priya Wahab

Abstract

The term 'women empowerment' can be associated with women's struggle for social justice and equality. But today the empowerment of women is a multi-dimensional concept. Ensuring equal rights to women is not enough, the prevailing thought that women are less capable than men in every field, has to be erased from the common mindset. Women are now coming out of the four walls of their homes and taking lead in all sorts of activities. The present paper aims at exploring the various aspects of literary writings by women writers on the empowerment of women. Among the many eminent women writers, one each from the pre-independence era, post-independence era and modern era has been chosen for exploring their writings on women empowerment. These well-known writers are Sarojini Naidu, Kamala Das and Anita Desai who have awakened the women of India through their extensive writings on women emancipation. They have been writing with immense concern, passion and understanding of their female characters presuming them to be liberated.

Keywords

Women empowerment, social justice, equal-rights, literary writings.

Introduction

More than a hundred years ago, some brave suffragette women such as Elizabeth Cady Stanton, Emmeline Pankhurst and Susan B. Anthony led

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8

Solid Waste Management in India

Merin K.Varghese

Abstract

One of the major problems which not only India, but majority of the countries is facing is waste management, which if not properly treated or disposed can lead to environmental degradation and health problems. It is of utmost importance to frame policies to manage the entire working of waste management. Government of India follows a decentralized method, i.e., the local authorities are in charge of collecting, transporting and disposing/recycling the waste. The public too places a very important role in reducing and segregating the waste at source. People should make sure that the surrounding they live in is clean and hygienic, being proactive in case of following reduce-reuse-recycle, the products they use is not harming the environment or any living being. Each and every person is responsible for safeguarding the environment.

Keywords

Solid waste, environmental degradation, government, decentralization, reduce, reuse, recycle, hygienic etc.

Introduction

Waste is defined as discarded materials or items, which are no longer in use nor can be used in future. Waste can be classified as liquid waste, solid waste, organic waste, recyclable waste and hazardous waste. These types of wastes are commonly found in our household and it's very important to segregate these wastes, so that they can be disposed off without harming the

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Implementation of fuzzy logic method for physical and mental illness by smart phones

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Abstract

Using Fuzzy logic, different types of complex systems can be analyzed more effectively than using traditional approaches. In revolutionary technology, a smart phone emits radio frequency energy which can be absorbed by the tissues in the body. Sleep dispossession is one of the common bad effects of using smart phones. Moreover phones produce light effects which can damage our eye sight. In this paper, we analyzed smart phoned are also found to have a negative impact on health issue.

Keywords: Fuzzy logic, MatLab, centroid, rule based system, fuzzification.

Introduction

This research work is giving some awareness and advice about health. Fuzzy logic is used to analyze and diagnose negatively affected person by the smart phone. Fuzzy has been deemed to be the best methodology, mainly because of its rule based system. To evaluate the performance and implementation of simulation model we used MATLAB tool software.

L.A. Zadeh[1] proposed fuzzy sets. Mamdani[2] suggested control of simple dynamic plant application using fuzzy algorithm. Zhen and Feng[3] suggested the design of neural network fuzzy controller in washing machine.

George and Yuan[4] developed fuzzy sets and fuzzy logic. Fuzzy logic based control system suggested by Kumar and Haider[5].

Mendel[6] presented fuzzy logic system for engineering. Hellendron and Thomas[7] proposed defuzzification in fuzzy Controllers. Yazid and Mouftah[8] suggested congestion control methods for BISDN.

Lohani and Hasan[9] suggested the design of an improved controller microchip for washing machine. Virkhare and Jasutkar[10] developed Neuro-Fuzzy controller based washing machine. Petricia Leema Roseline[11] developed unit consumption under fuzzy logic control of washing machine.

In this paper a model using fuzzy logic with three inputs and two outputs is proposed. This paper developed to analyze and diagnose the negatively affected person by the smart phones.

Structure of Fuzzy Logic System

Fuzzy inputs, outputs rule and defuzzification are given in Figure-1.

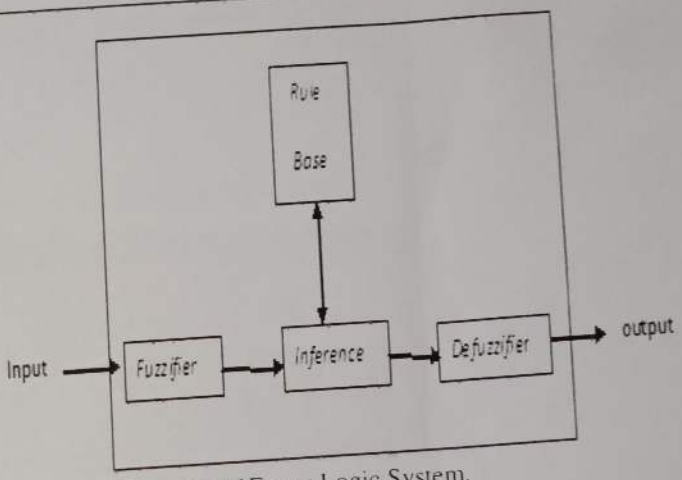

Figure-1: Structure of Fuzzy Logic System.

Fuzzy takes the system of information in normal language and converts it to values. The inputs are associated with membership functions are in form of words such as text, audio, video. Fuzzy system is determined the variables of input and output values such as less, more, extreme etc. are being selected. Afterwards, rules are being developed. Using those rules the relationship of input –output is developed. Using fuzzified inputs outputs are being produced and hence rules are determined. Those fuzzified outputs must be converted into real values. These real values can be used in real system. This system of procedure is known as defuzzification.

Proposed Design of Fuzzy Logic

Input parameters: i. Age group, ii. No. of hours, iii. Mode (Audio, Video, Text).

Output parameters: i. Physical illness, ii. Mental illness.

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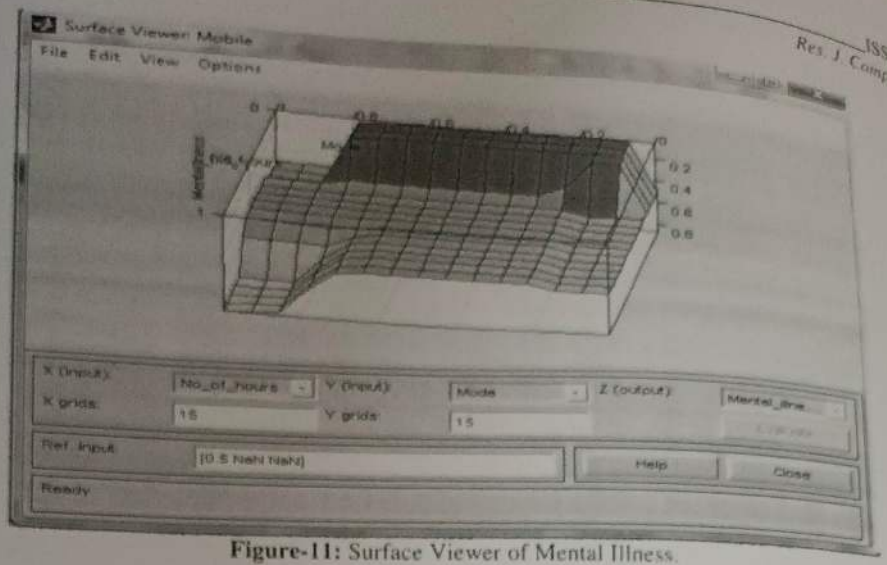

Figure-11: Surface Viewer of Mental Illness.

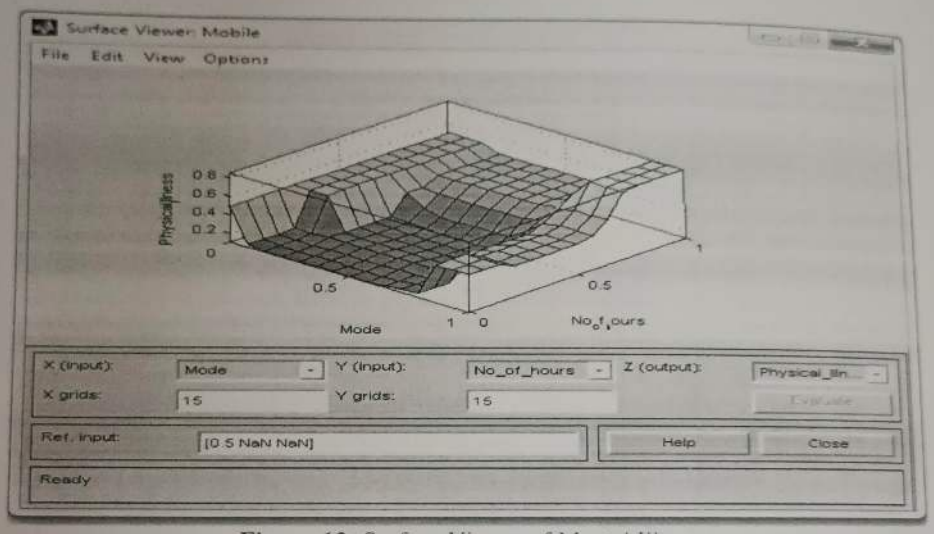

Figure-12: Surface Viewer of Mental Illness.

Conclusion

In this paper fuzzy rule based inference system for the negative impact of smart phone is analyzed with the input parameters type of age group, number of hours and different types of mode. The output parameter physical and mental illness is analysed using fuzzy logic Centroid method.

Using 18 fuzzy rules the three inputs i.e., age, No. of hours and Mode and the output value of physical and mental illness are calculated.

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Implementation of fuzzy logic method for physical and mental illness by smart phones

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Abstract

Using Fuzzy logic, different types of complex systems can be analyzed more effectively than using traditional approaches. In revolutionary technology, a smart phone emits radio frequency energy which can be absorbed by the tissues in the body. Sleep dispossession is one of the common bad effects of using smart phones. Moreover phones produce light effects which can damage our eye sight. In this paper, we analyzed smart phoned are also found to have a negative impact on health issue.

Keywords: Fuzzy logic, MatLab, centroid, rule based system, fuzzification.

Introduction

This research work is giving some awareness and advice about health. Fuzzy logic is used to analyze and diagnose negatively affected person by the smart phone. Fuzzy has been deemed to be the best methodology, mainly because of its rule based system. To evaluate the performance and implementation of simulation model we used MATLAB tool software.

L.A. Zadeh[1] proposed fuzzy sets. Mamdani[2] suggested control of simple dynamic plant application using fuzzy algorithm. Zhen and Feng[3] suggested the design of neural network fuzzy controller in washing machine.

George and Yuan[4] developed fuzzy sets and fuzzy logic. Fuzzy logic based control system suggested by Kumar and Haider[5].

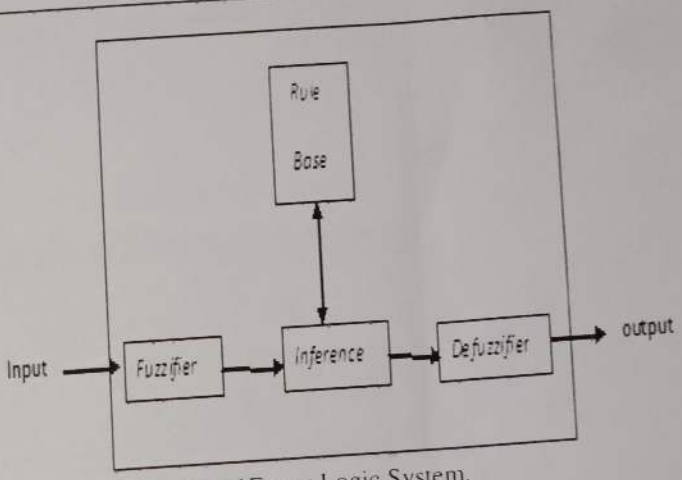
Mendel[6] presented fuzzy logic system for engineering. Hellendron and Thomas[7] proposed defuzzification in fuzzy Controllers. Yazid and Mouftah[8] suggested congestion control methods for BISDN.

Lohani and Hasan[9] suggested the design of an improved controller microchip for washing machine. Virkhare and Jasutkar[10] developed Neuro-Fuzzy controller based washing machine. Petricia Leema Roseline[11] developed unit consumption under fuzzy logic control of washing machine.

In this paper a model using fuzzy logic with three inputs and two outputs is proposed. This paper developed to analyze and diagnose the negatively affected person by the smart phones.

Structure of Fuzzy Logic System

Fuzzy inputs, outputs rule and defuzzification are given in Figure-1.



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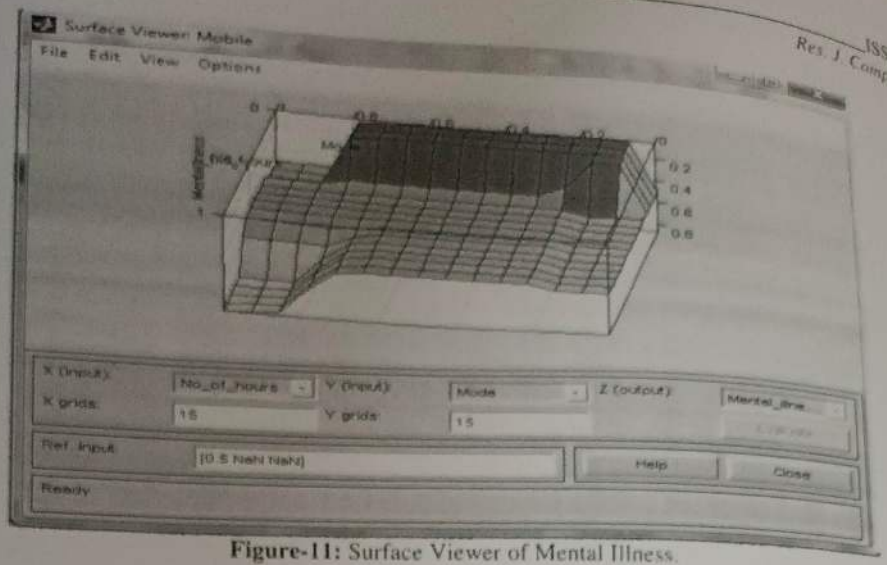

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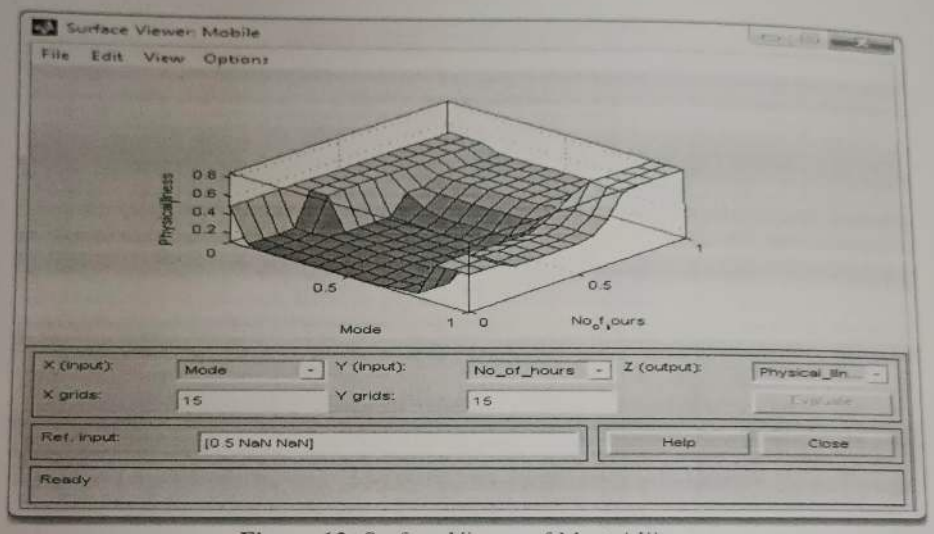

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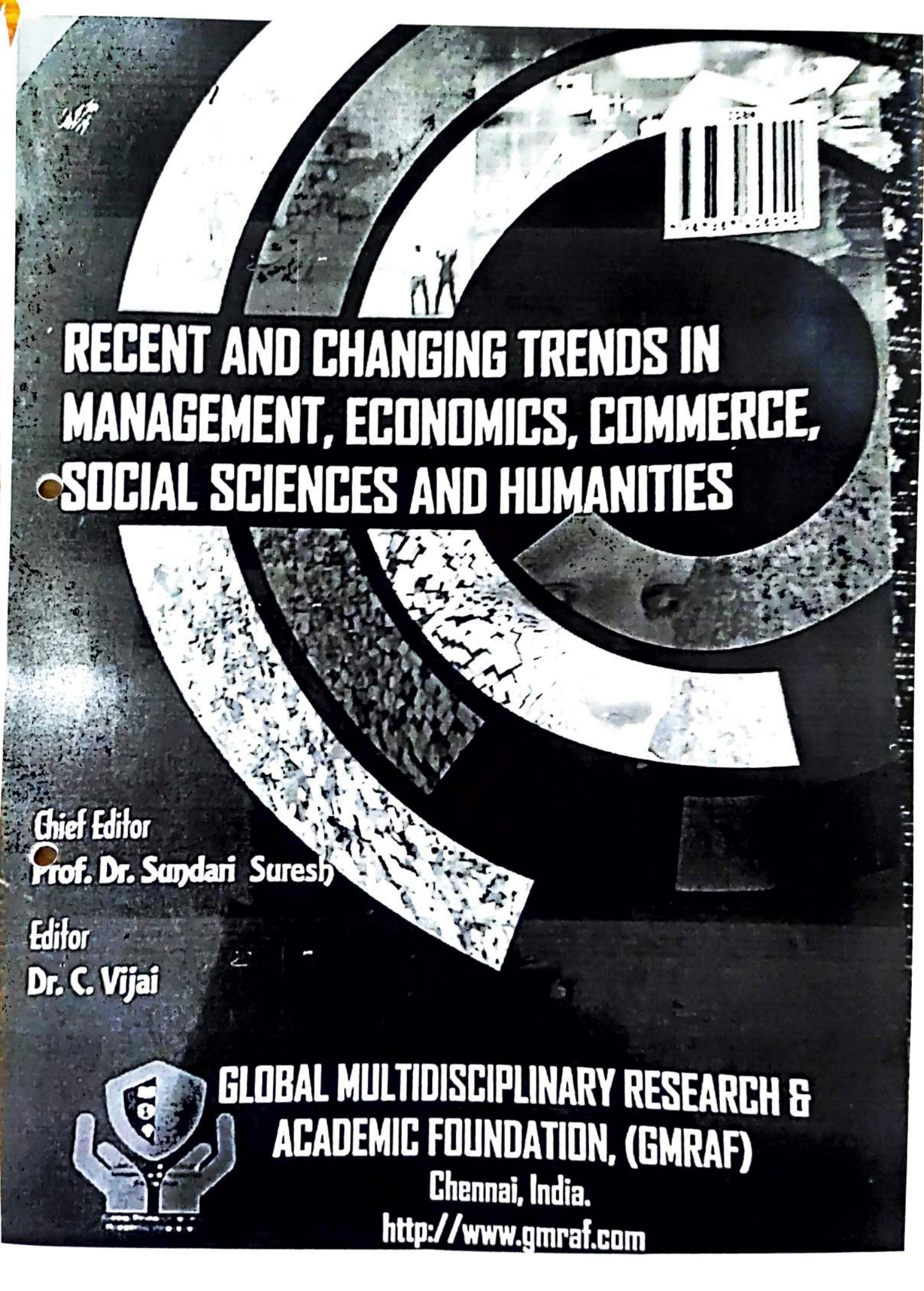

RECENT AND CHANGING TRENDS IN MANAGEMENT, ECONOMICS, COMMERCE, SOCIAL SCIENCES AND HUMANITIES

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© Prof. Dr. Sundari Suresh & Dr. C. Vijai

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CHAPTER - 19

ABILENE PARADOX - DECISION MAKING

Ms. N. Geetha Rajan
Assistant Professor, Christ College, Pune

Abstract

Decision making is the main function of management which nurtures the Planning function. It is a cognitive process which leads to a final choice, sometimes may or may not appropriate. In many cases we often land up in wrong decisions due to various reasons such as lack of knowledge, over confidence etc. One such reason is highlighted to create awareness among the readers that we all are unknowingly influenced by this psychological issue.

In Sociology, Man is termed as Social Animal, because he urges to go along with the society to survive in this competitive world. This psychology often misguides our opinions and accepts what all other wants. But there may be a situation that we human being goes with such decision which none of us want that. This could be the worst situation not only in the corporate world but also in our personal life, if this happens in decision making process. How does such a situation arise in the corporate world and what could be the possible remedies to overcome it. This is well explained with the help of a theory "A TRIP TO ABILENE PARADOX".

Keywords: A TRIP TO ABILENE PARADOX, corporate world, Decision Making Process.

Introduction

Today's business world is facing a great change and unpredictable future with a plenty of options. To overcome this problem, a leader has to work with a different mindset whereas to focus on possibility than on probability. He has to take the decisions fast from the huge amount of information where almost everything is more interconnected.

The decision making is a team process. It cannot be implemented and get the success without people. Even though man is termed as a social animal in sociology, it is quiet difficult to read the Hippocratic human mind. A decision taken by a single minded, over and self confident leader, who is supercharged by success and unchallenged by his team definitely, taking a trip to Abilene Paradox. Thus Abilene paradox is a situation where the people are fear to speak freely, which pulls down an effective evaluation of proper alternatives. Thereafter the potential problems are not identified. The Abilene Paradox was coined by Jerry B. Harvey, Professor Emeritus of Management at The George Washington University and author of the "The Abilene Paradox and other Meditation of Management.

Definition of Abilene Paradox: A circumstance where a group of individuals agree to a course of action based on the theory it is best for the group, despite going against the preferences of members of the group. This occurs when individuals feel their objections are not strong enough to support changing the minds of others in their group. Commonly referred to as "rocking the boat" or be a kill joy.

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DEMOGRAPHIC DIVIDEND: A CHALLENGE AND REALTY OF INDIA

Ms. Geetha Rajan N

Assistant Professor Christ College, Pune.

ABSTRACT

Demographic dividend is basically an economic benefit for any country. It refers to a growth in an economy which results from an effect of change in the age structure of population. It is a period where fertility rates fall due to significant reductions in child birth and infant mortality rates and thereby reducing the proportion of non-productive dependent population. The average life expectancy was increased marginally because of advancement in the medical field and awareness which results in increases in the ratio of the participation of working age population towards total population. It is an opportunity for any country to boon their economy if it grasps demographic dividend accordingly. This arise a need of proper planning and strategic formulation. According to statistic survey, the India will surpass china as the world's largest economy by 2025 with the large portion of those in working age category. This shows the opportunity what our country has to work towards. But if it is not executed, these assets can be a big liability for us. Therefore the objective of this paper is to study the term demographic dividend and its impact like a boon or bane for Indian economy. This paper discusses on challenges of our country with an emphasis on education and Skill Development and unemployability and all possible micro factors.

KEYWORDS: Demographic Dividend, Working age population, Education, Skill Development, Unemployability.

INTRODUCTION

A demographic dividend is the situation where in there is a decline in fertility and mortality rates boosts working population productivity. To receive the benefit of Demographic Dividend, a country must go through a demographic transition from rural agrarian economy to an urban industrial society. In the beginning stage the fertility rates fall, which leads to an increase in the labor force that is temporarily growing faster than the dependent population. This is the time the per capita income grows. This economic benefit is the first dividend received by the country through the demographic transition.

India is the second most populated county in the world and 50% of the population is below at the age of 25 and 62% is below at the age of 65. By 2020 the expected age population of India would be 29 years compared to 48 for Japan, 37 for China and 45 for Western Europe. **(WPA2015_Report).** This study shows that by 2025, India will be having 70% of Working Age Population.

What is Working Age Population?

The working-age population is the total population of an area which is considered capable to work based on the number of people in a predetermined age range. This measure will give an estimate of the total number of potential workers within an economy.

Demographic Dividend is a stage where a country is having more working age population and if it is executed in a proper planning, it could be a boon otherwise it could be a bane for any country. Therefore this paper focused on challenges what we may experience if we fail to convert assets into liabilities.

OBJECTIVES

- To discuss about the concept "Demographic Dividend" and it's expected challenges.
- To examine whether India will be able to harness the potential of demographic dividend and various policies and strategies taken by the Government of India in order to reap the benefits of demographic dividend.

RESEARCH METHODOLGY

This paper is based on purely secondary data and partly primary data. The different journals, newspapers, Reports and publications and related websites are consulted in order to enrich the study.

REVIEW OF LITERATURE

Assefa Hailemariam (Phd) (2012), "Demographic Transition and Demographic Dividend In Ethiopia: Opportunities And Challenges" this is a case study of demographic dividend in Ethiopia. The study claims Ethiopian time to start reaping the demographic dividend depends on the speed of demographic transition and the magnitude of the dividend, in their efforts to increase investment in human capital, design and implementation of policies to improve access to the labor market for young people, promoting industrialization in a sustainable way to improve job creation and strengthen

the financial sector to help and sustain investment. The paper concluded by quoting as the opportunities that are present now will not last long and will not be repeated, and pro policies take time to establish and effect, appropriate measures should be taken now. Any delay will effect on future prospects.

Aneesh K. A (2016), Harnessing Demographic Dividend: India's hurdles to Jump," this article discusses briefly the current unpreparedness of India for this nearing demographic dividend. Demographic dividend is a big task and he claims the biggest problem is the lack of skilled manpower in India and it is the only way to make the largely uneducated workforce capable of realizing benefits from the labor market. He concluded by saying that the developed India is not a dream, and for achieving that we should open all our doors open. Human capital is the main element to reap the benefit.

K.S. James, Srinivas Goli, Fall/Winter (2016) • volume xxiii, issue 1, Demographic Changes in India: Is the Country Prepared for the Challenge?, this paper focused on the future expected challenges of India to the demographic bonus. The authors stated that the country had to make great efforts to improve the quality of life in all age groups through improving human capital and improving skill level. Such efforts require a public investment in education and health.. They also addressed an issue like only 3 percent of the population is working in the formal sector and only a few are benefited by insurance based health care systems and bank sponsored education systems. They showed a poor fact that our spending towards health is on lower side compared to some poorer countries such as Malawi, Namibia, Tanzania, and Sudan. Investment in education also seems to be on lesser side.

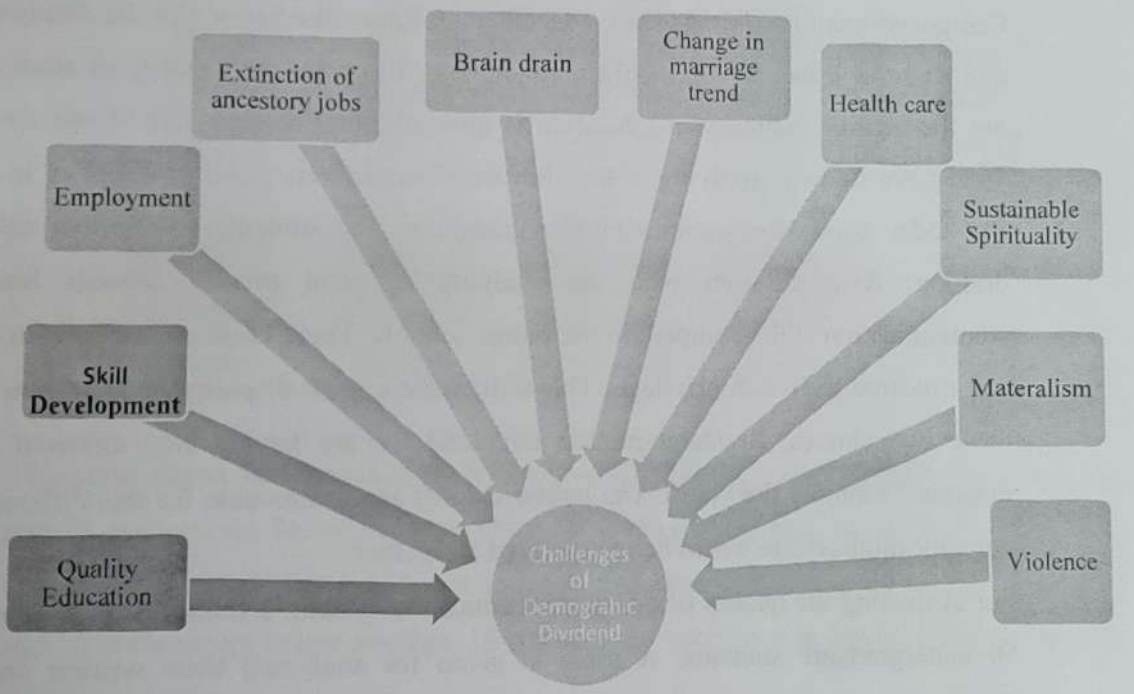
Chaitanya Talreja (2014) "India's Demographic Dividend: Realities and Opportunities", this paper explored the opportunities ad realities of Indian economy. He said the opportunity of demographic dividend cab be benefitted only if it is channelized in right direction. The challenge is to transfer the surplus labor capital from agriculture to industry to increase the productivity in both the sectors. The paper discussed about the skill gap and skill match. The study has an empirical analysis to show that the link between higher economic growth and favorable demographic conditions is not automatic and It is established through strong institutions, good policies, and investment in

healthcare and education, which can be recovered in the long run through higher productivity.

RESEARCH GAP

From the above said literature it is found that all the papers are just discussing about the concept of demographic dividend but there are very few empirical studies. The challenges from the education, skill development are mostly described in general. The gap identified is not all other possible challenges are discussed. In the present study, all possible micro problems are also discussed with suitable solutions.

EXPECTED CHALLENGES FOR DEMOGRAPHIC DIVIDEND



QUALITY EDUCATION

"EDUCATION IS THE PASSPORT TO THE FUTURE, FOR TOMORROW BELONGS TO THOSE WHO PREPARE FOR IT TODAY." As far as education is

concerned, needs to be an increased focus on both primary as well as higher secondary schools and colleges equally. A population with a quality education is strength of the country. The country like India has a huge disparity in providing primary education. India's dream of becoming a world leader in the 21st century is unlikely to be realized without an education system that keeps abreast with the needs of our future citizens. Today, India has approximately 250 million children going to about 1 million different schools. Out of this 30 to 40% students are enrolled in private schools. Around 20 back, only 18% students were enrolled in private schools and it is estimated that by 2025, 75% students will be studying in private schools. But in most of the other countries the state takes responsibility of primary education. For example UK 93%, USA 92%, FINLAND 98%, and in SRILANKA 97% of the children is enrolled in public schools.

Our government has initiated two educational reforms like Sarva Shiksha Abhiyan (SSA) and Right to Education Act (RTE), which helped to raise the quantity of admission but not the quality. Quality of education is quite abysmal through out of our country. In 1950's the students studying in the 10th standard had very good proficiency in English. But today even after completing the graduation, the students are finding difficult in drafting. Rich children who are studying in good private schools have great communication skills compare to the public schools. These children are born in privilege and expected to live in privilege. This will create a great disparity in education between these two classes. In Demographic Dividend we are having two different types of Personality among the youth. The public schools are accountable for this difference. And our education system has to be restructured altogether.

For evaluating the quality of our current education system, a small test is conducted with 50 undergraduate students. A topic is given for analyzing their writing and logical thinking skills. The evaluation was done on certain parameters like language proficiency, Grammar, Spelling and critical thinking. The topic was on current trend and it was given to them without any prior intimation. The study observed that all 50 got panicked initially for writing on a simple topic.

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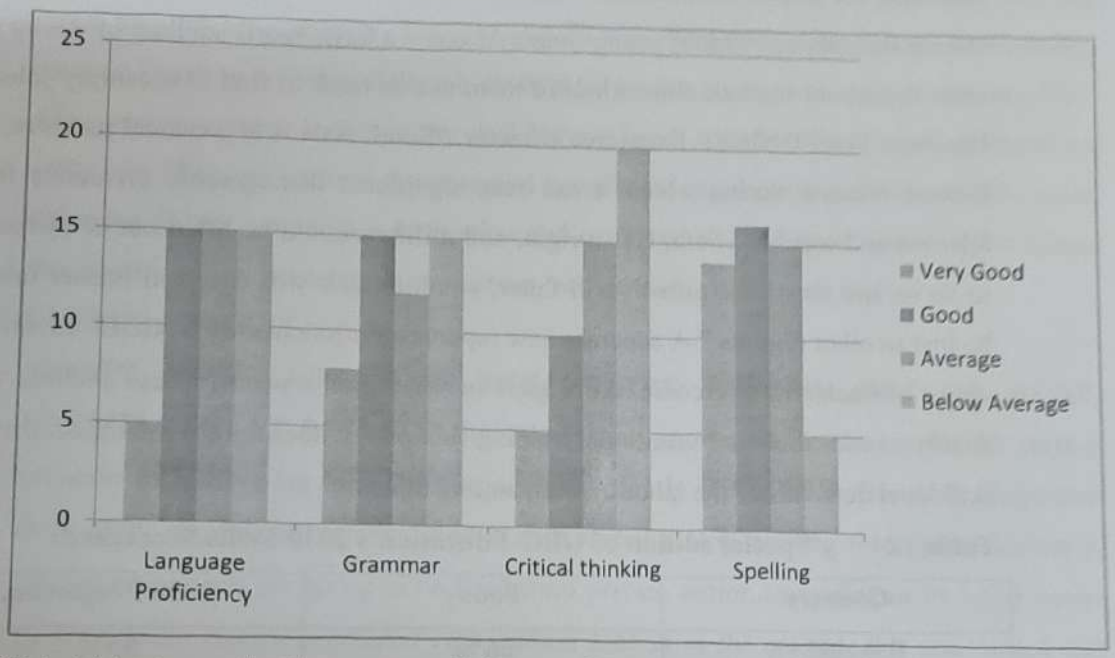
Table : 1 **Evaluation Sheet of Undergraduate students**

Particulars	Very Good	Good	Average	Below Average
Language Proficiency	05	15	15	15
Grammar	08	15	12	15
Critical thinking	05	10	15	20
Spelling	14	16	15	05



This table shows that language proficiency of 50 students only 5 students are very good and 15 students are below average. From 50 students, grammar skills of 08 students are very good and 15 students are below average. 5 students are very good in critical thinking and 20 students are below average. 14 students are very good in Spelling and 05 students are below average.

Critical thinking/ logical thinking is something which is self-monitored, self-disciplined, self-directed and self-corrective thinking a person should possess. This is the key area where our education system should focus.

Like USA, Canada and England we should have charter schools or academies, funded by the government but run by the private organization which will increase the accountability.

The National Assessment and Accreditation Council (NAAC) is an apex organization thatassesses and accredits the higher education of India. Similarly we need a third party to assess the quality of primary and secondary education.

SKILL DEVELOPMENT

"EDUCATION MAKES YOU SUFFICIENT, SKILLS MAKES YOU EFFICIENT"

"More than half of South Asian youth are not on track to have the education and skills necessary for employment in 2030" UNICEF REPORT

"Every day, about 100,000 young South Asians – a large sports stadium of young people enter the labour market, almost half of them not on track to find 21st century jobs," said Henrietta Fore, UNICEF Executive Director. "South Asia is at a critical juncture, with a limited window during which it can reap significant demographic dividends from its talented and capable youth. Get it right, and millions could be lifted out of poverty. Fail to do so and economic growth will falter, youth despair will rise, and further talent will be lost to other regions." A separate new report commissioned by UNICEF identified the major obstacles to overcome skills gaps in the region's youth. These include the low quality of education and vocational training suboptimal does not give students the desired skill level demands of the labor market, among others.

Table : 2 Special edition of GBC-Education's 2030 Skills Scorecard:

Country	Today	2030 Projection
Bangladesh	26%	55%
Bhutan	47%	81%
India	19%	47%
Maldives	16%	46%
Nepal	13%	46%
Pakistan	18%	40%

The rates of our school dropouts are still staggering. The low income people still feel that the high education will not fetch a good job for them. They are not aware about the many benefits what government is providing for them. Many of the reservation category people are still not benefited because of their unawareness. No long term goals are there for such destitute people in our country. To overcome this challenge, our education system should

follow experiential learning in the beginning stage itself. Introducing vocational education in the higher secondary level will be a great solution. A part form the basic subjects, a focus should be given to training like carpenters, cobblers, blacksmiths, nurses, tailors, weavers, real estate, construction, transport, textile, banking, tourism etc.

But today even though we are providing many short term courses in the label of skill development, it is need to be questioned whether it is fulfilling the objective. We are more focused on computer, Finance and Human Resources based skill development courses, ignoring the basic skills which definitely make a person to stand on his own foot. Our government has launched the skill India mission-Kaushal Bharat Kushal Bharat on 15th July 2015, to harness the potential benefits of demographic Dividend. The main objective of this campaign is to monitor and converge all skill development effort across the country and targeting to train 40 crores people with all different skills.

UNEMPLOYMENT

"WHEN MORE AND MORE PEOPLE ARE THROWN OUT OF WORK, UNEMPLOYMENT RESULTS" this quote is an appropriate statement for the current scenario as Infosys lay off 12000 employees recently in Oct 2019. Similarly Cognizant fired 5000 employees recently. And it is expected that in coming three years almost 6, 00,000 employees are going to be fired from private sector.*According to its latest report by ILO, India has an estimated 18.6 million jobless in the country this year which will increase to 18.9 million in 2019(India Today).*

To harness the potential of Demographic Dividend, our government should focus on improving the employability in the country. With an age population of 29 years, 3.5 percent of unemployment is not a good sign for welcoming demographic dividend. The unemployment is the challenging strategy in India as well as all over the world. Spending the major part of life say 10 to 20 years for education, and not getting the job is the worst part of life. The unemployment is affected to all from less qualified to highly qualified. Day by day, the economy is going down and all the field of private sector are affected such as Banking, Insurance, Automobile, Textile, Real Estate, and Telecom etc. ***In a bleak employment scenario, employment and its conditions when it comes to women in the country is even dimmer (Oxfam India)***This report claims on an average women are

paid 34 percent less than the similarly qualified male workers on performing the same tasks. Daily wages of a woman on an average is INR 105 and INR 123 in urban and rural settings is very less than the male worker (Based on NSSO (2011-2012) estimate). 2015 report says that 92 percent of the women and 82 percent of men were earning a monthly salary of Rs 10,000 which is very less than the seventh pay commission's recommended salary Rs 18,000. This report rightly observes that in spite of booming economy and increasing labor force the process of creation of job is extremely sluggish. The results on distribution of income and wealth are strongly depended to the processes in the labor market.

Government should come up with more jobs. The privatization in government field should be curtailed. All the possible steps have to be taken to boost the economy. Government should encourage the schemes like Skill India and startups. Educational institutes should also play an important role in creating awareness among the youths about such government schemes.

EXTINCTION OF ANCESTOR'S OCCUPATION

A FAMOUS SAYING "DON'T LOSE A DIAMOND WHILE CHASING A GLITTER"Demographic Dividend is a blessing for any country as there is an increase in the working age population in the economy. In this journey, the youth should be proud to accept the ancestor's job. There is no shame to continue the ancestor's occupation. Any job is having its own value. Many educated people leaving their executive job or after retirement, they are accepting their ancestor's occupation as a hobby or for uplifting the society. For example Mr. Sadashivam, Retired Honorable Chief Justice of India and Ex Governor of Kerala want to take up the farming as his ancestor's job. Mansukhbhai Prajapati, from Wankaner, Gujarat is a successful entrepreneur who relished the idea of modernizing the knowledge of pottery. Weaver, cobbler, pottery, farmer, coconut tree climber, blacksmith, goldsmith etc are the jobs getting extinct from the labour market.

BRAIN DRAIN

BRAIN DRAIN CAN BECOME BRAIN GAIN".The term brain drain was coined by Royal society, London in describing the movement of science and technological personal to the US and Canada.Ozden and Phillips (2015)describe 'brain drain' as the movement

of high-skilled labor from a less developed or developing country into a developed country; creating the phenomenon where the development of human capital occurs in one country and the utilization in another country. It is usually believed that migration of skilled workers cripples the economic development of the home country. This is one of the main challenges of demographic dividend. Qualified young professionals from our country immigrate to other nations in search of better prospects is really high cost involved. In order to diminish the high departure rate of human capital, favorable conditions have to be created for the IT skilled labor in their home countries. This can be taken care of by providing personal training and development coaching, more relevant and better opportunities with importantly, above par salary packages with extra benefits, so that the employees are assured of personal and professional growth. Hence, these developmental factors enhance the standard of living of the human capital which in turn motivates them to contribute for transforming India into a developed nation. And thereby there should be a mutual exchange of labor and technical skill between the countries benefitting both the parties. Thus Brain drain can become Brain Game.

CHANGE IN MARRIAGE TREND ***"THE HIGHEST HAPINESS ON THE EARTH IS THE HAPINESS OF MARRIAGE".*** In 2025, being a young populated country, another common problem what we are facing is a change in marriage trend. Nowadays both girls and boys are studying well and education span are extended and find a job after that again increased the no of years. By all these age of marriage also extended marginally.When they plan to seek marriage partner, there are so many conditions and demands are facing by both boys and girls. India has marginal gap in the ratio of boys and girls. Populations of boys are less compared to that of girls. Some of the families of girls are preferring boys with government jobs, qualification – equal or above than girls, financial background, same caste and religion, family background, preferably nuclear family etc.Families of boys prefer the girls, who are educated equal to boys, working girl; some of the boys are not ready to accept the odd jobs of girls like the jobs with night shifts, irregular timings, away from the residents etc. Many couples today do not want to get married or put it off as long as possible. Some do not want children because they consider that this would limit their freedomThe new dietary habits and environmental

factors, women are now experiencing menopause earlier, some as early as 35. Those who marry late are getting panic and are really fret about having kids. Women over 35 experience more complication in the pregnancy and the chance of miscarriage is also more. Male fertility also reduces with age. Apart from this, if they find difficult in getting a partner with age, there is a tendency to get desperate and choosing a wrong one. This leads to a failed marriage.

HEALTH CARE SYSTEM

HEALTH CARE IS A RIGHT, NOT A PRIVILIGE"Health systems as a whole should be strengthened, with expansion on primary health care; including vaccination Good health remains a key to reducing the vulnerability of young people, improving staff productivity and creating an environment for youth to develop their full potential. But in reality many young people engage in behaviors that are dangerous not only to their current state of health, but also endanger their health in the coming years. Despite improvements in health over the last thirty years, life is still being lost to early childhood diseases, inadequate newborn care and childbirth-related causes. More than two million children die each year from preventable infections. The health fitness among the youth is decreasing mainly because of unawareness, unhealthy lifestyle, more consumption of junk foods, frequent parties, unhealthy competition which leads to stress related disease etc.

According to the report by India Spend, a data-driven public interest journalism initiative, In the case of the death of a member of income, an average Indian has only about 8% of what may be necessary to protect your family from the financial shock, much lower compare to 84% in Taiwan, 44 % in Japan, and 67 % in Australia. Low awareness and the limited reach of insurance companies is the twin problems need to be addressed.

SUSTAINABLE SPIRITUALITY

"A CONSISTENT ETHICAL SYSTEM IS NEED OF THE HOUR."We dream a world without poverty, equity, dignity with respect, sacredness of the natural world, mutual care and love. This is an achievable dream, but the current development path is deeply flawed and not realizes our dream. The current scenario of the materialistic world, there is a

factors, women are now experiencing menopause earlier, some as early as 35. Those who marry late are getting panic and are really fret about having kids. Women over 35 experience more complication in the pregnancy and the chance of miscarriage is also more. Male fertility also reduces with age. Apart from this, if they find difficult in getting a partner with age, there is a tendency to get desperate and choosing a wrong one. This leads to a failed marriage.

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great need for strong ethical and spiritual framework of sustainable development to achieve this dream. To make a sustainable and world peace through spirituality; we must be emphasized to provide simple, spiritual and moral education to children and at an early age through schools, books, media, school curricula, etc. The introduction of seeds of ethical and spiritual values in the minds of children helps later in their life that leads to sustainable living. Similarly, measures should be undertaken to educate people at all levels to understand critical changes must be done in our economic, social, environmental and governance structure to work effectively in the process of sustainable development to create a prosperous future for all and generations unborn.Spiritual transformation of individuals is, change of mentality, exploring development centered on man, the realization of the carrying capacity of the nature, a vision of fullness in our relationship with nature and humanity, the voluntary application of spiritual and ethical value, etc., will no doubt make the spiritual movement a success. **MATERALISM** ***"MATERIALISM IS AN IDENTITY CRISIS"***Materialism dominates today's youth culture, today's young people are embracing materialism as any previous generation. It is a young generation without dreams, without love. Money, money and more money - this is a typical teenage dream. It reflects in every place, starting from education. Earlier the school was a place to acquire real knowledge, but now it is all about scoring high marks and mastering the technique of passing examinations. Attitudes towards careers have also changed. Earlier generation sought meaningful and challenging jobs for life. But today the priority is a comfortable lifestyle and a job with lots of perks. No one is daring to tale up challengeable jobs. Young people do not have the patience to climb the career ladder slowly - one step at a time. They want quick gainsTraditional values of love and family have been replaced by selfishness and greed. Today's teenagers are the centre of their own universe and have little time for their parents and family. We should develop the habit of "giving" and not "getting among the children. We should ourselves as a role model from restraining from material world.

VIOLENCE*"AN IDLE BRAIN IS A DEVIL'S WORKSHOP"*However, the demographic dividend has a huge potential drawback - many of the young men entering their 20s have little education and therefore have little chance of finding employment. What's more, many of them are lonely and sexually frustrated, as sex ratio has been

skewed for several decades with 37 million more men than women in India. According to official statistics, the youth unemployment rate in India in 2019 stood at 15 per cent for men and 11.3 per cent for women. Many are doing odd jobs for little money and with lots of time on their hands. There are a lot of social science research that finds a strong relationship between unemployment, especially among young people, and the incidence of violent crime. The youth spending their lion's part of life for education, job searching, searching for life partner, it will affect them very badly which leads them to frustration, stressed as well as depression. They also loose self-confidence and self-respect.

The unemployment is the main cause for terrorism and violence. The terrorist group will take this opportunity to brain wash the unemployed youths. The unemployed youths are easily target for terrorist groups, criminal activities, drug addicts, drug carriers, and smugglers.

CONCLUSION

The proportion of working age group in the total population is considered to be more important for framing the policies for economic growth and development. 1. Education and skills development have great promises to reap the demographic dividend. The quantity and quality of education needs to be improved. 2. Need professional training that expands so that young people can develop the skills necessary for productive employment. A workforce better trained and educated is a prerequisite for to harness the potential of Demographic Dividend. 3. As we have two different levels of youth population the government should take necessary steps to motivate the youngsters to take up their ancestors' jobs as well as should provide necessary measures to mitigate the brain drain from our country. 4. The change in the marriage trend is also need to be addressed by creating a proper awareness among the youth about the consequences of late marriage. 5. Health systems as a whole should be strengthened, with expansion on primary health care; including vaccination and importance of insurance based health care system. 6. It is very hard to imagine a consistent ethical system without spirituality. To attain sustainable spirituality, the government must be emphasize to provide simple, spiritual and moral education to children and at an early age through schools, books, media, school curricula, etc. 7.Materialism is an identity crisis and material happiness is an illusion that can never be achieved. This can be overcome by setting ourselves as a

role model from restraining from material world and the habit of "giving" and not "getting among the children. 8. Violence is a huge problem need to be addressed before its reach its helm. Our unemployment is too high. Our poverty is too high, our illiteracy is too high, and all of that extends and lends itself to crime and violence being too high. With these words would like to conclude by saying there are plenty of challenges which need to converted into opportunities in order to harvest the benefits of DEMOGRAPHIC DIVIDEND.

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GST: AN ATTEMPT TO TRANSFORM INDIA

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Abstract

Goods and Service Tax (GST) a comprehensive dual taxation system replaced the complex multiple tax structure, which prevailed in India on 1st of July 2017. GST is expected to achieve another milestone in the economic growth of India. The primary aim of GST is to submerge the scattered indirect taxation system of India and avoid its cascading effect. GST is expected to reduce the burden on manufacturers and encourage higher production. This study intends to exhibit in detail the features of GST and how it is going to affect the common man in India.

Introduction

GST, which stands for Goods and Service Tax, was passed in the Lok Sabha on 29th March 2017. It came into effect from 1st July 2017. The objective behind implementing GST was to follow – 'One nation one tax' system. GST is a single uniform tax levied on all goods and services I India. It has replaced the indirect taxes like excise duty, VAT, general sales tax etc. The main purpose of GST is to fix a single Revenue Neutral Rate (RNR) on goods and services so that the total revenue of the Central Government and the State Government remains the same. GST is an attempt to simplify the giant tax structure of India, which comprised of VAT (Value Added Tax), excise duty, service tax etc. India has adopted dual GST, which means GST will be levied by both Central government and State governments on supply of goods and services. According to the weekly update report by Goods and Service Council, the revenue in July 2018 was Rs. 96,483 crore and the revenue during July 2019 is a growth of 5.80% over the revenue of July 2018. Compared to 2018, in April- July 2019, the domestic component has grown by 9.2%, while the GST on imports has come down by 0.2% and the total collection has grown by 6.83%There are primarily four types of GST:

a) SGST- State GST, collected by the state govt.
b) CGST- Central GST, collected by the Central govt.
c) UTGST- Union Territory GST, collected by the Union Territories
d) IGST- Integrated GST, collected by the Central govt.

Tax structure before GST

The tax structure in India was divided into two parts – Direct and Indirect Taxes. Direct taxes were levied on income earned by the individuals and corporate firms, whereas indirect taxes are levied on sale and provision of goods and services respectively. The Central Government and State governments imposed most of the major taxes and the local authorities like municipality and local governments imposed some minor taxes.

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Types of Direct Taxes are:

- Income Tax
- Corporation Tax
- Interest Tax
- Wealth Tax
- Agriculture Tax
- Estate Duty
- Gift Tax
- Hotel Receipt Tax
- Expenditure Tax
- Others

Types of Indirect taxes are:

- General Sales Tax
- Union Excise Duties
- Stamp and registration fees
- Tax on vehicles
- Taxes and duties on electricity
- Service Tax
- Customs
- State Excise Duty
- Entertainment Tax
- Others

Research Methodology

The study is based on secondary data from various journals and research papers. It is a type of desc research paper.

Need of the Study

Taxation structure plays a very vital role in the development and smooth functioning of the economy study aims to understand better the new taxation system implemented by the Government of India.

Objectives of the study

1) To study in detail the newly implemented tax structure in India – Goods and Service Tax.
2) To identify the advantages and shortcomings of GST.
3) To do a comparative study between the old tax structure and GST.
4) To study the short-term impact of GST on consumer goods.

Literature Review ISSN: 2454-8499, Vol. V, Special Issue II, January - 2020, Impact Factor: 1.89(GIF), 0.679(IIFS)

Bansal and Gupta (2018) in their paper on Impact of GST on Indian Economy, states that GST would most likely eliminate multiplicity of administrative mechanisms and tax rates across different countries. It will have a positive impact on the retail market and will make supply chain more cost effective. They also mention that GST is a well-designed mechanism and is expected to bring in goodamount revenue for both central and state government.

Shanta and Nandargi (2018) in their paper on Impact of GST on Indian Economy and Consumer Experience says that Government India should protect the poor from the prospective inflation which might take place due to implementation of GST.

Dani in her paper on Impact of Goods and Service Tax (GST) on Indian Economy relates GST with IT and e-commerce. According to Dani, the GST structure will succeed only if the country has a strong IT networks. Mishra in her paper wrote that a single taxation system would benefit the manufacturing and service sectors and would encourage new businesses and entrepreneurs to engage more in these sectors. GST minimizes the complexity of taxation and helps in the free movement of goods across states. It will help in promoting new startups and businesses in India.

Short Term Impact of GST on Consumer Goods

Products that have become cheaper after GST

- Eatables like milk powder, curd, tea, rice, flour, spices, coconut oil, mustard oil, sugar, jiggery etc.
- Items of daily use like bathing soap, detergent, napkins, coal, kerosene, LPG domestic, LPG stove, toothpaste, etc.
- Stationary items like notebooks, pens, school bag, printers, etc.
- Healthcare items like insulin, X-ray films, diagnostic kits, medicine for diabetes, cancer, etc.
- Apparels like silk, woolen fabrics, khadi yarn, footwear below Rs.500 and apparel up to Rs. 1,000
- Other items like Helmet, electric transformers, cement, scooters, motorcycles, weighing machinery, tractor rear tyres and tubes etc.

Products that have become expensive after GST

- Luxury goods
- Tobacco products
- Mobile bills, insurance premiums, banking charges
- Internet, Wi-Fi and DTH services
- School fees, air tickets, courier services

Advantages of GST

- GST has simplified the Tax structure in India.
- As the cascading effect of a series of VATs has been removed, the products will be available at a cheaper rate.

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- Small scale businesses like service providers with turnover lower than Rs.20 lakh are exempted from paying GST and in the case of North Eastern sates the threshold is rs.10 lakh
- GST aims at reducing corruption and sales without receipts
- It reduces the need for small companies to comply with excise tax, service tax, VAT
- It provides accountability and regulation to unorganized sectors such as textile industry.
- GST has reduced taxes on certain good by 2% and others by 7.5%, such as smartphone and cars.
- As GST has replaced multiple taxes, the tax collected is likely to be distributed equally across the country
- The possibility of tax evasion is completely minimized with the introduction of GST
- GST brings in uniformity and transparency in the taxation process and allows centralized registration.
- GST aims to have a positive impact on India's GDP

Shortcomings of GST

- The software used for GST filing process leads to higher operational costs for businesses
- GST has been criticized and named as 'Disability Tax' as it imposes tax on braille paper, wheelchairs, hearing aid etc.
- Some manufacturers have suspended their reward programmes due to complexities in the taxation of products, which might affect the consumers.
- The GST transaction fees in the financial sector have increased from 15% to 18%.
- Insurance premiums have become more expensive with the implementation of GST.
- Petrol not being under GST fails the ideals of unification of commodities.
- As it is a new concept, it will take time for the people to understand it.
- If the seller increases his profit margin instead of passing on the benefit of GST on the consumer, the price of the product might increase.

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Difference between the old tax system (OTS) and GST

Parameter	OTS	GST
Structure	Under the old taxation system, the central taxes applicable were custom duty/central excise duty, central sales tax on commodities and services. The state taxes included state VAT, entertainment tax, luxury tax, tax on gambling, betting and lottery, sales tax deducted at source.	Under GST, all the central and state taxes will be subsumed and a single tax will be levied on all commodities and services
Basis of Levy	Tax will be levied at the place where goods are manufactured or sold, or the place at which services are rendered.	Tax will be levied at the place of consumption, like a destination-based tax.
Filing of Returns and Tax Collection	Service tax and central excise were uniform, and VAT varied from state to state.	The process is uniform and the dates for collecting or depositing tax and filing returns are common.
Excise Duty	Excise duty will be levied up to the point of manufacturing.	The excise duty will be replaced by Central GST and tax will be levied up to retail level.
Entry Tax	Entry tax is charged by certain states for inter-state transfers, detained as import in local area.	Entry tax is not applicable, but an additional 1% will be levied as tax on inter-state supply of certain commodities.
Cascading Effect	Credit between service tax and excise duty is available, but there is no set-off against VAT on excise duty.	Credit available on the whole amount of taxes up to retailer.

Conclusion

Taxation system is the backbone of an economy. The implementation of GST in the year 2017 brought in lot of dilemma and paranoia. The motive behind GST is to achieve an organized tax structure. It is not a tax concession scheme as such but is an attempt to fix a single revenue system. The government expects to receive higher revenue so that more developmental activities can be done for the public. It aims at minimizing tax evasion and corruption, which was quite prominent in the old taxation system. GST is fairly a new concept in India and it is too soon to analyze the impact of GST.

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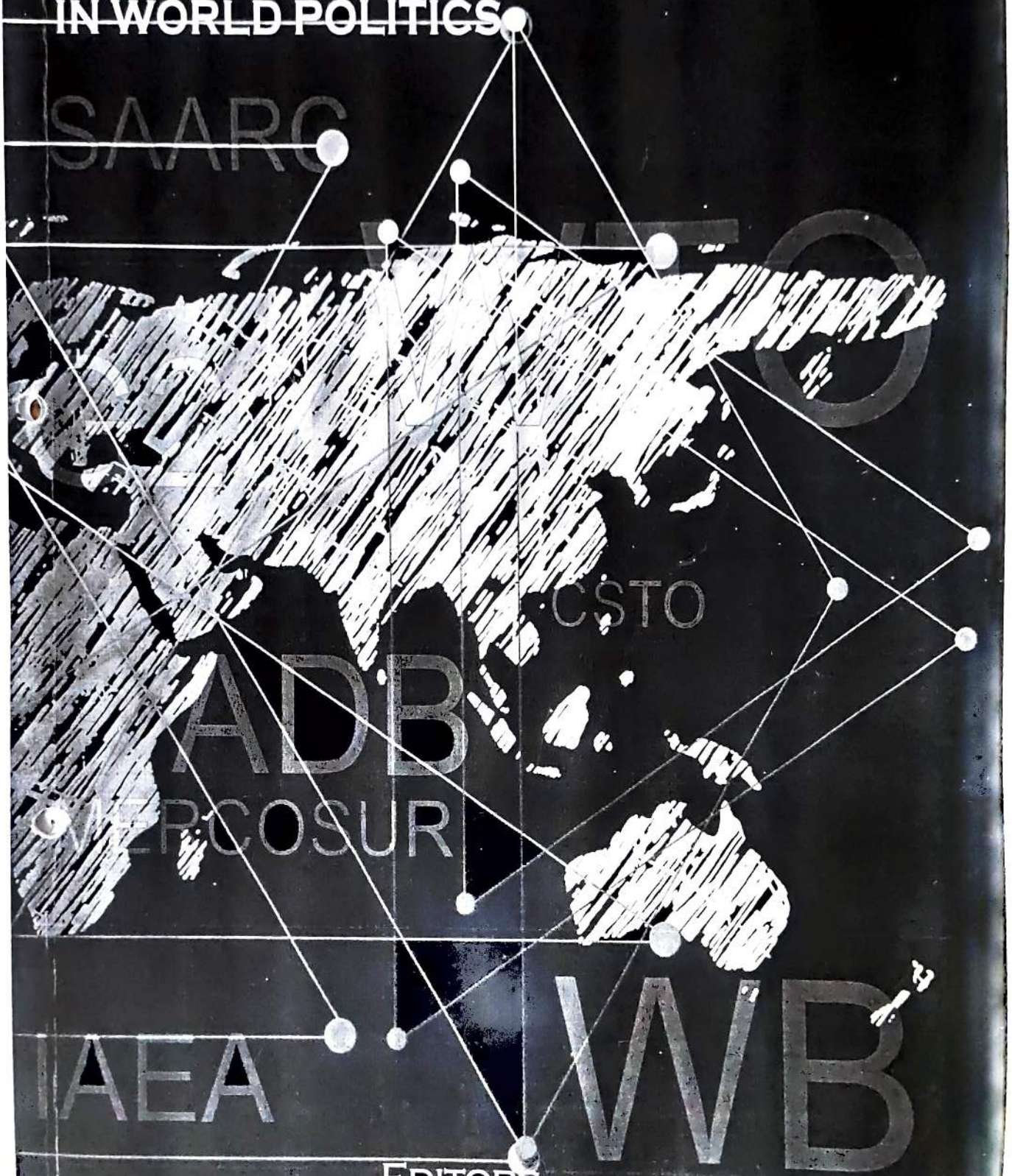
ROLE OF INTERNATIONAL ORGANIZATIONS IN WORLD POLITICS



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BISWANATH CHAKRABORTY
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ROLE OF INTERNATIONAL ORGANIZATIONS IN WORLD POLITICS

Edited by

Biswanath Chakraborty

&

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PREFACE

Understanding global politics does not confine only to the study of diplomatic relations and economic relations among the Nation States. It also encompasses all kinds of state as well as non-state actors. From the very beginning, scholars in the study of International Relations primarily emphasized on state, international actors and few non-state actors. After the First World War the development of International Relations as an academic discipline had received greater attention primarily on legal, institutional, philosophical as well as state centric approach. However, with the failure of League of Nations to establish peace and tranquility in the World order, the scholars in the study of International Relations like E H. Carr had challenged against the ongoing thinking of International Relations. Carr and other scholars in the study of International Relations had argued that International Relations cannot avoid power politics. Therefore, the study of International Relations basically deals with the study of diplomatic and military relations of the nation states. The experience of World War II gave a lesson to the scholars of International Relations that there might be wide scope for study of International Relations. However, in reality it had remained confined only in the military and diplomatic relations in the hands of realist school. After the devastating War, Hans J Morgenthau being a realist had challenged the age old thinking in the study of International Relations. Morgenthau had claimed whatever might be the ultimate objective in the study of International Relations, power is the ultimate aim that every nation states wants to maintain and secure. The Cold War politics in International arena after 1945 has also helped to foster the dynamics of realist politics as a dominant of thinking in World politics. From 1945-1960 apart from nation states, the UNO as universal actor had played an important role. In addition to that some important non state actors such as NATO, Warsaw, NAM and Economic actors like World Bank, IMF, had played an important role in World Politics.

However, in the mid of 1960 of the last century, the tight Bi-polar system in World politics has turned into loose bi-polar World System. As a result, many non state actors had started to play crucial role in World Politics. For example, the environment related matter the Green Peace as a non state environmental actor begun to play important role in World environmental policy programme. Several Human Rights Organizations started to play crucial role in shaping Human Rights policies and programmes globally. Likewise, the Women Organizations and various types of NGOs involved in the different developmental works throughout the World had also emerged as important actors in the World politics. The Regional organizations such as European Union, Oil & Petroleum Exporting Countries (OPEC), AU, ASEAN, OAS, OIC, etc. have also started to play important role in World politics. The cultural organizations as well as organizations relating to the games and sports such as Federation of International Football Association (FIFA), International Cricket Council (ICC), and Olympics Association are some of the examples who started to play important role in shaping the International Relations. As a result the combination of all the factors, the nature of Internal Relations has received a new dimension.

The state centric approach to the study of International Relations has changed with the rise of multiple actors in the arena of internal politics. Pluralistic thinking started to get momentum in the late 60s in the field of International relations. Side by side the economic factor also emerged as an important factor in the study of International Relations. Based on economic factors a large number of the scholars primarily Latin American were engaged in the project where they considered historically the World Capitalist System divided this World into two halves: the industrial developed Northern sphere which they termed as the 'Centre' or 'Metropolis' and underdeveloped Third World countries of the Southern sphere which they termed as 'Periphery' or 'Satellite.' In this discourse they emphasized the crucial role of Multinational Corporations (MNCs) or Transnational Corporations in World politics. From the beginning of the 70s in the last century International Politics under the System Analysis further gained popularity in the study of International Relation especially, in the hands of Kenneth Waltz.

After 1980s with the spread of Globalization, Liberalization and Privatization the study of International Relations got a new impetus. Along with age old state centric diplomatic approach, institutional and market centric approach became more prominent in the study of the International Politics. Globalization, Liberalization and Privatization has provided a new look in the field of International Relations. Market became more important than past. Non State actors got more important than earlier times. The age old myth developed by the Realist School that State as the unitary and integrated actor in World politics has become blurring. Economic actors in World politics have emerged as dominating actors among the other actors involved in World Politics. At the same time, the non state actors in the field of Human Rights, Environment, Sports and Culture as well as Militant activities are also becoming crucial in shaping of World politics, today.

Keeping in view, the importance of state, non-state as well as universal actors in World Politics the present edited volume **'The Role of International Organizations in World Politics'** have been organized to deal with all the major organizations which are directly or indirectly influencing International Politics at

Dedicated to

Professor Sukharanjan Chakraborty,

Former Professor of Jawaharlal Nehru University and
Founder of Centre for Sub- Continental Studies, Kolkata

&

Professor Asimananda Goswami,

Scientist, Saha Institute of Nuclear Physics, Kolkata and
Founding Member of Centre for Sub- Continental Studies, Kolkata

Both of our beloved Mentors have passed away in recent time who always encouraged us to remain in the field of research work.

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THE NON-ALIGNED MOVEMENT IN THE PRESENT WORLD ORDER

R Radhakrishnan
&
Assunta Mendonca-Sandhir

The Non-Aligned Movement (NAM) which had emerged as an alternate path when the world was shaped by binary opposite camps has traversed a long journey from Third World solidarity to a guarded pragmatism in a globalizing world.

Genesis

The Non Aligned Movement (NAM) which emerged in the aftermath of independence of many colonies was a ray of hope amidst the advent of the cold war, which had emerged after the end of the Second World War. It is perceived that throughout its history, the Non-Aligned Movement had played a significant role in the preservation of world peace and security.

The establishment of the United Nations Organisation (UNO) provided the much needed glimmer of hope in the world that was trying to resurrect itself to normalcy, while the Marshal Plan and COMECON were trying to rebuild the European countries ravished from the wars, along ideological lines. [i]

The NAM came at a very critical juncture when the UNO was under tremendous pressure in stabilizing the international order by assiduously seeking to uphold the elements of internationalism, nurturing and promotion of solidarity and called for cooperation and harmony among the erstwhile rivals in the wars fought over imperial gains and ideological supremacy.

At this juncture the Nehruvian worldview was perceived as a timely intervention that clearly sought to insulate India and other newly emerged independent nations from the vagaries of bloc politics and future conflicts and wars.

In 1960, in wake of the developments attained in the Africa Asia conference in Bandung, the move to usher in the Movement of Non-Aligned Countries got an impetus on the sidelines of the Fifteenth Session of the United Nations General Assembly, during which 17 new African and Asian countries were admitted. Leaders like Gamal Abdel Nasser of Egypt, Kwame Nkrumah of Ghana, Shri Jawaharlal Nehru of India, Ahmed Sukarno of Indonesia and Josip Broz Tito of Yugoslavia, who later became the founding fathers of the movement and its emblematic leaders, played an important role in this development.

The NAM was established based on a broader vision transcending the globe with its First Summit Conference held on 1-6 September 1961 in Belgrade. The Conference was attended by 25 countries. The principles that would govern relations among large and small nations, known as the "Ten Principles of Bandung", were proclaimed at that Conference. These principles were later incorporated as the main goals and objectives of the policy of non-alignment. The accomplishment of these principles was made as an essential criterion for seeking the membership of the Non-Aligned Movement. It was also known as the "quintessence of the Movement" until three decades ago. [ii]

The NAM filled in the vacuum created by the cold war that clearly sought to polarize the world into two diametrically opposite camps of free market economy and socialist economy, military pacts of NATO and WARSAW. These camps rekindled the rivalry between their camp followers in the new setting, through other means which manifested itself in the political, economic, social and psychological fields.

So the emergence of NAM was a timely intervention, as it provided equidistance from both the camps and insulated the newly emerging countries from the bloc conflicts. The NAM began with a committed policy to ensure that the masses of the hitherto suppressed nations due to European colonial rule will be able to assert their freedom and self determination. India, along with other Non-Alignment Movement countries, tried to shift the global political agenda away from the Cold War to the needs of the poorer countries.

Aims and Objectives

The NAM which had emerged as a anti colonial movement and had encompassed the third world has functioned as a moral political force within the domain of the international relations.

Its primary objectives have been to respect human rights, non interference in the internal affairs of the nations, respecting the sovereignty of nations of Asia and Africa, peaceful resolution of international conflicts and promotion of goodwill and mutual interest. These objectives had emanated from the historic Bandung Conference also known as Asian African conference, which is considered to be the precursor to the NAM conference. They are considered to be very relevant since there were apprehensions about the usage of collective defense agreements to the detriment of smaller powers whilst benefiting that of the great powers. In other words the general consensus was to insulate the newly emerged independent nations from the vagaries of the super power rivalries and to instill a spirit of solidarity and partnership which paved the way for the Non-aligned Movement.

The NAM which was formed in 1961 following the initiative taken by India, Egypt and Yugoslavia, had demonstrated its capability in furthering the aspirations of the humanity and had contributed immensely towards decolonization, thus gaining enormous visibility and presence in the global politics.

The NAM started off with non-alignment wherein it was emphasized that the countries must have space to operationalize their respective foreign policy independently based on sovereign equality and peaceful coexistence. Secondly it also stressed that the countries adhering to the principles of NAM must perennially support liberation movements across the globe in order to accelerate the decolonization process, across the world.

The NAM held great promises to change the world view and influence structural changes in the way the international order had been functioning. So in other words it was an age of "belief, confidence and idealism". [iii]

Milestones of NAM

The NAM countries played a significant role in the 1970s and 1980s, for seeking a new international economic order. In this context a demand was made in the Algiers Summit in 1973 and was reiterated during the Sixth Summit in Havana. This was in line with the earlier move during the Belgrade Summit wherein a conference was sought under the aegis of the United Nations for the deliberation on the economic problems faced by

the most heterogeneous and eclectic group, struggling to have a consensus. The NAM has not made any remarkable progress especially when it came to intervening and facilitating a dialogue and resolution of conflicts in the Global south. The failure of the NAM in addressing the Iran-Iraq war or the conflicts in Afghanistan and the Syrian crisis vindicate this observation. It has been ambivalent towards the Arab spring movements, which questioned autocratic regimes and sought realization of democratic aspirations of the masses. The acrimony between China and India and India and Pakistan which have the potential to vitiate the stability in South Asia, has not drawn the attention of NAM.

The most noticeable development is the loss of India to Japan in the polls for the non permanent seat in the Security Council in 1996, when India could only muster support from only 42 countries voted for India. This loss exposed the chinks in the NAM unity and bonhomie and was viewed as a major setback to the Afro Asian solidarity. In 2010 while there were no other contenders for the Asia seat which Japan had vacated India was able to garner the support of 187 out of 192 countries, for its entry into the UNSC. This was its seventh time since 1946.[viii]

Shifting tectonics of power

The cold war had ended with the demise of the Soviet Union in 1991 and the NAM is seen as a club with Summits rather than an effective group which could alter the world politics, upholding the cause of justice and equity. The NAM originated under particular circumstances when the world was confronted by binary opposite camps based on ideological differences. The USA and Soviets had tried to change the world by promoting their respective contending worldviews and it was NAM which had by and large insulated itself from the vagaries of the cold war. The demise of the bipolar system has accelerated the dominance and hegemony of a single superpower.

Francis Fukuyama had attributed to the emergence of one single ideology in the global affairs to the end of history. He states further that free market economy would play a vital role in ameliorating various nations irrespective of their economic clout.

Capitalism would be the natural element and it will be deeply embedded with the liberal revolution in economic thinking would ultimately pave way for the establishment of political freedom around the world.[ix]Hence all the countries are likely to opt for a homogenous system since they would gradually standardize themselves leading to uniformity on the basis of a centralized state that could lead to urbanize, transform customary forms of social organisation like tribe, sect, and family with economically rational ones based on function and efficiency, and provide for the universal education of their citizens. Moreover, the developing countries are still subjected to a global economic system dominated by the former colonial countries. What is yet to be ascertained is if the process of globalization which is a global political-economic process that facilitates the movement of people, information, capital and goods (capitalism), ushers in an egalitarian world order.

The second scenario is the rise of identity based mobilization and conflicts around the globe. This observation was highlighted as a hypothesis based on the study of certain pattern of developments. Samuel P. Huntington had argued that the humankind is affected due to various divisions on ideological, economic and cultural differences and the dominating source of conflict will be cultural. While the nation states will remain the most powerful actors in world affairs, the principal conflicts of global politics will occur between nations and groups of different civilizations. So the clash of civilizations will dominate global politics. The fault lines between civilizations will be the battle lines of the future[x]. This assumes importance since the last three decades which had witnessed the gradual decline and withering away of the socialist system across the globe, has also been witness to spurt in primordial conflicts[xi].

Present Scenario

The world is at the cusp of a new order and there is a need for the NAM to address issues concerning the disproportionate control over finance, technology, communication, biodiversity resources and weapons of Mass Destruction.

As witnessed in the history of world politics the elementary changes in the balance of power is directly proportional to the changes in the world order. The end of the Cold War period which was as a result of the collapse of the Soviet economy, state and ideology leading to the conclusion of the bipolar system in international relations. This had led to a scenario in which the USA sought to substitute the world order with an U.S.-led unipolar world order.[xii] The post 9/11 order has gravitated towards a polycentric order and has drawn focus towards countries like India, Brazil and South Africa and their growing relevance as regional partners in maintaining the world order.

The advent of terrorism that has afflicted the globe is also a source of concern. This assumes significance after the 9/11 attacks in the USA, since it merely vindicated the global face and reach of non state actors. This issue was raised by Atal Bihari Vajpayee, the Indian Prime Minister in February 2003, during the NAM Summit in Kula Lumpur.[xiii] This Summit had also raised concerns about proxy wars, money laundering, ethnic nationalism, xenophobia and violation of human rights.

The NAM has to retain and reinvent its relevance based on the unity and consensus among its member states, as it is the largest forum for developing nations outside the UNO. NAM can be retained as an instrument of foreign policy of states, in order to democratize the international order.

The NAM should strive to avert any kind of arms race among its members and seek to reign in those countries that seek to explore direct or indirect defense ties with major powers. The NAM must explore seeking a broad consensus among its members for the promotion of democracy, human rights, rule of law, good governance and transparency.

Some scholars perceive the nomenclature of "Non-Alignment" as a misnomer and have opined that it could be renamed as a solidarity movement of the southern states. NAM could explore the possibility of setting up a permanent secretariat in any of its original member states. The NAM member nations must be open to engagements with the Western powers and other powers in terms of a constructive engagement. The movement must maintain its autonomy and safeguard South–South collaboration for control over natural resources

within the region. The rich and resourceful states of the region should be encouraged to invest within the member states.[xv]

Finally the NAM should seek some remedial measures based on the declaration in the 17th Summit at the Margarita Island, Venezuela.[xvi] The key points include democratization of the UN, democratization of communication, defending self-determination, dialogue among Civilizations for promotion of religious tolerance and promote and preserve cultural diversity, support environmentalism of the south and redressed of the plight of the refugees.

In International Relations, it is the interests that remain permanent and the notion of friend and foes remain ambiguous, similarly the NAM is at crossroads and is subject to various contestations. It would be a herculean and an intractable exercise to subject its role and relevance through the prism of the 'end-of-ideology' thesis or the 'end-of-history' thesis.

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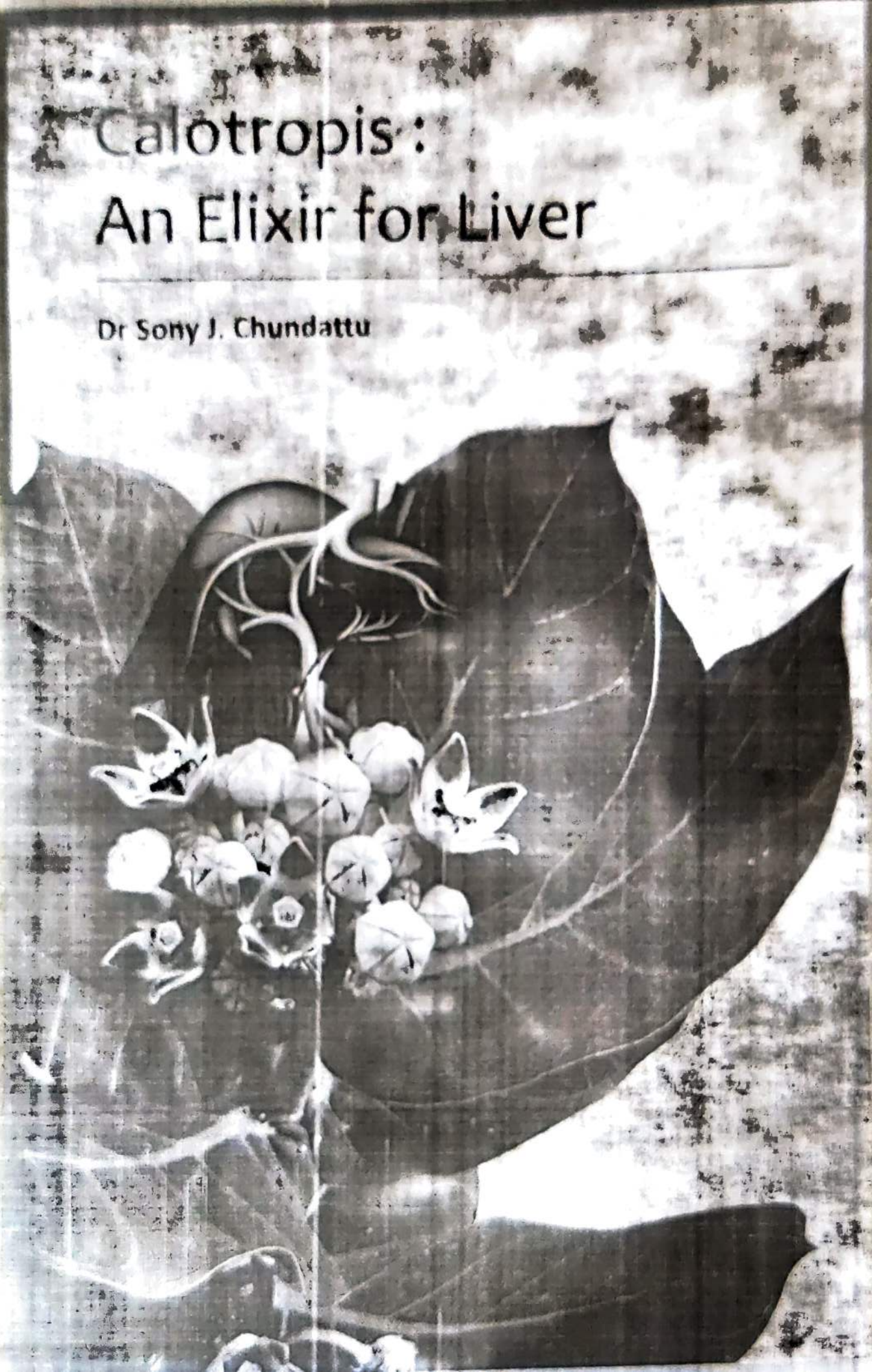
Calotropis : An Elixir for Liver

Dr Sony J. Chundattu



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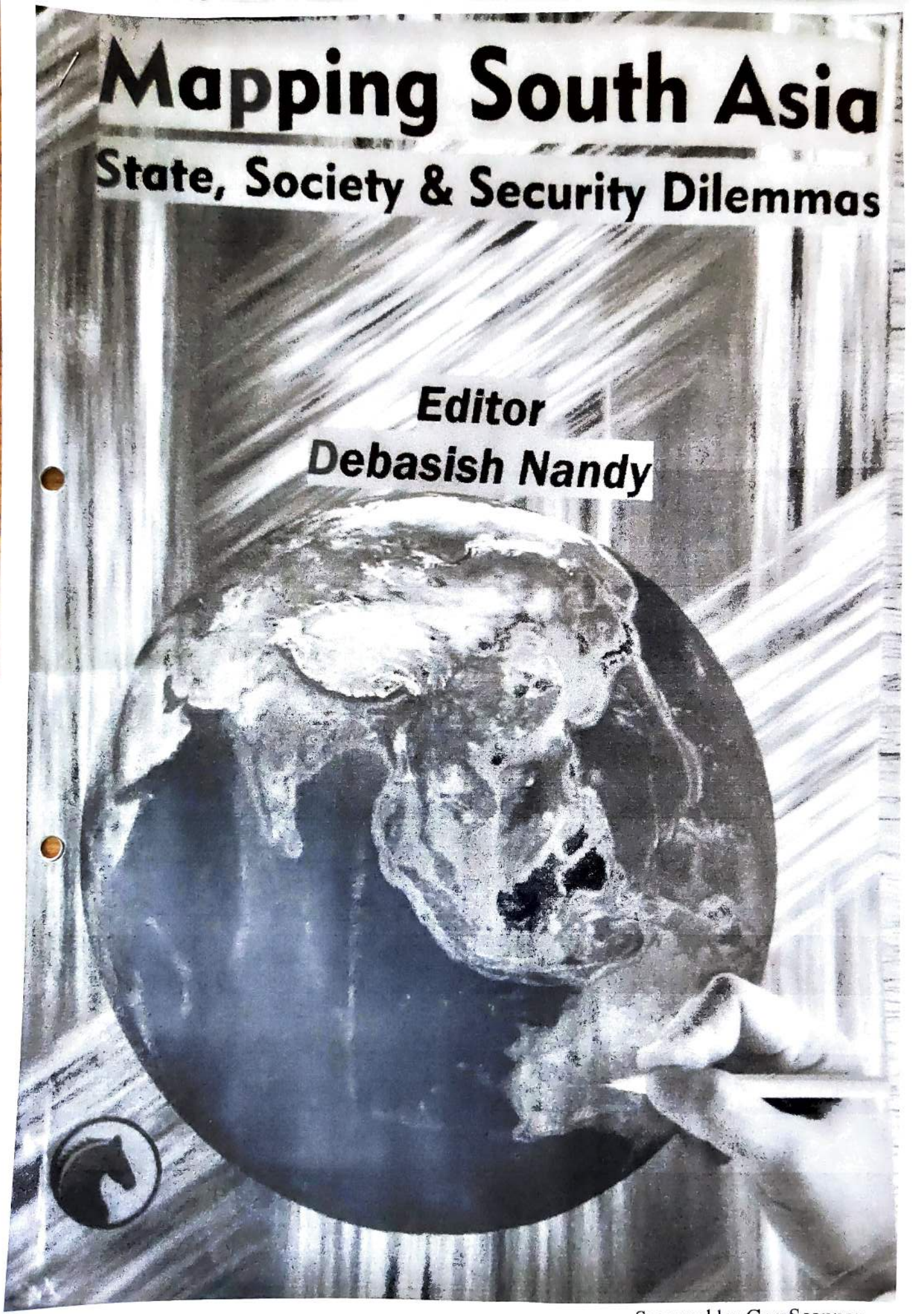

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Dedicated in memory of

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Former Professor ,Department of Political Science The University of Burdwan, West Bengal, India from whom I had been taught in P.G. and M.Phil. Courses

&

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Understanding Non Traditional Security - Some reflections on Human Security

-Assunta Mendonca-Sandhir

Threats to mankind has alarmingly increased, as such one can consider themselves to be secure only if one is safe from harm, but no one is or can be totally secure, assuming that accidents can happen, resources could get depleted, individuals may lose their jobs or there can even be an outbreak of wars that will threaten world peace. Undoubtedly the need to feel secure is a core human value, but it is also considered to be a prerequisite to enable one to lead a decent life.

In a long line of neologisms that included global security, common security, cooperative security and comprehensive security, the concept of human security has witnessed transformation. As of today, this very concept of Human Security has not only sought the attention of the academic community but also the policy makers to redefine and broaden the meaning of security, perceiving that International Security is something more than the military defense of state interest and territory.

What needs to be observed that "there has been a paradigm shift in this very concept of human security from an orthodox international security analysis that saw the state as an exclusive primary referent object and now very strongly asserts that the focus be primarily given to human beings and their complex social and economic relations along with or over states.[1] "Traditionally, security meant protection of the sovereignty and territorial integrity of states from external military threats, this being the essence of the concept of

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SHORT TERM IMPACT OF DEMONETIZATION ON INDIAN ECONOMY

Merin K. Varghese
Christ College- Pune

Abstract

On 8th of November 2016, India witnessed a groundbreaking event, which affected thousands of people of this country. The Indian Prime Minister announced the ban of Rs. 500 and Rs. 1000 notes from use as legal tender commonly termed as Demonetization. The main objectives for demonetization was to restrain corruption, to curb the use of high denomination notes for terrorist activities, destroy fake currency and making black money to flow into the main stream of the economy by declaring their income to the tax authorities. But this action caused chaos in the economy as people had to face the deficiency of liquid cash for day-to-day transactions resulting in long queues in front of the ATM. The informal sector and the primary producers were highly affected by demonetization. This paper aims to analyze the short-term effects of demonetization on Indian Economy.

Keywords:

Demonetization, fake currency, black money, informal sector, Indian economy

Introduction

One of the major problems of any country is the existence of black money. Black money is the income, which is unaccounted for, as the government does not know the sources of the income or it is the income, which is not declared for tax purposes. To tackle the situation of circulation of black money in the economy the government can take drastic steps like demonetization, which will force the holders of black money to deposit their money in banks or forgo their wealth. Before 2016, India has gone through the situation of demonetization twice. First time in the year 1946 and second time in the year 1978. In the year 1946, Rs. 1,000, Rs. 5,000 and Rs. 10,000 notes were discontinued from circulation. But all the three denominations were reintroduced in the year 1954. Then again in the year 1978, the ruling party deemed Rs. 1,000, Rs. 5,000 and Rs. 10,000 notes as illegal for the second time.

In the year 2016, Rs. 500 and Rs. 1,000 notes we demonetized which comprised nearly 86 per cent of value of notes in circulation (RBI annual report Aug 30, 2017). The main focus of RBI was then directed towards printing of new currency and transfer it from the printing presses to then currency chest and from there to the bank branches and ATMs in the least possible time. The Reserve Bank of India induced 23.8 billion pieces of bank notes, which values up to Rs. 5,540 billion in circulation from November 9 to December 31, 2016. By March 31, 2017, the notes in circulation increased close to 74 per cent of the notes in circulation as of November 4, 2016. In table 1, we can see that in the year 2016 the total currency in circulation was Rs. 16,415 out of which notes of Rs. 1,000 denomination accounted for 38.6 per cent (Rs. 6,326 billion) and Rs. 500 accounted for 47.8 per cent (Rs. 7,854 billion)

For the process of demonetization to be successful, proper management system is required. It will help the economy in various ways like, improve economic growth, achieve a stable inflation rate and change the financial holdings of the citizens. Without a proper management system the result of demonetization can be catastrophic.

Banknotes in Circulation						
Denomination (₹)	Volume (million pieces)			Value (₹ billion)		
	Mar-15	Mar-16	Mar-17	Mar-15	Mar-16	Mar-17
1	2	3	4	5	6	7
2 and 5	11,672	11,626	11,557	46	45	45
	(13.9)	(12.9)	(11.5)	(0.3)	(0.3)	(0.3)
10	30,304	32,015	36,929	303	320	369
	(36.3)	(35.5)	(36.8)	(2.1)	(1.9)	(2.8)
20	4,350	4,924	10,158	87	98	203
	(5.2)	(5.4)	(10.2)	(0.6)	(0.6)	(1.5)
50	3,487	3,890	7,113	174	194	356
	(4.2)	(4.3)	(7.1)	(1.2)	(1.2)	(2.7)
100	15,026	15,778	25,280	1,503	1,578	2,528
	(18.0)	(17.5)	(25.2)	(10.5)	(9.6)	(19.3)
500	13,128	15,707	5,882	6,564	7,854	2,941
	(15.7)	(17.4)	(5.9)	(46.0)	(47.8)	(22.5)
1,000	5,612	6,326	89	5,612	6,326	89
	(6.7)	(7.0)	(0.0)	(39.3)	(38.6)	(0.7)
2,000	-	-	3,285	-	-	6,571
	-	-	(3.3)	-	-	(50.2)
Total	83,579	90,266	100,293	14,289	16,415	13,102
Note: Figures in parentheses represent the percentage share in total volume/value.						

Table: 1- Source: RBI Annual Report, Aug 30, 2017

Objective of the Study:

This paper seeks to study the impact of demonetization on Indian economy. This paper also tries to analyze the success rate of demonetization.

Literature Review:

Litvack and Vigne (2017) in their paper "Demonetization in India and Emerging Challenges" states that demonetization had an impact on many areas of the economy. It had a major impact on the labor market and resulted in increased uncertainty and decrease in employment elasticity. Businesses had to shut down, farmers faced trouble buying seeds, auto and rickshaw drivers didn't have any other payment option other than cash, employers couldn't pay their employees, hospitals denied service to patients who had old bank notes, fishermen couldn't sell their fishes on time, families had to face the difficulty of purchasing food and weddings were cancelled throughout the country.

According to Kushwaha, Kumar and Abbas (2018), demonetization was an effort to handle the threat of illegal money, corruption, terror funding, and counterfeit currency. This action was considered as a surgical strike against the undeclared money in the history of Indian economy. Demonetization caused severe cash shortages with harmful effects on various small business, agriculture and transportation. The positive impact of demonetization is it encouraged digital mode of payment like e-wallets and apps, online transactions using e banking, usage of plastic money etc. The benefits of demonetization will be seen in the medium and long term.

Veerakumar (2017) argues that demonetization was a big shock to Indian society. This drastic step was taken for several measures such as tax evasion, counterfeit currency and funding of illegal activities. People who have deposited money in their bank accounts beyond the specified limits, showed unaccounted income and was subject to higher tax and other penalties. This action helped the economy to move efficiently towards cashless infrastructure with alternative payment methods like e-wallets, online transactions and usage of debit and credit cards.

Kumar and Bumra (2017) says that demonetization caused short-term implications for cash intensive sectors like consumer market and agriculture sector. Due to shortage of cash with the farmers there was delay in payments, which affected the related companies for a short period of time. As the economy shifts towards a cashless platform, demand will be back from the unorganized sector to the organized sector.

According to Swami and Jain (2017), demonetization was a right move taken by the Prime Minister Narendra Modi, but the execution could have been better. The supply of the new currency should have been faster so that people could have adjusted to the new system easily. The rural class especially the women who were not accustomed with the system of electronic cash had to face the major problems of liquidity crunch. Another problem was the high denomination of Rs. 2,000 which was difficult for the public to use that note purchases as no one was willing to give back change to the customer.

Research Methodology

The study is descriptive in nature. It uses descriptive method of analysis based on genuine secondary data taken from government reports, online journals, research papers, newspapers etc. The data collected on various aspects of demonetization are mostly in the textual and tabular form. Each point has been explained separately to get a clear analysis of the work done.

Impact of Demonetization on:

Black Money: One of the major objectives of demonetization was to fight back against black money and corruption in the economy. Out of the total currency in circulation, which is Rs. 17 lakh crore, approximately Rs. 3 lakh crore is black money (Kushwaha, Kumar & Abbas, 2018). People are holding cash in hand to avoid tax payment. Demonetization was a step taken to counter tax evasion. An article in India Today on demonetization (August 30, 2018) says that according to RBI report, out of Rs. 15.41 lakh crore of demonetization currency notes of Rs. 500 and Rs. 1,000, Rs. 10,720 crore was not returned to the banks or the RBI, i.e., only 0.7 per cent of the demonetized currency notes were junked in the process. These figures might also indicate that black money held in the form of cash was very low or the execution of demonetization was not effective because of which most of the banned notes were laundered back into the banking system.

Terror Funding and Counterfeit Notes: Prime Minister Narendra Modi said that demonetization has destroyed the terrorist activities, drug mafia, human trafficking and the underworld instantly (GN Bureau, November 23, 2017). Finance Minister Arun Jaitley also said in an interview to Doordharshan that demonetization resulted a reduction in terror activities. The number of stone-pelters on the streets of Kashmir also reduced post demonetization and similar situation was seen in Chhattisgarh (The Hindu, November 08, 2017). Counterfeit currency or fake currency formed the major chunk of terrorist funding which turned worthless after demonetization. Thus, demonetization led to a positive impact on reduction of terror funding and violence in the country. Table 2 shows a significant rise in the number of counterfeit notes detected in the year 2016-17.

Denomination-wise Counterfeit Notes Detected in the Banking System (April to March)			
(No. of pieces)			
Denomination (₹)	**2014-15**	**2015-16**	**2016-17**
1	**2**	**3**	**4**
2 and 5	0	2	80
10	268	134	523
20	106	96	324
50	7,160	6,453	9,222
100	181,799	221,447	177,195
500 (MG series)	273,923	261,695	317,567
500 (New design)	0	0	199
1,000	131,190	143,099	256,324
2,000	0	0	638
Total	594,446	632,926	762,072

Table 2. Source: RBI Annual Report August 30, 2017

Taxes: According to Crisil, India witnessed an increased growth from 8.2 per cent in FY16 to 26.8 per cent in FY17 and 21.0 per cent in FY18. The corporate tax also increased in the corresponding years from 5.7 per cent to 7.0 per cent to 16.3 per cent respectively (Financial Express, May 29, 2018). An increase in the number of new tax filers has also been recorded due to demonetization. According to a report assessed by Press Trust of India (PTI) there has been a sustained increase in the income tax returns (ITRs) filed in the last four financial years as compared to 3.79 crore ITRs filed in 2013-14 fiscal and it increased to 6.87 crore during 2017-18 fiscal, i.e., by 81 per cent (The Times of India, PTI Dec 3, 2018). The report said that it was a positive impact of demonetization. A report of Central Board of Direct Taxes (CBDT) said that the Income Tax Department has collected more than Rs. 6.63 lakh crore for the fiscal year 2018, which is recorded as high growth rate of last seven years.

Gross Domestic Product: Around 6 months after demonetization the Gross Domestic Product (GDP) growth rate, which is a measure of all the final goods and services produced in a period of time, mostly annually or quarterly, slumped down to 6.1 in the January - March 2017 period, which is the lowest in more than two years. This figure was not expected as the Central Statistics office had estimated a growth of 7.1 per cent as compared to 7.6 per cent in 2015-16. With March 2017-quarter report, India lost its status as world's fastest growing major economy.

Digitalization: Another major objective of demonetization was to enhance digital or cashless transactions. According to RBI annual report, August 30, 2017 the total volume of retail electronic payments averaged around 37 per cent during April to October 2016, and it shot up to nearly 70 per cent in November and up to 123 per cent in December 2016. After this the growth rates have been moderate but it remained high. Demonetization led to the growth of digital payments and according to RBI it would continue its efforts to move towards a cashless economy as well as ensuring safety and enhancing the online payment system. Demonetization thus helped the country to take a step forward towards cashless economy. This will help to solve the economic and security problems usually associated with cash.

RBI's printing of notes: According to RBI's annual report Aug 30, 2017, the total amount spend for printing of notes during 2016-17 was Rs. 79.65 billion compared to Rs. 34.21 billion in 2015-16. The supply of higher denomination notes in the year 2016-17 was 13,702 million pieces as against 5,268 million pieces supplied in 2015-16, which is higher by 160 per cent. Due to urgent supply of notes across the country during demonetization, notes were also remitted by air, which resulted in increased expense on freight charges.

Common man: As demonetization was announced it caused a disturbance in the minds of the people thinking about the consequence of this action taken. People had to face liquidity crunch, which means they were not able to get sufficient volume of cash for day-to-day transactions. A high percentage of the population uses currency to meet their daily needs. The informal sector of the economy, like daily wage earners, laborers, small traders etc. use cash frequently. These sections lost their income at the time of liquidity crunch. In a series of tweets by former Finance Minister P Chidambaram said that 100 lives were lost and 15 crore daily wage earners lost their livelihood for several weeks. Many Small and Medium Sized Enterprises (SMEs) were shut down and many jobs were destroyed (India Today, August 30, 2018).

Conclusion:

The news of demonetization came as a shock to the people of India. Though the intension of this action was noble, its impact on common man was rough. It did help the economy to curb black money and counterfeit currency and decrease terrorism but it was not fully successful, as most of the black money is not held in the form of cash. If it was implemented in a better way, the common man and mostly the poorer section of the society who deals with cash for their day to day transactions wouldn't have suffered the problems due to lack of liquid cash. Demonetization was definitely a step towards cashless economy, which will make our lives more convenient and secure. Other strong and effective policies should be implemented to achieve the targets of the government. The short-term impact of demonetization was bad for the economy of India with fall in GDP and consumer demand and most importantly, the sufferings of the common man.

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Depreciation of Indian Rupee: Causes and Impact

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Abstract

India is facing a huge problem of currency depreciation and this has led to a volatile, uncertain and risky economy. This study focuses on the studying the impact of depreciation of Indian rupee against US dollars and also tries to learn the downfall of Indian rupee. Indian rupee once which was equal to the USD, over the course of few years showed a steady depreciation which has turned all the plans and strategies upside down. Stock markets have become uncertain and flexible, current account deficit is affected which leads to balance of payment and inflation is rising up. RBI tries it difficult to come up with a strategy to encounter this massive fall of Indian rupee.

Keywords: Depreciation of Indian rupee, Current account deficit, Indian economy, Inflation

1. Introduction

With the recent depreciation of Rupee, it has hit a life time low against the US Dollar. The Rupee has weakened nearly 8per cent since May 2013 and 15per cent since March 2012. Since the launch of economic reforms in 1991, the Indian rupee has undergone a series of devaluations and is getting weaker in recent times. Inside the Keynesian theory, money depreciating helps monetary development by advancing net exports, aggregate demand, and yield through the multiplier impact. Notwithstanding, this is just an important condition for degrading to be expansionary, not an adequate one.For example, by making imported inputs more expensive, devaluation can contract the aggregate supply curve, and cause inflation. For a similar reason, it can likewise bring down venture, a key part of total demand. Moreover, if it is a crisis-driven devaluation, it could damage the country's reputation and lead to a flight of capital and destabilize the economy.

` Depreciation of rupee has become a big worry for the Indian government because it has depreciated to an all-time low with respect to the US Dollar. Trade rates assume an indispensable job in a nation's level of exchange or, in other words relatively every free market economy on the planet. Therefore, exchange rates are among the most monitored analysed and manipulated economic measures by the government. This study aims at exploring the dynamics of exchange rate mechanism, rupee

journey against dollar since independence, factors influencing the fluctuation of Indian Rupee and finally modelling the exchange rate through multivariate analysis (OLS).

India isn't the main rising country encountering a lofty decrease in its money's esteem. A few developing business sector countries are additionally encountering sharp deterioration over the possibility of inevitable decreasing of the US Federal Reserve's arrangement of quantitative facilitating (QE) program. The South African Rand and the Brazilian Real contacted four-year lows against the US $ in June 2013. But the Chinese Yuan and Bangladeshi Taka, most Asian monetary standards have seen sharp deterioration since the start of 2013. In any case, the Indian rupee has fared more poor than other rising country monetary standards in light of its twin deficiencies – current record and financial shortfalls.

2. Objectives of the study

1. To know the reasons of continues downtrend of Indian rupee value against US Dollar.
2. To study the effects of depreciated Indian rupee on economy.

3. Review of literature

Harberger (2003) in his study in explained the impact of economic growth on real exchange rate and also stressed upon that there is no relation between economic growth and exchange rate.

Due and Sen (2006) found out that the relation between exchange rate , capital and volume of flow of capital, fiscal and monetary policies have been conducive.

Kim and Ying (2007) comment that with financial liberalization and improvement in information technology, devaluation may be more contractionary than before as it worsens the balance of payments of countries with heavy foreign currency liabilities.

4. Discussion

A. Downtrend of Indian Rupee

The rupee has dove to an unsurpassed low against the dollar and its fall has turned into a subject for discussion. The typical talks on the fall of the rupee raise large scale financial matters, for

example, moderate monetary development, enormous current record shortage, rising imports and so on.

At the time of independence when India had no foreign borrowings the rupee was at standard with the dollar. With the presentation of the 5 years plan and the subsequent necessities for outside ventures the dollar gradually climbed. In 1985, as soon as the Rajiv Gandhi government was pulled out due to Bofors scam, the US dollar equalled to Rs. 12.35. And later since economic liberalisation and privatisation, the dollar rate steadily increased to a rate of Rs. 26.40. In early 2000 the rupee lost its value much as it hit a low of Rs. 45 per dollar. In 2013 it hit a historical low till then of Rs. 65.42agaist USD and never the less this steady depreciation continues even in 2018 where USD has hit a low of Rs. 70 per USD. Central bank and economists are trying to figure out what strategy could be implemented to put a halt to the depreciation of Indian rupee.

Source: www.moneycontrol.com

This circumstance is to a great degree unsafe for the economy and a nation all in all. Swapping scale influences exchanging connections between countries. The swapping scale is a vital fiscal strategy apparatus for rising economies like India. India has received expansion focusing on and has less adaptable swapping scale game plans. The devaluation of rupee influences purchasing power of income, interest rates, inflation and current deficits. The swapping scale of the cash decides the genuine return of the portfolio that holds the main part of its speculation. It influences purchasing power of income and capital gains derived from returns, income factors such as interest rates, inflation and even capital gains from domestic securities.

Since 2008 after the global financial meltdown, rupee has been devaluating constantly against dollars. The Indian markets opened up during 1990 which helped India to grow economically and also in terms of opportunities for being one of the super powers in the world. However, the devaluation of rupee against dollar in recent years has ripped apart the economy causing heavy destruction. The after effects of the downtrend of rupee value against the dollar have caused our nation a huge loss.

B. Depreciation and Economy

- **Increases in the CAD:** Due to the depreciation of Indian rupee, the gap between a nation's foreign earnings and its payments would be more. Imports become more costly which will increase the outflow of foreign reserves. Since India faces current account deficit, it is predicted that with the largest import of oil, this deficit can widen the gap. Withdrawal of funds by investors also leads to outflow of foreign reserves and this also leads to deficit widening. The central bank will have very less control on the policy cut which may be a burden for the borrowers and thereby increasing its current account deficit (CAD). A widening gap in the current account deficit can be a challenge for the growth of an economy.

- **Inflation:**

  Depreciation of rupee can boost the inflation adversely as the inflation rate in the economy is already close to two digits. As the imports go expensive the fuel price would be inflated considerably. Rise in fuel prices will impact the prices of raw materials and other products which will lead to a steady inflation in the economy. This will create a huge pressure on the developing economy in terms of growth. Curbing inflation will become a tedious task for the central bank and government.

- **Costlier Imports:**

  Cost of imports will be relatively high as the depreciation of rupee is considered. India imports more than 70% of its crude requirement from various nations. Oil prices relatively will be inflated as the cost of importing fuel rises. Per barrel crude will rise as Indian rupee continues to depreciate. However it needs to be noted that the exporters will be in profits as they will generate more revenue considering the exchange rate fluctuations. Hence this will boost exports

in India. DK Joshi, chief economist, Crisil, told *Business Standard* that the benefits of the depreciating rupee would be neutralized by the hit to exports. Still, sector wise, experts believe information technology exporters will be among the biggest beneficiaries.

- **Corporate Competition**

  As an impact of costlier raw materials, Indian companies may face challeges in the global market as the cost of production of such companies will be relatively higher than that of other foreign competitors. This will affect the revenue generation of Indian companies compared to their foreign competitors. As per a study conducted by Morgan Stanley, gross revenue margins of companies are at decade lows, mainly due to higher raw material costs and tougher competition. This will encourage companies to shrink the remuneration packages or human labour.

- **Affects foreign investments**

  Falling rupee will also create panic among the Foreign Institutional Investors as they have a huge uncertainty amount the performance of the economy with the continuous downfall of the currency. This may lead to FIIs withdrawing their investments from Indian markets or booking their profits early. The recent plunge in the rupee rate has let the FIIs pull out their investments from the Indian markets which lead to an economic correction. The markets collapsed and it will affect those investors who have invested their money in the market as the market is highly volatile. On the contrary NRIs and other Indian nationals who earn abroad will be interested to invest in India as they will be able to make an advantageous deal with the fall in the Indian rupee.

- **Education and tourism**

  Education and tourism part has additionally been influenced with the dip in the value of Indian rupee. Students, who are studying abroad, will be burdened with the effect of devaluing rupee. Costs towards the college/university expense and additionally that of living will build, which will expand the weight of education. Indian rupee depreciation will likewise influence the tourism. The charges of travel and accommodation will be more costly and this will pull down the interest impressively which will again cause a negative effect.

5. Conclusion

A nation focuses on selling goods and services to other nations in surplus than that of purchasing from other nations. Hence, maintaining the required amount of foreign exchange in the economy for the growth and development of the country. However in case of India, the deficit has been increasing and this was due to major setback on the depreciation of Indian rupee. It is assumed that the fallen in rupee to a certain extent is good as it helps the market to correct itself in the course of time. The movement of the value of a currency will not cause much effect on the economy if it is for a short run. The regulatory bodies in India have taken various corrective measures to curb the volatility of the currency fluctuation. However one needs to wait and see how much it can bring some kind of relief to the economy. It is inferred that the fall in the estimation of Indian rupee has a few outcomes which could considerably affect Indian economy.

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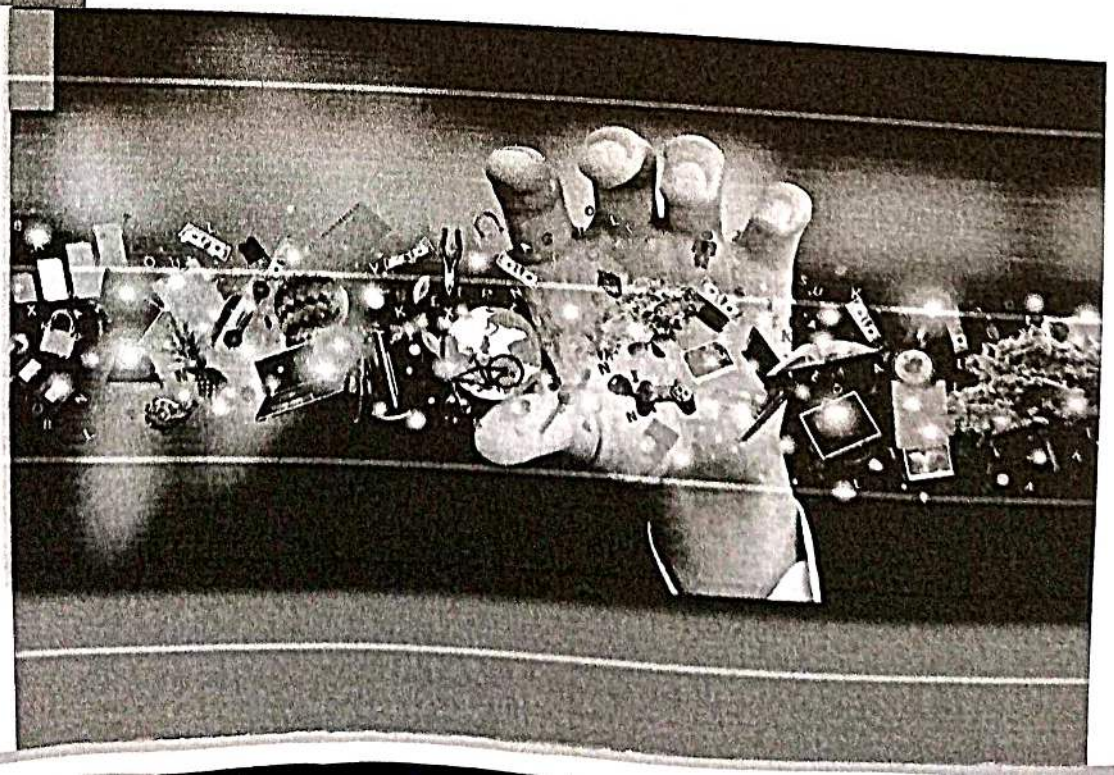

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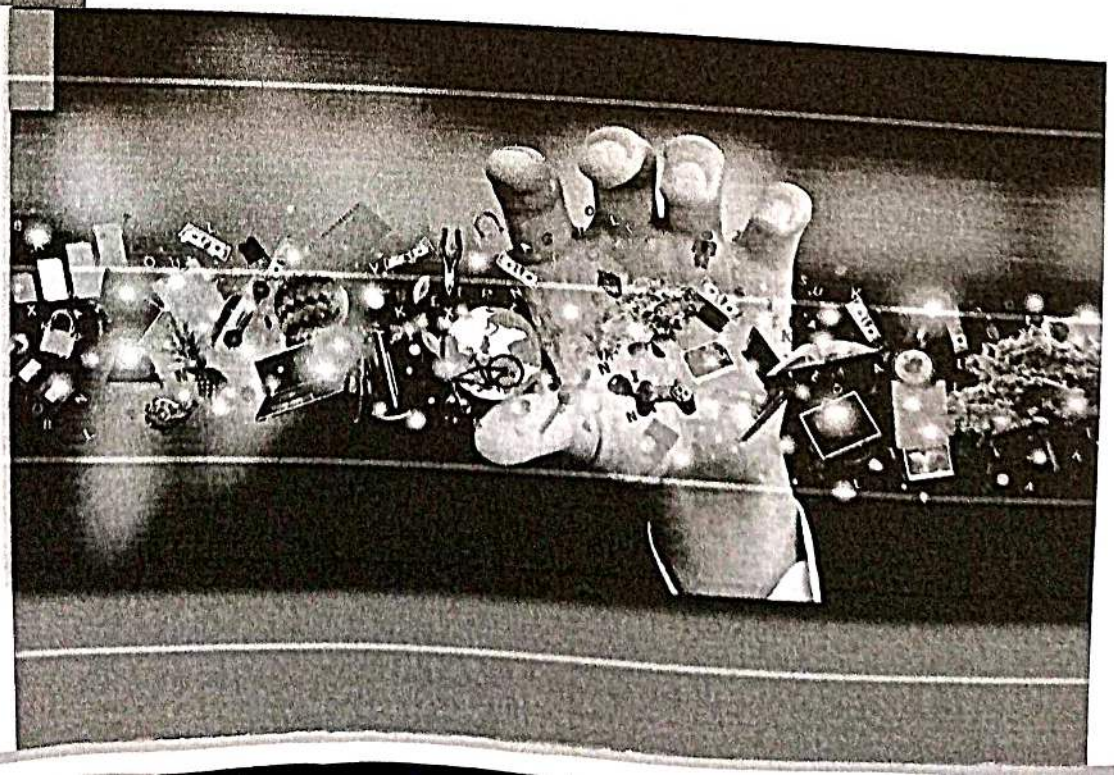

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Mrs. Leena Sanu[1] and Ms.Alphonsa Varghese[2]
[1] Christ College , Pune.
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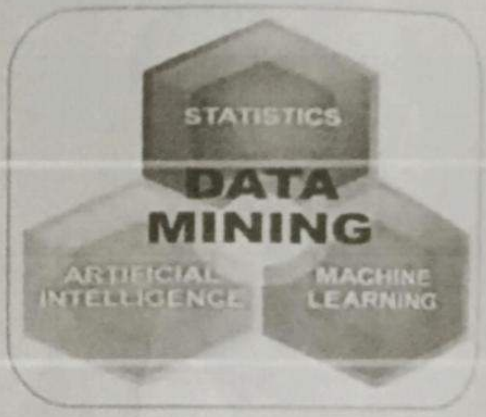
ABSTRACT:-

Payment services areundergoing a rapid change driven bytechnological development. At the onset of demonetization there has been a tremendous increase in the use of different online applications for transactions which involve different payment platforms. Through these transactions large amount of data is generated ,which is used for analyzing the behavior of consumer and also to detect different hidden patterns. Data Mining deals with extracting hidden knowledge, unexpected pattern and new rules from large databases.



Data mining techniques are used for this analysis where relevant data is extracted from datasetsand converted into knowledge as per the business requirement. In dealing with such payment platforms a consumer may face lots of threats and challenges related to the security of data. This paper aims at identifying the various data mining techniques used in different payment platforms and the challenges and threats involved in using such interfaces.

KEYWORDS— *Data Mining,Payment Platforms,Data Mining Techniques,Challenges.*

INTRODUCTION:

Technological advancement and innovation have led to the transition of payment methodology which had become a more efficient and easier way of transaction for the consumers.Here comes the relevance of the term FinTech which enables consumers to perform financial services that uses smart phones for mobile banking, investing services and crypto currency.Customer expects transactions to be quick, seamless and personalized which the non banking players like Google,Apple,Facebook have taken advantage as they have valuable information regarding customers requirements and preferences in payment methodology.

OBJECTIVES OF THE STUDY:

- To study the various data mining techniques
- To analyze the various threats and challenges involved in the use of payment platforms

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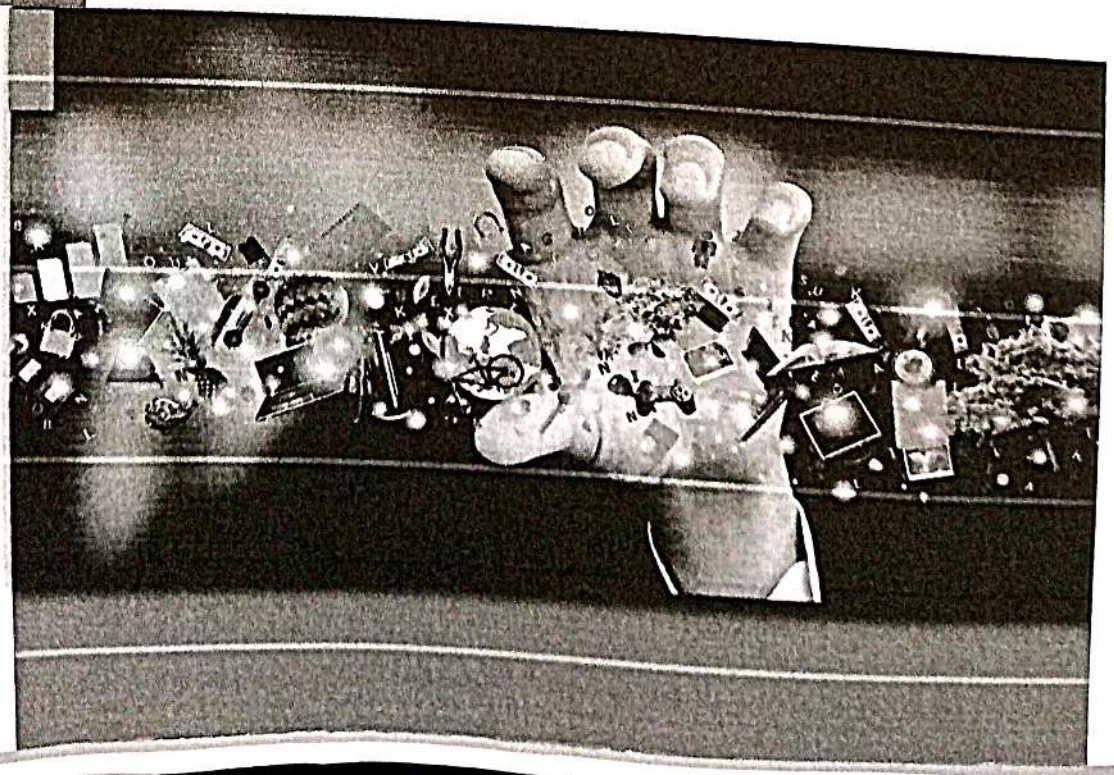

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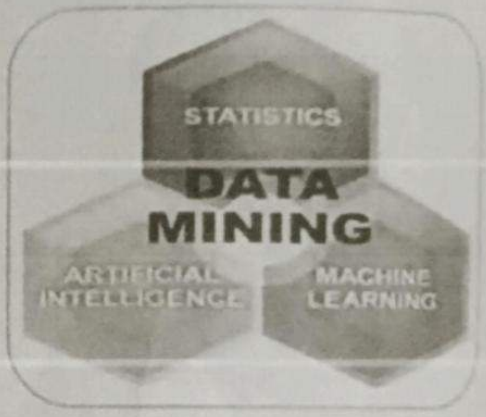

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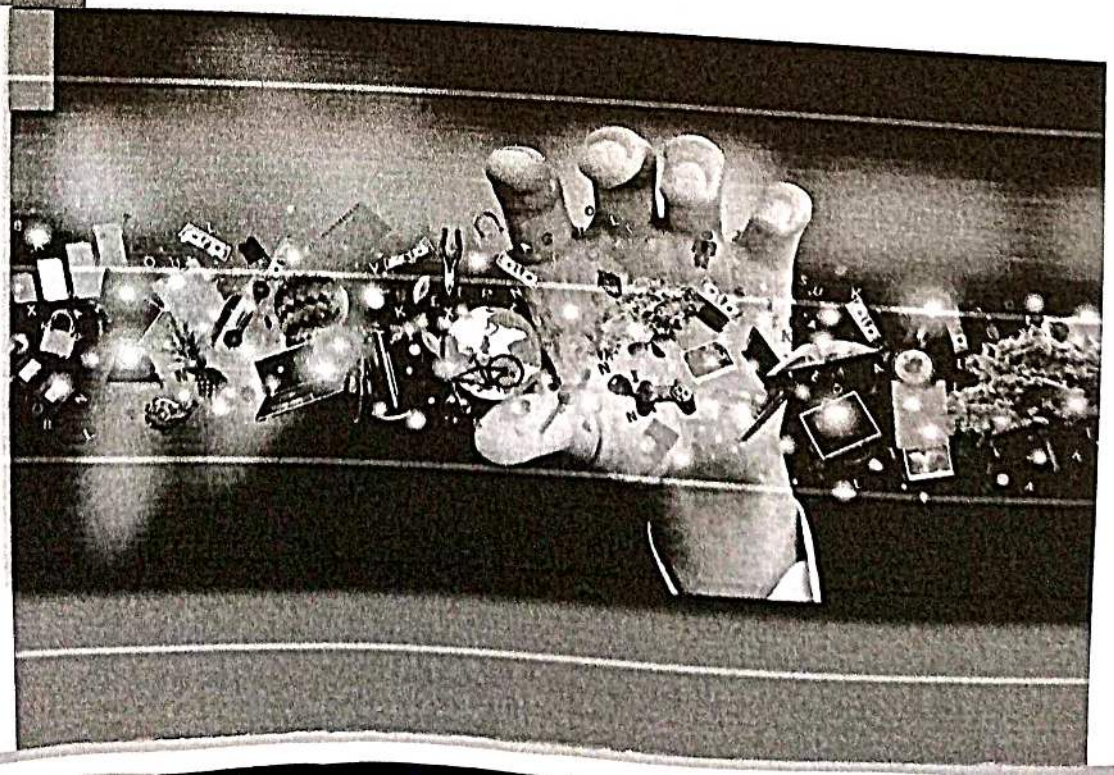

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GROWTH OF MUTUAL FUND INDUSTRY IN INDIA

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Abstract:
The not so systematic banking industry in case of investment has been a major cause of promoting savings and investments through financial markets in India. After the opening up of markets in the year 1991, there has been progressive growth in the capital market in India. The new economic and liberalization policies gave way to a subsequent push for the financial services in India. Small and medium scale investors have been encouraged to bring in their investments into financial markets through the introduction of mutual funds. This is an exploratory study on the mutual fund industry in India and its growth since a decade. This study is conducted based on secondary data analysis only.

Keywords:
Mutual funds, mutual fund industry, growth, Asset management, Investment, savings

Introduction:
Indian financial system is pillared in the financial set up of financial institutions, markets and instruments. Each one of these pillars plays a vital role in building up the industry as a genre. These helps in the smooth functioning of the system along with the regulatory authorities like RBI, SEBI, IRDA and AMFI. Indian Financial system ought to provide convenient, smooth and efficient financial services to the stakeholders. In the recent past Mutual Fund investment has had a progressive growth in Indian financial set up. This has made investors more comfortable with the capital market systems in India. The service of a fund manager helps the investor in a comfortable investment even though the investor need not be aware about the complex capital market structure. Mutual Funds are pooling of money from desired investors having common financial goals. These funds collected are invested by the fund managers in the stocks diversified according to the nature of return expected. The fund manager analyses the stocks and selects the debt and equity instruments to invest the corpus thus collected.

On an adverse note, mutual funds affect the economies of scale of research and investment. Each scheme has its own identified features and diversification depending upon the need of the investors. There are various categories of funds in mutual funds and each of it possesses unique characteristics. Broadly mutual funds are classified as equity funds, debt fund, balanced funds and hybrid funds. Each method of investments including Mutual funds bears risks and uncertainty. Uncertainty or risk here means the financial loss an investor will bear due to market fluctuation. Risk and potential returns are connected. More the risk, there is a higher chance of earning higher returns. At the same time more the risk higher is the chance of making a loss. Hence the factor of risk is a key player in making a higher return on the investments through mutual funds.

Objectives of the study:
India is a developing nation and gross investments and savings plays a vital role in the continuous development of the economy. Capital and money markets have been in trend since years, However there has been a tremendous growth in mutual fund investments in India within the time frame often years. This has not only increased the GDP but also brought in considerable increase in the per capita income. In this study

we look forward to learn the mutual funds and its industry in India with special reference to its growth in the past decade. Hence the objectives of the study are as follows:

1. To learn the Mutual fund industry in India in a nutshell.
2. To understand the growth of Mutual Fund industry in India.

Hypothesis:

The following hypothesis has been set for the study

1. Mutual Funds are growing as an investment instrument in India
2. There is a considerable and steady growth in the Mutual fund industry during the past decade.

Review of Literature:

Umarani (2012) explained and evaluated the development of assets and recommended certain means that should be taken by every one of the stakeholders of the business in a thorough way to make an environment that will take the Indian Mutual Fund industry to its next dimension of development.

Kale and Panchpagesan (2012) studied on the framework of Indian Mutual Fund Industry and the purposes behind its poor entrance, which is because of the absence of target explore. It was discovered that the business keeps the whole worldwide industry as its benchmarks.

Panigrahi (1996) lays her study on the development of Mutual Fund Industry in India. She inspected that why shared assets have flourished over the most recent couple of years, what is the degree of development and whether the administrative structure for their tasks mirrors the adjustment in condition.

Scope of the Study:

There has been growing need and scope for Mutual Funds in India over the years. Choices were very few when the Asset Management Companies started to offer mutual fund investment strategies. However, later during the recent past, diversified and object oriented mutual funds were introduced which confidently pulled a lot of investors to mutual funds which helped the Mutual Fund industry the modern world become so wide. Hence, this study provides a platform which throws light into the Mutual Fund industry and the growth of Mutual Funds in India

Limitations:

Few limitations of the present study are:

1. Only secondary data is considered for the study. Primary data hasn't been incorporated.
2. The study pertains mainly into the recent past of the Mutual Fund Industry in India.

Discussion

1. Mutual Fund Industry in India

The historical backdrop of Mutual Fund Industry in India can be followed back to 1963, with the dispatch of the Unit Trust of India by the Government of India under an Act of Parliament. In 1978, the administrative and authoritative control of UTI was exchanged from the Reserve Bank of India to IDBI (Industrial Development Bank of India). The principal common store conspire that was presented in India by UTI was in the Unit Scheme (1964). UTI had Assets under Management worth Rs. 6,700 Crores, before the years over 1988.

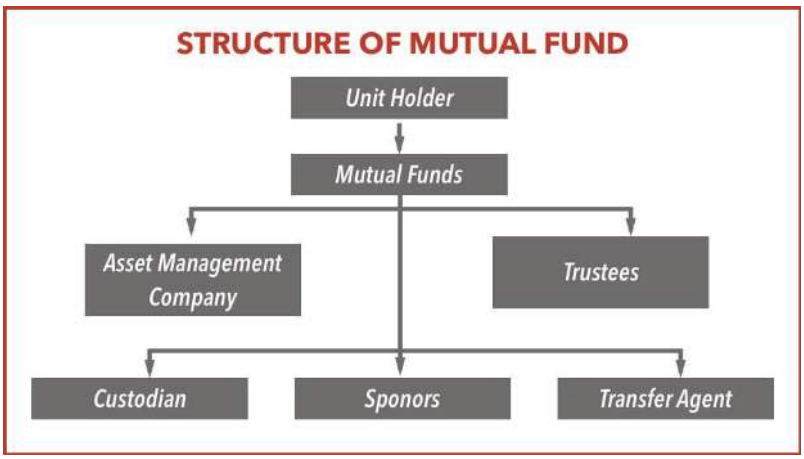
In 1987, public sector companies like, State Bank of India, Punjab National Bank, etc. and other non-UTI enterprises like General Insurance Corporation of India (GIC) and Life Insurance Corporation of India (LIC) entered the market and built up open market mutual funds. From the year 1993, private division reserves were built up in the shared store industry. Around the same time, Mutual Fund Regulations were presented in India under which every shared store aside from UTI must be enrolled. In 2003, the Unit Trust of India Act 1963 was revoked and was isolated into 2 separate substances – the

UTI Mutual Fund, which is supported by Punjab National Bank, State Bank of India, Life Insurance Corporation of India and Bank of Baroda and the second element is the Specified Undertaking of the Unit Trust of India.

The Indian Mutual Fund Industry has recorded Rs. 13, 460 billion Assets Under Management (AUM) by December 2015 and is announcing a consistent development till date. At first, corporate had been the principle supporter of AUM however soon, the retail portion ended up being the section that contributed the most to AUM development in India.

Structure of mutual funds in India

In India, the mutual fund industry is controlled and regulated with an overview of operational and transactional transparency and investor's protection. The organizational structure of Mutual funds is decided by Securities Exchange Board of India. These regulations compel the funds to be registered under the Indian Trust Act, 1882 as a trust to function in the financial markets. With the association of the RBI and SEBI, the mutual fund industry has developed allowing speculators a chance to benefit as much as possible from this venture road. In India a mutual fund operates through four pillars of structure they are a sponsor, Board of Trustees, an asset management company and a custodian.



Source: AMFI

Sponsor: A sponsor is a body corporate who sets up a fund. It might be one individual acting alone or together with another corporate body. Be that as it may, if any individual holds 40% or a greater amount of the total assets of an AMC, he will be considered to be a sponsor and will be required to satisfy the qualification criteria indicated in the mutual fund regulation.

Custodian: The Mutual Fund is required by law to ensure their portfolio securities by setting them with a custodian. About every fund utilize qualified bank custodians. Just an enlisted custodian under the SEBI control can go about as a caretaker to a mutual fund.

Board of Trustees: A mutual fund must have a Board of Trustees, independent trustees of the organization holds the property of the fund in trust to serve the unit-holders. They are in charge of securing the unit-holder's advantage.

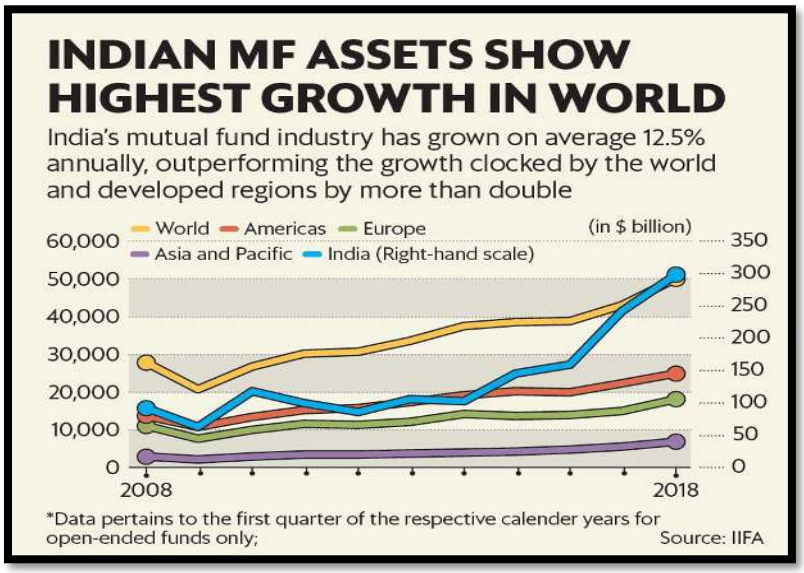
Asset Management Company: The job of an AMC is very noteworthy in the shared store activity. They are the fund managers and hence care for the regulatory elements of a mutual fund for which they charge the executives expense.

2. Growth of Mutual Fund industry in India

Since 1970 the mutual fund industry in India had reached out to a maximum of 361 funds with a combined asset corpus of 47.6 billion dollars and 10.7 million stakeholders. Rising interest rates, market vulnerability and policies affected the growth of mutual fund industry in India adversely. The importance of new hybrid and innovations in the new models of mutual funds were initiated. These resulted in booming the industry in early 1990s along with the financial reforms of Liberalization,

Privatization and Globalization. The new concept of 'No Load' funds reformed and triggered a change in Mutual Fund Industry. It started attracting small and individual investors to the concept of mutual funds.

- **Technological Advancements:**
  AMCs have actualized different innovative measures so as to improve client commitment from different parts of the general public. Two projects that have been presented by AMCs are the Investor Awareness Program (IAP) and District Adoption Program (DAP). Different innovations that are being utilized by AMCs are web applications, portable applications and web based life. These organizations have enhanced their online interface for straightforward entry and use. Web based life stages have been perceived as a compelling channel to use innovation that would aid venture and portable applications to connect more clients from various sections.
- **Rise in Savings**
  The nation's mutual fund industry has a colossal development potential as Indian families' investment funds add up to Rs 20-30 lakh crore. AMCs in a report analyses that there is a gradual shift in the savings of Indian households from traditional savings such as gold, bank deposits, etc to financial savings such as Systematic Investment Plans and Mutual funds. The Industry's AUM had crossed of ₹10 Trillion in the May 2014 and in a short span of about three years, the mutual fund AUM doubled to an amount of ₹ 20 trillion and ₹22.86 Trillion for the in the month of August 2017 and December, 2018 respectively.



Indian Mutual Fund industry's Average Assets Under Management (AAUM) stood at ₹ 24.09 Lakh Crore (INR 24.09 Trillion). Average Assets Under Management (AAUM) of Indian Mutual Fund Industry for the month of December 2018 stood at ₹ 24,08,538 crore. Assets Under Management (AUM) as on December 31, 2018 stood at ₹22,85,912 crore. The AUM of the Indian MF Industry has grown from ₹ 4.13 trillion as on 31st December 2008 to ₹22.86 trillion as on 31st December, 2018, about 5 ½ fold increase in a span of 10 years!! The MF Industry's AUM has grown from ₹8.26 trillion as on 31st December, 2013 to ₹22.86 trillion as on 31st December, 2018, more than 2 ½ fold increase in a span of 5years[1]

As per a report by the Association of Mutual Funds of India and worldwide examination firm Crisil, over the recent 10 years, India's Mutual Fund industry has become 12.5% every year by and large, beating the development timed by the world and created areas by more than double. The total financial savings and Mutual Fund corpus in the nation, has expanded over the most recent three years to 14% in March 2018 from 10% in 2016. In the debt market, banks and insurance agencies still proceed to command and contribute a major piece of their cash in government securities..

[1] https://www.amfiindia.com/indian-mutual

The absolute number of records (or folios according to common store speech) as on December 31, 2018 has crossed a milestone of 8 crore and remained at 8.03 crore (80.3 million), while the quantity of folios under Equity, ELSS and Balanced plans, wherein the most extreme speculation is from retail fragment remained at 6.76 crore (67.6 million). This is 55th continuous month seeing ascent in the number of folios. The absolute number of folios with 42 fund houses in the nation ascended by 11.45 lakh in October and hit a record-breaking high of 7.90 crore, as indicated by information given by the Association of Mutual Funds in India.

Month-on-Month change in folios

Category	No. of folio as on October 2018	No. of folios as September 2018	Absolute Change	Change (%)
Equity	48995450	48117778	877672	2%
ELSS	11137153	11012590	124563	1%
Balanced	6336101	6278352	57,749	1%
ETFs other	814875	777110	37,765	5%
ETFs gold	317700	320328	-2,628	-1%
Income funds	9778815	9800968	-22,153	0%
Gilt	63,731	63,867	-136	0%
Liquid	1468406	1401496	66,910	5%
Fund of funds inves	119268	114010	5,258	5%
Total	79031499	77886499	1145000	1%

Source : SEBI

Folios are numbers assigned to singular financial specialist accounts; however a speculator can have various records. The ascent in the quantity of folios mirrors a developing inclination for mutual funds among financial specialists. In the course of the most recent couple of years, the quantities of financial specialist accounts have expanded fundamentally due to vigorous commitment from retail speculators, particularly from smaller towns.

Equity funds, Income funds and Gilt funds witnessed outflows for few consecutive months. Investors are redeeming income funds as their returns are disappointing. The yields are expected to to rise due to fear of rising prices. Balanced funds saw rise in net inflows to Rs2, 630cr amid August 2018 contrasted with Rs287cr in earlier month. Arbitrage funds witnessed net outflow of Rs2, 452cr in August 2018 compared to net inflows of Rs1, 133cr in July 2018. The indicators of macroeconomics identifies the pressure due to rise in the prices of crude oil, rupee dollar rate fluctuation, federal rate at US, inflation, trade wars, etc has set the market downfall.

Findings:

From the study few findings that were noted are as follows:

1. Mutual Fund Industry is well controlled and monitored by the regulators and has provided utmost importance to investor protection.
2. In the recent decade, Mutual Fund Industry in India has been growing considerably and has become one of the huge attractions for a huge investors group.
3. Economy and per capita savings has gone up higher compared to previous decades and savings has been substantial which has contributed to the growth and development of Indian economy and GDP.
4. Along with investment the investors has been fetching higher rates of returns on mutual funds which is comparatively higher than bank deposits and or other investments in debt instruments

Conclusion:

Mutual funds are most favored investment instruments. Average earners would prefer investing in mutual funds as it caters them to earn returns much better and higher than the other investment avenues available with them. It is much safer as the mutual funds are sheltered and certain funds have close to zero risk. It is imperative to increase great, be that as it may comprehension of shared reserve speculations, as clients can without much of a stretch be misinformed by the advertisements and offers advanced by money related organizations. The mutual fund is controlled by AMCs who is in charge of offer and buy of interests in understanding of reserve's targets. In India there is parcel of extension for development of mutual funds organizations gave support fulfills everyone's needs and enhancements in administration principles and divulgence. The vast majority of the speculators are not completely mindful about different elements of mutual funds.

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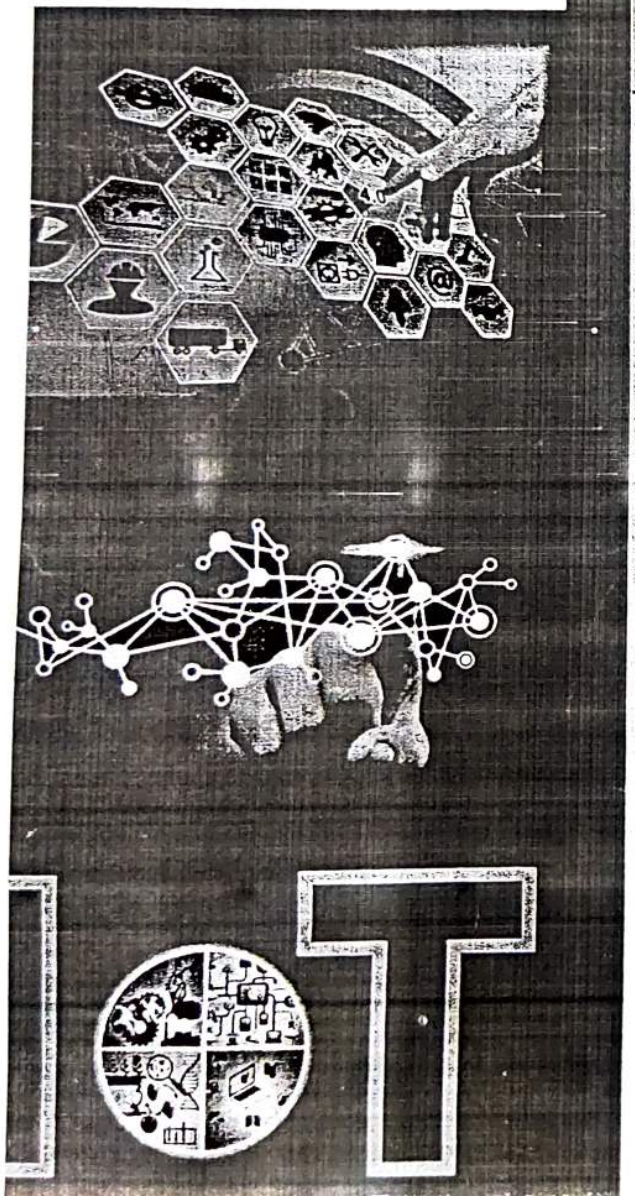
Effect of Suryanamaskar on minimum muscular fitness of sedentary life peoples of Aurangabad city

Dr. Abhijit Kadam, Shaunak M Pagi, Sagar P Dhikale, and Dr. Shashikant Pardeshi

Abstract
The Purpose of this study was to keep fit or physical fitness a need of the days in human society. In this age lack of muscular fitness leads towards the various disorders in human body. The muscular fitness of sedentary people's more over today's medical sciences has also given graters importance towards the development of muscular fitness. Various reports also revealed that practice of Suryanamaskar result for improvement of muscular fitness. Hence it was therefore considered appropriate by research scholars to investigate the effects of Suryanamaskar on minimum muscular fitness of sedentary life peoples. The aim of this study was find out the "effect of Suryanamaskar on minimum muscular fitness of sedentary life peoples of Aurangabad city". Whole population (N=100) of this study were tested and selected a total number of 60 male sedentary peoples from administrative employee in the age group of 30-50 years who had failed into three to six test items with the help of purposively method of sampling from Vivekanand college Samarth nagar Aurangabad. And the subjects were divided into two group by randomly method i.e, 30 experimental group and 30 control group. Sedentary peoples of the experimental group were given the 30 minutes Suryanamaskar, but the controlled group did not part in Suryanamaskar program. The duration of Suryanamaskar program was six weeks which was given six days in a week. Pre and post test conducted on the both the groups and data was analyzed by independent sample t-test. The results showed that the descriptive statistics gain of pre and posttests of subjects on Test 1 Abdominal and Psoas muscles, Test 2 Abdominal muscles without Psoas, Test 3 strength of Psoas and Lower Abdominal muscles, Test 4 strength of the Upper Back muscles, Test 5 strength of the Lower Back muscle and Test 6 strength of Back and Hamstring muscles as experimental group was (7.33 ± 5.2, 7.33 ± 5.8, 7.00 ± 5.3, 7.33 ± 4.4, 7.67 ± 5.0 and 8.33 ± 3.7) respectively and the descriptive statistics gain of pre and posttests control group was (1.33 ± 5.0, 1.67 ± 5.3, 1.33 ± 4.3, 1.33 ± 5.0, 0.33 ± 3.1 and 1.00 ± 3.0) respectively. The calculated 't' value of subjects in Test 1 Abdominal and Psoas muscles, Test 2 Abdominal muscles without Psoas, Test 3 strength of Psoas and Lower Abdominal muscles, Test 4 strength of the Upper Back muscles, Test 5 strength of the Lower Back muscle and Test 6 strength of Back and Hamstring muscles was 4.52, 3.93, 4.50, 4.84, 7.34 and 8.25 respectively. To determine the effect of Suryanamaskar on minimum muscular fitness of sedentary peoples of Aurangabad city, independent sample t-test was used at 0.05 levels of significance in relation to pre and post-test of Kraus – Weber's minimum muscular fitness tests. A significant level at 0.05 ($p = 0.001$) effect of Suryanamaskar was found to improve the minimum muscular fitness of sedentary peoples, and it was also concluded that the beneficial effect of Suryanamaskar can be applied to all sedentary peoples to improve the fitness, physical health and sports activities of the sedentary peoples.

Keywords: Suryanamaskar program, sedentary people and Kraus - Weber's minimum muscular fitness test

Introduction
The wealth of the nation resides on the health and vitality of its people. Every nation is becoming increasingly concerned about physical fitness of its men, women and children; recognized physical fitness is fundamental and useful living in any capacity. Physical fitness is the basic of all activities in our society. Physical fitness is defined as asset of attributes that people have or achieve that relates to the ability to perform physical activity (USDHHS, 1996). Physical fitness means different things to different people and may include muscular strength, muscular endurance, cardio respiratory endurance, body composition and flexibility. The association between the muscular fitness and quality of life is impressive.



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E-COMMERCE – A CATALYST TO THE DEVELOPMENT OF INDIAN ECONOMY

Prof. Blesson James, Prof. Manjari Parashar
Christ College Pune, India

ABSTRACT-This paper focuses into the likely impact of e-commerce industry on the development Indian economy. Post globalization and expansion of internet accessibility through various gadgets, Indian economy has shown a steady growth. This growth was mainly because of the contribution of Ecommerce. Business deals and trades through the internet have become more common and flexible which now caters to the need of the customer as per his/her needs. Ecommerce has become a catalyst to the development of Indian economy in all the aspects. The traditional approach of business and trade are now being replaced with online shopping and trade. New innovative ideas and conversion of such ideas into service providers has enabled the Indian economy to place its development much faster than expected. There is a considerable increase in the customers preferring the services of urban clap, infibeam, book my show, yatra, amazon, Flipkart, Paytm, etc.in recent times.

KEYWORDS: **E-commerce, Indian economy, Internet penetration, Technology, Development**

I INTRODUCTION

Over the past decades Indian economy has shown considerable growth and this has enables the economy to have a strong base in the field of technological advances. The Indian ecommerce industry is growing in a commendable way that by 2034 India is expected to outshine United States of America and become the second largest ecommerce industry after China. The Indian e-commerce industry is expected to reach US$ 64 billion by 2020 and US$ 200 billion by 2026 from US$ 38.5 billion as of 2017. This kind of expected increase in the ecommerce industry is due to the high internet penetration in India. A transformation is seen in the business dealings with the introduction of Ecommerce in India. As per the reports of telecommunication ministry the digital transformation in the country is expected to increase India's total internet user base to 829 million by 2021 from 445.96 million in 2018. Since 2014, the Government of India has announced various initiatives namely, Digital India, Make in India, Start-up India, Skill India and Innovation Fund. The timely and effective implementation of such programs will likely support the e-commerce growth in the country.

II OBJECTIVES

The objectives of the study are as follows:

1. To study the growth of Indian Ecommerce industry.
2. To identify the contribution of Ecommerce in the recent developments in Indian Economy.
3. To learn the future opportunities in Ecommerce industry.

III REVIEW OF LITERATURE

Kumar, A, 2015 in his research specifies that Indian economy at present may be at a lagging stage compared to other economies but in the near future has a great potential because of high level of internet penetration and population factor.

Mirmiran & Shams, 2014 in their study states that the developing nations has wider scope of possibilities of benefitting ecommerce than that of developed economies as the developing economies have more room for reducing inefficiency and increase in production.

Dr.S. Hariharputhiran,2012explains that the security and privacy concerns in ecommerce remains as a concern for customers.

Rathad & et.al, 2011in their finds explains that a consistent internet growth in a positive way can bring in boom in the ecommerce industry. They also specifies that convenience and comfort are the most important factors that needs to be worked upon as security and privacy concerns are looked into by software companies and banks by providing improves internet security.

Terminologies used:

- **E-Commerce:** 'E-commerce' means buying and selling of goods and services including digital products over digital & electronic network.
- **E- Commerce Entity:** A company incorporated under the Companies Act 1956 or 2013 or a foreign company engaging in the e-commerce business.

IV DISCUSSION

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E- GOVERNANCE IN INDIA – IMPACT ON CUSTOMERS

ISSN: 2394-2517

Prof. Manjari Parashar
Christ College Pune, India

Prof. Blesson James
Christ College Pune, India

ABSTRACT

E- Governance has been through plethora of innovations during the past decade. It enables the government to deliver information and services to the nation at large with the help of ICT (Information and communications technology) tools much faster and accurate. There are many dimensions to e-governance. This paper is a descriptive study which focuses into the e-governance (G2C) practices in India and the initiatives taken by the government for implementing electronic medium into information and services in the country. This study is conducted based on secondary data analysis only.

Keywords: E- Governance, G2C, ICT.

I INTRODUCTION

E-governance, extends to electronic governance, is the combination of Information and Communication Technology (ICT) in every one of the procedures, with the point of upgrading government capacity to address the requirements of the overall population. Government Business Process Re-building using IT to upgrade trades is the most essential for change across over government and thusly ought to be executed by all administrations/divisions. The essential reason for e-governance is to disentangle forms for all, i.e. government, subjects, organizations, and so on at National, State and local levels. India is very late in getting into the e-governance system. Be that as it may, one must not overlook that India is a tremendous nation and we are looking at serving a population of something beyond than 1 billion and different substances including various organizations, partner elements taking into account more than 35 dialects and 150 languages. The focal E-Governance is to make organizations productive, open and advantageous. The utilization of E-governance is to overcome the limits. That is of a conventional paper-based framework. It is the improvement of current government. What's more, it likewise gives better government services to the resident. Subsequently, E-governance conveys simple, moral, accessible and responsive government.

II OBJECTIVES OF THE STUDY

India is a developing nation and technology plays a vital role in the continuous development of the economy. Transformation of manual or printed mode of operation into digital information and data is being implemented. However, the expected penetration of the digitalized operation in public sector is not been satisfying. In this study we look forward to learn the present situation, challenges and way ahead of E-governance in India from the point of view of customers. Hence the objectives of the study are as follows:

1. To understand the concept of E- Governance and its Initiatives.
2. To learn the impact of E-Governance on customer.
3. To identify the limitation of E-Governance.

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Sustainable Development Effective Crisis Management Practices to tackle unexpected epidemics by 2030 with reference to a case study from Nipah outbreak.

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Sustainable Development: Effective Crisis Management Practices to tackle unexpected epidemics by 2030 with reference to a case study from Nipah outbreak

Ms. Geeta Rajan N (M Com Coordinator)
Asst Professor, Christ College, Pune.

Abstract

Vision India 2030: Agenda for Sustainable Development, with its commitment to leave no one behind, it enhances modern transportation, job opportunities, education and quality of life as well as international migration. It is a fact that all development invites economic growth but on the other side there may be some negative impact as well. One such negative impact what this paper is highlighting is unexpected epidemic diseases which can affect the people as well as economy of a nation. There is an imperative today to promote sustainable development. A vision of what this encompasses is presented in the new sustainable development agenda that aims to end poverty, promote prosperity and wellbeing of people while protecting the environment by 2030. As an arm of the United Nations for Development, UNDP has a key role to play in supporting countries to make this vision a reality: putting societies on a path of sustainable development, managing risks and improving resilience, and promoting prosperity and well-being.

The objective of this paper is to find out what strategy is implemented by a small state Kerala to tackle the most deadly virus Nipah. The study focused on one such Nipah Virus outbreak witnessed in May 2018, at a small district Kozhikode, Kerala. It is a deadly virus which takes a patient into coma within two days. The only treatment is giving proper supportive care. According to the statistic survey, the last outbreak what the Kozhikode witnessed could have taken the lives of more people. With the help of efficient management techniques, the Kerala state managed to control the death toll to seventeen. Therefore the main sole purpose of this study is to learn an efficient crisis management to tackle unexpected as well as unknown epidemics outbreaks. A comprehensive outbreak response is always complex, comprising many elements that should be harmoniously coordinated. Such Four elements are studied in this paper with reference to Nipah Virus outbreak.

Keywords: Nipah Virus, Crisis Management, Epidemics outbreaks, Sustainable Development, Infodemic.

Introduction

Our history gives ample evidence of how much deadly the threat of contagious diseases can cause us irreversible damage. During 15th century, Europeans used this as a weapon to conquer American continents. They intentionally brought the diseases and spread out among the natives. This disease travelled even faster than the Europeans and was the reason why many native civilizations wiped out. Five hundred years from that time, today world is more interconnected with each other. Therefore such outbreaks can carry potential to travel oceans, Mountains, continents with even greater speed.

In September 2015, the Sustainable Development Goals (SDGs) were officially adopted by UN member states. They replace the expiring Millennium Development Goals (MDGs). 17 SDGs are universal and focus on five key elements that must be achieved by 2030. These include People, Planet, Peace, Prosperity, and Partnership. SDGs view development as economically, socially and environmentally sustainable. Actions need to be taken in various problems at the national and international level for the actualization of the SDGs. The World Health Organization (WHO) has developed a joint global public health approach to accelerate progress and meet ambitious global targets set for **2030** for HIV, TB, malaria, hepatitis and NTD in the era of Sustainable Development Goals (SDGs). Concentrated global attention is needed in unexpected epidemic

diseases, such as the Zika Virus, Nipah Virus, Swine Flu, Ebola Virus etc which requires a response that is enlarged for a long time to end the disease.

Nipah virus was first identified in 1999 among the pig farmers of Malaysia and Singapore causing respiratory illness. This outbreak caused a mild illness among the pig, but nearly 300 people were affected out of which 100 deaths were reported. After this outbreak, no Nipah cases are reported in either Singapore or Malaysia.

In 2001, Bangladesh and Siliguri, in India experienced Nipah virus, one with a strain different in the original outbreak happened in 1999. Unlike the Malaysian outbreak, India and Bangladesh experienced Nipah Virus outbreak several times.

Recently in 2018, there was a third outbreak in Kerala, which was localized in two districts Kozhikode and Malappuram. This had claimed the lives of 17 people. But in similar case on Swine flu outbreak, a deadly virus had taken the lives of more than 1800 people up to march 2015. Therefore what helped the Kerala state to control the virus is an interesting subject to study.

Objectives

- To study a best approach of infection control practices for any epidemic disease.
- To study the importance of crisis management in controlling epidemic diseases.
- To identify the efficient infection control practice and supportive care of Kerala Government towards Nipah Virus.

Review of Literature

Stella R.Quawh and Lee Hin Peng et al. (2014) the research was conducted on Crisis Prevention and Management on SARC Outbreak, Singapore. SARS is Severe Acute Respiratory Syndrome, infection which may come from normal contact with anyone. Four public health issues were considered such as prevention measures, self health evaluation, SARS knowledge and appraisal of crisis management. The research was conducted on 1201 adults whose age is less than or equal to 21. They found that sex, age and attitude were associated with practicing preventive measures. This study improved the understanding of the social dimensions of infectious disease outbreaks.

Benjemani champman,phd., Audrey Kreseke, Phd., and Doug powel Phd. 2012. Article on "Crisis Management: How to handle outbreak Events" claims that the crisis management cannot be readily learned in the classroom. The skills can be horned through the observation of successful and failure stories. And even the simulation helps to sharpen the skills.

A report by the National Restaurant Association about crisis management Primer for food borne illness: emergencies are unpredictable. For food industry the profit cutting challenges comes when the workers are absent, equipments were broken, and if the deliveries are not on time. But the actual crisis occurs when the wellbeing of customers and employees endangered. They claim that the crisis management plan cannot work miracles; it had to be preplanned and well prepared. The plan should be according to the needs and limitations of company's particular operation.

Alexander S. Kekulé, 2015, a research paper on learning from Ebola Virus: How to prevent future epidemics claimed that WHO is incapable to control outbreak of infectious diseases in the world less developed area. This paper analyzed the reasons for the failure of the international response and important measures to improve resilience, not only early detection but also proper prevention.

Research Gap

Form the above mentioned review of literature, a gap identified is only failures, limitations were studied. This paper is explaining a proper crisis management technique which helped a state ministry to arrest a deadly virus completely. This strategy will be helpful for tackling any outbreak in future. The scope of the

study is vast because India like highly and densely populated country should always be alert about such diseases.

Research Methodology

The paper is based on secondary data from various sources like Newspaper, Website, and Magazines etc. And some primary data is collected by making telephonic interviews with the concerned people in Kerala.

Reason for the Catastrophe

On 2nd may 2018, the first victim of Nipah virus, Mohammed Sadiq was taken to the Kozhikode Government Medical College for further treatment, where he later succumbed to death. By the time Mohammed Sadiq died, his brother Mohammed Salih had already been infected. He was admitted in Baby Memorial Hospital, Kozhikode with suspected viral infection same as that of his brother. The samples were tested at Manipal Institute of Virology and National Institute of virology in Pune. At both the places the samples tested positive. Sadiq had passed the virus to more and more people, increasing the total count of infected to 18. In the first week, 10 deaths were reported, which includes a nurse who had taken care Sadiq before the diagnosis without any preventive measures. The outbreak first started in Kozhikode which moved later to Malapuram District. Near about 2000 people were quarantined and were kept under observation in both the districts. The reason of this outbreak in Kerala is still unclear since from Siliguri as well as even from Bangladesh in west Bengal, where the last outbreak was witnessed is too far from Kozhikode, for the virus to land. No similar cases have ever been reported in Kerala Previously. The experts were faced with such an unprecedented condition for to find out the causes of deadly attack Nipah. The Nipah Virus was not found in the samples of bats from the well at the house of the first victim, Mohammed Sadiq in Perambra.

To control such outbreak in future, it is really very important to find out the source of this deadly virus. But the crucial question "How did the deadly virus reach Kerala?" is still unanswered. The experts had come across with a various assumptions such as a famous ornithologist Dr. K. Sugathan Claims that the virus lives in the solution which are very high in sugar content, like the saliva of fruit bat, therefore it cannot be found in insectivorous bats.

An Outbreak Anatomy: Nipah Virus

The way Kerala deals with the Nipah virus outbreak holds important lessons for all of India on how deadly viruses are being handled by an alert administration.

Prompt Diagnosis

Four deaths from one family made the doctors impatient to identify the cause. Without wasting a single time the doctors at Kozhikode medical college had vigorously tried to rule out the common causes of encephalitis.Neurologists knew at the time that they were looking at exotic viruses, which had never been seen in Kerala before. They took the help of Virologist at Karnataka's Manipal Centre for Virus Research (MCVR), about 300 km from Kozhikode. The victim's samples were dispatched to MCVR.They ruled out common causes of encephalitis such as Japanese encephalitis virus, Herpes Simplex virus and Leptospira bacteria. Only one pathogen seems to be able to cause same symptoms and cause disease among several family members simultaneously. "That's Nipah," they declared.

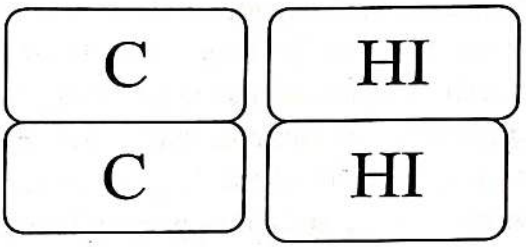
Meanwhile within the span of 21 days, Nipah claimed the lives of 17 of the 19 people infected in Kozhikode, the death rate was 89%. But without an immediate diagnosis, it can get worse. In Bangladesh, some of the first outbreaks killed dozens in the Meherpur and Naogaon districts, but were not recognized as Nipah until after they ended. In the 2001 Siliguri outbreak, investigators discovered that it was a Nipah virus that infecting people only six months later. At that time, 60 people were infected and 45 people had died.

India has a poor record of outbreak investigations. About 10,000 people develop symptoms similar to encephalitis each year, but never receive a diagnosis. Some regions, such as Gorakhpur in Uttar Pradesh and Muzaffarpur in Bihar, saw thousands of deaths in repeated annual outbreaks before causes were established.

In this context, the discovery of an exotic pathogen in the second patient affected by an outbreak, as was the case in Kozhikode, has few precedents.

A comprehensive outbreak response is always complex, comprising many elements that should be harmoniously coordinated. These elements are organized into four main blocks.

Coordinating responders



C : Coordinating responders
HI : Health Information

C : Communicating risk
HI : Health Interventions

Outbreaks by definition are extraordinary events that often require humans and humans financial resources and may also depend on additional partners, institutions and other sectors. Strong coordination is very important at all times to ensure that all resources and partners work effective together to control the outbreak. WHO is often expected to lead internationally Respond to supporting national health authorities.

Once the MCVR shows exactly the Nipah virus, they must move quickly. Under the 2005 International Health Regulations, India was obliged to report an emerging infectious disease outbreak to the World Health Organization.Therefore, the MCVR team must be sure of their findings. The only way to do this is to ask the NIV, Pune, to run a diagnostic test for Nipah in the second set of samples of the Perambra family. Instead of waiting for NIV's Confirmation, they started isolating the people immediately.The State health-care machinery did not wait for confirmation to start the operation. Kozhikode's district medical officer put together a team of entomologists and visited Salih's Perambra home to collect mosquito samples and fogged the area, just in case the mosquitoes were the disease vectors.

The state medical association and government doctors were already on high alert. An Ebola outbreak protocol instructed State doctors in infection control measures - isolating patients, using surgical masks and decontamination surfaces. That is a very fast response to any size.

The prompt coordination among the Ministry, Administration and Medical Department had succeeded in arresting the further growth of the deadly Nipah Virus.

Health Information (HI)

In any case, the information is necessary to monitor it, measure the impact of the interventions and Guide decision making throughout the crisis.There are two particular types of information: disease surveillance and information on interventions (process and outcome indicators), which shows the coverage and the impact of the interventions carried out.Surveillance provides information on the number of cases and deaths per period and place (people, time and place). The information about the interventions allows knowing about which ones are performed and what is their coverage and impact.

Kerala government had closely monitored on the outbreak regarding people, time and place to control the death toll as well as spreading of the pathogen. They had a clear picture about how the pathogen moved from the zero patient to the seventeenth one. The chart 1 explains that from patient Zero the disease had transmitted to three people from the same family within three days. Thereafter these three had taken

medication from Perambra Taluk hospital where in the disease had transmitted to more 4 people. The affected people had taken treatment from Kozhikode medical college had again spread the disease to further more 11 people taking the death toll to 17.

Meanwhile the state authorities had started with the preparation for second wave of infection.On the one hand, infection control measures were established by day 20, and more than 1,400 people who came into contact with the 19 confirmed cases are being monitored closely. On the other hand, unlike other viruses such as measles in the Paramyxovirus family to which Nipah belongs, Nipah does not spread efficiently and only transports people within a meter of very sick patients.

The number of confirmed new cases coming up each day has slowed after the first wave from patient zero. But the Nipah virus can incubate in the body about 21 days, health officials cannot take it easy for a while. Only after 42 days, or two incubation periods, pass after the last confirmed case will the State be declared Nipah-free.

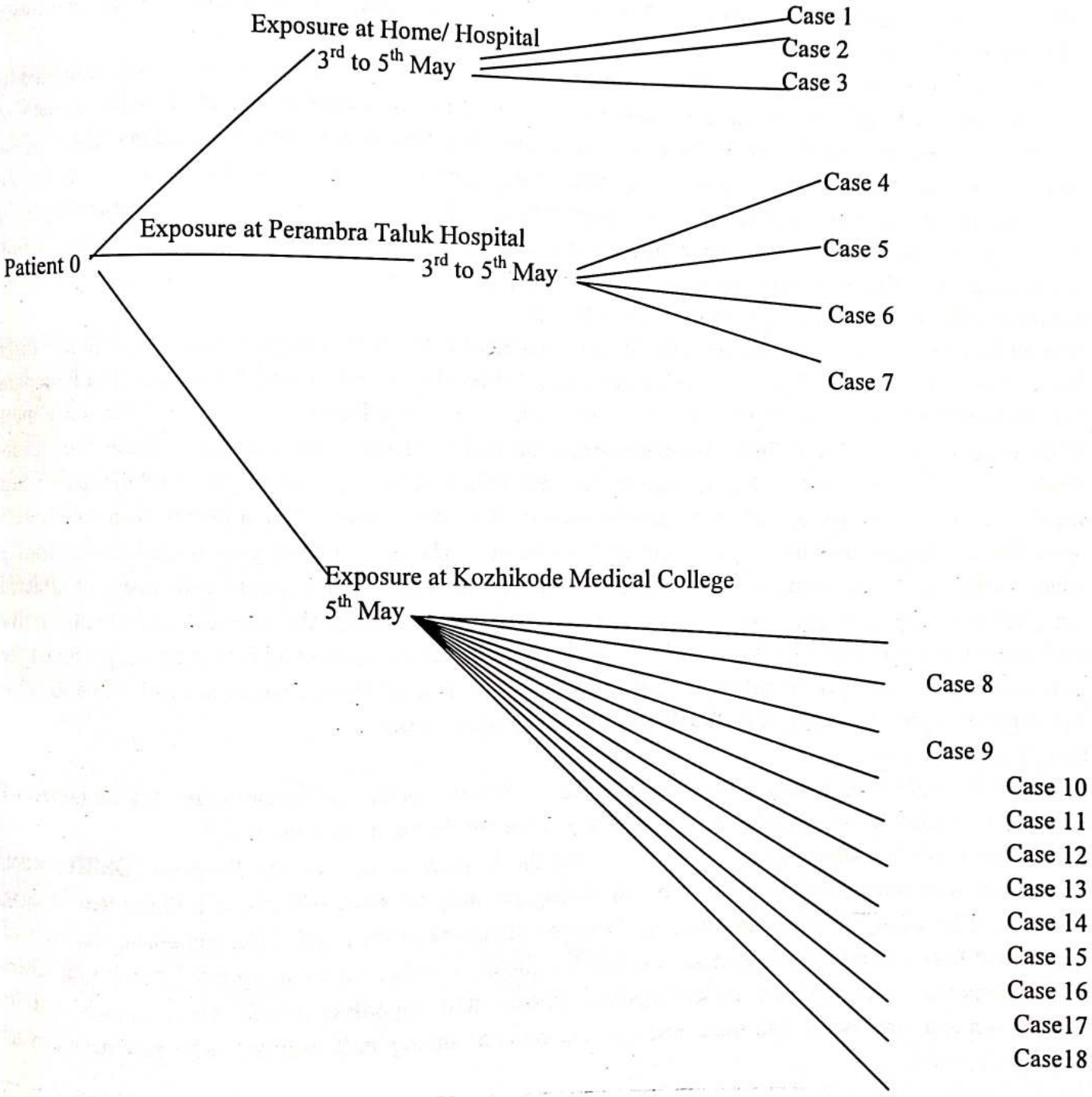


Chart 1

Communicating risk (C)

During the evolution of any major outbreak, cases and deaths will inevitably increase. An epidemy is the rapid spread of infectious diseases to a large number of people in a given population within a short period of time. Similarly, there may be another type of epidemic: the rapid spread of Information of all kinds, including rumors, gossip and untrustworthy information. We describe the phenomenon as "infodémic".

The infodemic, like the epidemics, can be managed. Field epidemiology is an important part of Response to the outbreak. It covers three main areas: (1) health monitoring and identification threats, (2) investigation of outbreaks, and (3) actions for mitigation and control. Similarly, successful of the management of the infodemic will be based on (1) its follow-up and identification, (2) the analysis of them, and (3) control and mitigation measures.

Risk communication is an essential intervention in any response to disease outbreaks, and is equally necessary to manage the infodemic. The communication of risks in epidemics implies two ways Communication that is dynamic and evolves as the outbreak develops.

Immediately after the confirmation of the diagnosis and the first death, on May 19, the State intensified its actions of public health control. He immediately informed the Center, which in turn alerted WHO. A team of experts from the National Center for Disease Control (NCDC), New Delhi, complemented the efforts of the State to quickly move behind the root cause, undertaking public outreach activities in parallel. Despite the fact that the government installed a 24-hour control room and published authentic public communiqués, a section of the media went crazy reporting non-scientific and exaggerated news. Social networks, mainly WhatsApp, were flooded with rumors that created social panic. The rumor that Nipah was caused by fruit severely affected the entire supply chain beyond Kerala.

The rumors in the village of Changaroth, 45 km from Kozhikode, took people to the edge, driving them away from their homes at night and leaving those affected for fear of contracting the disease. The Perambra Taluka Hospital, which normally has up to 100 patients, wore all day long with a deserted look, since most of the patients who were not Nipah sought discharge due to fear.Effective risk communication at three levels finally paid off. Fears among villagers who by then moved unnecessarily with masks could dissipate when their Panchayat leaders along with teams of doctors and Delhi staff contacted them in their own homes with objective information and their assessment of the situation. The district-level government administration educated the general public on what should and should not be done, while addressing the needs of affected families, including safe cremations for the deceased. At the state level, the political and administrative leadership was updated through frequent informative meetings for the media and reassured people about the general situation with specific messages. The health minister herself urged citizens not to believe in What App rumors as part of a strong and effective narrative to mitigate fears.

Health Interventions

An effective health intervention is required to arrest any disease such as (a) transmission, (b) the impact on Health systems, (c) severe mortality and morbidity, (d) the political and other sectors.

The district's medical officer, Kozhikode, entrusted the District Mental Health Program (DMHP) team, Kozhikode, with addressing the panic and fear among the hospital staff, which had initially treated these patients without taking sufficient precautions. Who were unaware of the Nipah infection during that period? The DMHP team visited these hospitals to assess the situation and plan an appropriate intervention. There was widespread confusion and misconceptions among staff regarding Nipah virus disease and its transmission and infectivity. And there was genuine concern among staff members who had close contact with these patients.

The intervention strategy included a general plan for health education with respect to the disease and the nature of disease transmission and infectivity, together with specific interventions to treat associated anxiety

and psychological stress. A general health education class was provided to staff as a group. The individual problems were evaluated. Most of them had symptoms of anxiety. A fraction of them had insomnia. Benzodiazepines were prescribed for some of them. Relaxation techniques were taught. Follow-up psychological support was provided and provided over the phone.

As the news about infections spread and the Department of Health decided to keep the contacts of the infected patients isolated for observation, the general public became alarmed and generalized panic. In this situation, the psychological intervention required an extension and a reorientation. The Director of Health Services convened a meeting at which representatives of the Department of Psychiatry of the Medical College of the Government, Kozhikode; Institute of Mental Health and Neurosciences (IMHANS), Kozhikode; and the Government Mental Health Center, Kozhikode participated. It was decided to launch a mental helpline from Nipah to deal with the new situation. This was considered the best option since people did not trust to go to hospitals for fear of contracting the disease.

DMHP Kozhikode coordinated the program. The helpline provided the public with three phone numbers they can call to get psychological help related to Nipah's viral illness, between 9 am and 5 pm. The numbers were publicized. The psychiatrists handled the telephone calls in these three institutes.In addition to the telephone helpline, an outpatient advice and assistance (OP) clinic was launched in the Psychiatry Department of the Medical College Hospital to address psychological problems among health professionals and staff.

Two survivors were also visited at the end of the quarantine period and psychological support was provided.

Conclusion

Mankind has faced many brutal epidemics which has not just claimed the lives of thousands but also gave us unforgettable pain to suffer. Epidemic disease is still prevalent in this modern era even though we have most advanced health care. Vision 2030 should aims to counter such outbreak in future. Countering such outbreaks is one of the goals of United Nations Sustainable Development Goals. Sustainable Development Goals 3 targets by 2030 to terminate the epidemics of AIDS, tuberculosis, malaria and neglected tropical diseases and combat hepatitis, water-borne diseases and other communicable diseases. It also supports the research and development to arrest such outbreaks. This paper presents a case study on Nipah Outbreak and its Crisis Management which will act as a model to be followed in handling future outbreaks. The four elements explained in this paper clearly shows how efficiently the Kerala Government managed to arrest such a deadly virus attack. They themselves set an example of role model for everyone in the history.

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BRAIN COMPUTER INTERFACE (BCI) DEVICE AND ITS ROLE IN STRESS MANAGEMENT OF LOCO PILOTS, INDIAN RAILWAYS.

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ABSTRACT

Employee wellness and safety is an important part of a successful workplace. An Employer has a key role to play in preventing work-related injuries and diseases. An organization must recognize the importance of healthy, happier and safe workplaces for their employees. The objective of this paper is to prove Brain Computer Interface devices (BCI) would be of a great service to Indian Railways in handling Stress Management of Loco Pilots.

BCI is an external device and has a direct communication between the brain, bypassing the traditional pathway of peripheral nerves and muscles.

A mobile BCI has the advantage of ultimate portability as well as a low system cost derived from using customized Electroencephalogram (EEG) recording and signal processing modules. To implement a mobile BCI with online processing, a mobile terminal such as a mobile phone or a PDA might be an ideal platform for data transmission, signal processing, and feedback presentation.

This research aims to study the feasibility of BCI as a Tool for Stress Management of Loco Pilots.

Keywords: Brain Computer Interface (BCI) Loco Pilots, Health & Safety, Stress Management, Electroencephalogram.

INTRODUCTION

Indian Railways had seen a lot of highs and lows and Loco Pilots have made a huge contribution in it. The Loco Pilots duty requires it to be executed with perfection without a slight margin of error. Any negligence or even a small mistake can lead to fatal accident putting in risk not only the Loco Pilots but also thousands of passengers travelling in the train. Even though Railway has largest workforce of our country, Loco Pilots problems are still unattended and neglected.

BCI is a relatively new field of science with a seemingly limitless range of applications. It is a communication system that translates the activity of brain into commands to an external device such as mobile or computer. In other words, the user can react to the situation only by brain activity and without nerves and muscles. This device is very helpful for the disabled people to interact with others.

The BCI device can monitor workers' vital signs like heart rate, skin temperature, stress level, attention and fatigue, and provide real time analysis if the worker is in danger. The BCI can lead to more efficient and accurate work, as well as safer and healthier workforces. The gadget integrates a wearable and wireless EEG system with a mobile

phone to implement a visual-evoked potential (VEP) based BCI, which can be used to directly make phone calls based on users' EEG or raise an alarm under high pulse rate, high stress level which can lead to strokes or accidents at workplace. Therefore, this can be effectively used to improve the safety measures of Loco Pilots in Trains.

REVIEW OF LITERATURE

T.V.Saranya and M.Shalini (2017) BCI for security application based on inter subject information for locker safety by implementing a protective opening of locker with the help of brain signal. This paper claimed that the brain waves are unique for each individual and does not match in any case. The locker can be opened only when the programmed person think about it. Therefore it is highly secured.

B. Rebsamen, E. Burdet, C. Guan, H. Zhang, C. L. Teo, Q. Zeng, C. Laugier, and M. H. Ang Jr. (2016), discussed in their article about a wheel chair system which activate with the help of BCI. This could be useful in various hospitals with comparatively low cost. They proved the brain programmed Wheel chair is very effective to navigate inside a typical office or hospital environment.

Huli, P. R. (2014) He had studied on Stress Management in Adolescents. The objective of his study was stress can be caused due to many factors and can alter the relationships dynamics in the family. He had observed that stress during adolescence were because of bothered family dynamics, inability to cope with studies, peer anxiety, drug abuse, lack of skill. One of the important trends which were being observed was getting minute satisfaction from the electronic media. He was concluded that Proper care should to be taken in helping to take the right decisions which may affect their future.

Devi, U. T. (2011), researched on "A Study on Stress Management and Coping Strategies with Reference to IT Companies". This research aims at identifying the level of stress among the IT employees to suggest the strategies for coping with the stress. It was identified that employees experienced pressure from heavy work load followed by competition, fear of job loss and others. The study identified various strategies like Stress management programs, Physical activities, Stress-audit, Life style modification programs etc. to cope with the stress.

RESEARCH GAP

From the above review of literature, the identified gap is that stress management can be effectively done with the help of modern technologies. With the advancement in technologies, many gadgets and devices have been developed and studied by innovators and researchers and are being used in western countries for workers, disabled, locked-in patients etc. This paper is giving an innovative idea to Railway Management to control the stress of Loco Pilots by using BCI.

OBJECTIVES OF THE STUDY

1. To understand the level of stress among the Loco Pilots.
2. To understand the technology underlying BCI.
3. To study how the BCI can help in stress management of Loco Pilots, Indian Railways.

HYPOTHESIS OF THE STUDY

H0: There is no significant relation between quality and problems of work life with the stress level of Loco Pilots.

H1: There is a significant relation between quality and problems of work life with the stress level of Loco Pilots.

RESEARCH METHODOLOGY

The study used both primary and secondary sources. Primary data was collected using a questionnaire schedule among 42 Loco Pilots. The sample selection was on Stratified Convenience Sampling. The questionnaire and structured interview was developed to assess their stress level. Secondary sources included books, journals, articles and reports from newspapers, weekly's, magazines, etc. The data collected from both the primary and secondary sources were quantified and analysed in qualitative terms. For testing of hypothesis **Chi Square test is used.**

SAMPLING METHOD

Stratified convenience sampling method adopted for data collection from all the categories of Loco Pilot Mail, Loco Pilot Passenger, Loco Pilot Goods and Asst. Loco Pilots are as follows:

Sr. No	Category	Population	Sample
1	Mail LP	39	16
2	Passenger LP	8	3
3	Goods LP	36	6
4	Asst. LP	76	17
Total		159	42

The total size of the sample is drawn from the four categories i.e., 16+3+6+17 = 42. **Therefore 42 is the sample size for the study.**

RESULTS AND DISCUSSION

- Loco Pilot's ergonomic is the Loco Cabin in which they work. The control stand comes first and 79% of Loco Pilots claim that the design of control stand is not uniform. There are about 10 types of Diesel locomotives and almost 20 types of Electric Loco motives. The control stand differs from loco to loco and it leads confusion in day to day working.
- The Loco Pilots frequently use the operating handles to accelerate the speed and to apply the brake for controlling the speed. Almost 92% of Loco Pilots are stating that operating handles are not easily accessible to them.
- There are so many gauges and displays on locomotives such as speedometer, various air pressures gauges and oil pressure gauges. The Loco Pilots should continuously concentrate on the gauges on run. Almost 80% of Loco Pilots claims that it is difficult to read the gauges and displays during night on run since it were not illuminated properly.

- The Loco Pilots are working with both the electric and diesel locos. There is no problem for front visibility in electric locos since both side Loco Pilots cab is provided. In diesel locos only one cab is provided. If the cab is leading is known as short hood driving and if the cab is trailing it is known as long hood driving. In case of long hood driving front visibility is not clear especially on curves.
- The front visibility of locos plays a vital role in driving because Loco Pilots should constantly watch on track and signals. In case of signal passing at danger, severe punishments are imposed on Loco Pilots even though there is no lapses, accidents etc. It is stress to drive with improper visibility. Nearly 88.09% claim sometimes no proper front visibility.
- The lookout glasses should be clear for driving and for that proper maintenance should take place. During rain and fog, working of wipers should be important for driving. Nearly 93% of Loco Pilots claim that both the lookout glasses and wipers are not working properly.
- During night driving head light focus should be clear. As per railway rule, the focus should be minimum 250 Meters. Above 88% feels sometimes head light focus is not sufficient.
- The speedometer is having a vital role in driving. The speed of the different section, different locos, different coaches are varies. The Loco Pilots are supposed to run at maximum speed whatever the speed given in their manual to minimize the late running of trains. In any cases they are not supposed to cross the maximum speed. In case of over speeding, maximum penalty will be imposed to them. Almost 80% respondents are not happy on the existing system and they prefer Digital Speedometer.
- The Loco Pilot's seat quality is poor. It is enough to have Loco Pilots shuffling uncomfortably in their seats. All the basic necessary of the seat like height adjustment, side adjustment, forward and backward movement, back rest, hand rest, proper cushion and foot rest etc are not provided properly in all the times. In electric locomotive cabs, seat is very small and it is wall mounted without any adjustment. It is not comfortable for a Loco Pilot to drive while seated.
- The working space of loco motive is congested. 95.23 % of Loco Pilots claim that there is no enough space. There is no air conditioning and 95.32 % agree that there is no effective cooling and heating system. The Loco Pilots cab is not pollution free which can affect the health of lungs. Almost 96.45 % agree that cab is not pollution free. The temperature in Loco Pilots cab is more than atmospheric temperature during summer season.
- 100 % of Respondents claim that their cab is not sound proof. Everyone is finding difficulty with the position of the horns of the locos. According to them the horns are not located far from the Loco Pilots cab. The sound decibel of the horns normally exceeds the normal limits.
- A Loco Pilot in his each duty on an average crosses 100-200 level crossing gates. As per the railway rule he has to sound the horn from 600 meters before the level crossing gates continuously.

- In addition to that a Loco Pilot should sound the horn on curves, cuttings, approaching tunnels, running through a station, where the visibility is restricted and for warning the trespassing public. This facts show they are very much exposed to sound pollution on entire duty hours.
- Most of the Loco Pilots attended the run over cases on their duty. In the run over case if the victim is survived they have to carry the victim to the nearby railway station towards their journey where the medical facility is available. For example the medical facility is available in rear station within few kilometers, but there is no provision to back the train for immediate medical attention. In case of, the victim is killed, the Loco Pilots have to collect the body parts and clear the track for free movement of other trains. They have to hand over the body to the available railway servant nearby. In case of cattle run over they have to stop the train and remove the carcass and clear the track for the free movement of other trains. They are mentally disturbed after attending such run over cases and cattle run over. But immediately they have to continue their journey without any time gap till their destination. Even after reaching destination there is no counseling given to cope up the mental disturbance.
- This clearly shows that Loco Pilots working environment is not conducive for proper working with good health. These analyses fulfil the objectives such as to study the various causes of stress and its level.

HYPOTHESIS TESTING

NO	PARAMETERS	ALWAYS	SOMETIMES	NEVER
1	Continuous concentration during the entire duty	39	03	0
2	Palpitation during untoward incidents like run over cases	34	06	02
3	Feeling irritation while long honking	30	10	02
4	Possibility of relaxing in duty hour	0	05	37
5	Feeling sleepy after a continuous Shift	32	08	02
6	Back Pain while working	20	15	07
7	Numbness and tingling after continuous working in one position for more than approximately six hours.	0	39	03
8	Headache during day time trains on summer season	0	38	04
9	Holding back the Natural Calls for long hours on entire duty	39	03	0

Based on this table the calculated value of Chi Square Test is 397.66 and the table value of chi square at 5% level of significance for 26 degree of freedom is 21.792. So the calculated value of chi square is much more than table value.

Therefore H0 is rejected and H1 is accepted. Thus it is proved that-

"There is a significant relation between quality and problems of work life with stress level of the Loco Pilots".

BRAIN COMPUTER INTERFACE (BCI)'s ROLE IN STRESS MANAGEMENT OF LOCO PILOTS

BCI can be used to analyse the stress level of Loco Pilots which in turn helps the Railways to understand and provide quality service for better involvement of Loco Pilots. These data and analysis can be used to personalize Stress relief programmes for Loco Pilots in the form of special Yoga or Meditation exercises, music which can give dedicated, motivated and passionate workforce for Railways. Stress Management maybe an insignificant activity but if it is implemented correctly, it is beneficial for both the Loco Pilots and the Railway by reducing manual error.

BCI with all its advancement carry the potential to provide a link between human brain and device systems. BCI can be used for providing a connection between Loco Pilot's brain and his cabin which can lead Loco Pilots using his Brain Signals to operate in his cabin. This can lead to reduction in reflection time a very important measure which really omits the accidents in time. Every seconds of the delayed action will cause a severe impact because the train will cover the distance of 1km within 33 seconds.

BCI can also reduce the confusion and the Stress which Loco Pilot goes through while operating different types of cabins. For instance If a Loco Pilot sees an obstacle on his path it may take him a few seconds to realise the danger and to take the necessary actions. These lapses of seconds can lead to a fatal accident. But suppose a Loco Pilot is operating with BCI, the moment the BCI realises the Stress Level of Loco Pilots and receives signal from brain it takes the control of train in which it applies brake and horn some seconds before Loco Pilot will actually responds to apply it. BCI is a part of machine learning it will understand the behaviour of Loco Pilots with time, which in turn may leads to even faster results leading to efficient services. By reducing the reflection time of Loco Pilots, we can avoid a major fatal accident or minimize the impact of accident by reducing the train speed.

If the BCI able to monitor the alertness of the Loco Pilot, it can give audio and visual warnings when the Loco Pilots loses his concentration/ alertness/falling asleep. This will help them to avoid Signal Passing at Danger (SPAD). Since the Loco Pilots are facing more than hundreds of signal on his duty shifts, if he passes any one of the signal at danger for one foot also, draws major punishment i.e., removed from service even though there is no loss of Railway property, causalities and no time loss. Avoiding major punishment for minor human lapses may definitely trigger the stress level on duty. BCI will be a major relief for Loco Pilots on this case.

CONCLUSION:

The problem of stress is unavoidable and inevitable in any occupation. The present study concludes various important acumens related to stress of Loco Pilots that is very essential for the Indian Railways to identify since driving job comes under a safety cadre. BCI is a futuristic technology that Railways is yet to explore in various areas. This pilot study proved that Stress of Loco Pilots is significant that they required a proper stress management strategy. The primary objective of this research is to highlight the advantages of BCI and also suggests ways to deploy such methods to analyse the stress of Loco Pilots in Indian Railways.

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BEHAVIORAL FINANCE: IT'S ROLE OF PSYCHOLOGICAL FACTORS IN FINANCIAL DECISIONS

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Head of Department, Department of Commerce, Christ College Pune

Abstract

Behavioral finance is the study of the influence of psychology on the behavior of financial practitioners and investors and its subsequent effect on markets. behavioral finance is of interest because it helps in explaining why and how markets might be inefficient. Behavioral finance is part of finance that seeks to understand and explain the systematic financial market implications of psychological decision processes. It covers the areas of cognitive psychology, social sciences and anthropology to explain irrational investor behaviour that is not being captured by the traditional rational based models. This paper analyses the development of behavioral finance, reviews stock market and property market behavioral literature and identifies issues in the property market that can be better understood and explained using behavioral models. This paper is a focus on the behavior and attitude of the investors towards market. The risk taking ability of the investors is assessed based on the probable difference in gender. The study examines the buying behaviour of investors in the financial market in terms of its implications and extends of impact Understanding Behavioral Finance can provide insights into how an investor invest, but isn't necessarily a cure for all. Understanding the proper techniques doesn't automatically mean that the investment options that unfold will be better: but the reasons put forward, is in a belief that it's likely that it will be better. Success of long-term investment is not just about developing investment skills, it is also about discipline. We might have to defer short term instantaneous gains for long term sustainable gains.

Keywords: Behavioural Finance, Stock Market, Investment

INTRODUCTION: Behavioral finance is the study of the influence of psychology on the behaviour of financial practitioners and investors its subsequent effect on markets. Behavioral finance is part of finance that seeks to understand and explain the systematic financial market implications of psychological decisions and processes. Investing is a difficult and complex decision making task. It is not only an art but also as a science. Like any other commodity, investment in financial products, available the financial market, are purely psychological in terms of the decision involved. The investors makes mistakes, gets confused and faces numerous other problems when they are dealing with financial products, more specifically securities of various companies. A normal consumer, even to purchase a regular commodity, compares product features and prices it with many others. But in the financial market, the comparison is quite difficult and complex as the options are wide and the market is integrated with high end technology, the consumer must educate himself for the same and do substantial research with the huge data and supportive analysis frame work available which is not in the case of regular commodities. Buying procedure and buying decisions for both markets are more or less similar but for the financial market it seems to be complex due to:

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BEHAVIORAL FINANCE: IT'S ROLE OF PSYCHOLOGICAL FACTORS IN FINANCIAL DECISIONS

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Abstract

Behavioral finance is the study of the influence of psychology on the behavior of financial practitioners and investors and its subsequent effect on markets. behavioral finance is of interest because it helps in explaining why and how markets might be inefficient. Behavioral finance is part of finance that seeks to understand and explain the systematic financial market implications of psychological decision processes. It covers the areas of cognitive psychology, social sciences and anthropology to explain irrational investor behaviour that is not being captured by the traditional rational based models. This paper analyses the development of behavioral finance, reviews stock market and property market behavioral literature and identifies issues in the property market that can be better understood and explained using behavioral models. This paper is a focus on the behavior and attitude of the investors towards market. The risk taking ability of the investors is assessed based on the probable difference in gender. The study examines the buying behaviour of investors in the financial market in terms of its implications and extends of impact Understanding Behavioral Finance can provide insights into how an investor invest, but isn't necessarily a cure for all. Understanding the proper techniques doesn't automatically mean that the investment options that unfold will be better: but the reasons put forward, is in a belief that it's likely that it will be better. Success of long-term investment is not just about developing investment skills, it is also about discipline. We might have to defer short term instantaneous gains for long term sustainable gains.

Keywords: Behavioural Finance, Stock Market, Investment

INTRODUCTION: Behavioral finance is the study of the influence of psychology on the behaviour of financial practitioners and investors its subsequent effect on markets. Behavioral finance is part of finance that seeks to understand and explain the systematic financial market implications of psychological decisions and processes. Investing is a difficult and complex decision making task. It is not only an art but also as a science. Like any other commodity, investment in financial products, available the financial market, are purely psychological in terms of the decision involved. The investors makes mistakes, gets confused and faces numerous other problems when they are dealing with financial products, more specifically securities of various companies. A normal consumer, even to purchase a regular commodity, compares product features and prices it with many others. But in the financial market, the comparison is quite difficult and complex as the options are wide and the market is integrated with high end technology, the consumer must educate himself for the same and do substantial research with the huge data and supportive analysis frame work available which is not in the case of regular commodities. Buying procedure and buying decisions for both markets are more or less similar but for the financial market it seems to be complex due to:

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Effect of Monetary Policy on Economic Growth of India

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Abstract

The Reserve Bank of India, to achieve certain objectives frames the Monetary Policies. It mainly deals wit the money supply, which includes money in circulation and credit creation by banks. The main objective o Monetary Policy is to achieve price stability and economic growth. The money supply should be controlle in such a manner that it enhances growth but does not lead to high inflation. Thus Monetary Policy plays very important role in stabilizing the economy and achieving appropriate balance between the demand fo money and the output of the economy. I have found that Monetary Policy is indeed the backbone of a economy and these policies are essential for sustainable economic growth.

Keywords: Monetary Policy, Money Supply, Inflation, Price Stability, Economic Growth

Introduction

Understanding the effects of Monetary Policy is important in studying macroeconomics and the policy framework in India. Monetary Policy in any developing country aims at economic growth and stable price level in the economy. To achieve this objective first and foremost the money supply should be in line with the demand for money. Inflation is an important indicator to analyze the Indian economy, other key indicators being Gross Domestic Product (GDP), Money Supply, Credit Availability, Interest Rates, etc. There is a major conflict between stabilizing prices and enhancing economic growth. Aiming for a stable price level can reduce economic growth as there is an inverse relationship between inflation and unemployment rate. That is higher the inflation rate, lower will be the unemployment rate, which is also known as the Phillips Curve. Therefore to increase the employment rate there should be an increase in the inflation rate, which causes a rise in the price level of the economy. So a gradual inflation in the economy is good as it increases the employment rate and hence the output also increases.

Objective of the Study

This paper seeks to study the relationship between Monetary Policy and Economic Growth. This paper also tries to understand the dynamics between monetary policy and money supply and how it affects the interest rate, savings rate and investment.

Literature review

According to Jhingan (2007) Monetary Policy by influencing the cost and availability of credit and also controls inflation and maintain equilibrium in the balance of payments. It plays a very important role in the economic growth of an underdeveloped country. It aims at controlling the credit for controlling inflation and stabilizing the price level, stabilize the exchange rate, achieve balance of payment equilibrium and promote economic development.

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Sikdar(2007) says that the ultimate target of the monetary authority is to control the major macroeconomic variables like unemployment, inflation and growth of real GDP. The RBI chooses to control the intermediate targets which are the monetary aggregates (M1, M2, M3) where

M1= CU+D

M2=M1 + savings deposits with post office savings bank

M3 = M1 + net time deposits of banks (interbank deposits are netted out)

Here CU is the total currency held by the public in the form of coins and notes, D is the demand deposits of the public in banks (interbank deposits are not included) or the interest rate. Regulating the money supply helps in regulating the rate of inflation and changing the structure of interest rates affects growth by altering the incentives for investment in financial assets and physical capital.

According to Ahuja (2007) Monetary Policy deals with the distribution of credit between uses and users and also with the interest rate for both lending and borrowing. It regulates the money supply, the cost and availability of credit in the economy. It plays a major role in promoting economic growth in developing countries. Monetary Policy needs to ensure price stability along with encouraging economic growth, as excessive inflation will have an adverse distribution effect and hinders economic development also.

According to Chaturvedi (2005) Monetary Policy refers to the policy measures pursued by the Reserve Bank of India to control, administer and regulate the money supply, foreign exchange rates, the cost and availability of credit in the country. Through Monetary Policy the central bank can alter the money supply and the rate of interest so that the monetary system is fully benefitted. Monetary Policy can provide sufficient financial resources for economic development as well as help the developing countries to expand their output levels, employment and income.

Research methodology

The data collected on the various aspects of Monetary Policy are mostly in textual form. Sections have been made to give a clear understanding of the work done.

Instruments of monetary policy

(a) Bank Rate: It is the rate at which the Central Bank lends short term loans to the Commercial banks against securities. When the interest rate is low, the borrowing power of the commercial banks increases, which encourages them to borrow reserves and with which they can advance loans. This improves the credit creation of the commercial banks. Whereas when the interest rate is high it discourages borrowing and is followed by rise in the rate that the commercial banks charge on their loans. This decreases the demand for credit by household and firms.

(b) Cash Reserve Ratio: It is the percentage of money that the commercial bank has to deposit with the Reserve Bank of India. When there is an increase in CRR the loan able funds available with the commercial banks reduces and the money supply contracts. The banks might also have to recall the loans to meet the higher reserve requirements. The opposite happens when CRR is reduced. The ability of the banks to create deposit money increases. CRR variation is not very frequently used, as it will destabilize the system.

(c) Open Market Operations: It deals with the purchase and sale of securities in the open market. An open market sale contracts the money supply and a purchase increases the money supply in the economy.

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Targets of monetary policy

(a) High and Stable Employment: One of the major goals of all governments is to reduce t unemployment rate and bring it down to a tolerable level as it causes economic imbalances a sufferings, along with loss of potential output.

(b) Economic Growth: Another important goal of Monetary Policy is to achieve a stable economi growth, which means there should be a steady rise in the National Income, which is essential fo improving the standard of living of a country. For new investments and infrastructural developmer adequate credit should b available along with proper management of monetary and credit policies t assist in the process of growth and development of the economy.

(c) Price Stability: Inflation has an adverse effect on the social welfare of a country. As the price level o the economy rises it hits hard on the people especially the poorer sections of the society. Therefore price stability is a major goal of all economic policies.

(d) Financial Stability: It is very important that the funds from the savers reach the business firms with profitable investment opportunities. Financial crisis can cause major disturbances in the economy leading to low economic welfare. A stable financial system is required to absorb economic shocks, which is an important goal of the Central Bank.

(e) Stability in the foreign Exchange Market: Another important goal of the RBI is to maintain a stable foreign exchange rate, i.e., to avoid sharp fluctuations in the value of rupee. Appreciation and Depreciation of a currency makes it difficult for traders to plan and trade in the foreign exchange market.

Impact of monetary policy on gdp

Gross Domestic Product (GDP) is defined as the value of all the goods and services produced in an economy in a given time period, such as a year.

$$GDP = C + I + G + (NX)$$

C- Consumption

I - Investment

G - Government Expenditure

(NX)- Net Exports

- Consumption means spending by households on goods and services. Good include durable goods like automobiles, appliances and nondurable goods like food and clothing. Services like medical care and haircuts are intangible items.

- Investment is purchasing goods that can be used in future to produce more good and services. It includes purchases of capital equipment and inventories.

- Government Expenditure is spending by the local, state and central governments on goods and services. It also includes the wages of government workers and spending on public works. Transfer payments which is a Social Security Benefit is not included in the calculation of GDP as the payment is not made in exchange for a currently produced good or service.

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- Net Exports is equal to exports, that is, purchases of domestically produced goods by foreigners minus the imports, that is, purchases of foreign goods. The term net refers to the fact that imports are subtracted from exports.

There are two types of GDP:

(a) Nominal GDP: It measures the value of the goods of a given time period with the prices of the same time period.

(b) Real GDP: It measures the changes in the physical output in the economy of different time periods by valuing all the goods produced in the year at constant prices. By calculating real GDP, i.e., by multiplying today's physical output by prices of 2000(for example) is the worth of today's goods if it was sold at the prices of 2000.

By calculating Nominal GDP we get to know the value of the goods produced in 2018 at the market price that prevailed in 2018. Nominal GDP changes with the changes in output and prices, therefore it cannot give a correct measure of the growth of the economy. Hence Real GDP is preferred to Nominal GDP to compare the output and growth of the economy in different years.

According to the Survey of Professional Forecasters on Macroeconomic Indicators (2018) - The GDP is likely to grow at 7.4 per cent in 2018-19 from 6.7 per cent in 2017-18 and it might further increase by 20 basis points(bps) in 2019-20 by the support from private consumption and investment.

The Real Gross Value added is expected to grow by 7.4 per cent in 2019-20 with the support of industry and service sectors.

Monetary policy and money supply

Money supply is defined as the total stock of money along with other liquid instruments circulating in an economy at a particular point of time. It consists of paper currency, coins, demand deposits, cheques and demand drafts. Analysis of the money supply in an economy is important to frame policies and control the interest rates, which leads to increasing or decreasing the amount of money circulating in the economy. An increase in the money supply lowers the rate of interest, which induces investment and more money is available with the consumers, thus increasing the consumer demand which ultimately results in economic growth. Increase in spending leads to increase in demand for goods and services therefore increasing the production, that is, the supply in the economy increases resulting in increased economic activity.

The opposite is the case when there is decrease in the supply of money. The interest rates increases resulting in decline in investment and less money in the hands of the public. This reduces the consumer spending and the demand for goods and services falls reducing the production in the economy and the economic activity declines.

The monetary authority controls the money supply in the economy through tools of Monetary Policy. Monetary Policy can be expansionary or contractionary. Expansionary Monetary Policy means that there is increase in money supply with reduced rate of interest. Contractionary Monetary Policy means decrease in the money supply with high rate of interest.

Role of monetary policy in economic growth

Economic growth is the expansion in production and capital stock in the economy with increase in the real national output or income. Accelerating the rate of savings and investment in the economy can fasten the process of economic growth.

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(a) Monetary Policy and Savings: Certain policies can be undertaken to raise the aggregate rate saving. For instance, a high real interest rate (nominal rate of interest minus rate of inflation) pol induces savings so that capital accumulation is speeded up. This argument is based on assumption that savings are positively elastic or sensitive to interest rates. Therefore in a developi country like India to induce more savings a relatively higher interest rate policy should be adopted that large sources are made available for investment in fixed capital stock. To promote econom growth in a developing country the Monetary Policy should aim at raising the rate of saving and should ensure a suitable real rate of interest to provide incentives for saving.

(b) Monetary Policy and Investment: Monetary Policy plays a very important role in increasing the lev of investment by making the available savings and resources mobilized by banks for purposes investment and production. The bank fulfills this duty by offering bank credit for investment business. When the money supply in the economy increases the interest rate falls, which induc investment. Early economists believed that the effect of money supply and rate of interest c investment is not very vigorous. The investment on fixed capital is interest inelastic. But accordir to credit-availability effect on investment, an increase in the supply of money enlarges th availability of bank credit for investment purposes, raising the level of investment in the economy.

Limitations of monetary policy

We have established that Monetary Policy is an important tool in the growth of an economy, especially in developing country like India. The policies are flexible and can easily be applied according to the changin market conditions. These policies can easily be altered, adopted or reversed in a short period of time and i free from political pressures. However there are few limitations or challenges of the Monetary Policy due t the following factors:

(a) Underdeveloped Capital Market: A systematic and an organized money and capital market i required to effectively implement the various credit control measures.

(b) Unorganized Sector: The central Bank fails to influence the market rate of interest by changing th bank rate as a huge part of the financial resources come from the unorganized sector.

(c) Banking habits of the People: In developing countries, most of the people are unaware of the importance of the banking system due to illiteracy, social causes, etc. People use money for most of the transactions and therefore bank fails to achieve the desired effects of business activities through credit control measures.

(d) Lack of Cooperation by Banks: Commercial banks play an important role in achieving the goals of Monetary Policy. In the developing countries, the commercial banks do not provide the required assistance to the central bank for implementing the Monetary Policy. For the Monetary Policy to be successful both should work together with proper coordination.

(e) Large number of Non-bank Financial Intermediaries (NBFI): The NBFI's operate in large scale in the developing countries but they are not under the control of the monetary authority, which limits the effectiveness of Monetary Policy in such countries.

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Conclusion

The biggest challenge of the RBI's Monetary Policy is to keep the money supply within reasonable limits without affecting the price level by making the necessary changes in the instruments of Monetary Policy. Failure of Monetary Policy can be very harmful for the economy. The adverse affects include high unemployment rate, hyperinflation, stagflation, recession, decrease in import and export and can even cause total monetary collapse. For a stable economic growth of an economy, efficient management of money supply and a suitable Monetary Policy is very important. Monetary Policy should mainly focus on improving the system efficiency rather than increasing liquidity into the system as a means of decreasing the interest rates.

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KALPA SAMEEKSHA

18th & 19th Dec 2018

This is to certify that Prof. / ~~Dr.~~ / ~~Mr.~~ / ~~Ms.~~ Rutuja Purohit

of Christ college, Pune.

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~~He~~ / She has presented / published a paper entitled A study on recent trends in online shopping and its impact on consumer behaviour.

Dr. Janardan Pawar	Prof. Chetan Wakalkar	Dr. Tarita Shankar
Principal In-Charge	Group Director - IGI & Chief Patron	Chairperson - IGI & Chief Patron

A STUDY ON RECENT TRENDS IN ONLINE SHOPPING AND ITS IMPACT ON CONSUMER BEHAVIOUR

ISSN : 2394-2517

Prof. Deepa Sujith
Asst. Professor Department of Commerce & Management, Christ College Pune

Prof. Rutuja Purohit
Asst. Professor Department of Commerce & Management, Christ College Pune

ABSTRACT:

With the development of technology and increased users of internet, a dynamic and convenient channel for shopping is developed since 1995 in India. Online shopping is the process whereby consumers directly buy goods and services from the seller over internet without any intermediary service. It has seen a remarkable growth, transforming shopping experiences of people making it more reachable, time saving and efficient. Due to fierce competition in this field, marketers find out different innovative techniques to grab interest of the customer. As such there are many changing and challenging trends coming up in this area of shopping. This paved way for drastic change in Indian consumer's shopping pattern and many reasons are accountable for the same. This change in consumer behaviour has put forth a challenge for the survival and growth of E-Commerce industry. It becomes an absolute necessity to understand and analyse the factors behind consumer behaviour- apparent and noticeable responses of consumers- while buying. This study focuses on the recent trends in online shopping and its impact on consumer behaviour in India.

Keywords: Online shopping, emerging trends, consumer behaviour

INTRODUCTION

Ever since E-Commerce industry entered into Indian market it saw a vast growth with growing internet users in the country. Change in Indian life style, availability of wide range of products, endless selection, convenience and comfort of buying, time saving etc are some factors that created an increased desirability in consumers to go for it. At the early stage it was a channel with limited facilities to do shopping by placing order and giving cash on delivery. As time went on, innovations and changes in the style of marketers brought in various trends in this field. They tried identifying the needs of customers and took efforts to convert them into purchase order.

Online consumer behaviour is a complicated socio-technical phenomenon. Digital environment has a huge impact on consumers mind and buying behaviour. Cultural differences and biases towards online shopping are prevalent with online stores being present in many countries around the world. Each society's value system is different and in the current scenario, it is the company's prerequisite to provide an atmosphere that gives confidence to the buyer which in turn aids them to take hold of the market seizing opportunities

Objective-

1. *To study the trends in online shopping*
2. *To analyse the impact of recent trends in the consumers' behaviour*

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This paper specifically aims to analyze the recent trends in Information technology and its resultant effects on employee behavior and the work culture.

Keywords: employee behavior, work culture, Information technology, employee experience

Introduction

The 21st century has been defined by application of and advancement in Information technology. Information technology has become an integral part of our daily life. According to Information Technology Association of America, Information technology is defined as "the study, design, development, application, implementation, support or management of computer based information systems."

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Objectives of the study:

- To study the various recent trends in Information Technology
- To study the impact of the changes in Information technology on employee behavior
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Methodology: The research Paper is based on the secondary data published time to time through reference books, journals, book reviews, newspapers and various websites and government schemes declared time to time. The information has been analysed and presented systematically to draw certain conclusions on the basis of the study.

Recent trends in IT

The key factor as we move into 2018 is going to be IT sustainability i.e the process of ensuring information technology development that meet the requirements of the current market but also safeguard the development potential of future markets.

1. Artificial Intelligence and Advanced Machine Learning: Applied AI and machine learning are composed of many technologies and techniques(such as deep learning,neural networks and neural networks and natural language processing. These technologies, which are different from traditional algorithms and programs ,make the machines intelligent.

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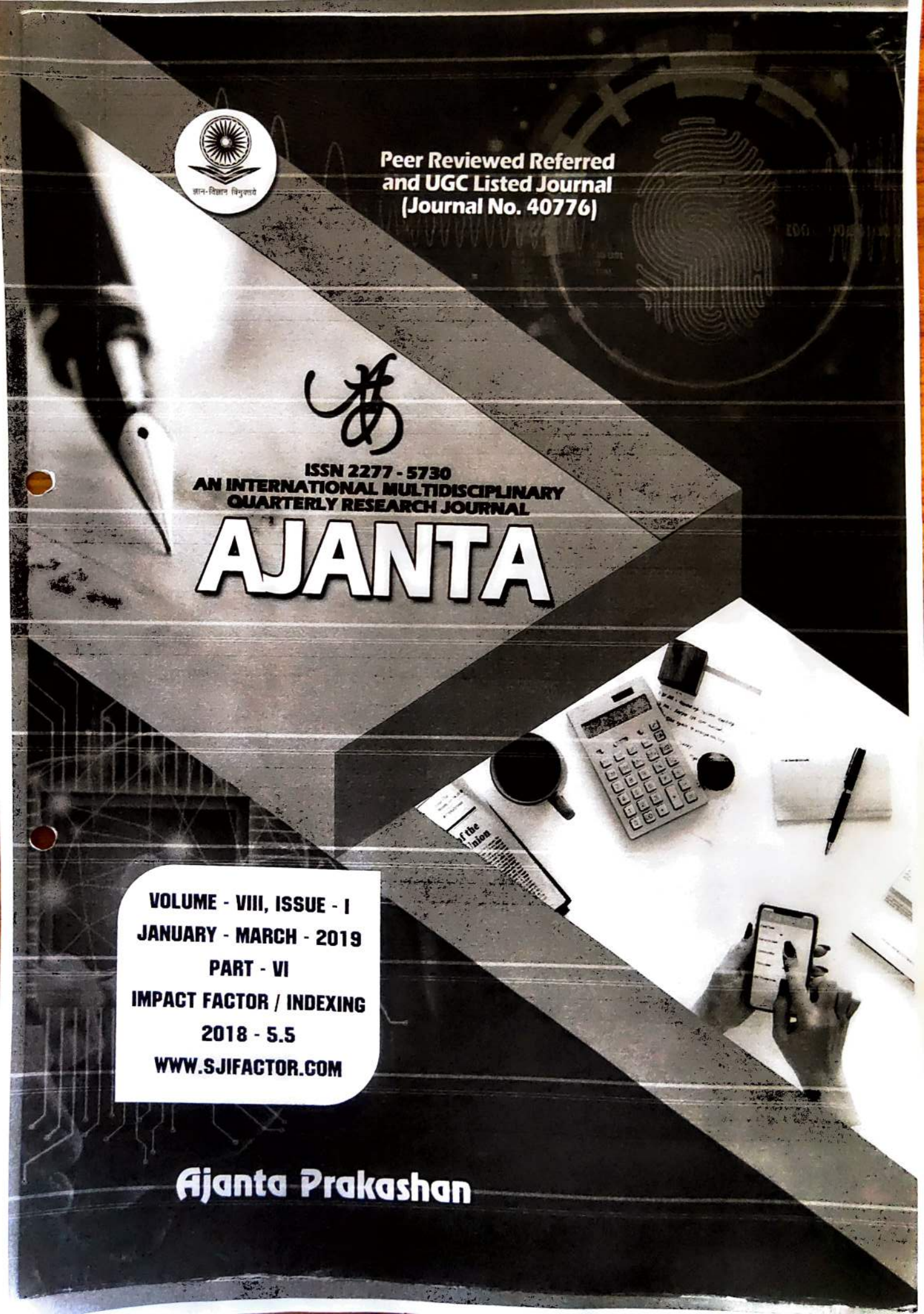

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22. An Analysis of Presence of Abilene Paradox in Decision Making

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Abstract

Decision making is the main function of management which nurtures the Planning function. It is a cognitive process which leads to a final choice, sometimes may or may not appropriate. In many cases we often land up in wrong decisions due to various reasons such as lack of knowledge, non willingness for change, overconfidence etc. In this paper one such reason is highlighted to create awareness among the readers that we all are by unknowingly influenced by this psychological issue.

In Sociology, Man is termed as Social Animal, because he urges to go along with the society to survive in this competitive world. This psychology often misguides our opinions and accepts what all other wants. But there may be a situation that we human being goes with such decision which none of us want that. This could be the worst situation in the not only in the corporate world but also in our personal life, if this happens in decision making process. This paper focus on how does such a situation arise in the corporate world and what could be the possible remedies to overcome it. This is well explained with the help of a theory **"A TRIP TO ABILENE PARADOX".** The data has been collected from the corporate leaders, educational leaders as well as the employees who regularly conduct and attend the meetings.

Keywords: A TRIP TO ABILENE PARADOX, corporate world, Decision Making Process.

Introduction

Today's business world is facing a great change and unpredictable future with a plenty of options. To overcome this problem, a leader has to work with a different mindset whereas to focus on possibility than on probability. He has to take the decisions fast from the huge amount of information where almost everything is more interconnected.

The decision making is a team process. It cannot be implemented and get the success without people. Even though man is termed as a social animal in sociology, it is quiet difficult to read the Hippocratic human mind. A decision taken by a single minded, over and self confident leader, who is supercharged by success and unchallenged by his team definitely, taking a trip to

Abilene Paradox. Thus Abilene paradox is a situation where the people are fear to speak freely, which pulls down an effective evaluation of proper alternatives. Thereafter the potential problems are not identified.

The Abilene Paradox was coined by Jerry B. Harvey, Professor Emeritus of Management at The George Washington University and author of the "The Abilene Paradox and other Meditation of Management.

Literature Review

Kathryn J. Deiss, (June 1999), Revisiting the Abilene Paradox: Is Management of Agreement Still an Issue? Issue 8, discusses some of the variables that affect group decision-making and suggests several facilitation tools. It concludes with how to avoid true loneliness by giving our thoughts and opinions voice and trusting the group with which we are working.

Svenson (1979) Claims that most of the decision problems were solved without a complete search of information. He explained that human decision-making cannot studied on final decisions. The perceptual, emotional and cognitive process involve in the decision making should be studied. He concluded by saying that decision making theory should be based on structural analysis data.

WILLIAM BOULDING (1194) a paper on strategic decision making of mangers tried to fill a gap between normative and actual decision making. He had distinguished strategic managerial decision making from other choices. They proposed a conceptual model of how managers deal with the decision making process. He concluded that the managers overreact to competitive actions and they are able to predict competitor's action than the customer response.

Research Gap

From the above review of literature, the identified gap is that decision making can be effectively done by avoiding Abilene Paradox. This paradox can be identified by asking similar questions in different way to find out the real opinion of a person on a specific matter. The variations in answers could definitely helps to conclude the presence of Abilene Paradox in decision making process.

Objectives of the study

- To enlighten the concept of Abilene Paradox.
- To find out whether the Abilene Paradox affect the Decision making process.

Hypothesis of the Study

H0: Abilene Paradox Plays a Significant role in Decision Making.

H1: Abilene Paradox does not Plays a Significant role in Decision Making.

Research Methodology

The study used both primary and secondary sources. Primary data was collected by using a questionnaire schedule among 48 respondents from various fields such as corporate, educational leaders and employees from various organisations. The sample selection was Convenience Sampling. The questionnaire and structured interview was developed to assess their stress level. Secondary sources included books, journals, articles and reports from newspapers, weekly's, magazines, etc. The data collected from both the primary and secondary sources were quantified and analysed in qualitative terms. For testing of hypothesis **Percentile method** is used to find out the presence of Abilene Paradox in decision making,

Definition of Abilene Paradox: A circumstance where a group of individuals agree to a course of action based on the theory it is best for the group, despite going against the preferences of members of the group. This occurs when individuals feel their objections are not strong enough to support changing the minds of others in their group. Commonly referred to as "rocking the boat".

The Abilene Paradox is a term associated with the decision making. In an Abilene Paradox a group of people take a decision which is counter to the decisions of many in the group. This is a thinking disorder that each member in a group believes that their preferences are counter to the group and not to object. In other words the human beings are very reluctant to act contrary to the group even if they are right. This study connects Abilene Paradox with the decision making agreement and to find out some solutions to overcome a Trip to Abiliene.

Back Ground Study

One of the Chinese sayings goes like this: "If you wish to plan for a year, 'sow seeds'; "If you wish to plan for 10 years, 'plant trees; "If you wish to plan for a life-time, develop men".

This study deals with Decision Making which is one of the unique and challenging talents for any manager. Similarly the Abilene Paradox clearly gives a picture of importance of decision making.

The story behind the Abilene paradox is that in 1971 the Prof Harvey and his family had a trip from Coleman to Abilene, Texas, on a hot Sunday to eat at a restaurant. But while returning no one in the family had actually wanted to do it all they agreed it because the others wanted to go. Thus the Abilene Paradox born.

One more similar story which can explain the Abilene Paradox in our Indian scenario is- A couple of years back, a man wanted to take his family out for dinner. He asked his wife where we can go. Knowing that he like Indian food, she replied: "let's go to a Gujarati Thali Restaurant".

Both his son and daughter nodded in agreement. On return the son said" I wish dad had taken us to some Chinese restaurant - he loves Chinese food". "Or at least for the wonderful south Indian food" added his daughter. "Yes, even I would have loved to go Chinese restaurant", the father replied.

His wife looked surprised, but didn't we all unanimously agree to go to Gujarati thali Restaurant" She asked.

He said sheepishly" I didn't want you to feel bad". And both of our children nodded in agreement.

Here were four people who of their own volition collectively agreed to go there. This situation happens in our day to day life and in corporate world. This is Abilene Paradox. Prof Jerry Harvey calls it "The Inability to Manage Agreement".

Perhaps we all have gone through such incidents in our life. Apart from that so many such similar incidents we can find in our society. So many corporate and political decisions are prime examples of Abilene Paradox.

Various decisions which the famous dictator, Hitler took in his reign as chancellor like attack on Poland, mass execution of Jews, operation sea lion, extreme rightist view of Nazi were example of Abilene paradox. This proves any dictatorship or a bureaucratic decision is definitely leads to a trip to Abilene.

Abilene paradox occurs due to lack of confidence to stand on our own views. Leaders/ Managers should particular to take a note of this, and create a supportive environment where Abilene paradox cannot take place. It is absolutely a situation of the failure to achieve a shared understanding of a problem before deciding on a solution.

A case study on Nokia explains the impact of Abilene Paradox

Nokai's ill judged restructuring circa 2003, decision taken to overcome the rapid change in the mobile phone market led to its failure in the mobile phone market industry. Nokia was a product oriented company which focused on developing only one or two models in a year. In response to the rapid change in the industry, its management decided to produce 40 to 50 models in a year, which was a quite terrifying number. Company knew that it would be difficult to make so many models from a product focused structure. Therefore they reorganize the company into different divisions of team for creating standard components like cameras which could be mixed and matched into many new models in a year. This decision split Nokia's all leading mobile phones into three units- Multimedia, Enterprise and Phones. The architect of this decision was most successful CEO, who steered the company from 1992 crisis, JORMA OLLILA.

Earlier where Nokia producer of a very few standard, after this division from 1st Jan 2004, became a producer of Dilbert material. The reason behind this situation was, the reorganization encouraged a competition between the staff and resources. The company became completely bureaucratic. Thus Nokia's decision of Restructure was a trip to Abilene Paradox, because it was not whole heartedly supported by the management. This case study of Nokia Explains How the Abilene Paradox situation led it to the stage of Failure.

Most of the projects are initiated as a solution for a perceived problem. Many projects fail because of inefficient front end decisions. Front end decisions are the decisions which taken in the initial stage, when the initiative is an idea. These decisions are quite difficult because it has to be made on the basis of ambiguous and incomplete information. Therefore such decisions incorporate the honest views and opinions of all the stake holders of an organization. First step is to ensure that all stakeholders who are in the decision making process should have a common understanding of the project/ problem. i.e., what is to be done? Second step is that everyone should have a shared understanding of how the projects can be successful achieved. A genuine dialogue is the only method to prevent the pointless journey to places that organization can ill afford to go.

Data Analysis and Interpretation

To analyze the paradox, two or more similar questions are taken to identify the psychology of the respondents by verifying the answers. If there is a variation then there exists the paradox.

- **Do you expressly state your objectives for the organization and specific objectives for each major project or decision? / Are objectives clearly stated in advance of the meeting?**

The Likert scale from I -5 (1 indicates we rarely states our objectives and 5 indicates we always states our objectives. / 1 indicates Objectives are seldom set out in advance and 5 indicates objectives are always set out in advance)

Out of the data collected, there are variations in the answers such as for the first question likert scale < 3 Scored 89.58% where as for the second question likert scale < 3 scored 31.25%. Even for the second question 2% of the respondents claims that the objectives are not stated in advance with a likert scale of 1.

Even if the objectives are stated but not in advance it will not help the employees to give a proper suggestion on major decisions. Therefore majority of the people will be confused in giving their opinions.

- **Are agendas circulated to all members in advance of the meeting? / Does everyone come prepared and ready to make decisions?**

The Likert scale from I -5 (1 indicates Agendas are rarely circulated and 5 indicates we always circulated / 1 indicates Unprepared and 5 indicates Generally Prepared)

For the question No: 1, whether the agendas are circulated to all members in advance of the meeting, the likert scale 1 scored 14% and likert scale 5 scored 37%. Whereas for the question No 2: does everyone come prepared for ready to make decisions the Likert Scale 1 Scored 12.5% and the Likert Scale 5 Scored 10.46%.

From the responses given in the question No 1 we can figure out that there is high bureaucracy in the organization which can pose a threat to creativity and innovation in the system. And the responses from the question no 2 shows that there is high chance of Abilene paradox in the organization since without preparation one cannot propose his views confidently and there is chances of the person remaining silent or distracting the process.

- **Do you feel that each person expresses his or her candid judgment on which choices would best advance the team's purpose without any fear or hesitation? / Are all members fully exchanging views, taking responsibility for action items and follow-up?**

The Likert scale from I -5 (1 indicates don't feel that they are open up to speak and 5 indicates they are quiet open up to speak / 1 indicates people hold back and don't take ownership and 5 everyone offers ideas and takes action)

For the question No:1 whether all the people are open up for the discussion the Likert Scale < 3 scored 25% and > 3 Scored 60.41%. For the question No 2: whether people hold instead of taking ownership the Likert Scale < 3 Scored 18.75% and the Likert Scale > 5 Scored 50%.

From question No: 1 we can conclude that each person in the team has a voice or say in matters prevailing the decision and there is a healthy, friendly environment which will encourage discussion. But from question No: 2 we get the result that members of the team are not willing to take responsibility for a new ideas which further opens up that there should be a risk taking culture in team where members should be encourage to take calculated risk and give positive results in return.

- **Do you listen to each person's opinions or their views about a topic to withstand the real issue(s)? / Are differences of opinion suppressed, or is conflict effectively used?**

The Likert scale from I -5 (1 indicates Seldom listen to people and 5 indicates always listen to people / 1 indicates conflicts isn't very effectively used and 5 Conflict is effectively exploited for new ideas).

For the question No: 1 whether we seldom listen to people, the Likert Scale < 3 scored 8.33 and whether we always listen to people, the Likert Scale > 3 scored 87.5. And for the question No :2 whether the conflicts isn't are very effectively used, the Likert Scale < 3 scored 10.45% and whether the conflicts are effectively used, the Likert Scale > 3 scored 43.75% and for this the neutral response, ie., the Likert Scale = 3 scored 31.25%.

The responses of question No: 1 claims that each and every persons view is given equal attention. But from question No: 2 the paradox can be clearly felt where most of the respondents claims that opinion are suppressed.

- **Does the group generally make good decisions at our meetings? / Do you think all the options/alternatives for a project or decision get out on the table?**

The Likert scale from I -5 (1 indicates tend to make poor decisions and 5 indicates tend to make good decisions / 1 indicates rarely gets decision out on the table and 5 always gets decision out on the table).

For question No : 1, whether tend to make poor decision Likert Scale < 3 scored 20.83% and whether tend to make good decision the Likert Scale > 3 Scored 68.75%. And for the question No : 2 to find out whether all the decision rarely gets out on the table < 3 scored 20.83% and to find out whether all the decision always gets out on the table > 3 scored 58.33% and the neutral response the Likert Scale = 3 scored 20.83%.

From question No: 1 it can be seen that all meeting lead to a decision which everyone feels are good. But a marginal section in question No: 2 say that all alternatives are not heard which proves that decision making is not a fully group activities but are dominated by strong individuals or characters in meeting.

Conclusion

This study proves that Abilene paradox is a significant factor in decision making. To achieve an effective decision making process we have to counter Abilene paradox. The Abilene Paradox is a symptom of a person who do not have the confidence, skill and security to voice their own needs in a group setting. All the organizations have to take decisions and make actions every day. Basically these decisions and actions are based on wrong agreement within the group. An agreement occurs when all the key stakeholders accept the decision and support it, even

though the final decision may not be the first preference of each team member. In other words, reaching to a general agreement is not simply voting. For that we need to raise members and group consciousness about the problem. Therefore it is necessary to build a strong dialogue, advocacy skills, inquiry and should know how to use appropriately to the situation. Thus everyone in a team should avoid a true loneliness by contributing thoughts and opinions as well as trusting the group where we are working.

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